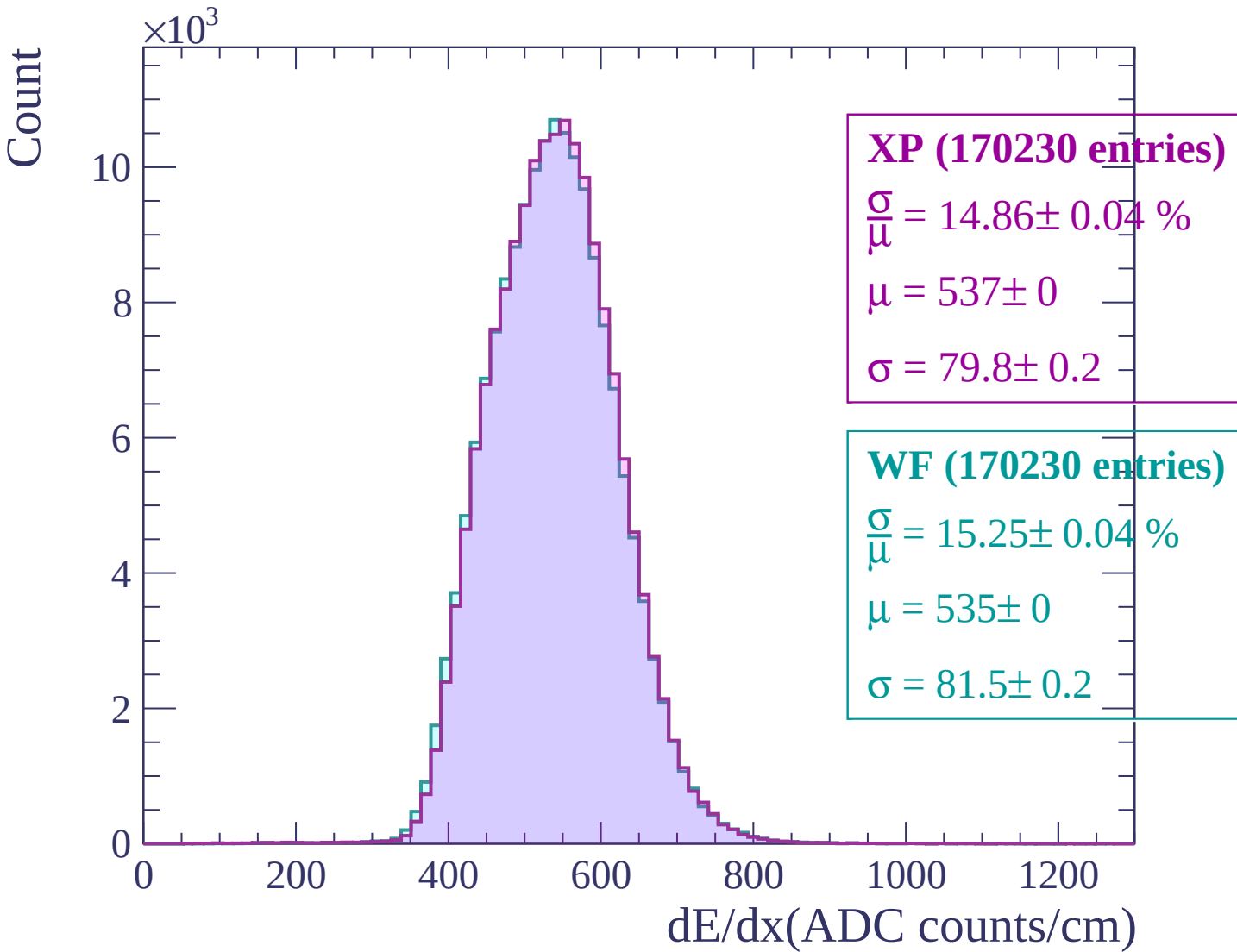
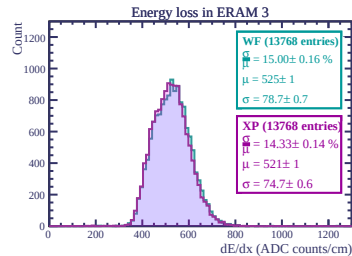
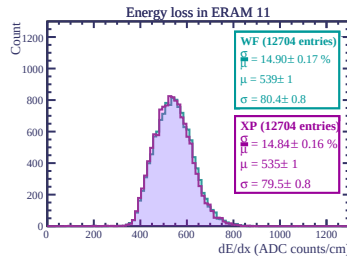
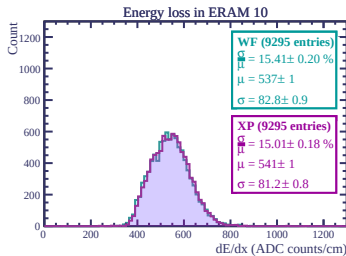
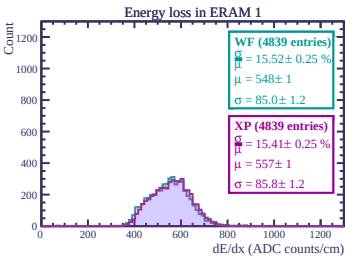
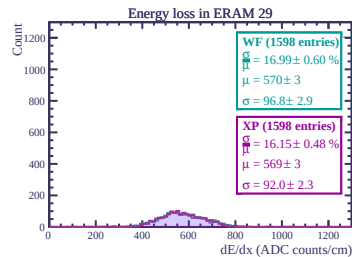
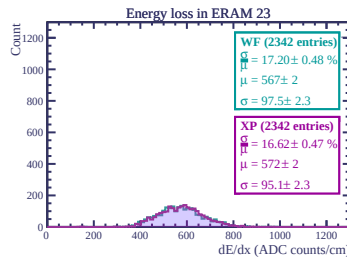
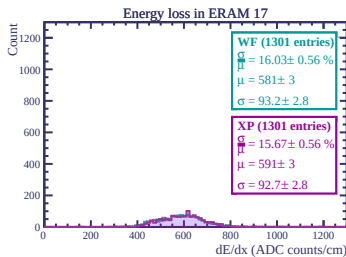
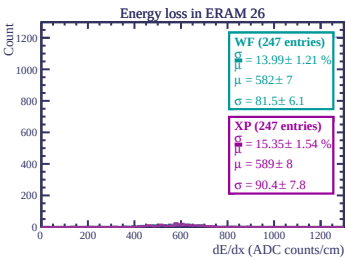
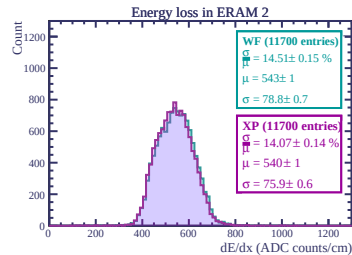
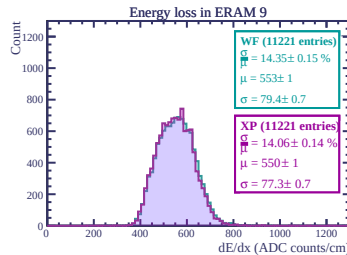
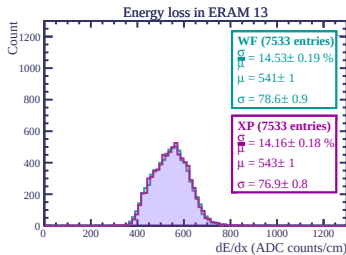
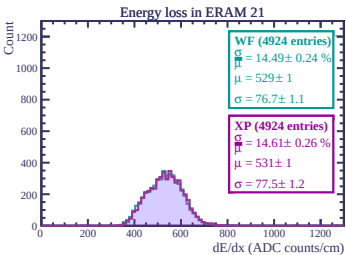
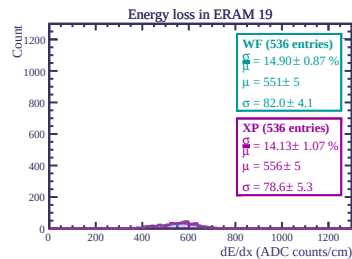
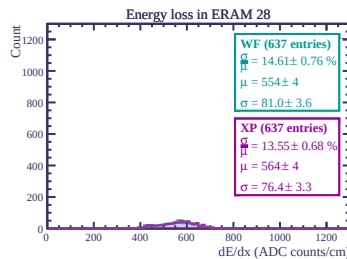
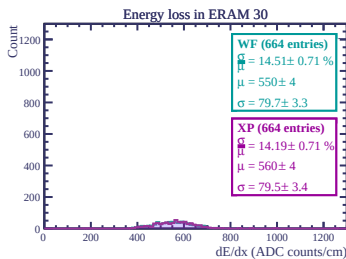
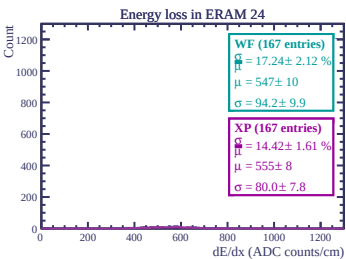
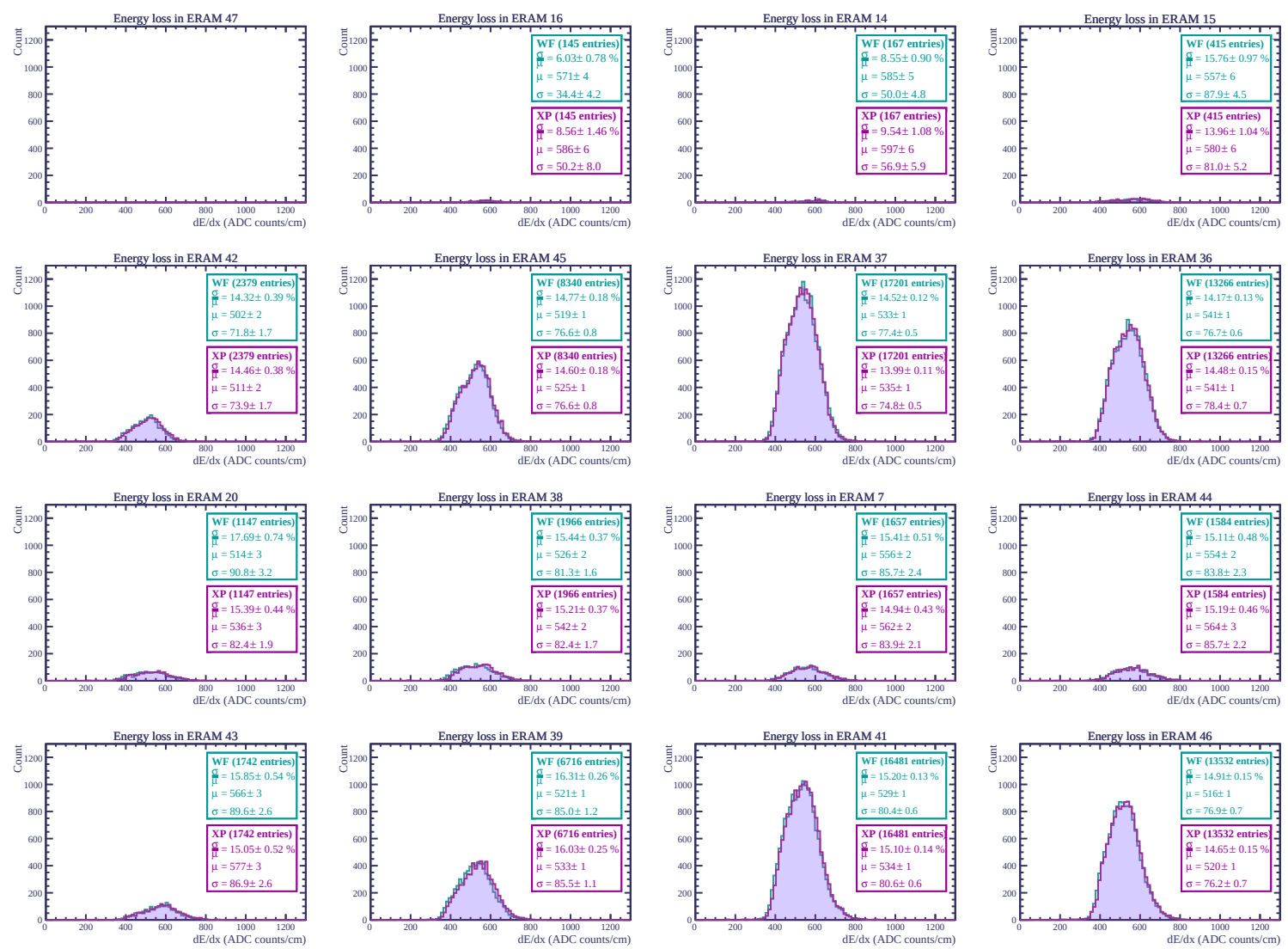
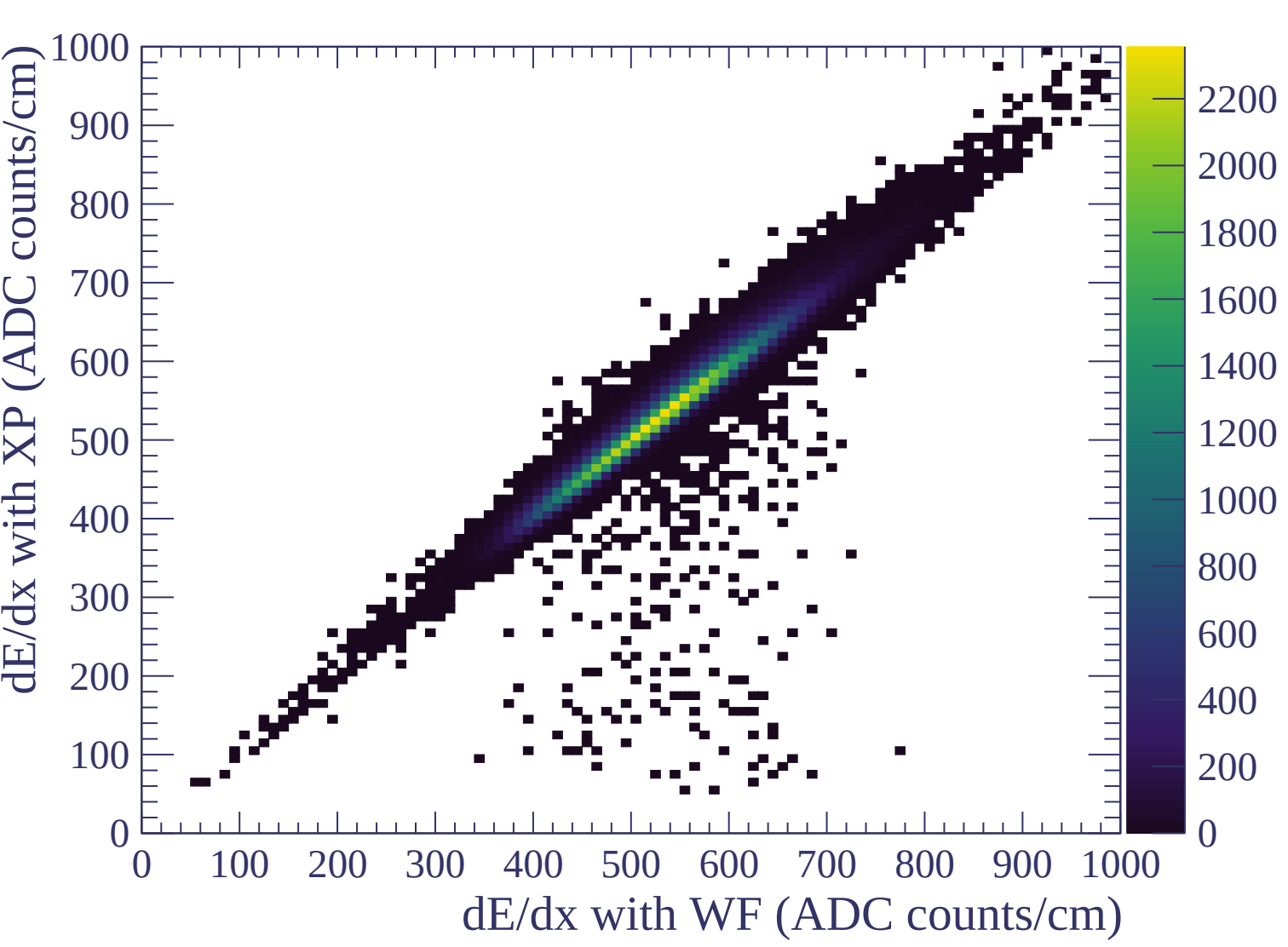
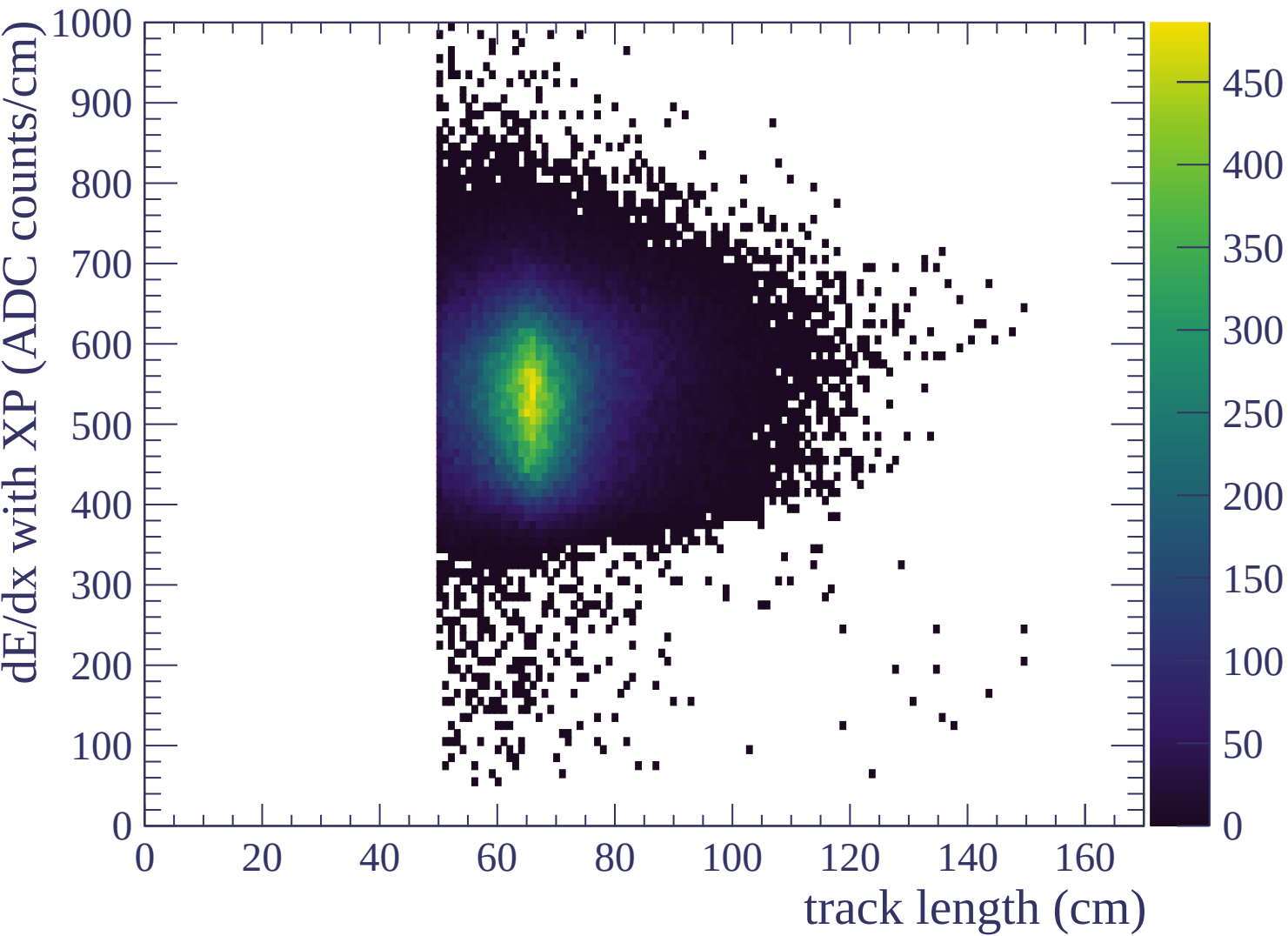


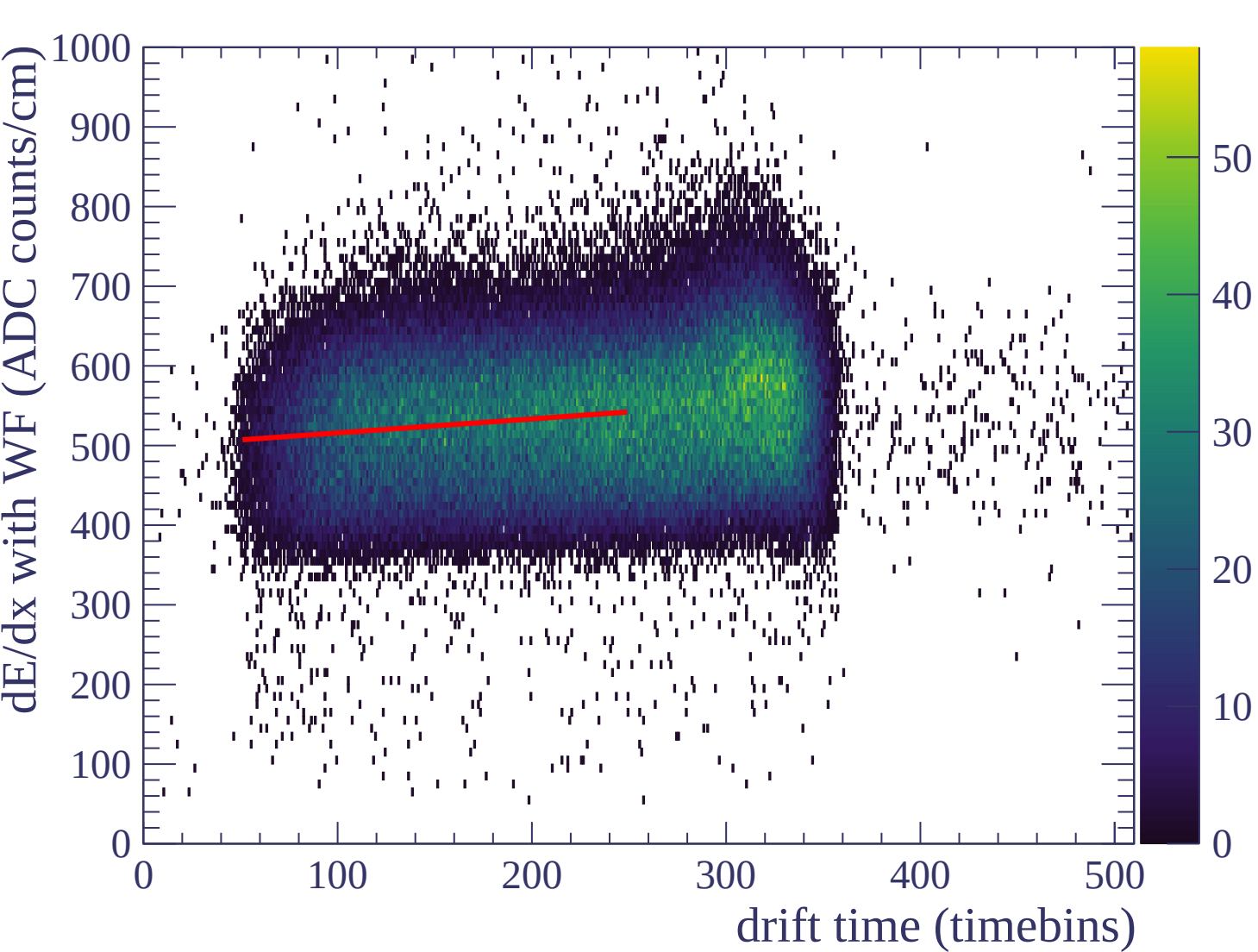


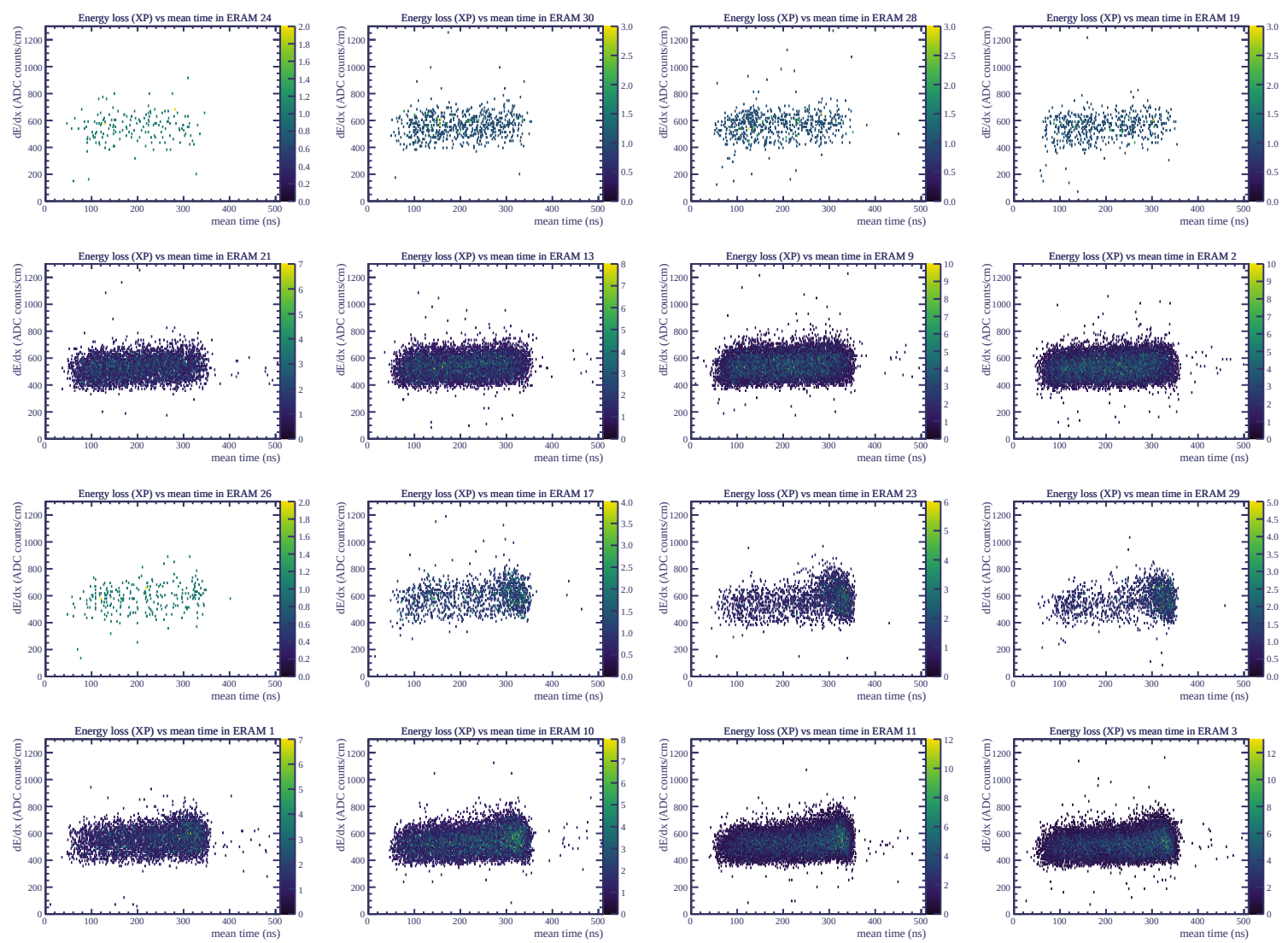


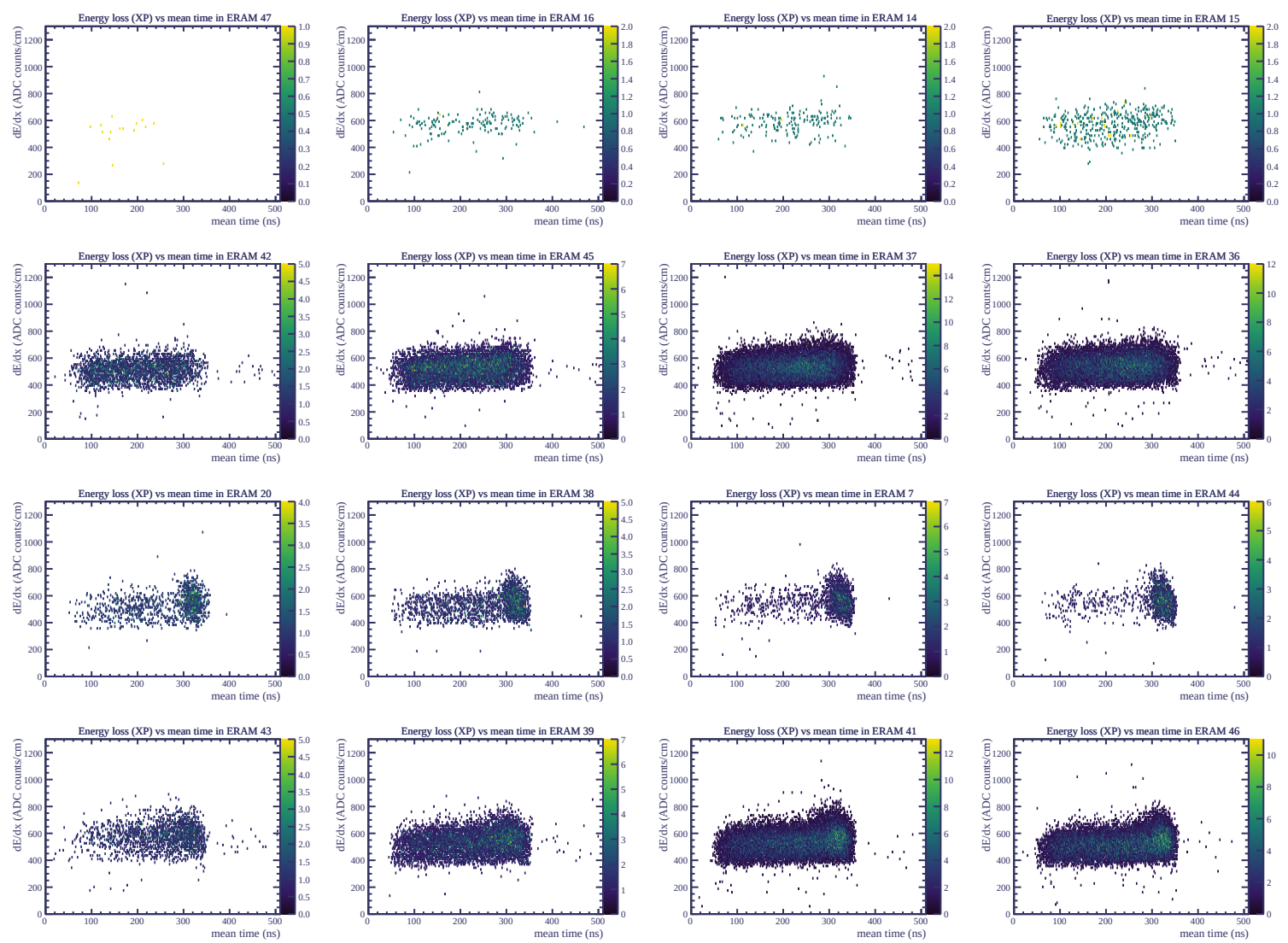




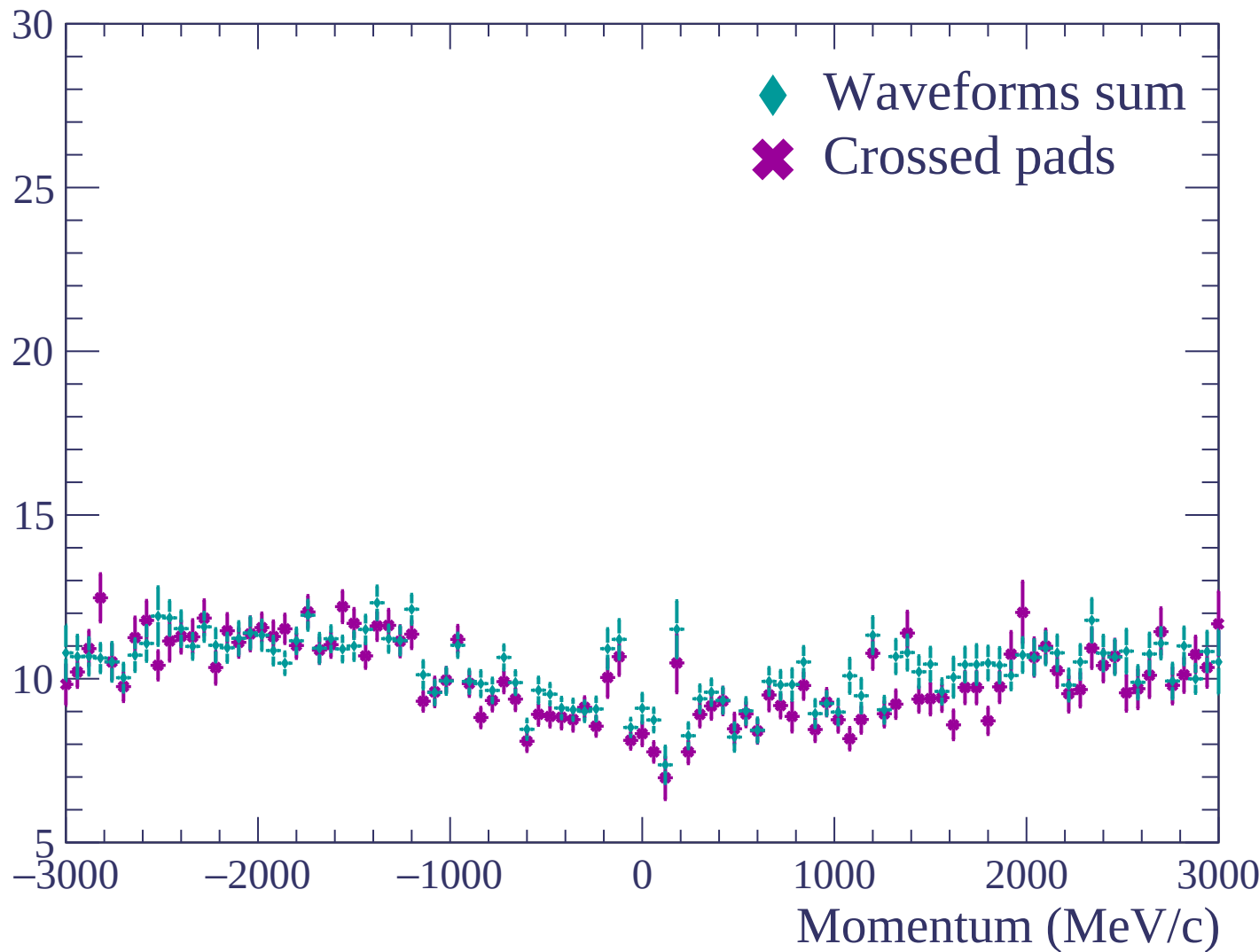


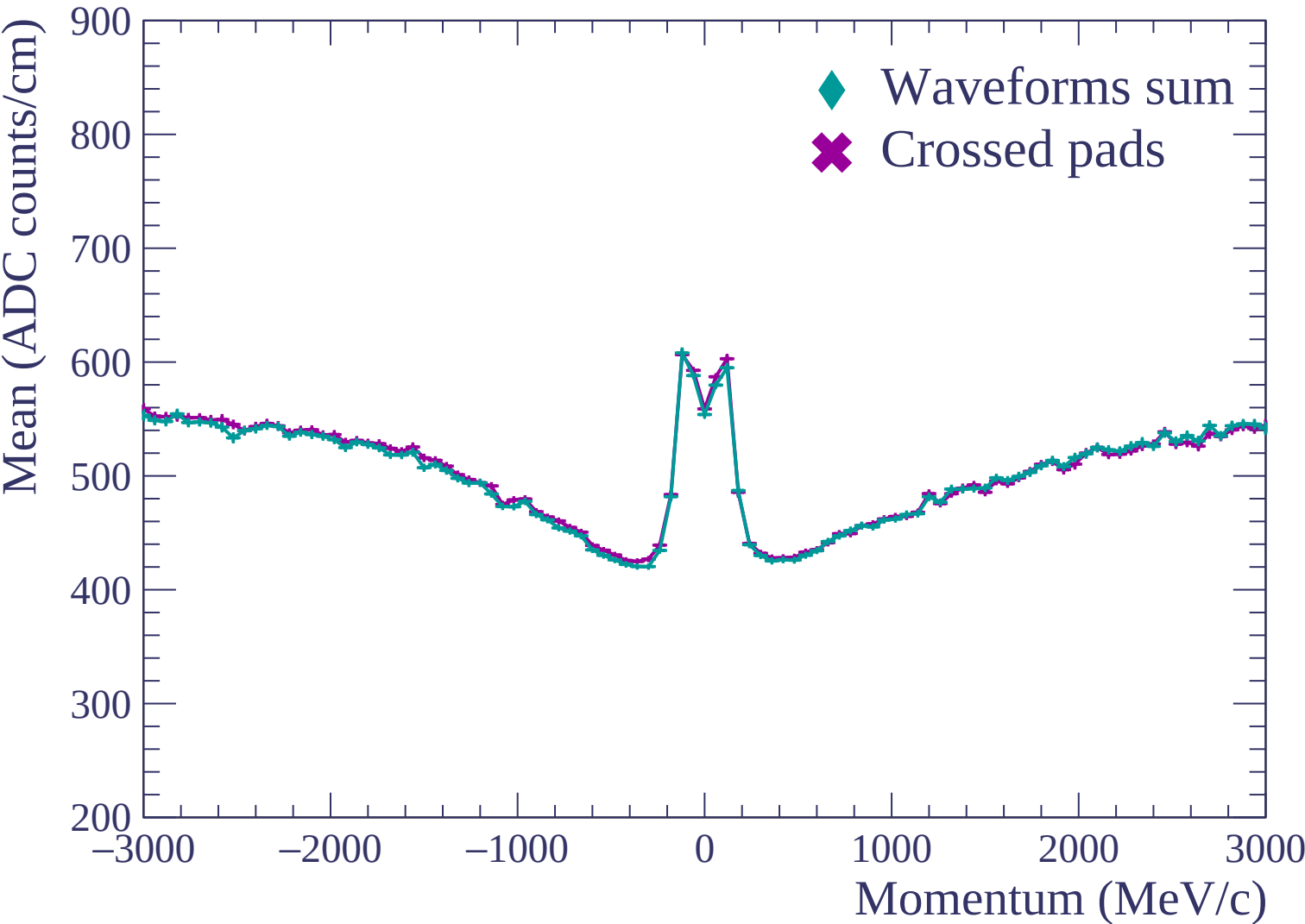


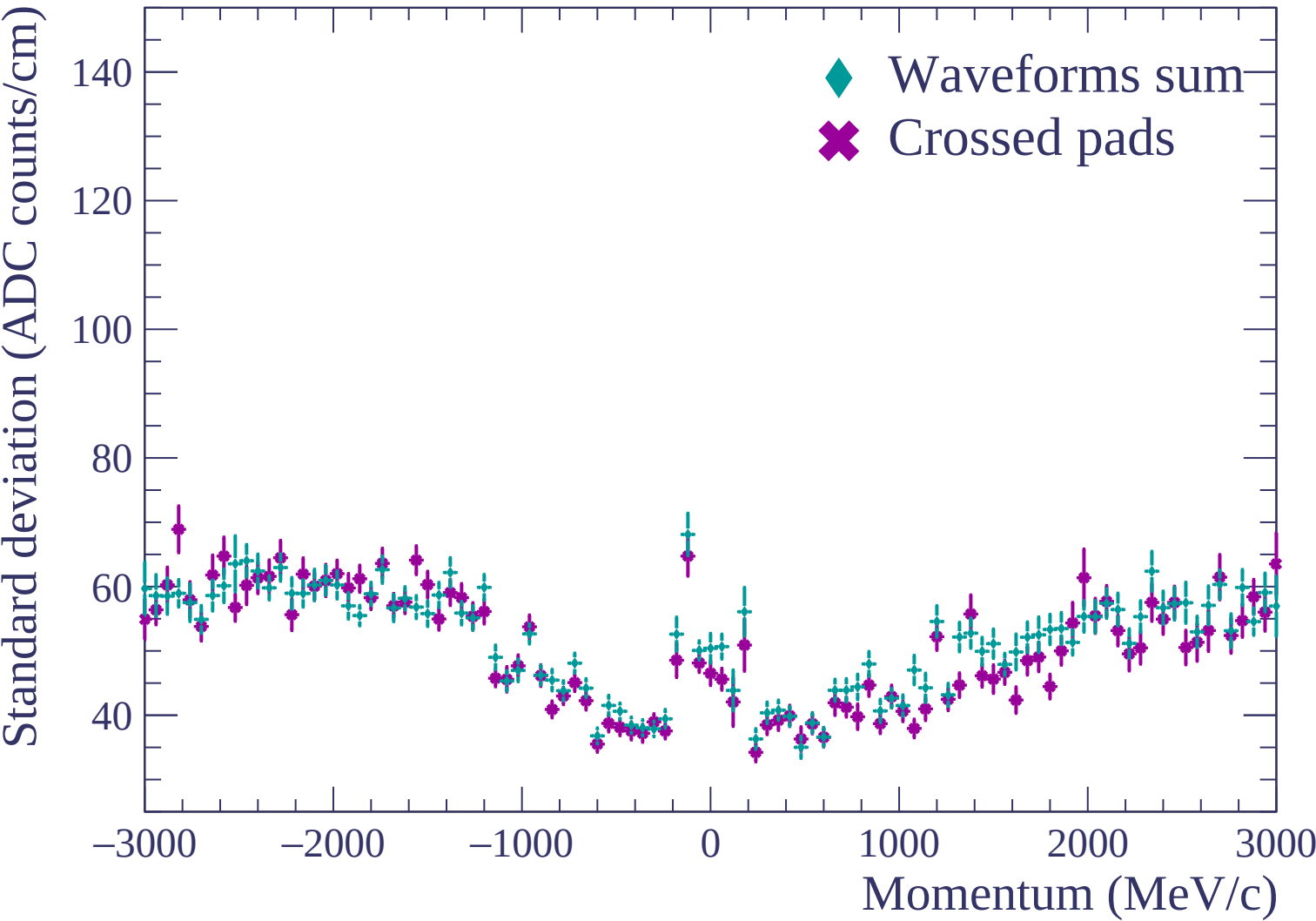


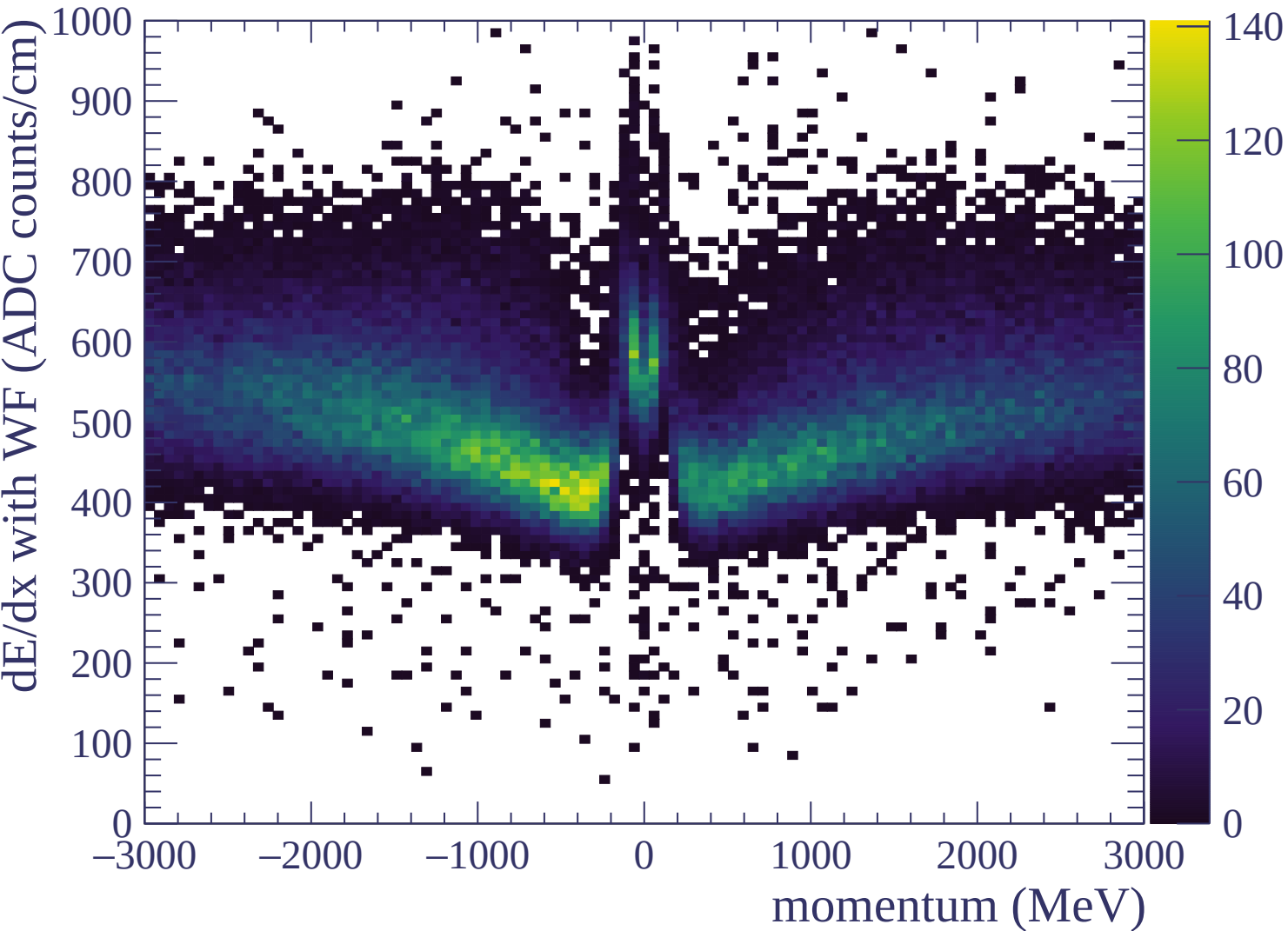


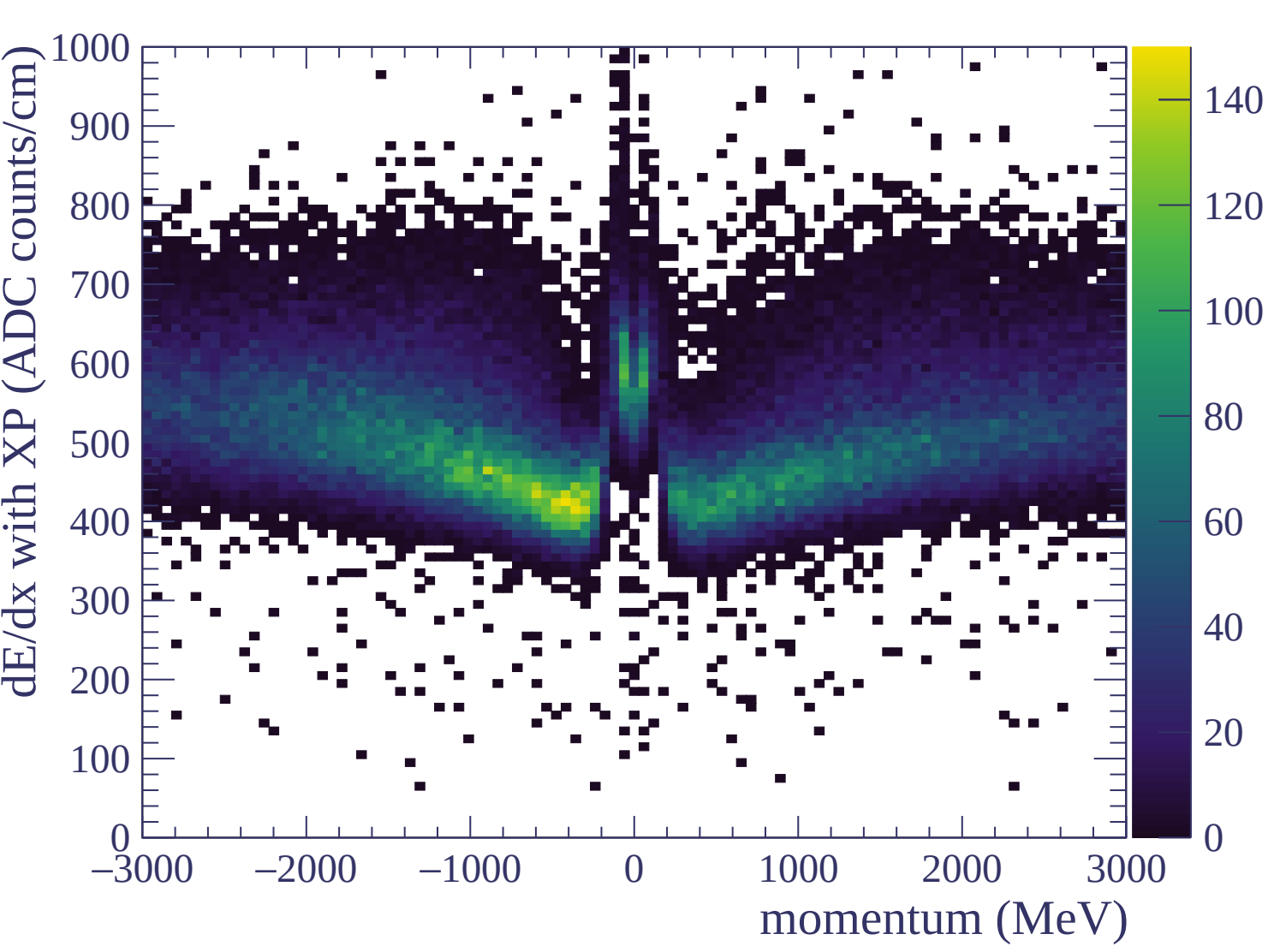
Resolution (%)

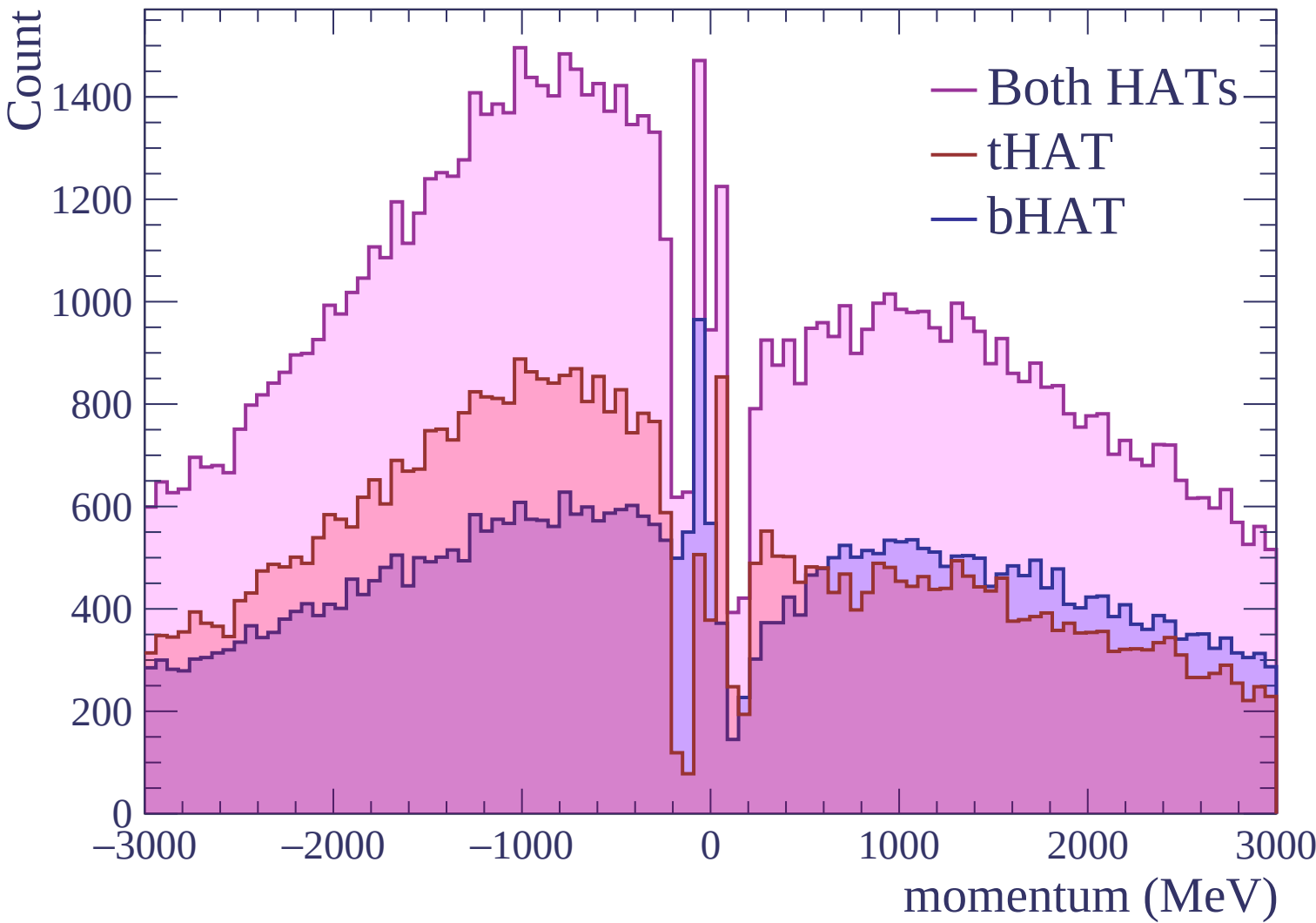






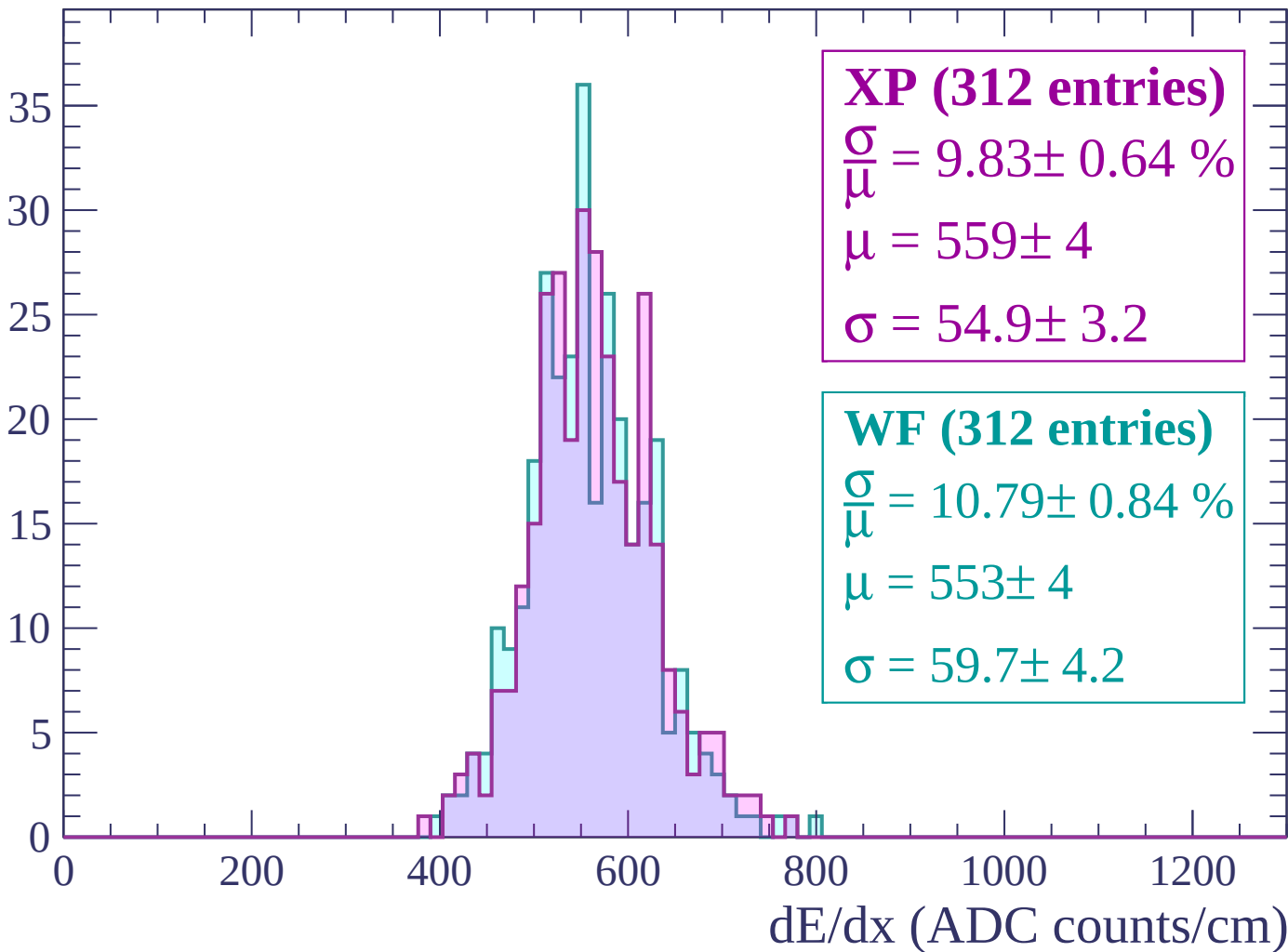






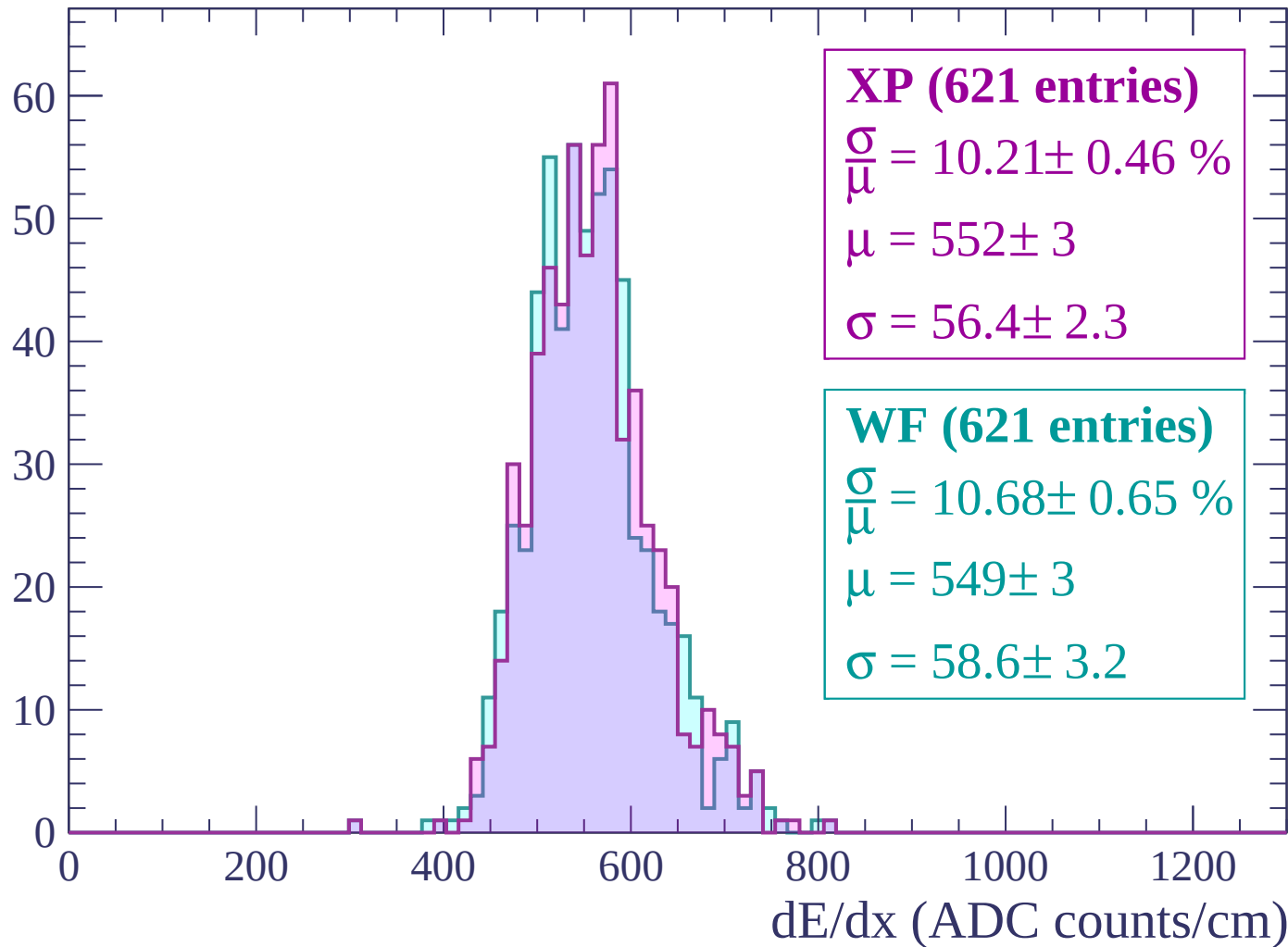
Energy loss | $-3000 < p < -2940$

Count



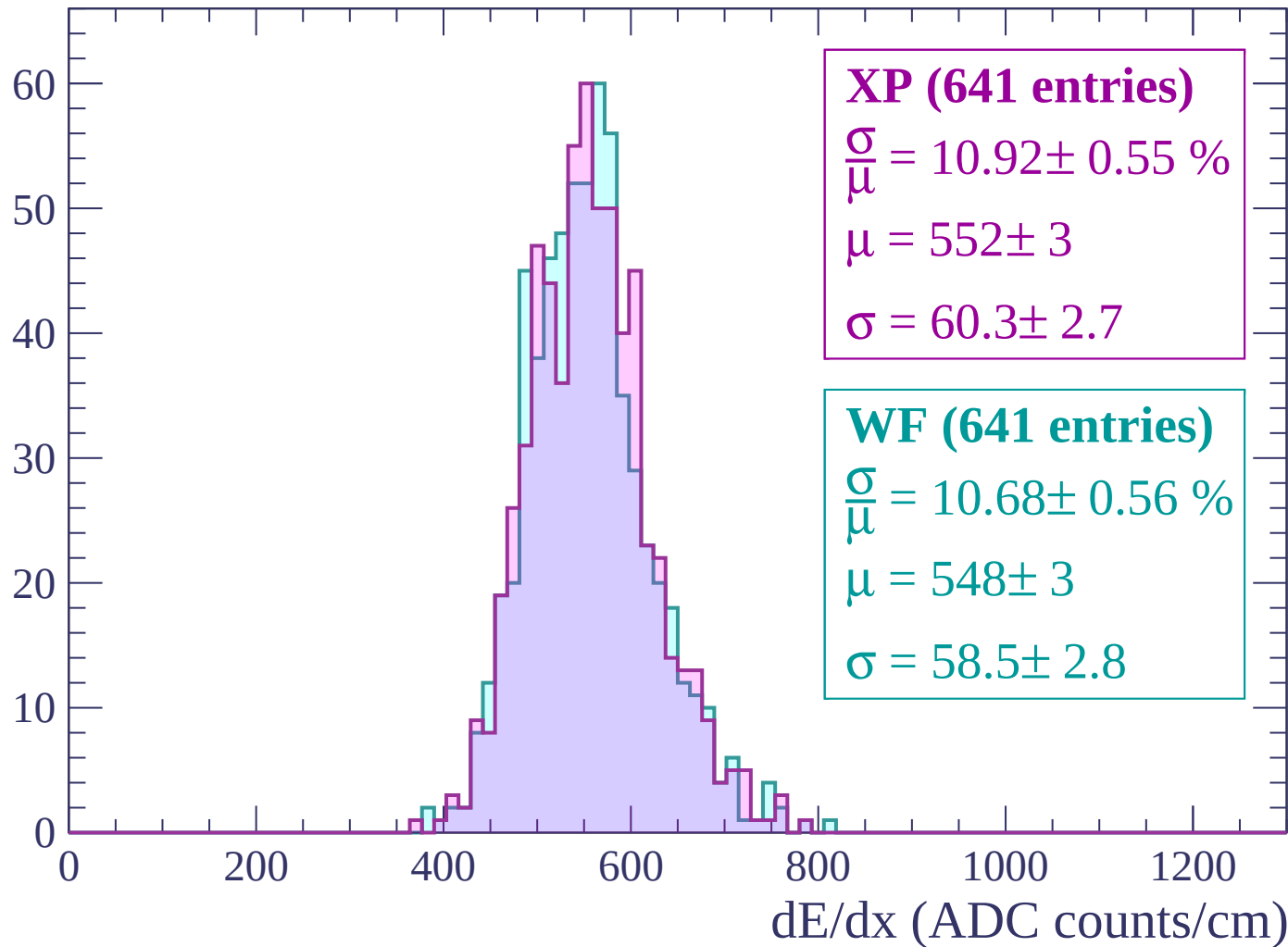
Energy loss | $-2940 < p < -2880$

Count



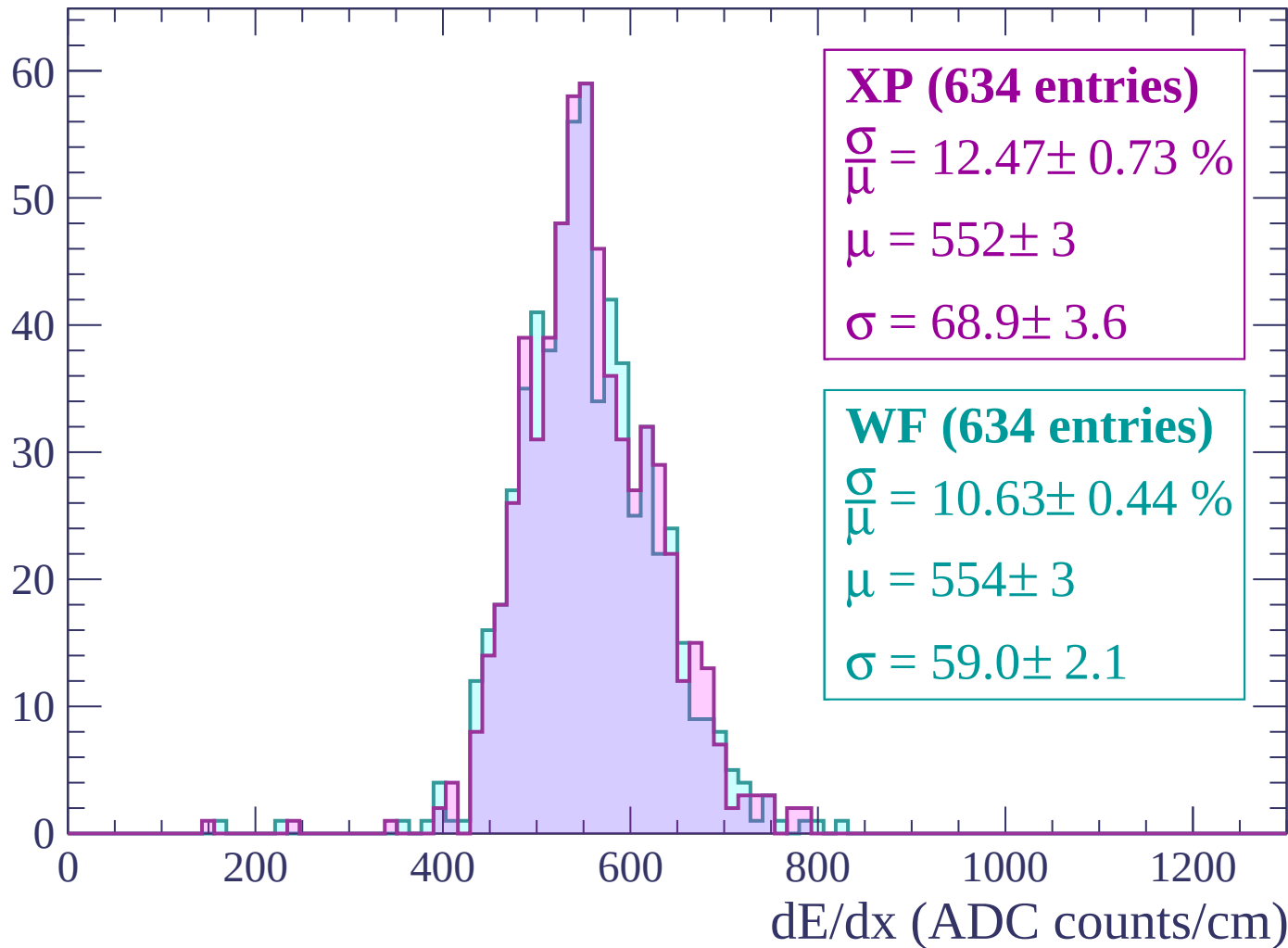
Energy loss | $-2880 < p < -2820$

Count



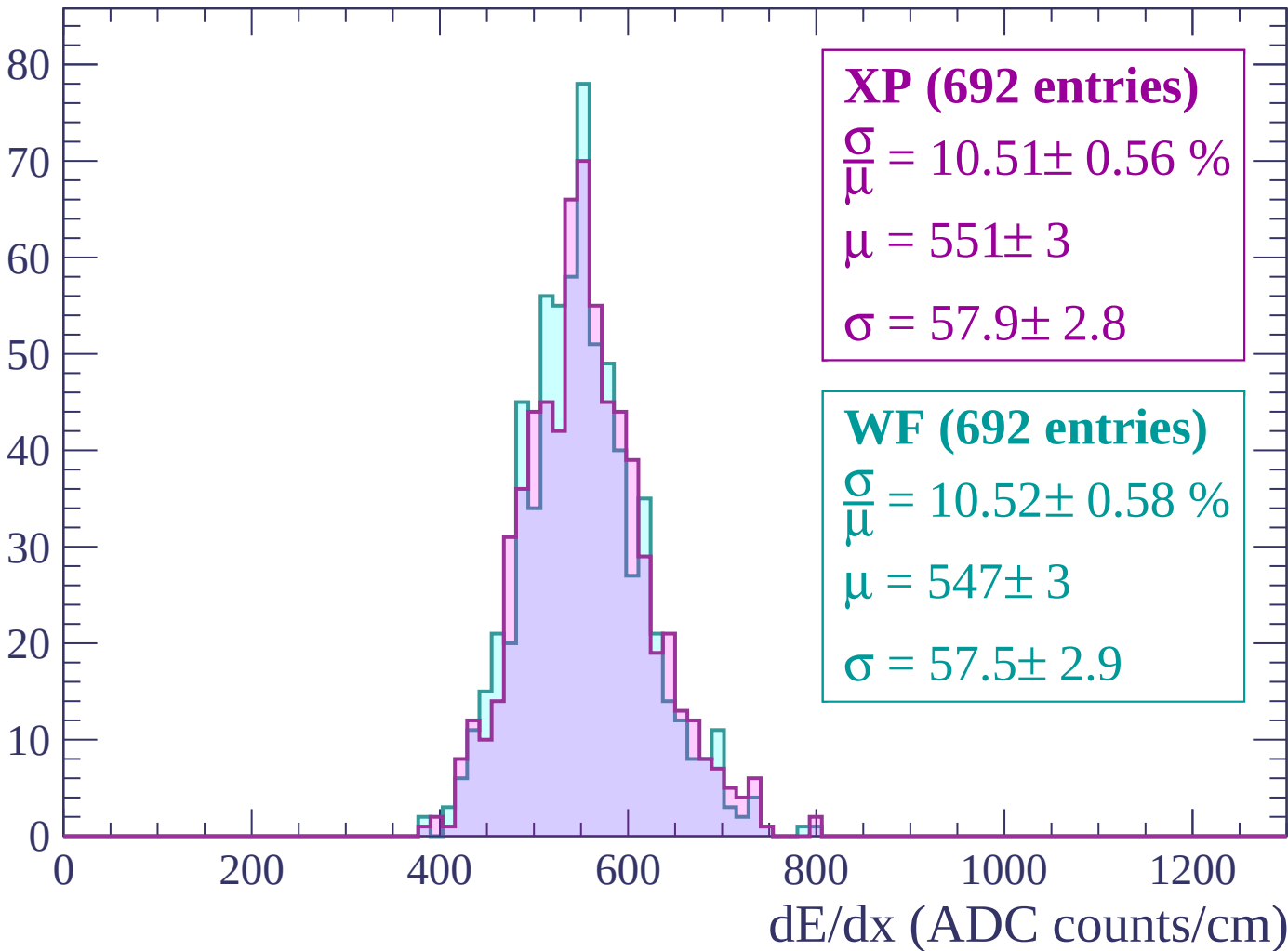
Energy loss | $-2820 < p < -2760$

Count



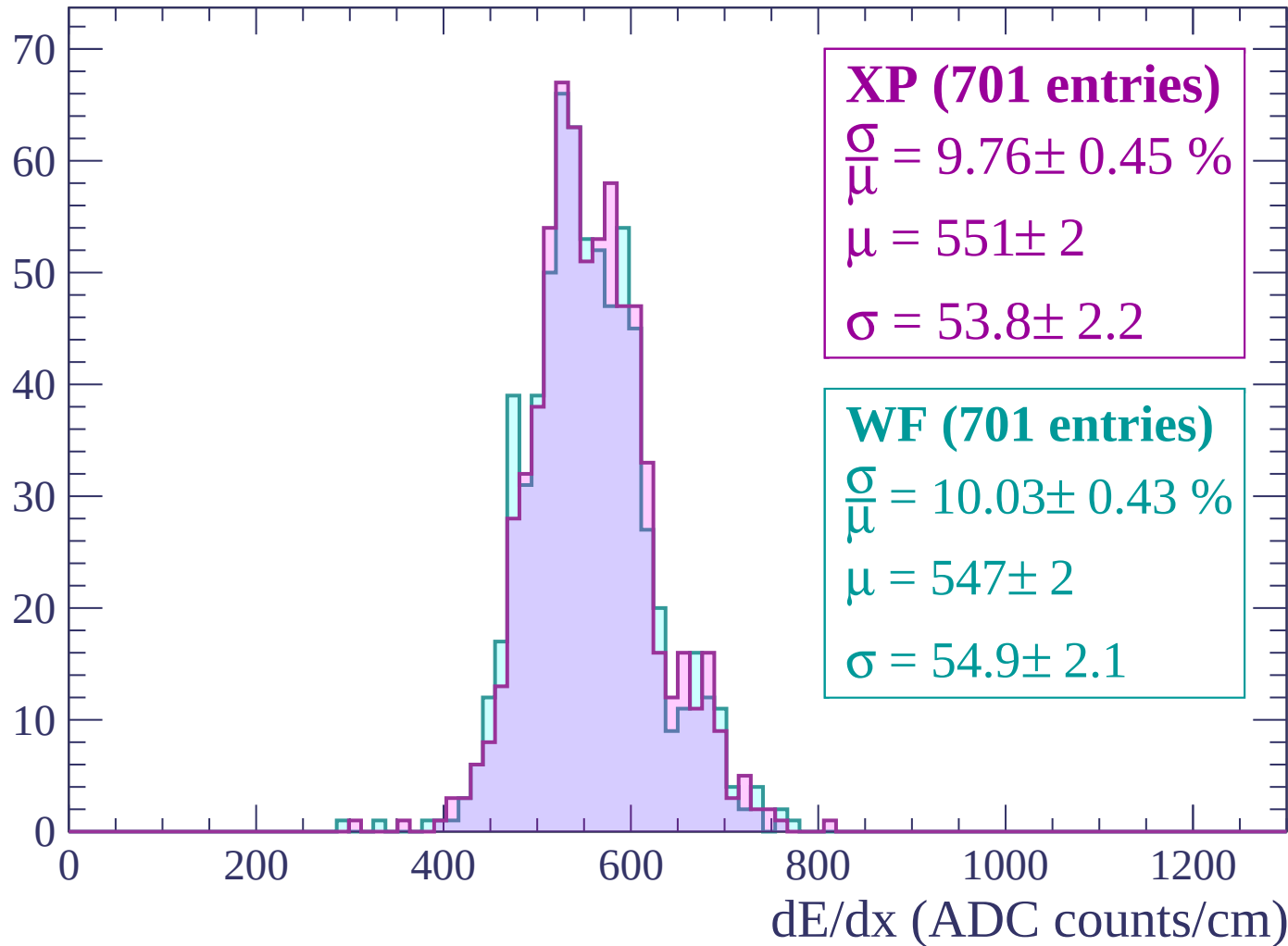
Energy loss | $-2760 < p < -2700$

Count



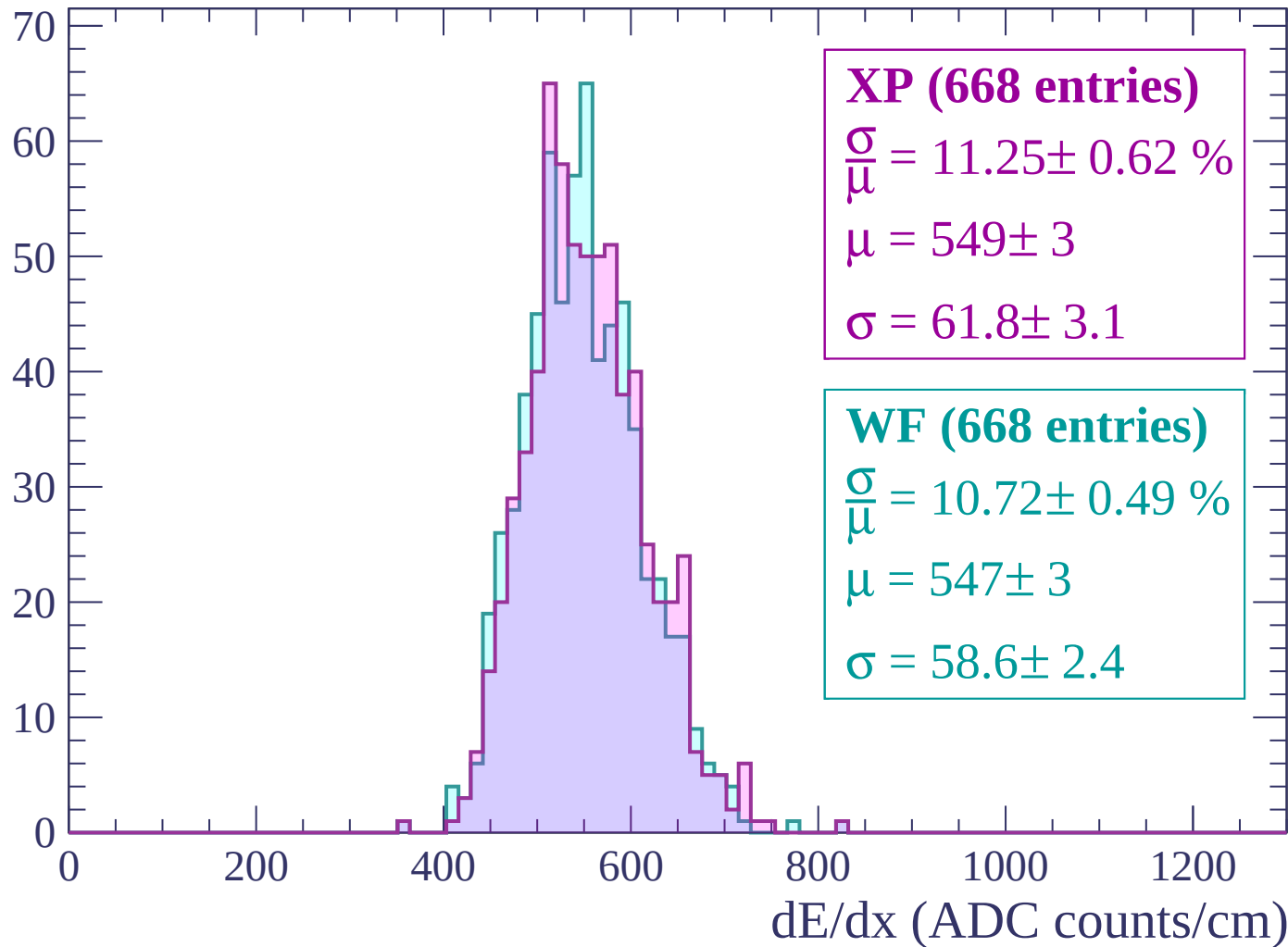
Energy loss | -2700 < p < -2640

Count



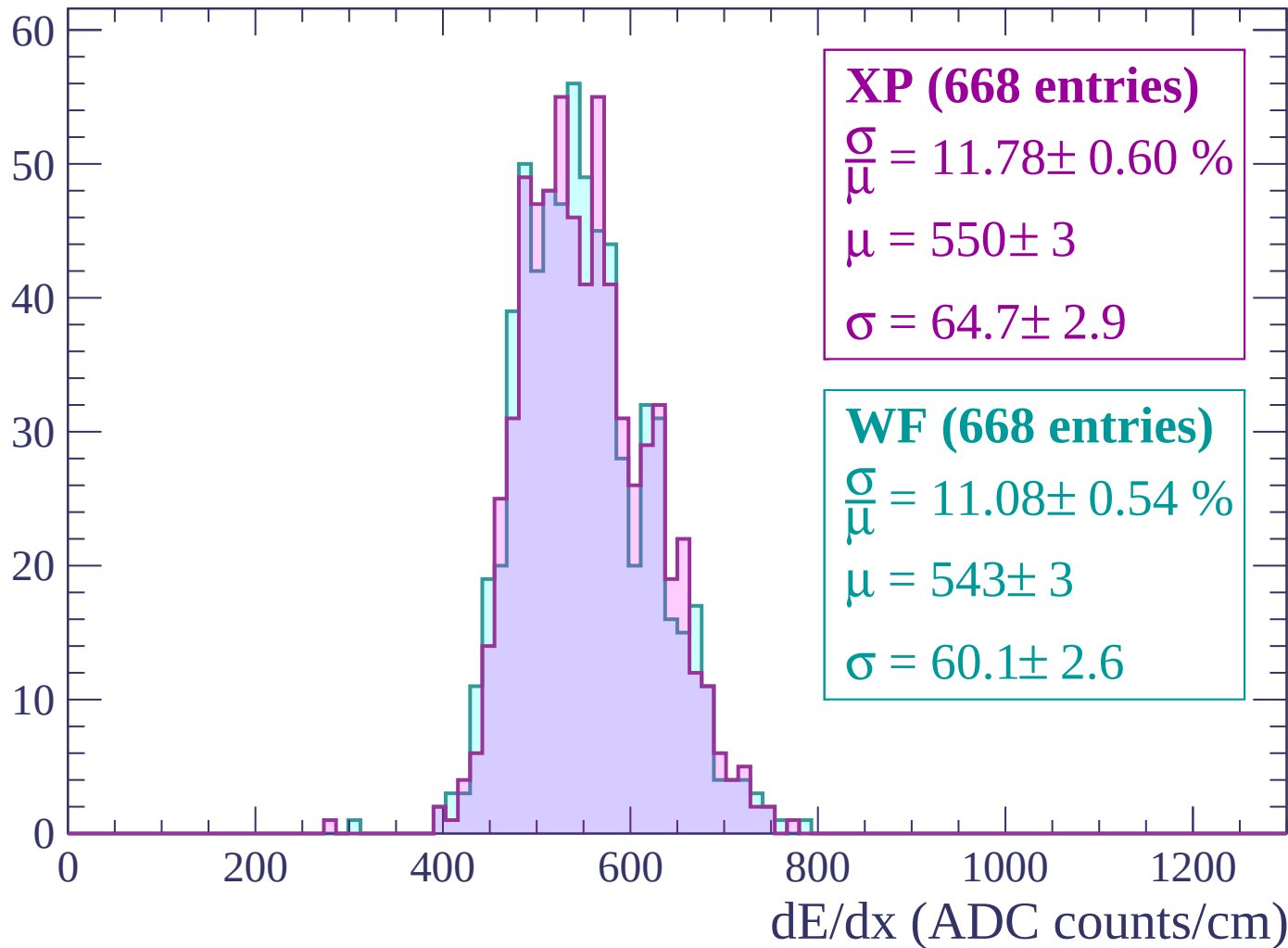
Energy loss | $-2640 < p < -2580$

Count



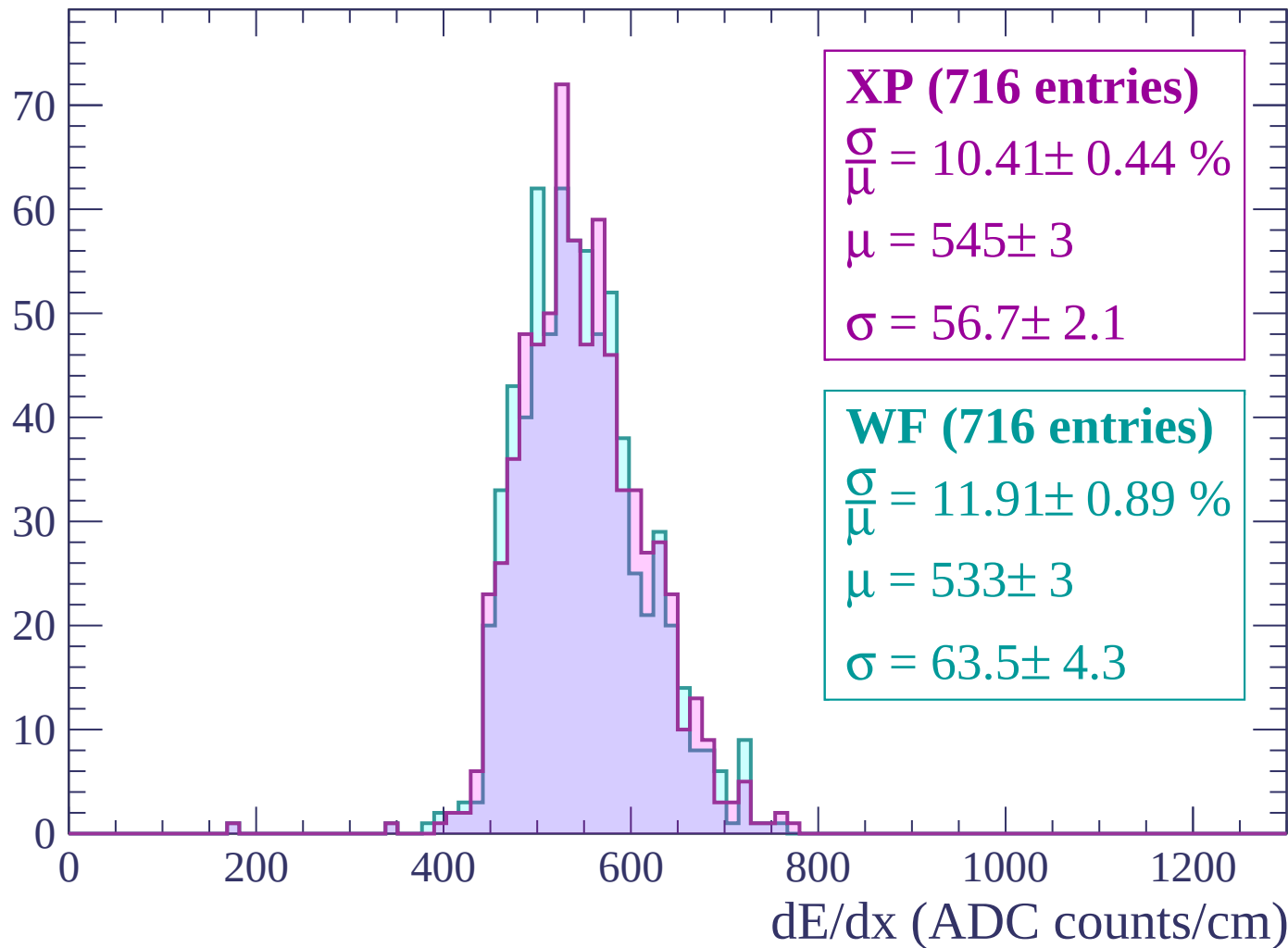
Energy loss | $-2580 < p < -2520$

Count



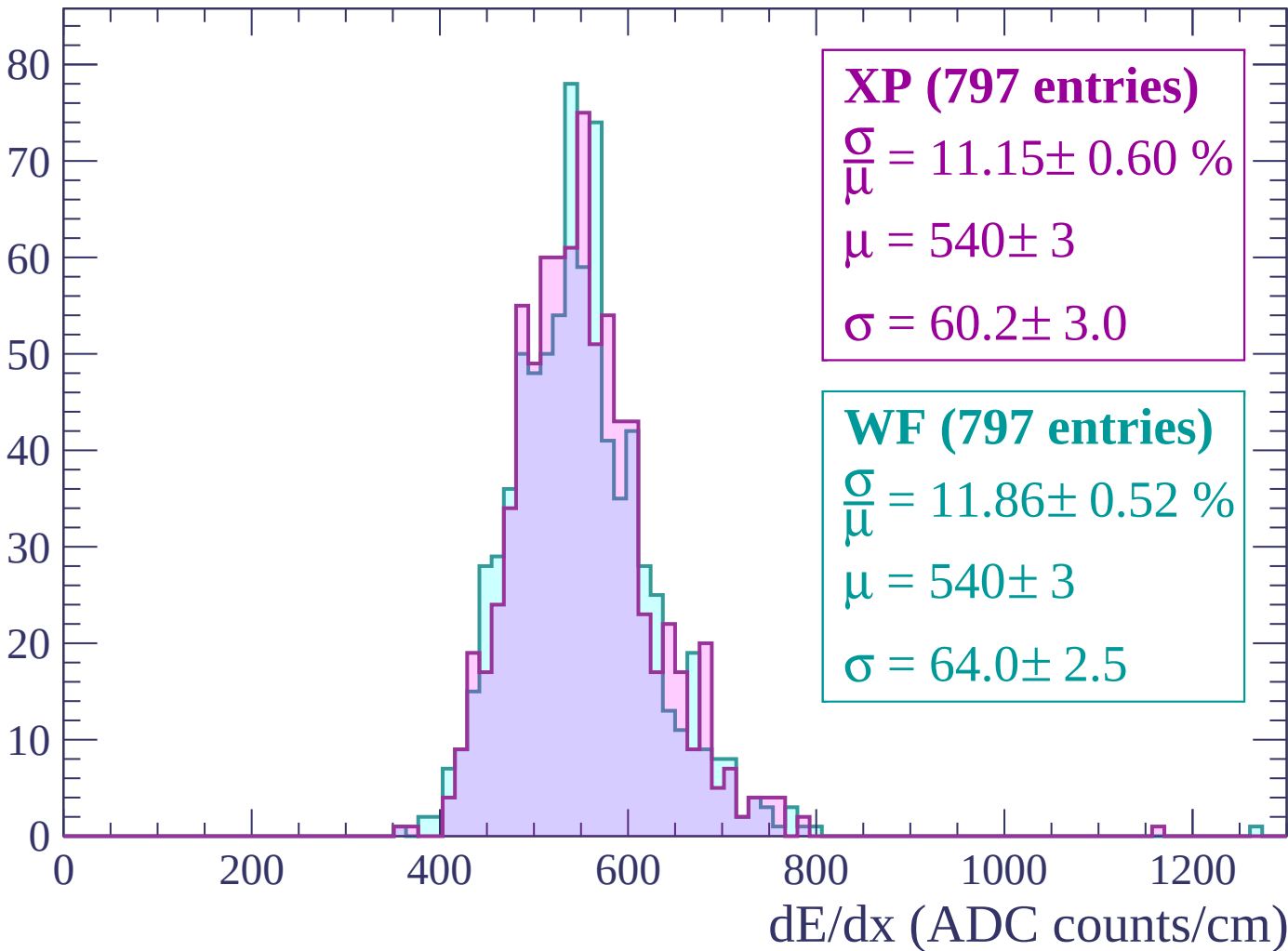
Energy loss | $-2520 < p < -2460$

Count



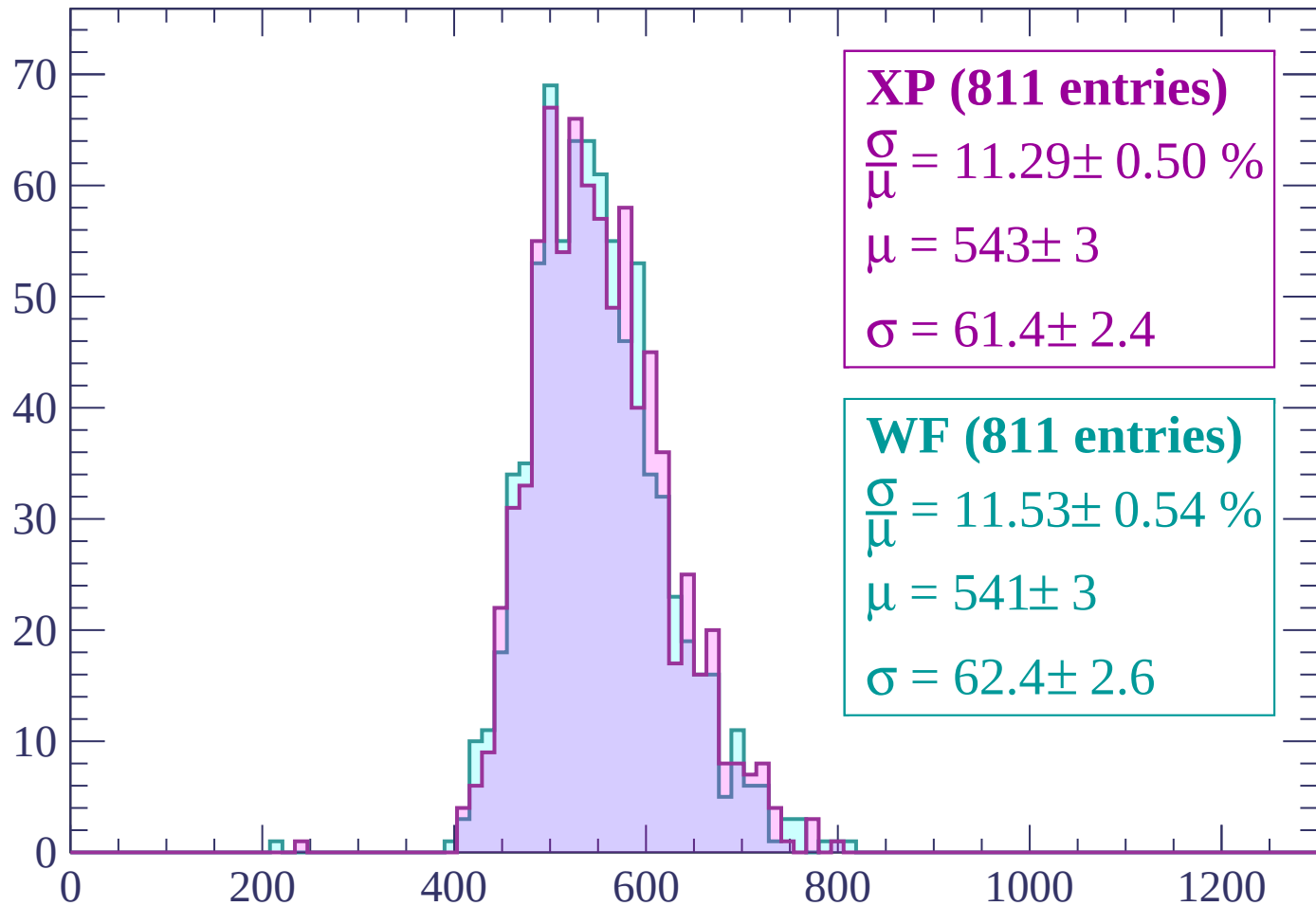
Energy loss | $-2460 < p < -2400$

Count



Energy loss | $-2400 < p < -2340$

Count



XP (811 entries)

$$\frac{\sigma}{\mu} = 11.29 \pm 0.50 \%$$

$$\mu = 543 \pm 3$$

$$\sigma = 61.4 \pm 2.4$$

WF (811 entries)

$$\frac{\sigma}{\mu} = 11.53 \pm 0.54 \%$$

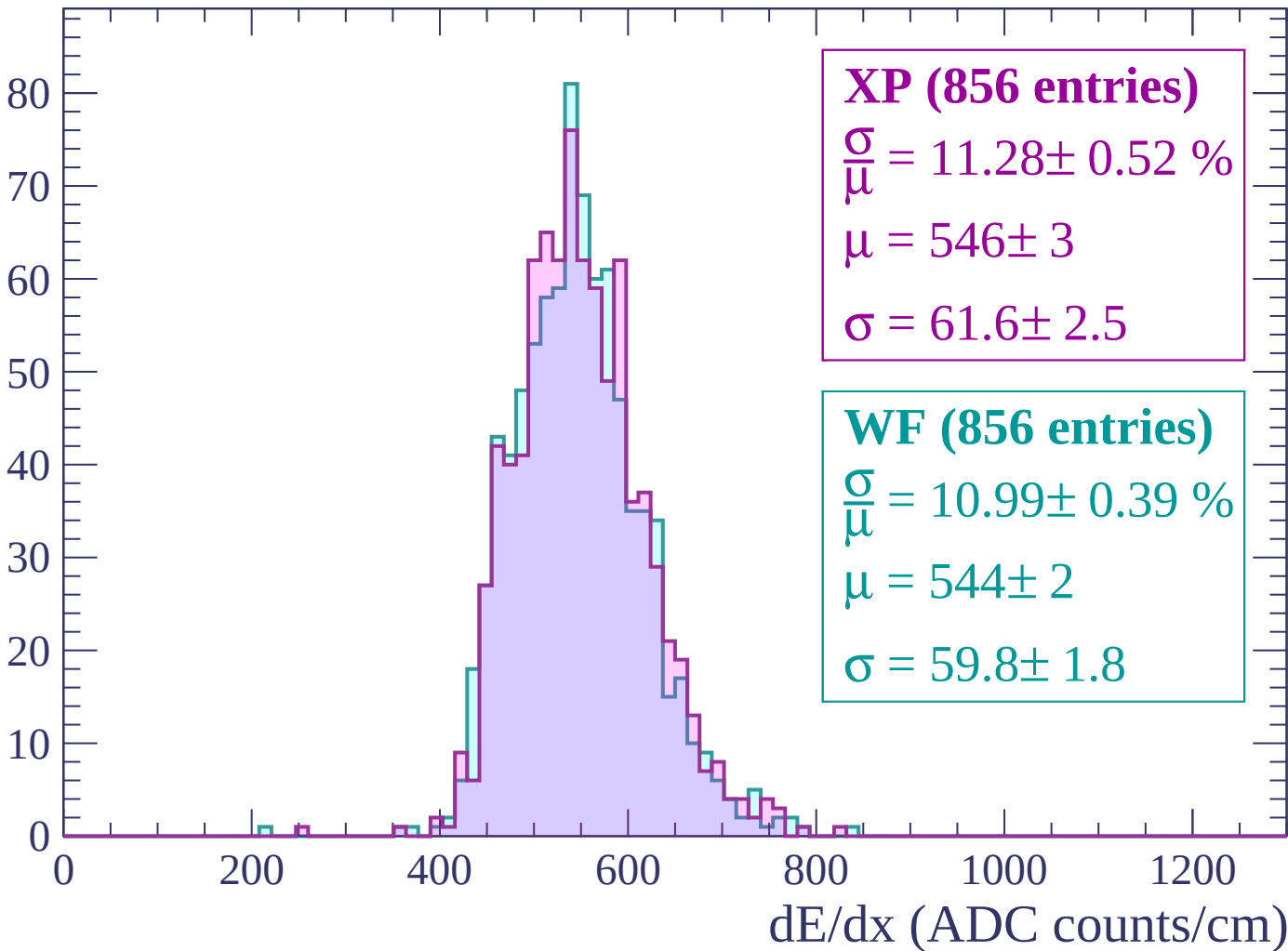
$$\mu = 541 \pm 3$$

$$\sigma = 62.4 \pm 2.6$$

dE/dx (ADC counts/cm)

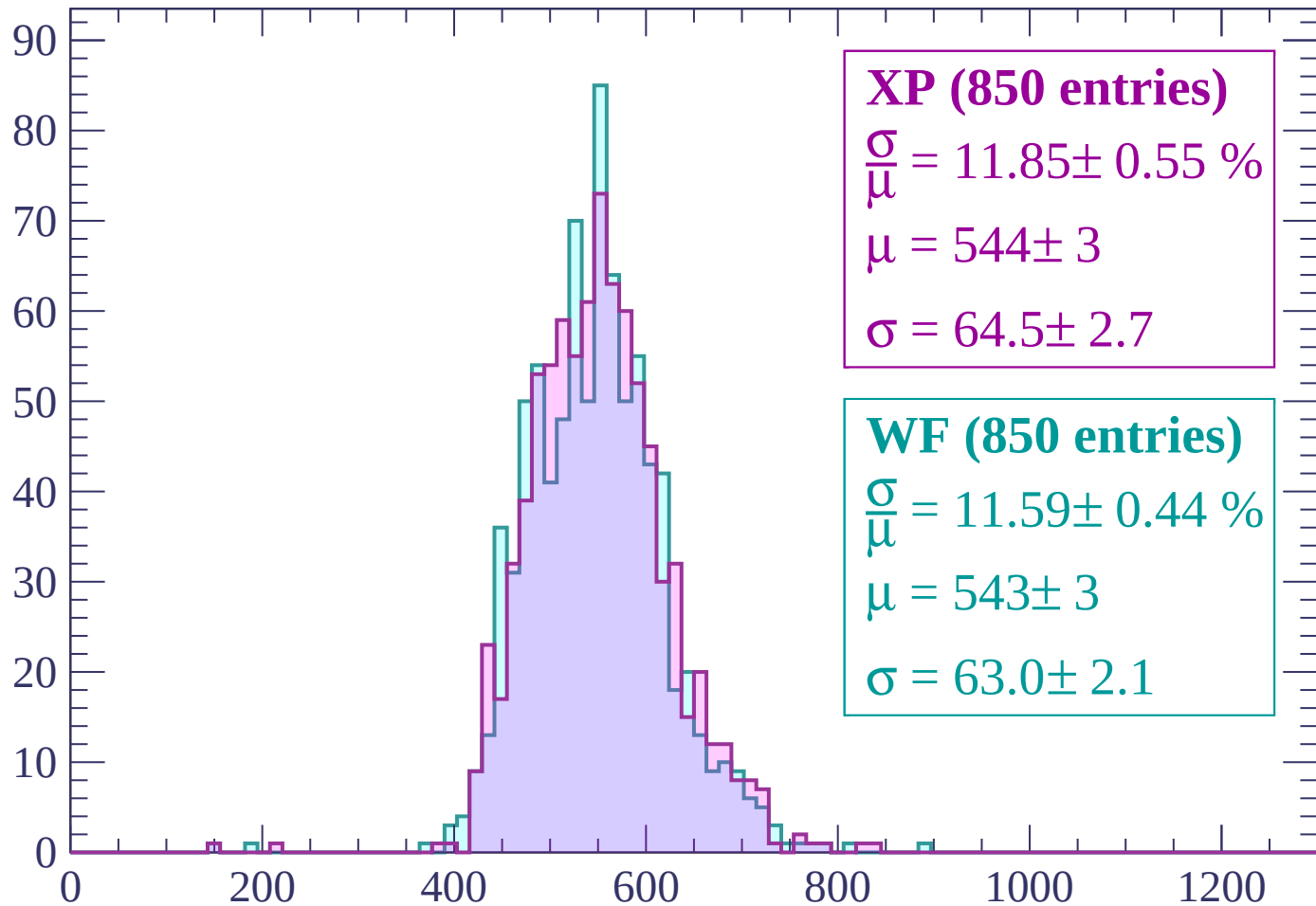
Energy loss | $-2340 < p < -2280$

Count



Energy loss | $-2280 < p < -2220$

Count



XP (850 entries)

$$\frac{\sigma}{\mu} = 11.85 \pm 0.55 \%$$

$$\mu = 544 \pm 3$$

$$\sigma = 64.5 \pm 2.7$$

WF (850 entries)

$$\frac{\sigma}{\mu} = 11.59 \pm 0.44 \%$$

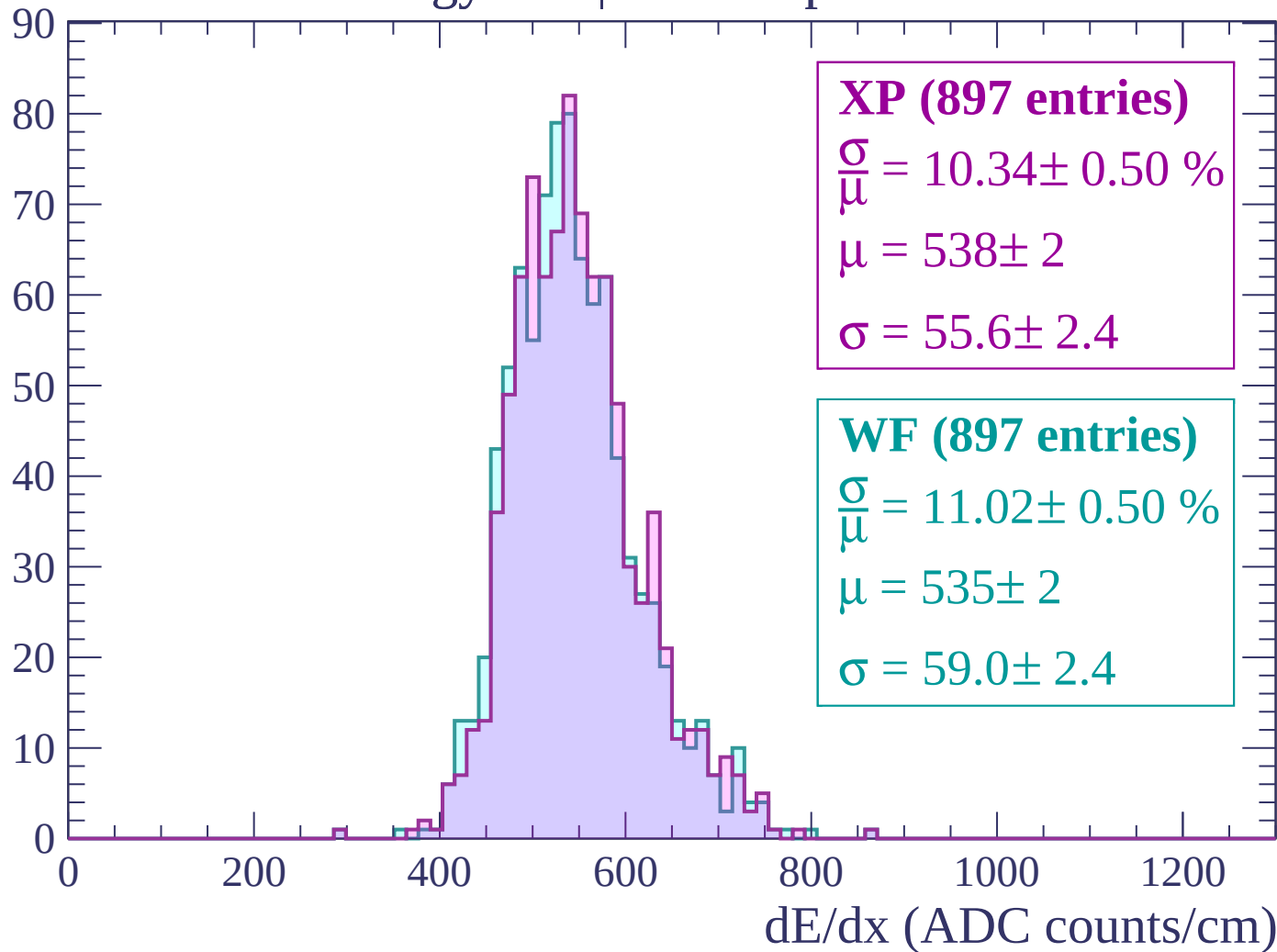
$$\mu = 543 \pm 3$$

$$\sigma = 63.0 \pm 2.1$$

dE/dx (ADC counts/cm)

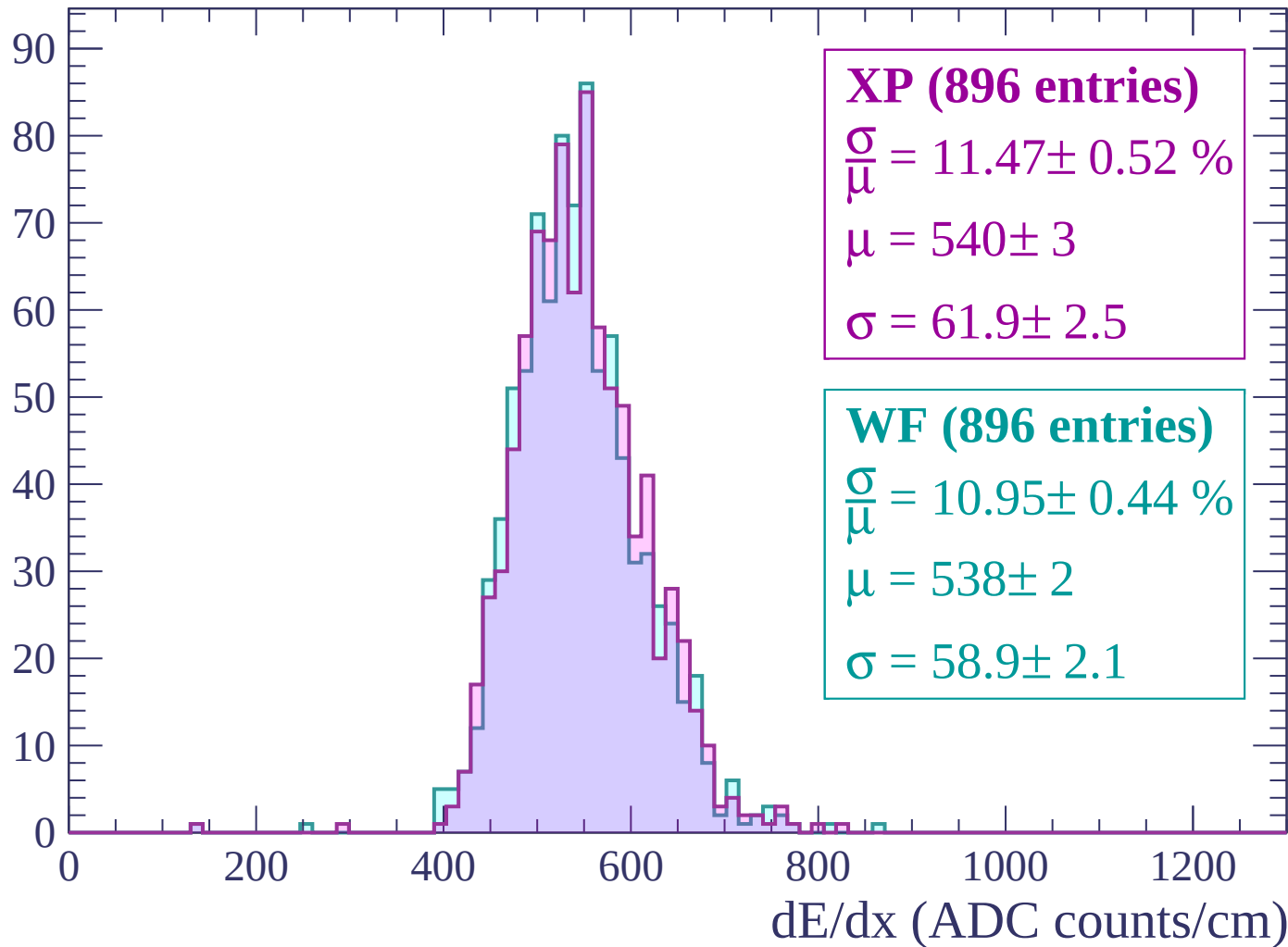
Energy loss | -2220 < p < -2160

Count



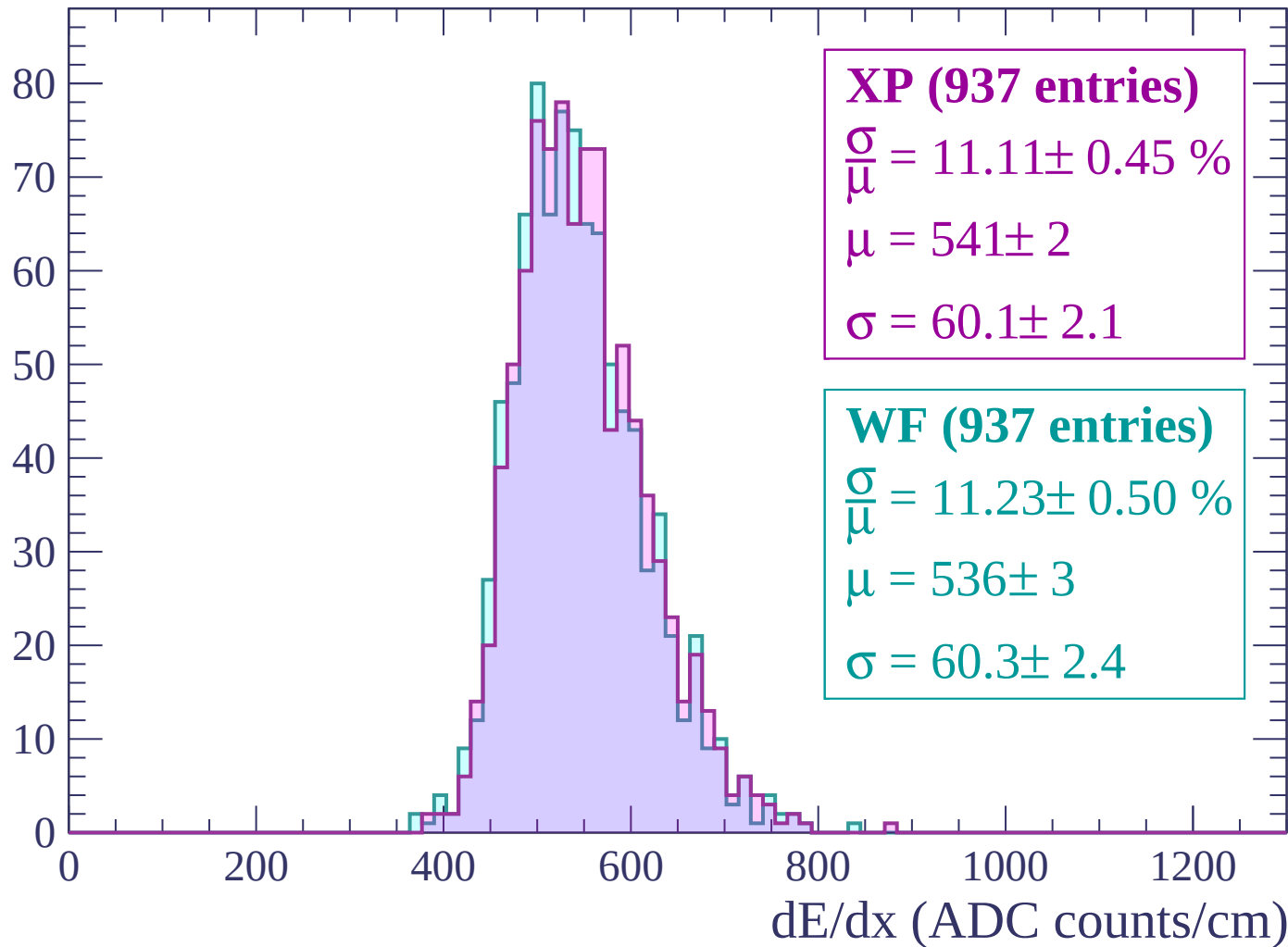
Energy loss | -2160 < p < -2100

Count



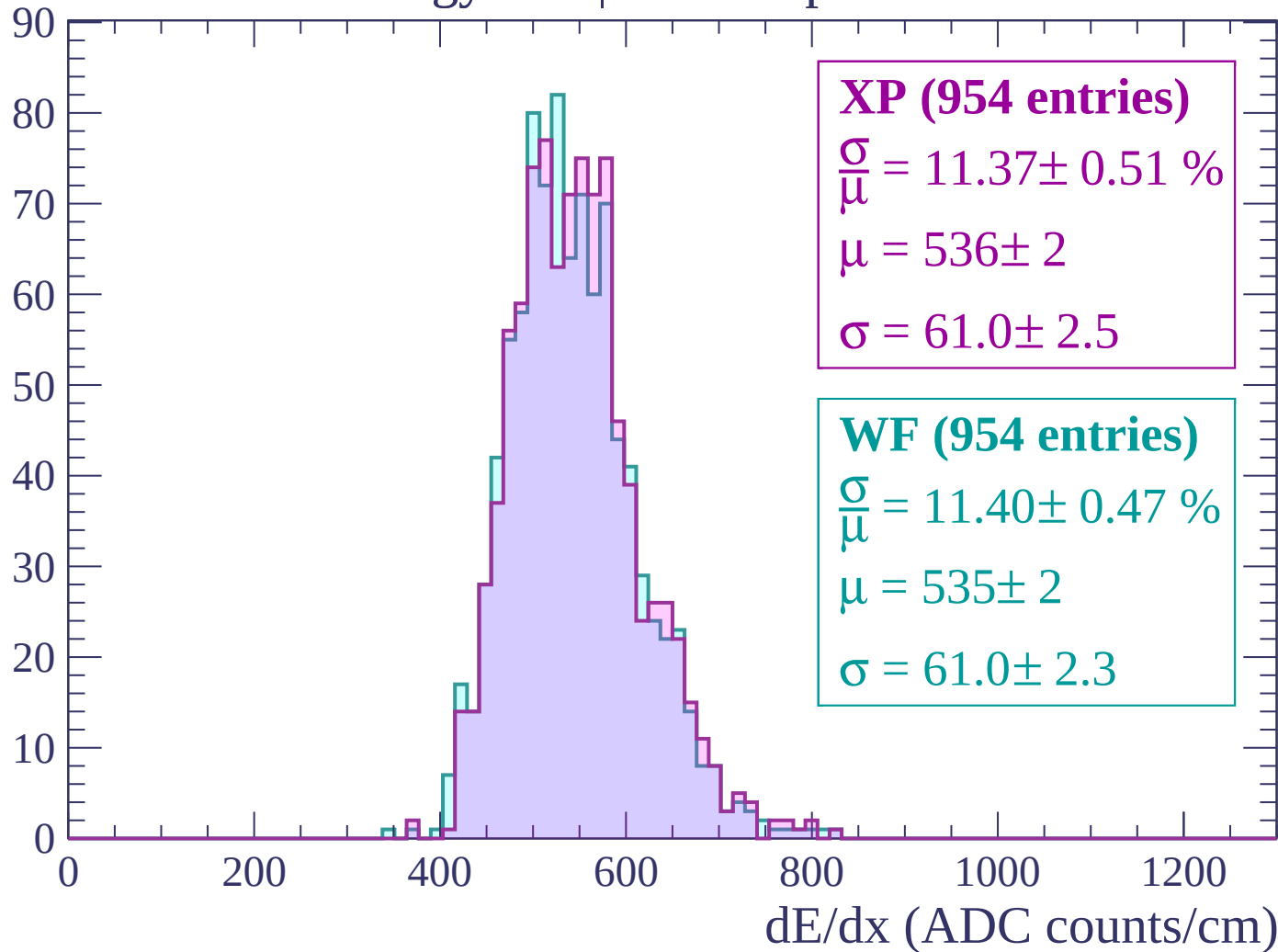
Energy loss | -2100 < p < -2040

Count



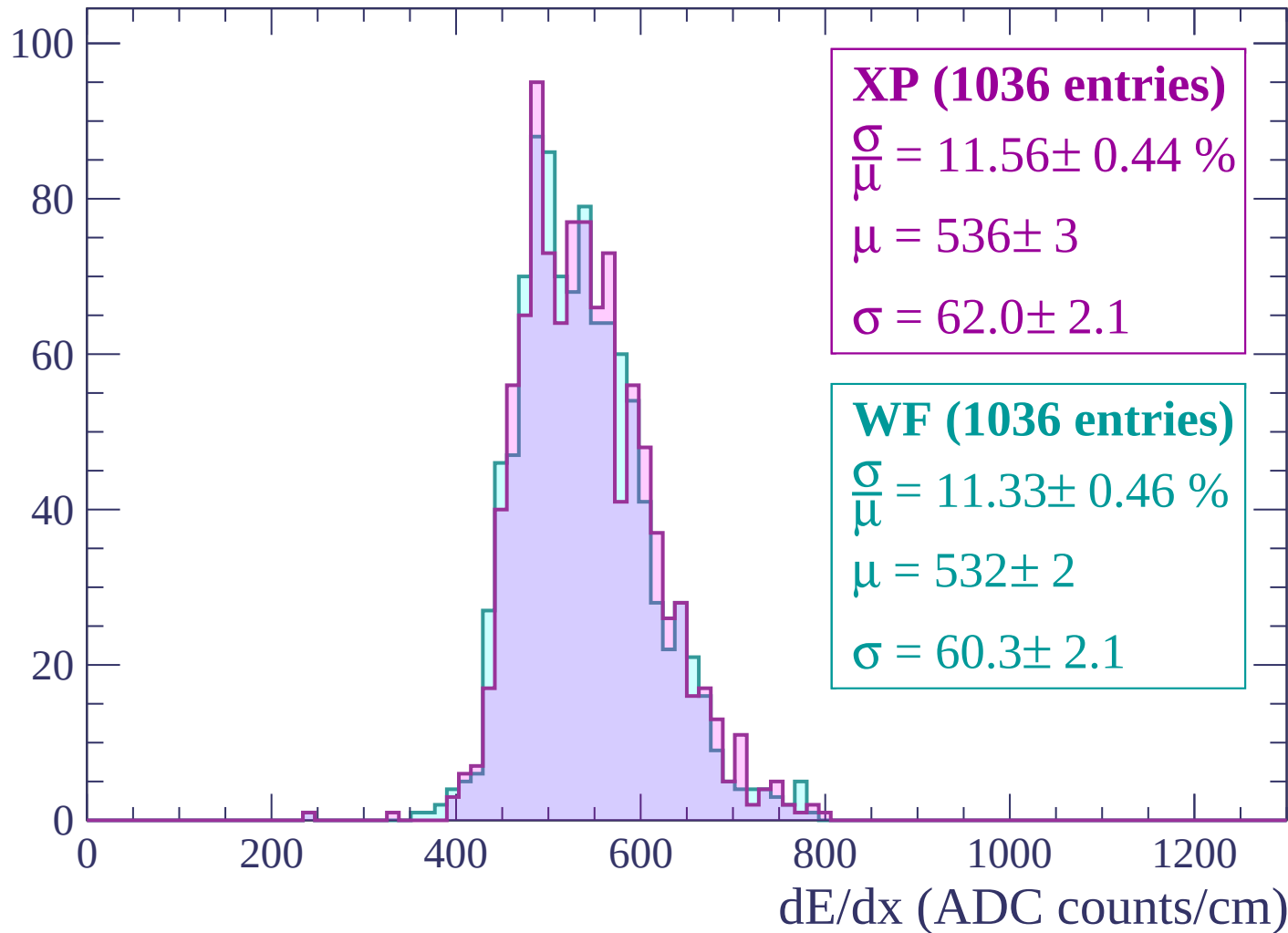
Energy loss | -2040 < p < -1980

Count



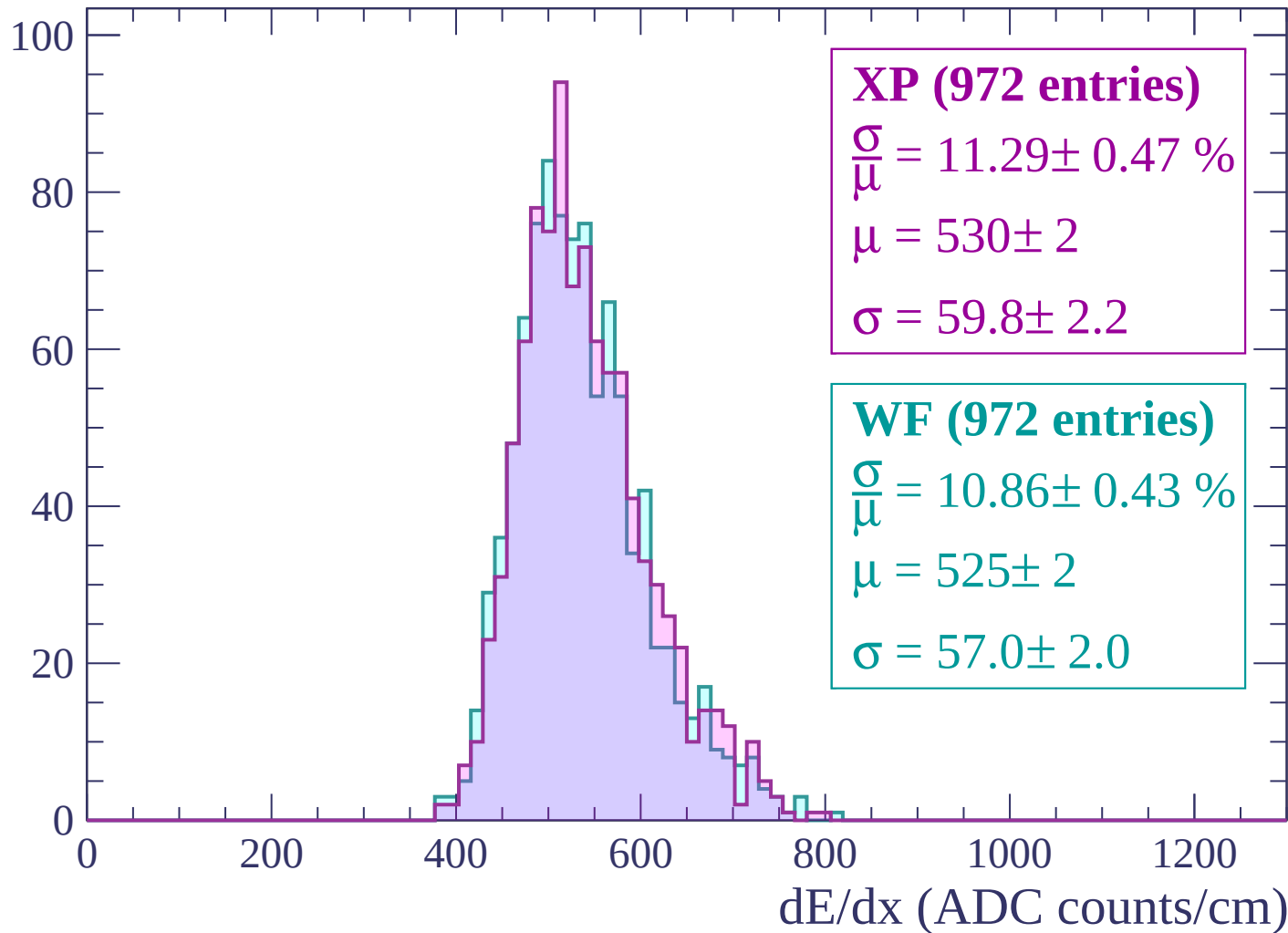
Energy loss | $-1980 < p < -1920$

Count



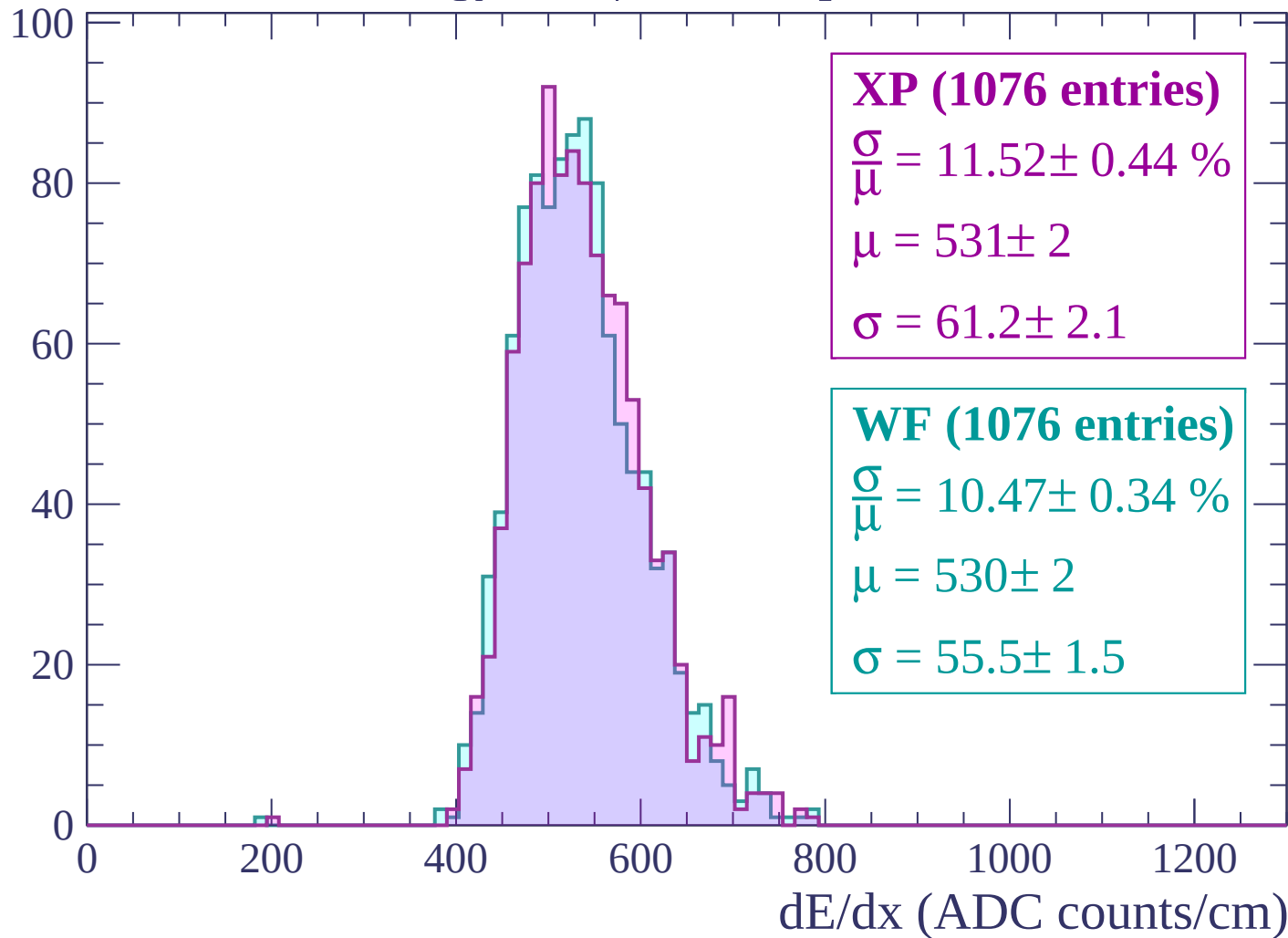
Energy loss | -1920 < p < -1860

Count



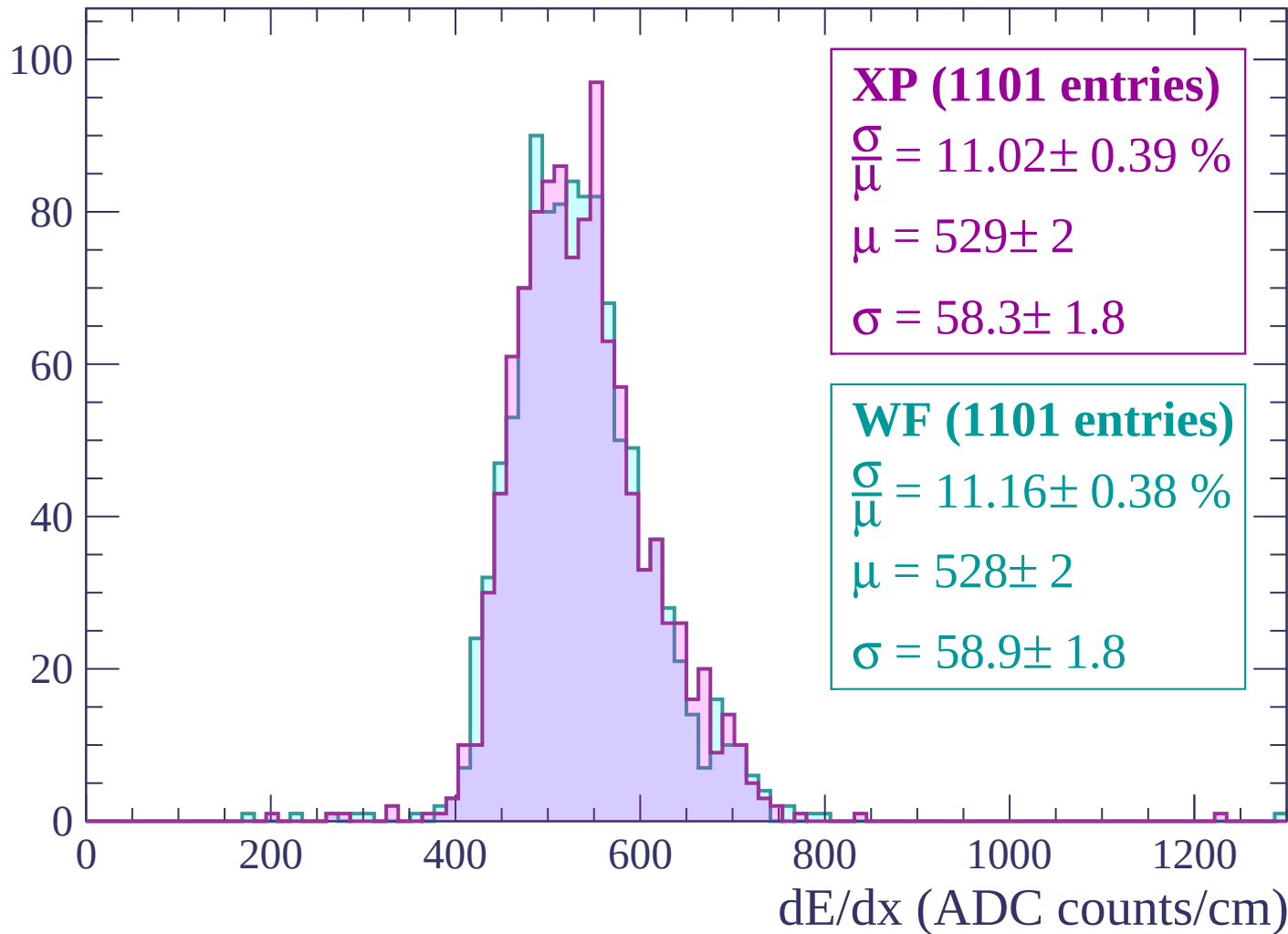
Energy loss | -1860 < p < -1800

Count



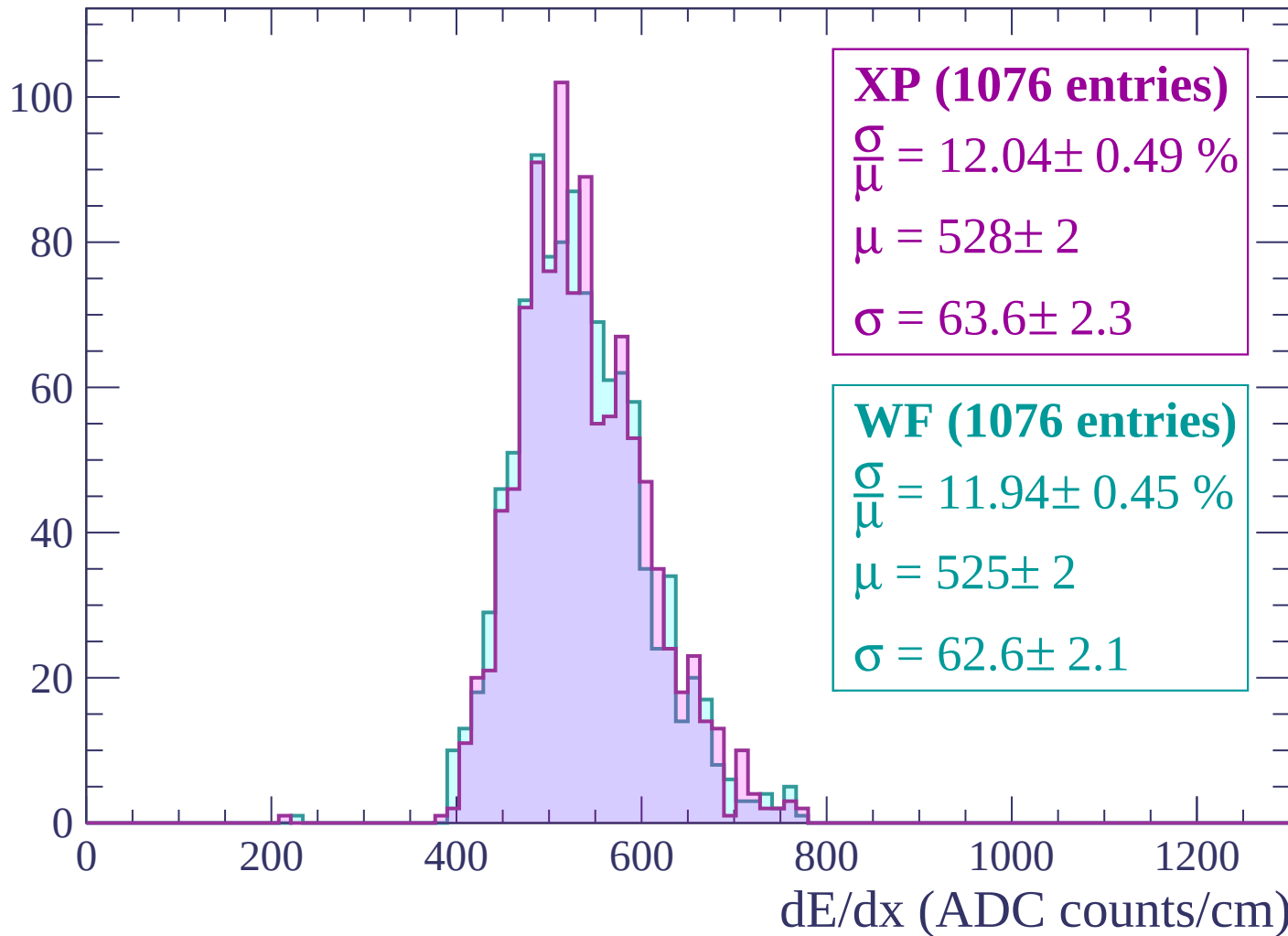
Energy loss | $-1800 < p < -1740$

Count



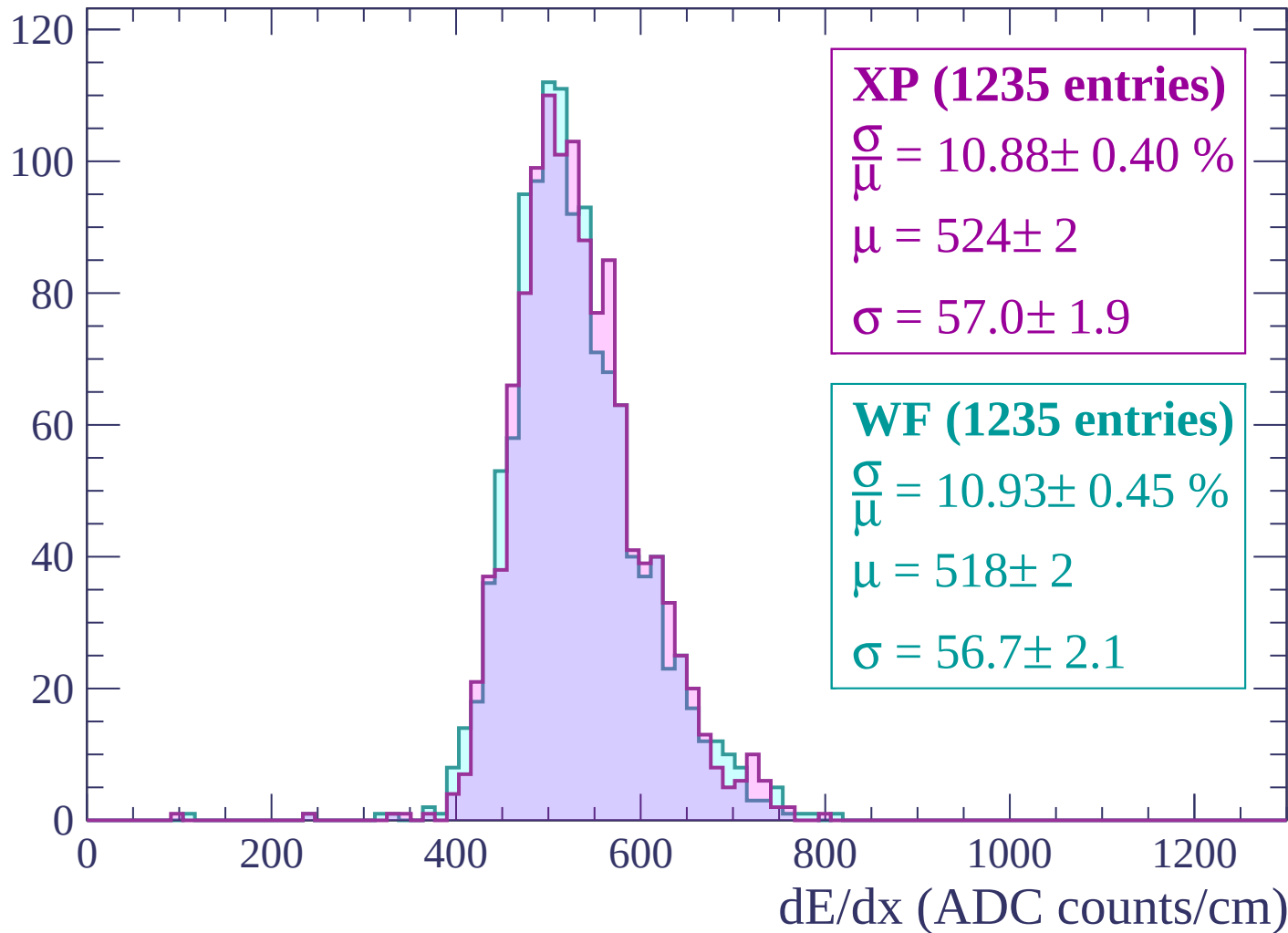
Energy loss | $-1740 < p < -1680$

Count



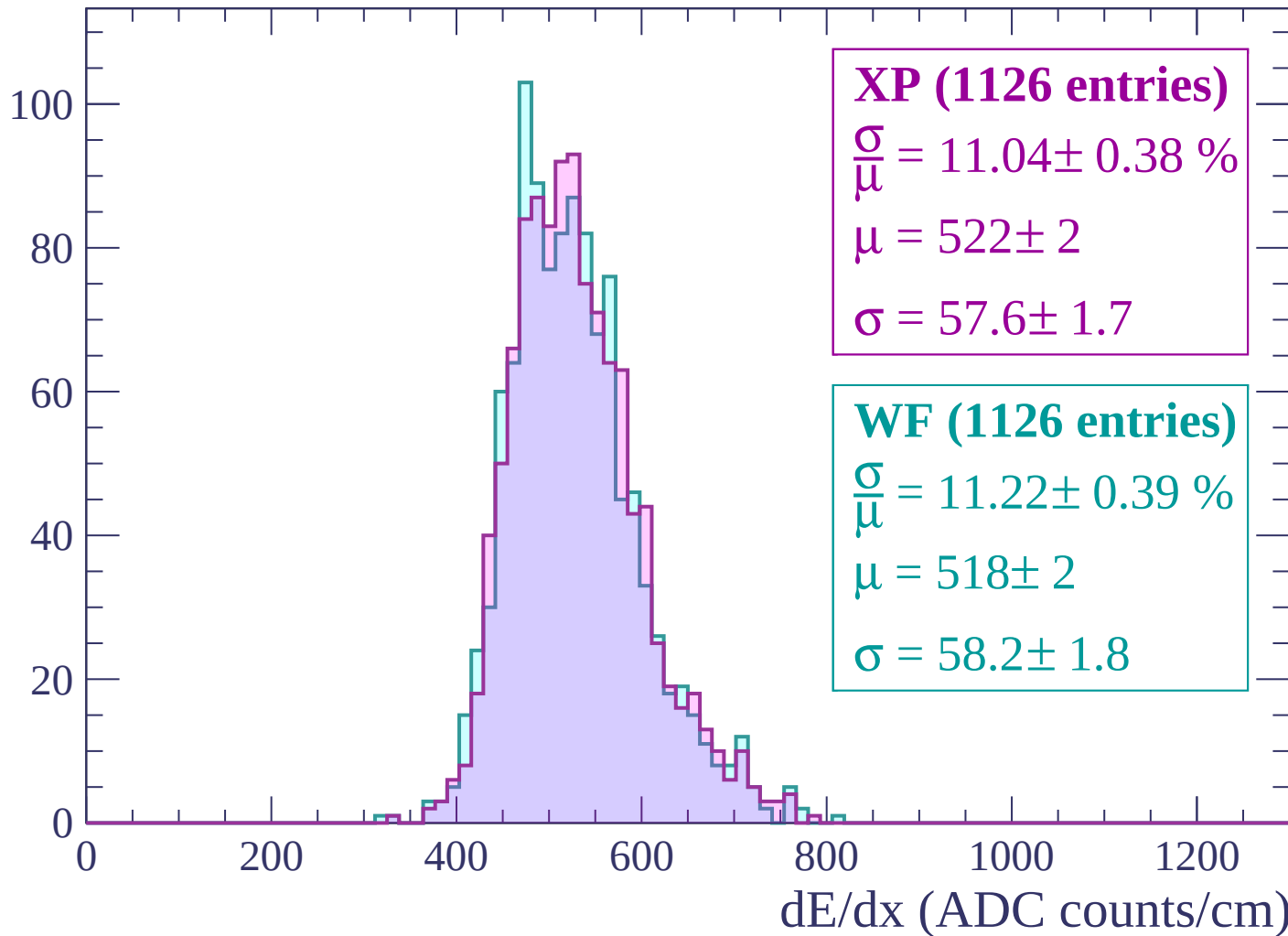
Energy loss | $-1680 < p < -1620$

Count



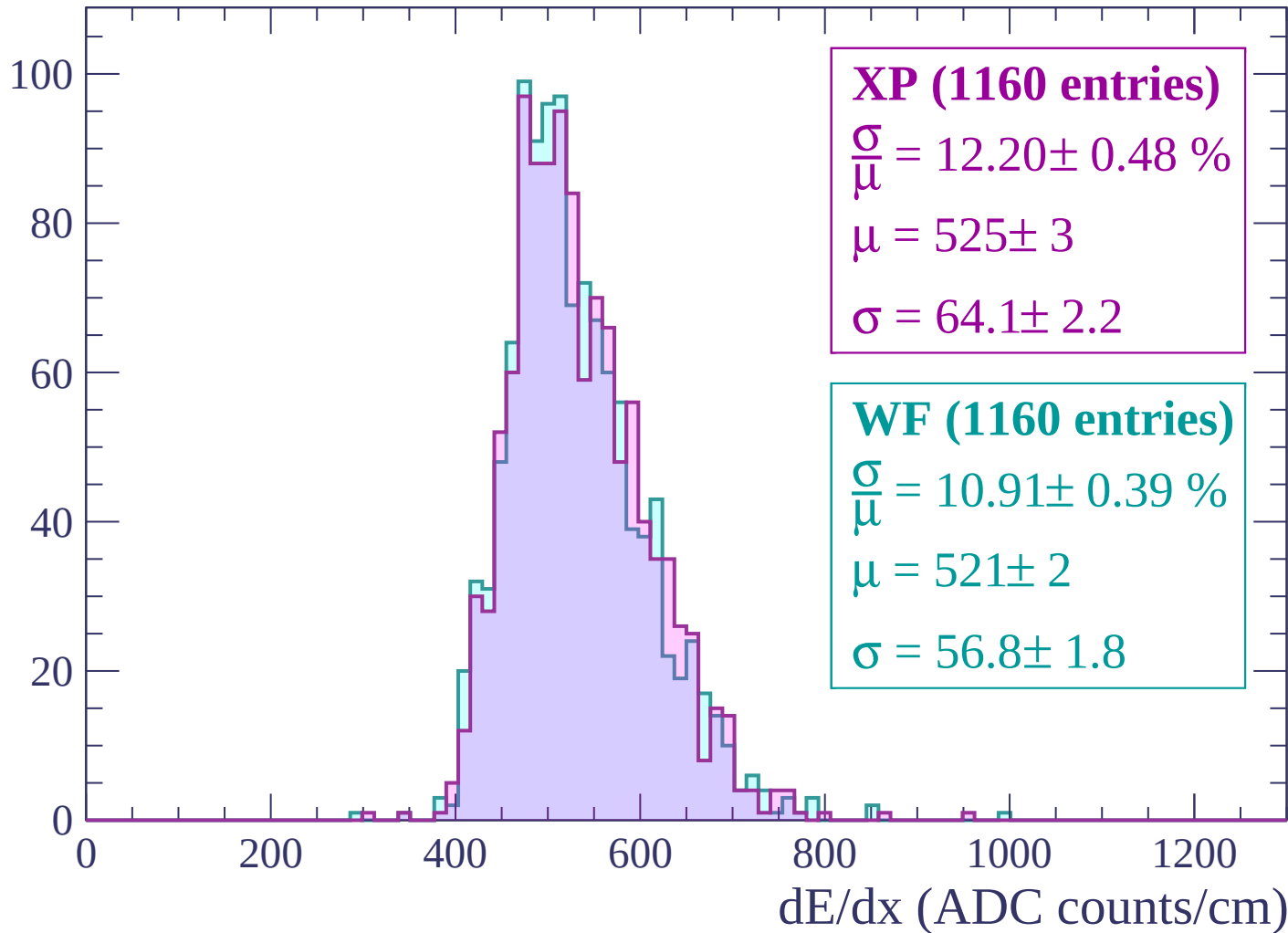
Energy loss | $-1620 < p < -1560$

Count



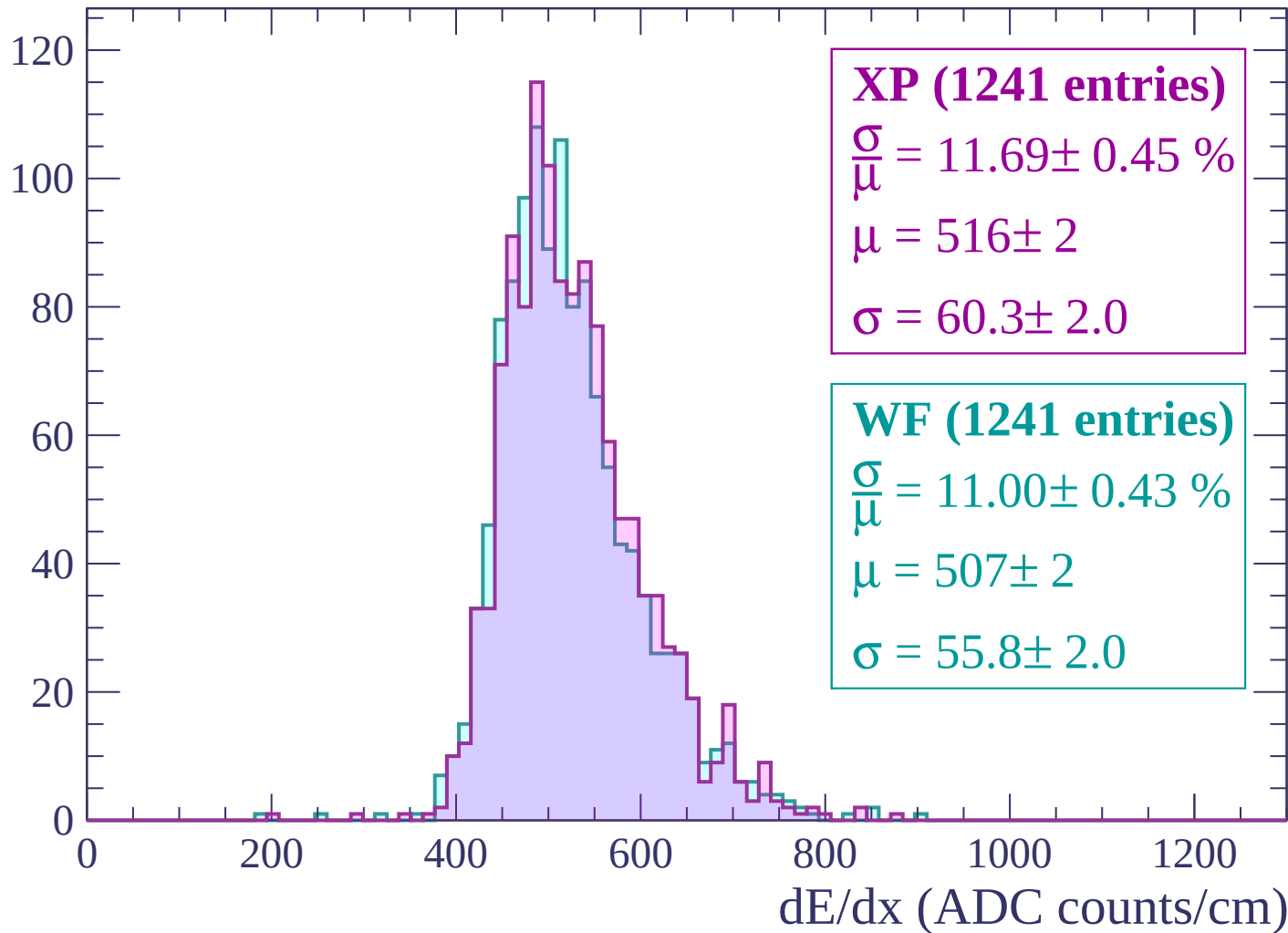
Energy loss | -1560 < p < -1500

Count



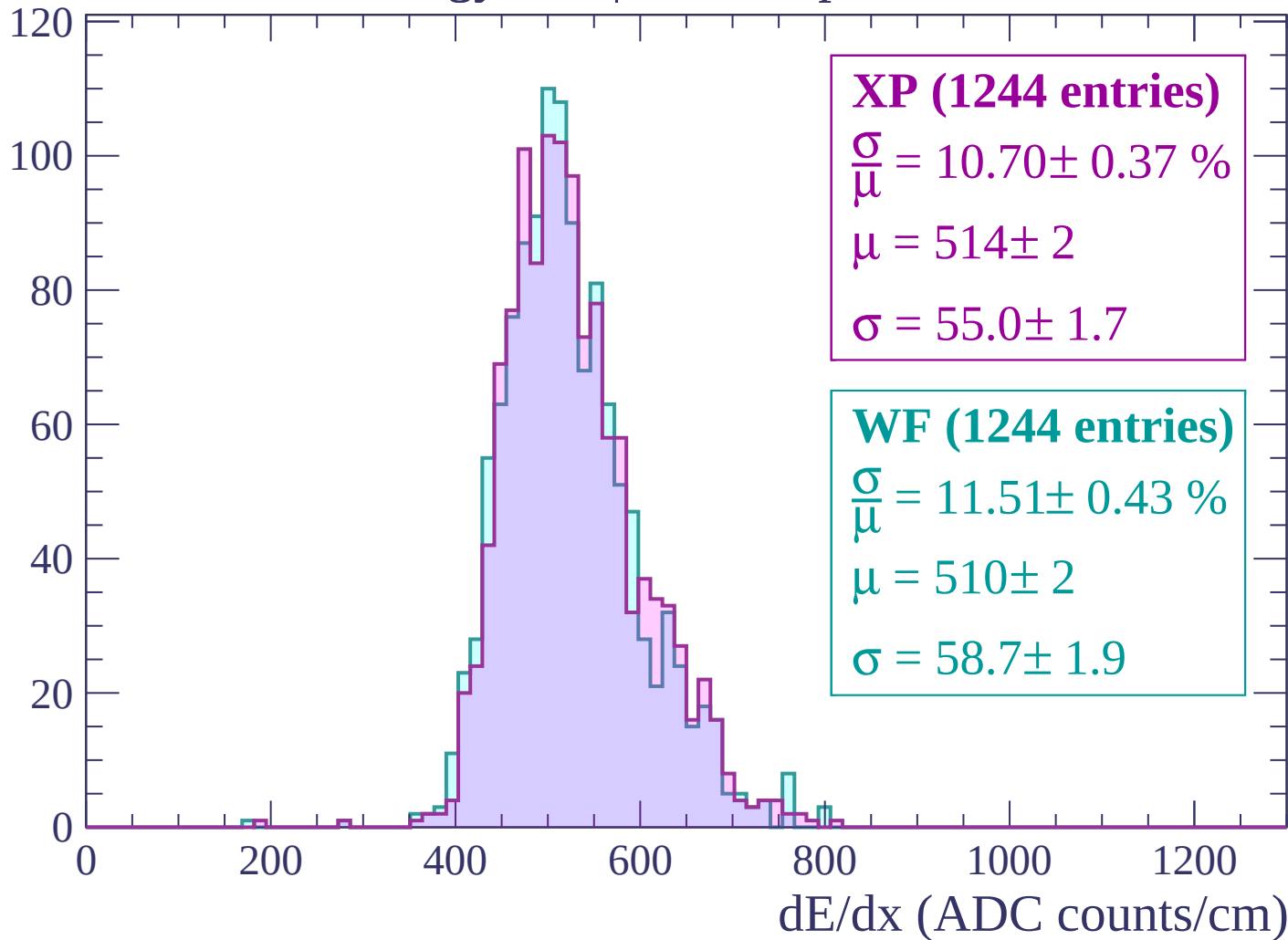
Energy loss | $-1500 < p < -1440$

Count



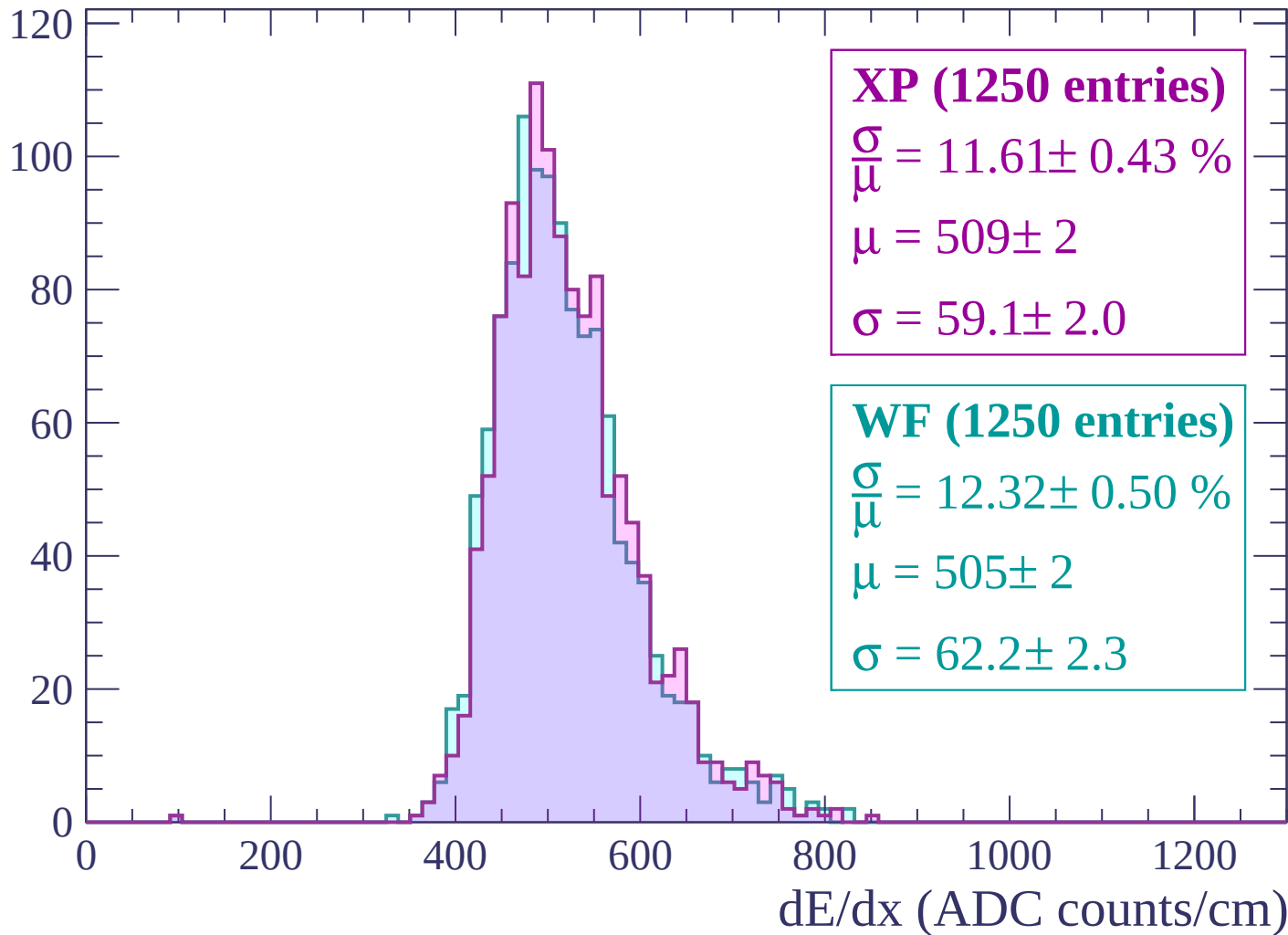
Energy loss | -1440 < p < -1380

Count



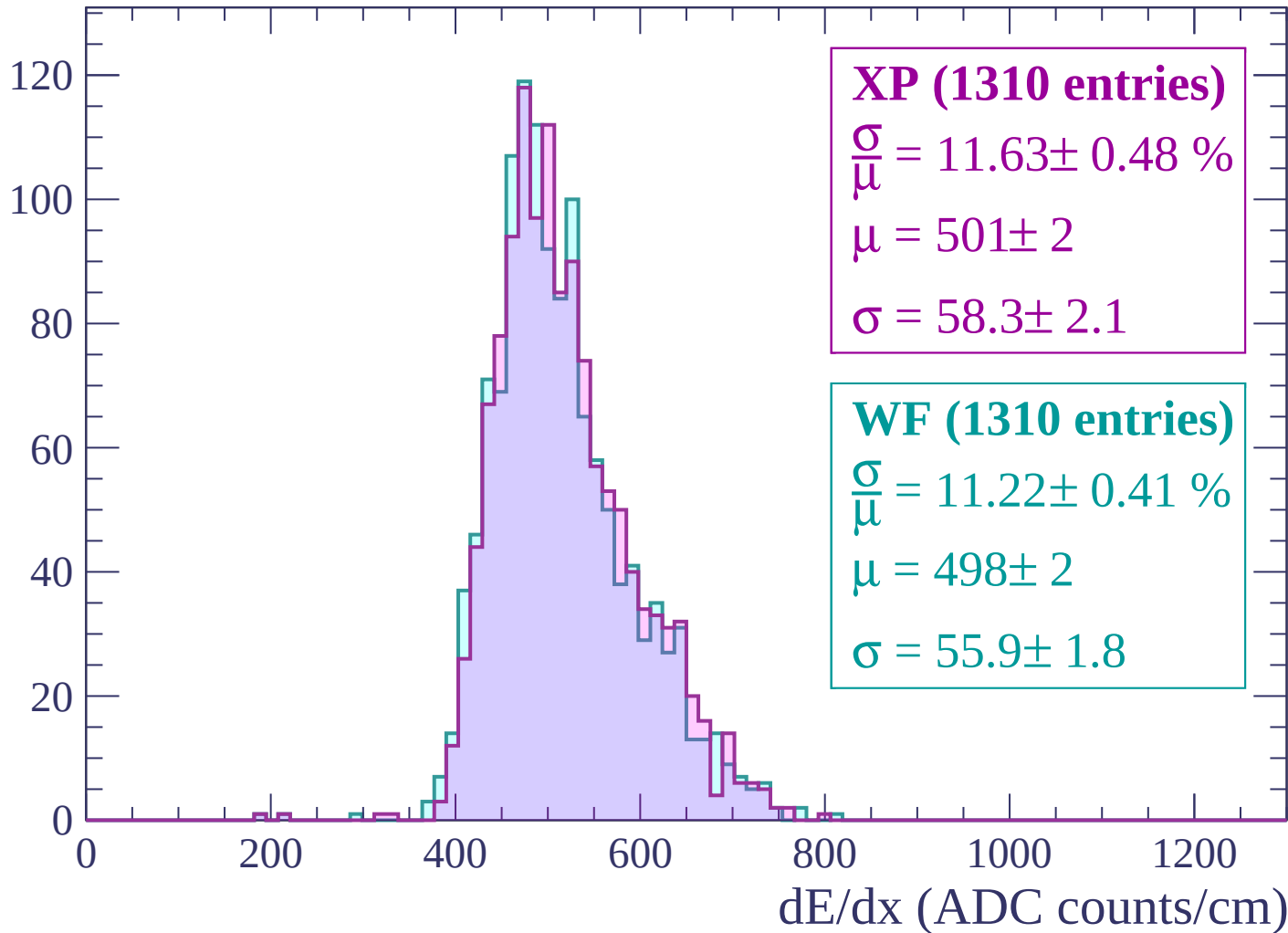
Energy loss | -1380 < p < -1320

Count



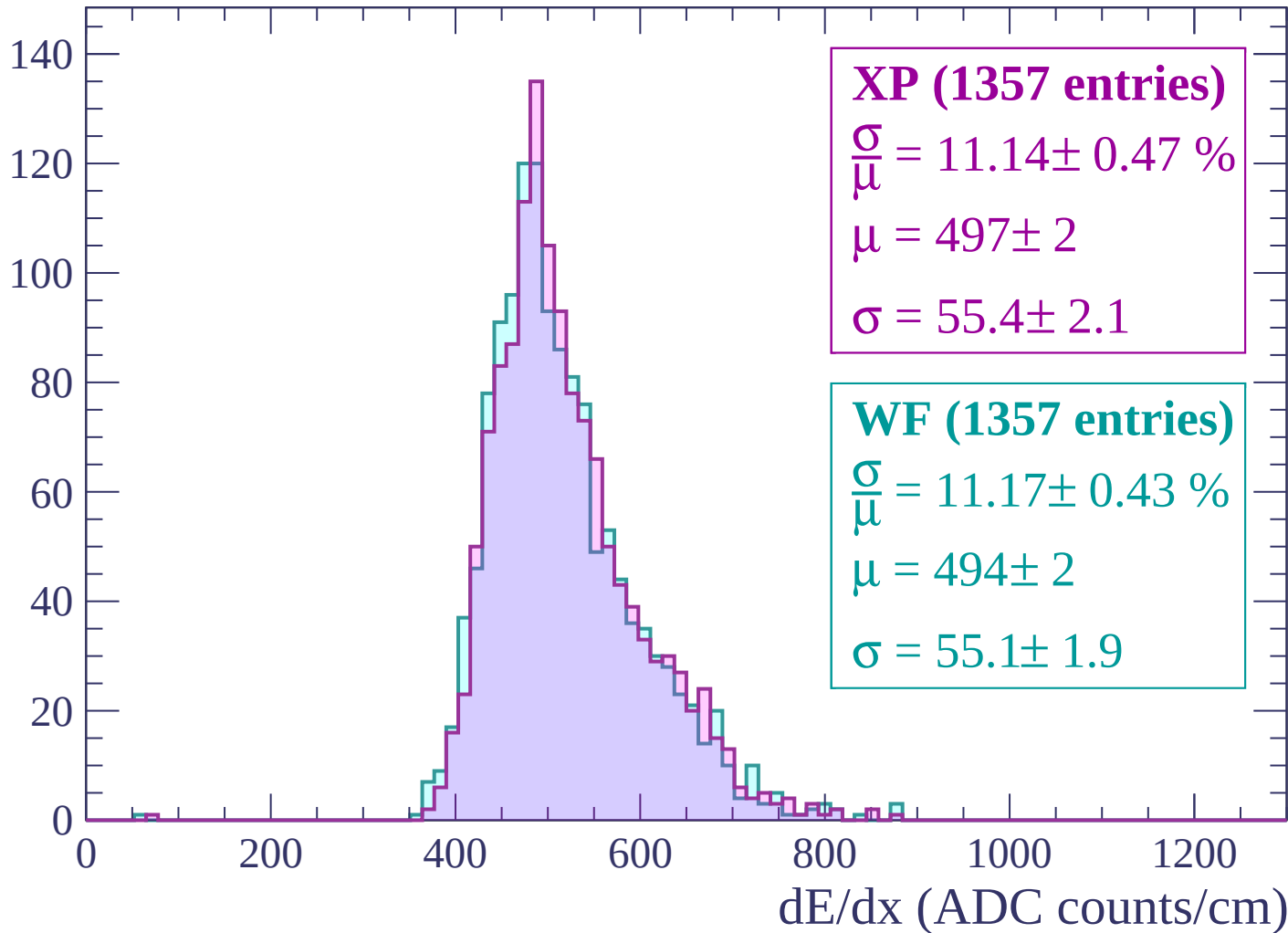
Energy loss | -1320 < p < -1260

Count



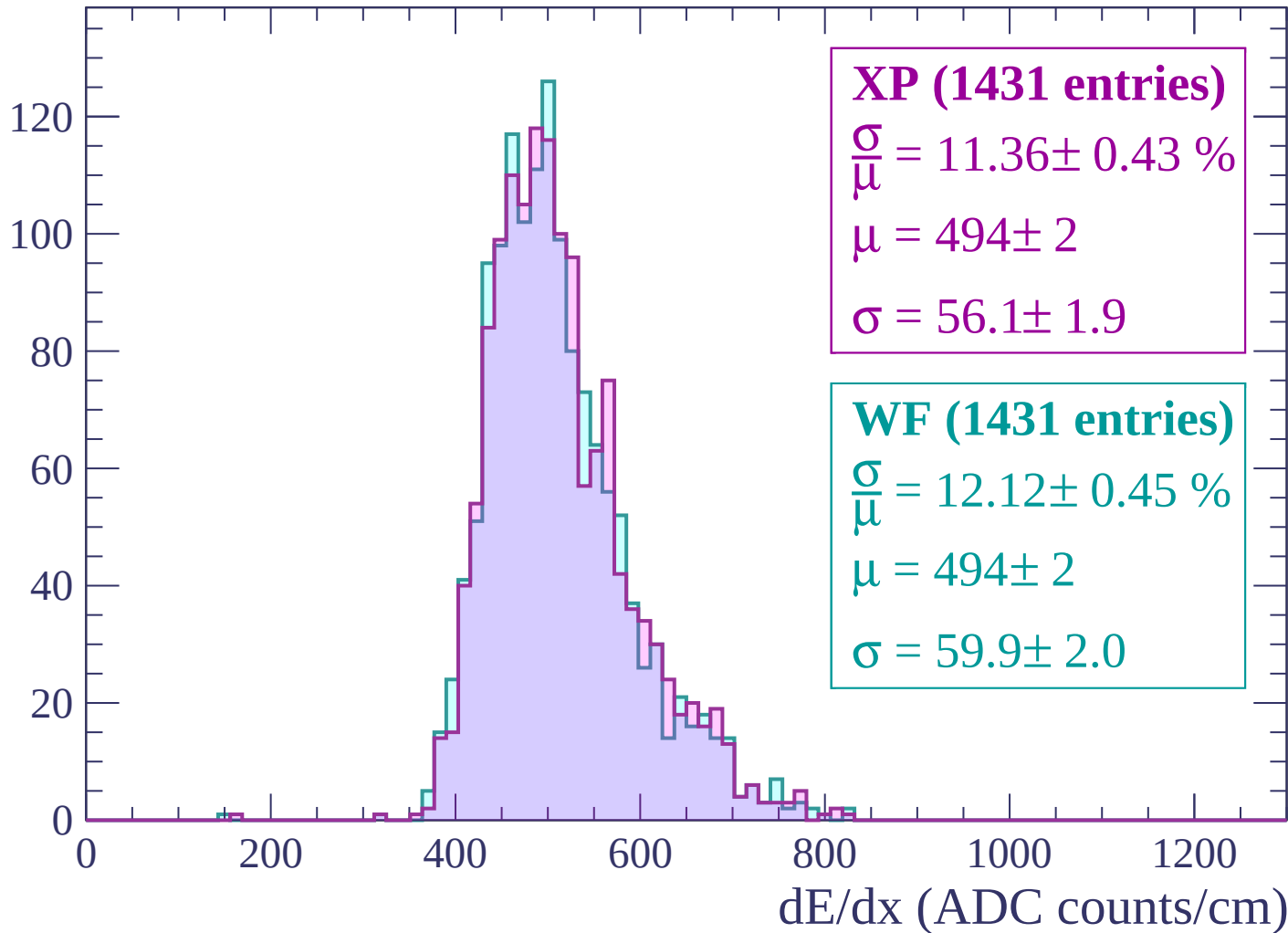
Energy loss | $-1260 < p < -1200$

Count



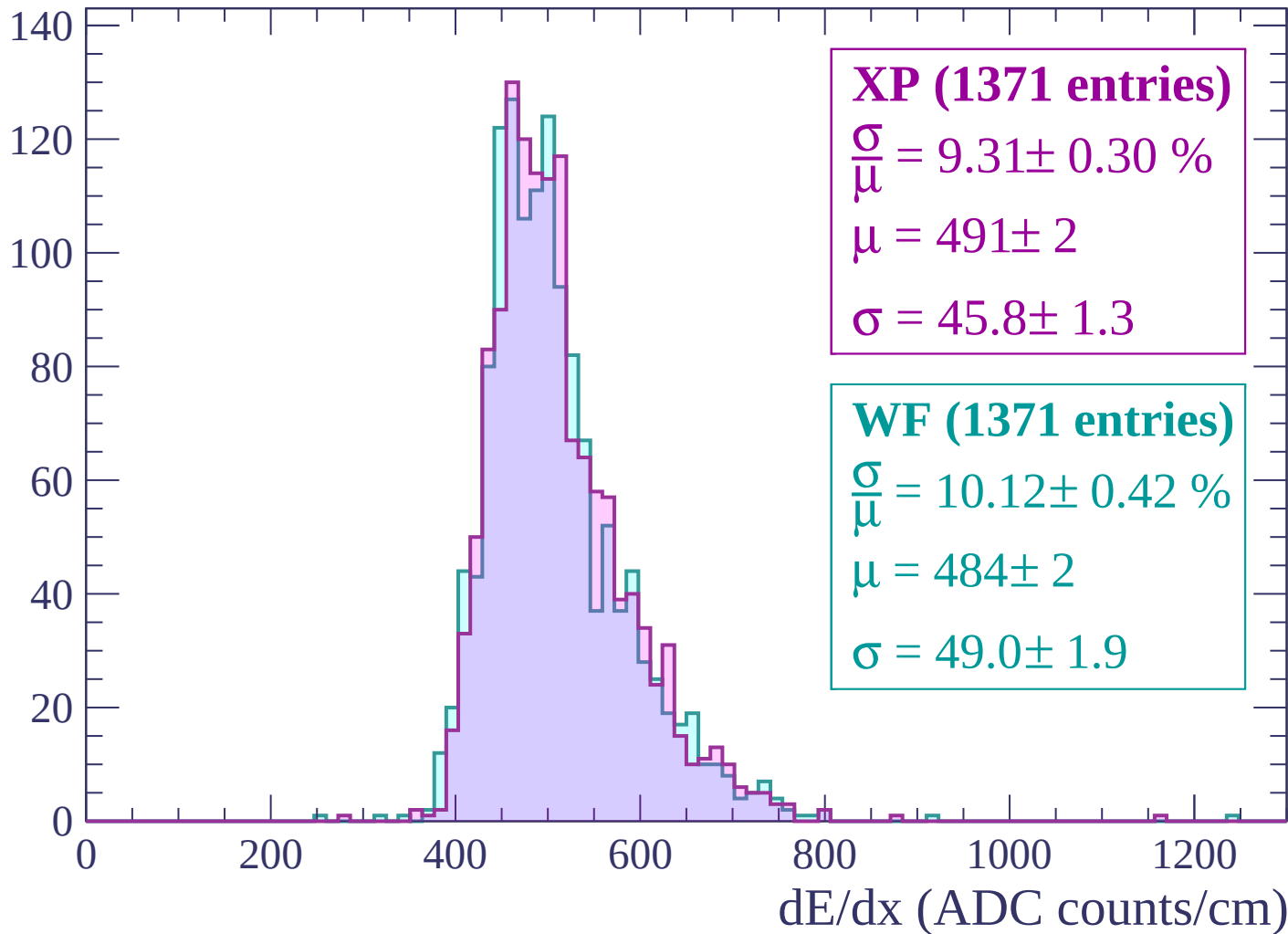
Energy loss | -1200 < p < -1140

Count



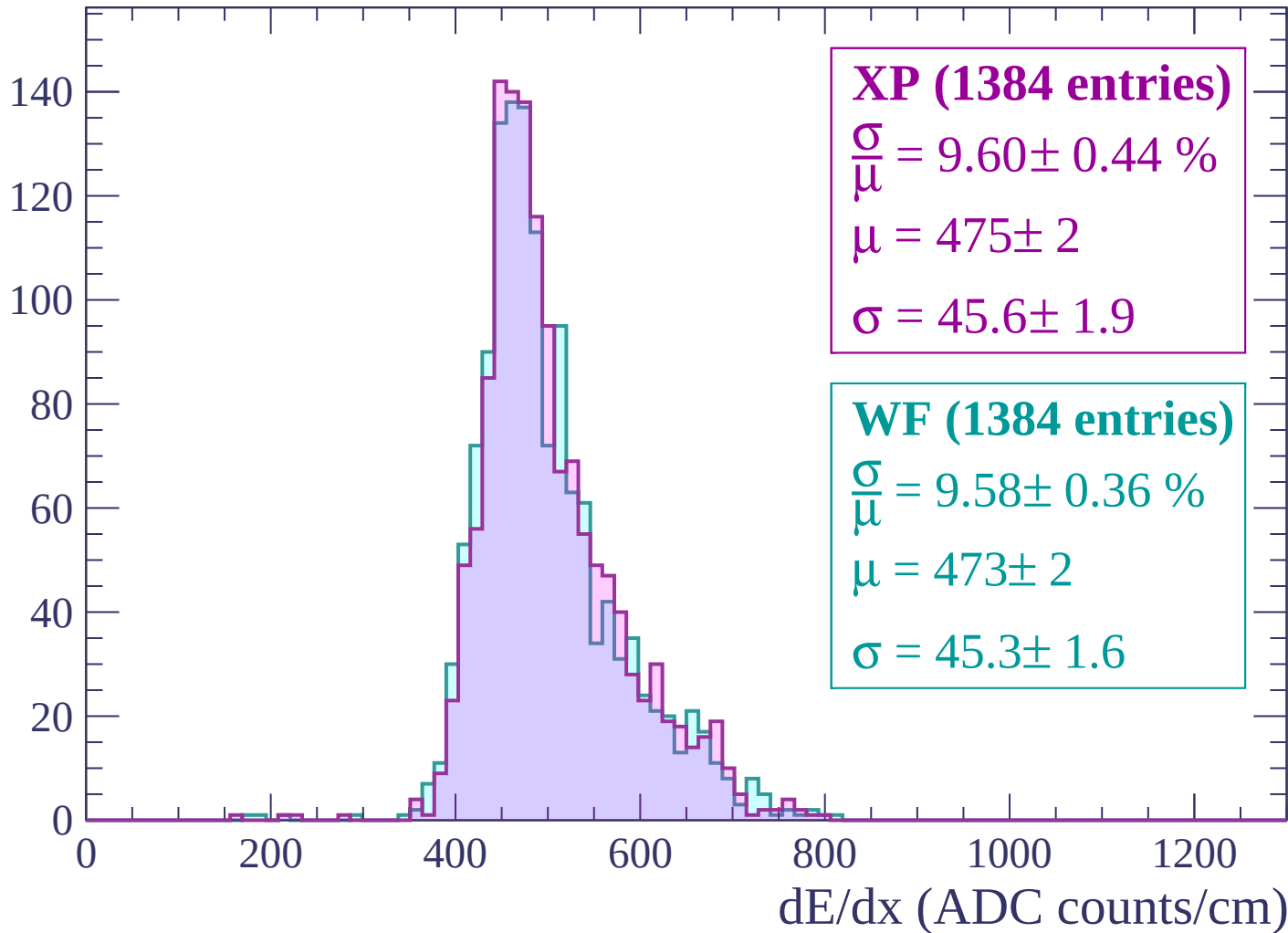
Energy loss | $-1140 < p < -1080$

Count



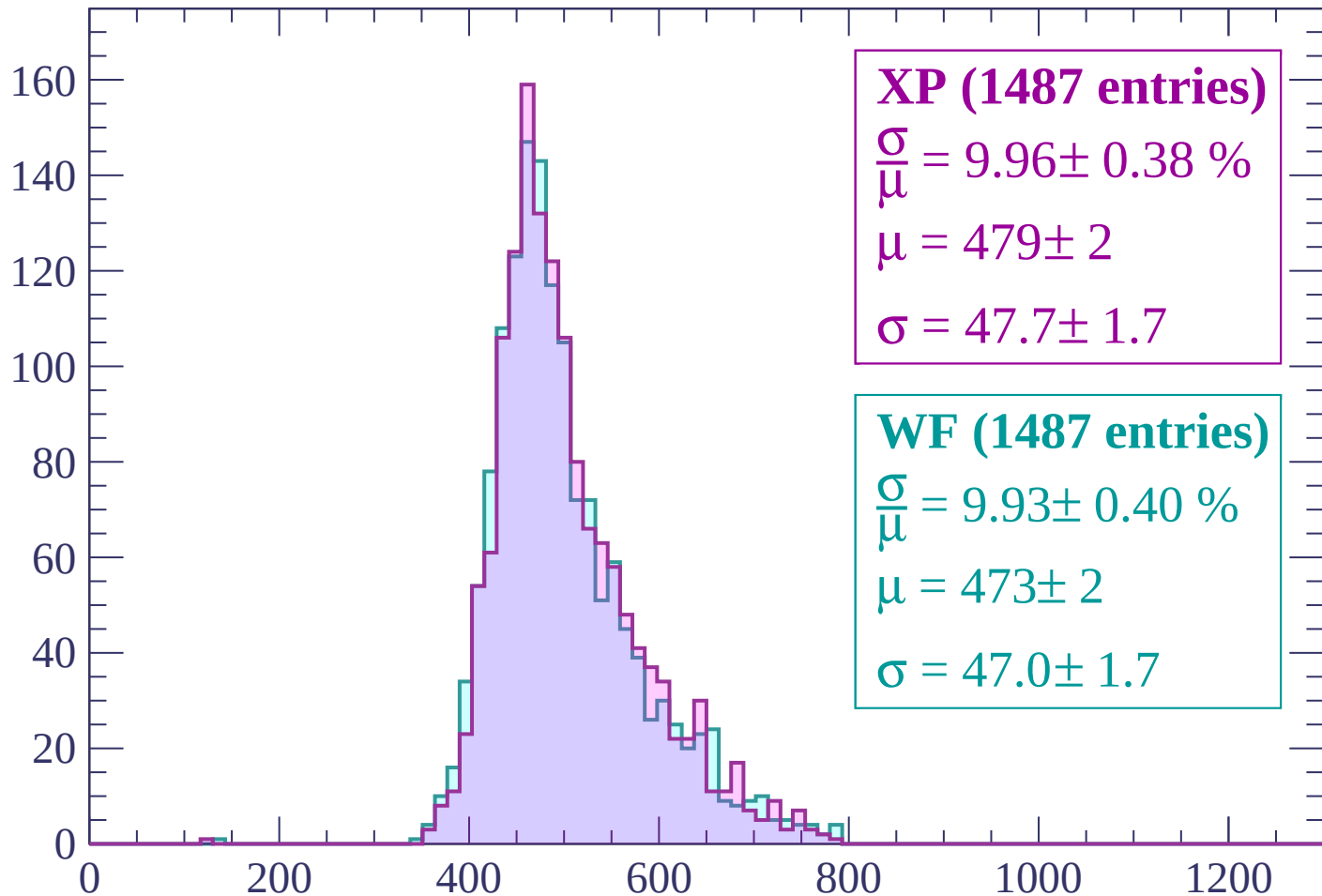
Energy loss | -1080 < p < -1020

Count



Energy loss | $-1020 < p < -960$

Count



XP (1487 entries)

$$\frac{\sigma}{\mu} = 9.96 \pm 0.38 \%$$

$$\mu = 479 \pm 2$$

$$\sigma = 47.7 \pm 1.7$$

WF (1487 entries)

$$\frac{\sigma}{\mu} = 9.93 \pm 0.40 \%$$

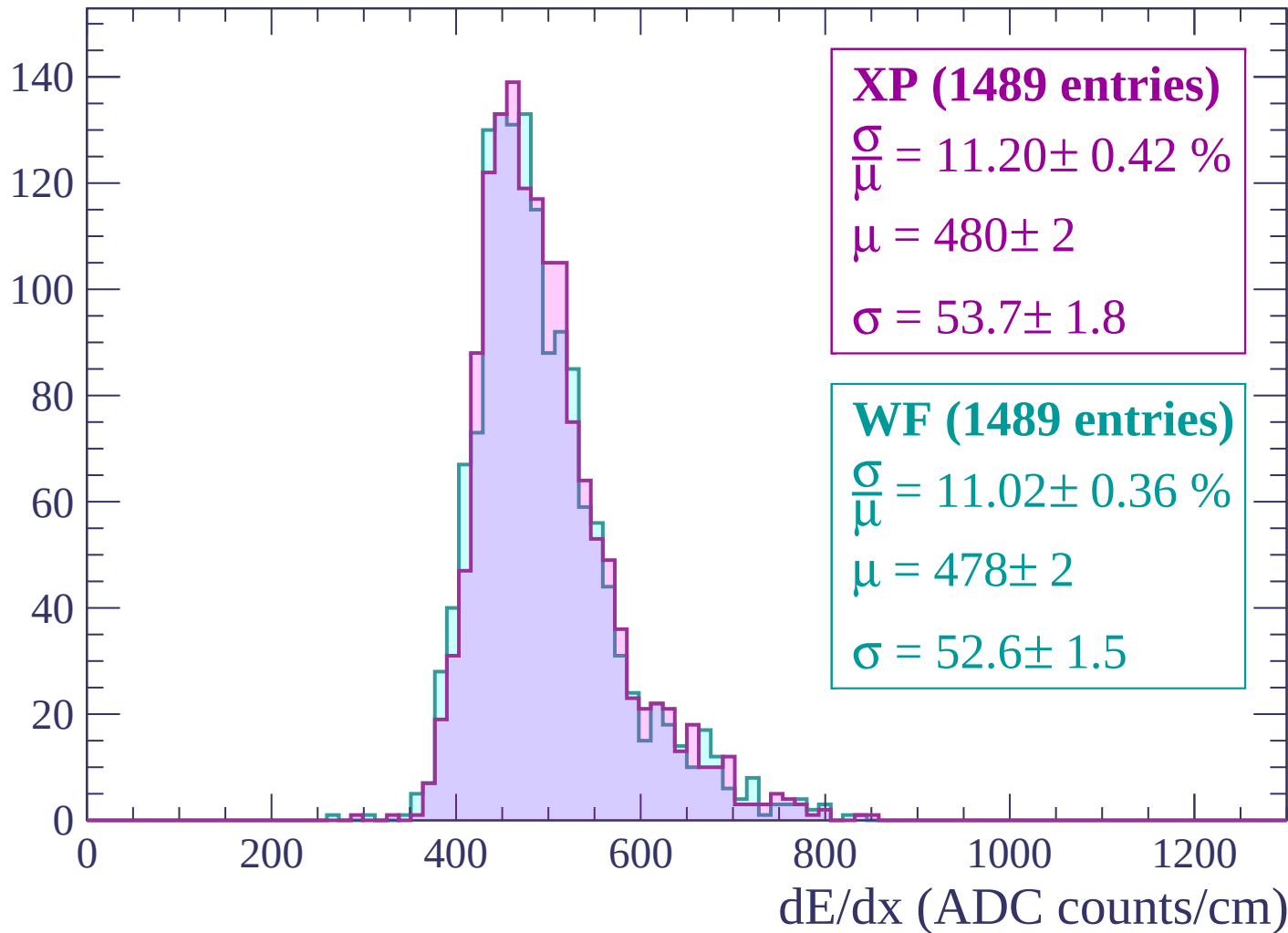
$$\mu = 473 \pm 2$$

$$\sigma = 47.0 \pm 1.7$$

dE/dx (ADC counts/cm)

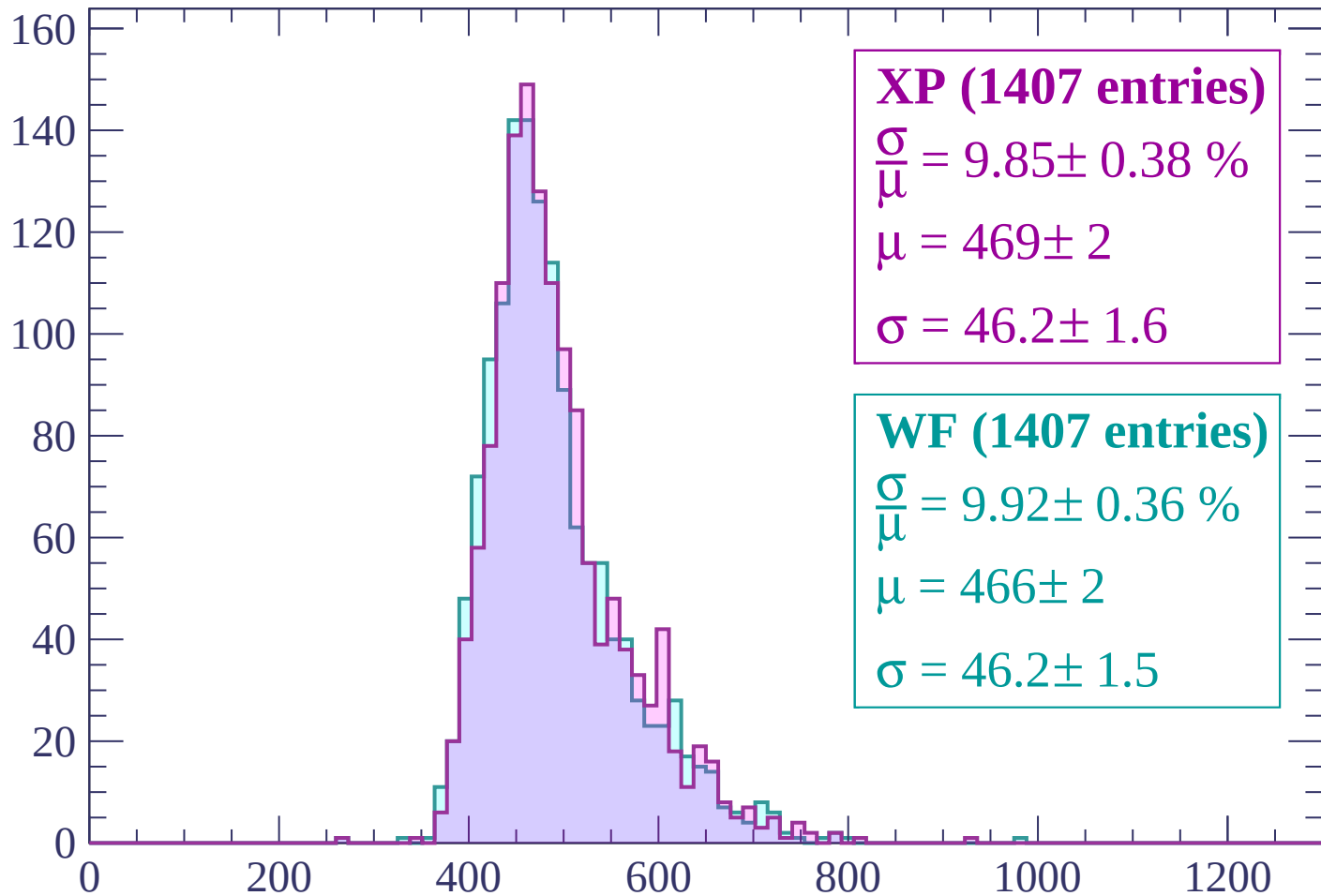
Energy loss | $-960 < p < -900$

Count



Energy loss | $-900 < p < -840$

Count



XP (1407 entries)

$$\frac{\sigma}{\mu} = 9.85 \pm 0.38 \%$$

$$\mu = 469 \pm 2$$

$$\sigma = 46.2 \pm 1.6$$

WF (1407 entries)

$$\frac{\sigma}{\mu} = 9.92 \pm 0.36 \%$$

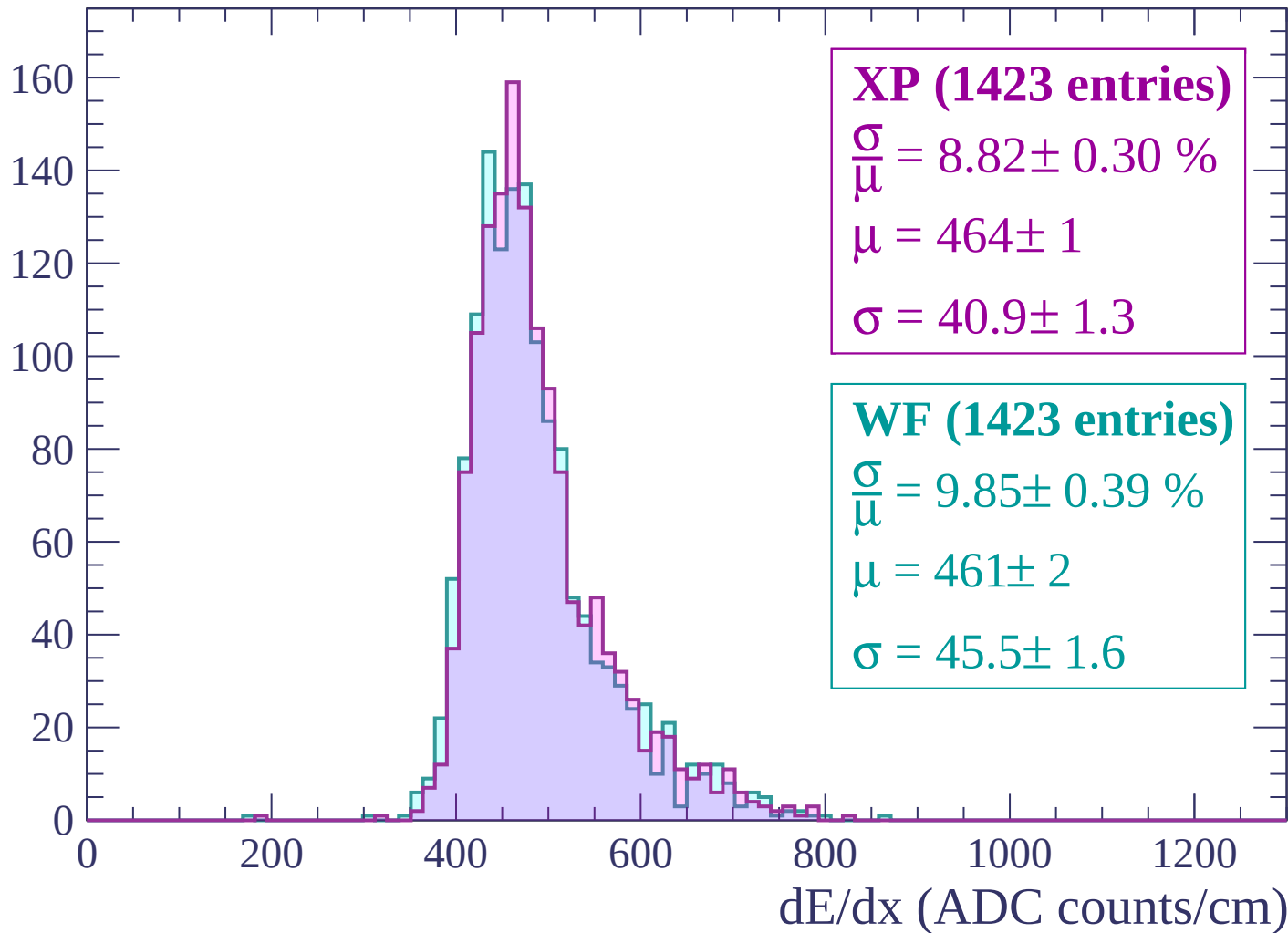
$$\mu = 466 \pm 2$$

$$\sigma = 46.2 \pm 1.5$$

dE/dx (ADC counts/cm)

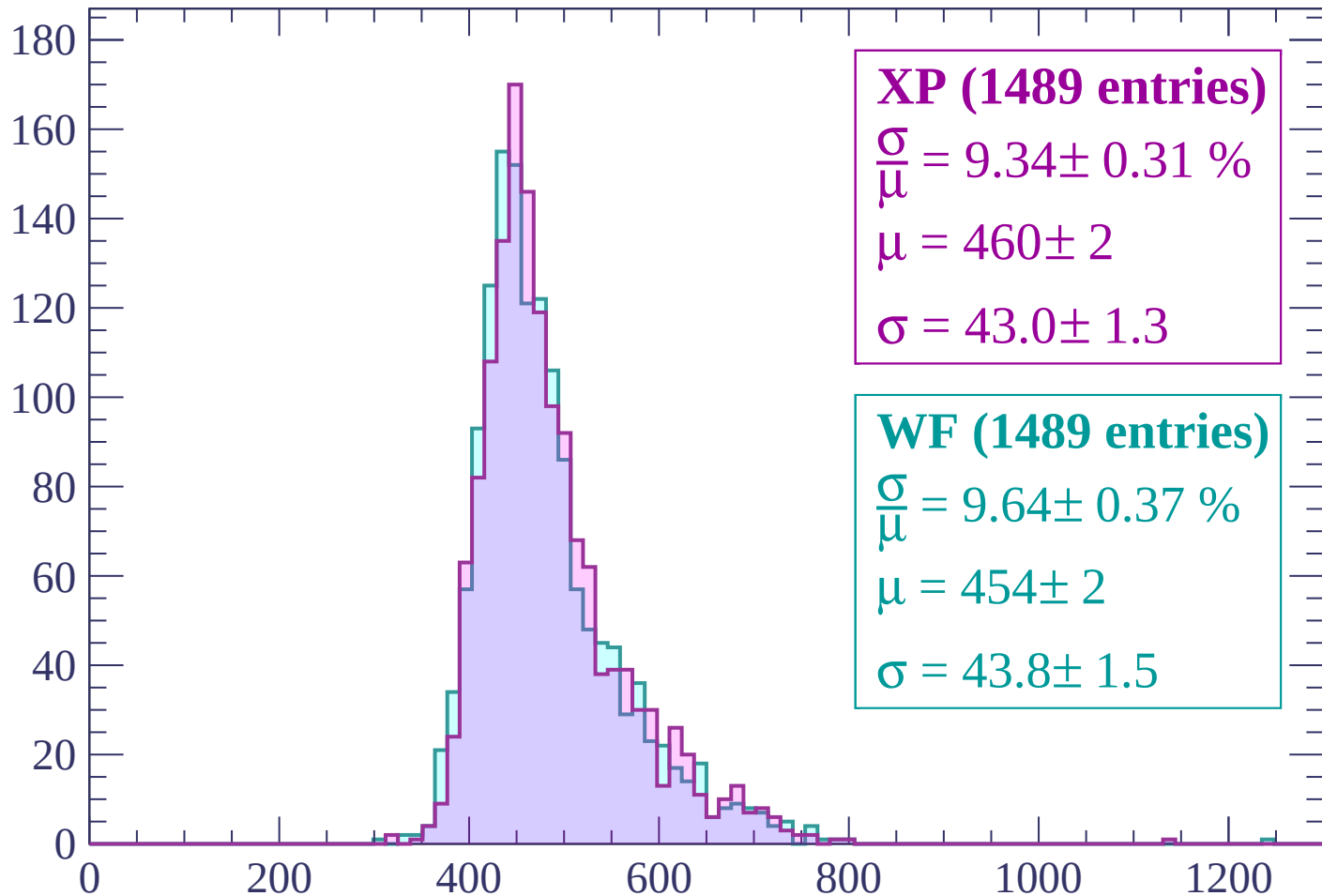
Energy loss | $-840 < p < -780$

Count



Energy loss | $-780 < p < -720$

Count



XP (1489 entries)

$$\frac{\sigma}{\mu} = 9.34 \pm 0.31 \%$$

$$\mu = 460 \pm 2$$

$$\sigma = 43.0 \pm 1.3$$

WF (1489 entries)

$$\frac{\sigma}{\mu} = 9.64 \pm 0.37 \%$$

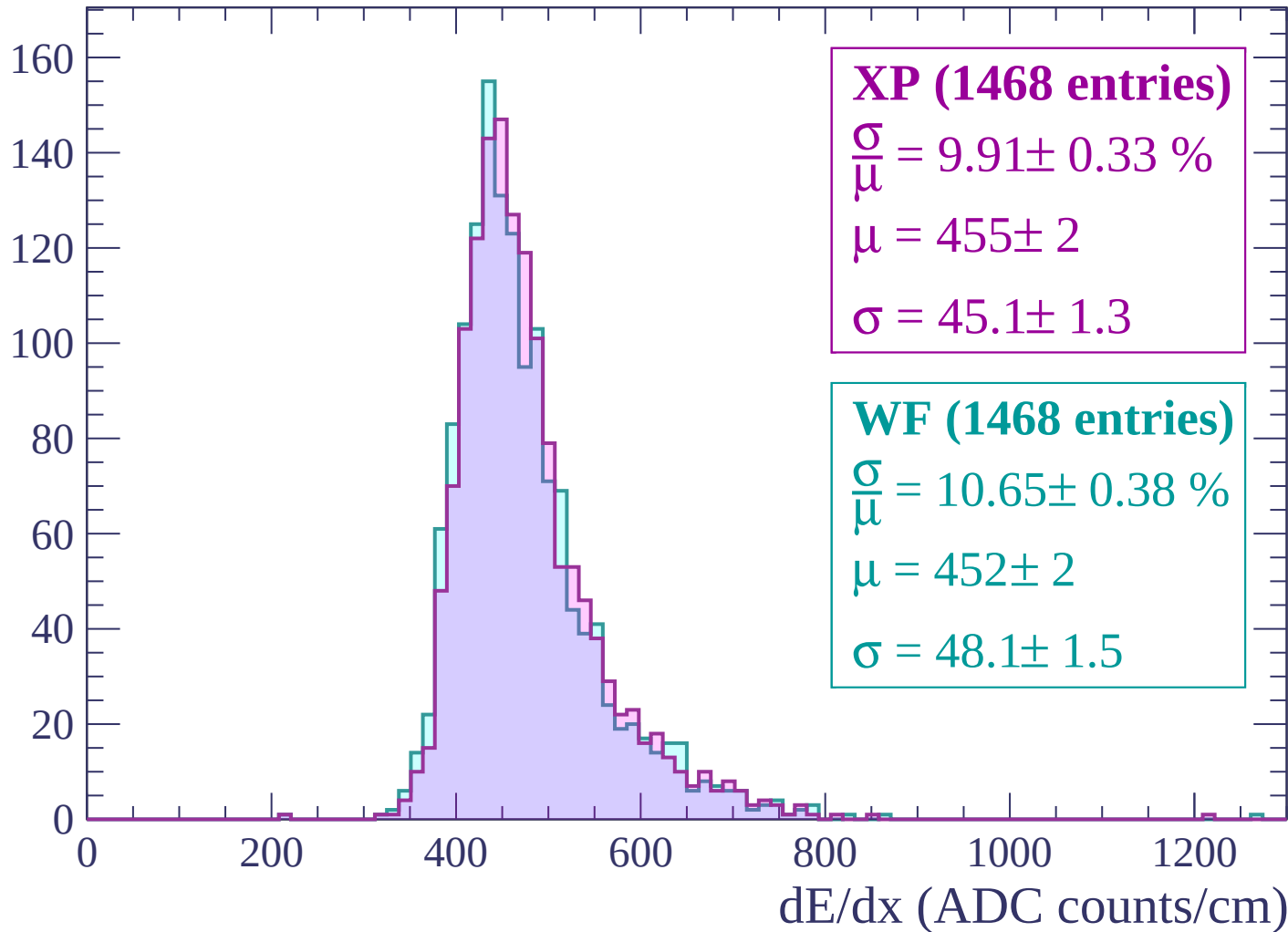
$$\mu = 454 \pm 2$$

$$\sigma = 43.8 \pm 1.5$$

dE/dx (ADC counts/cm)

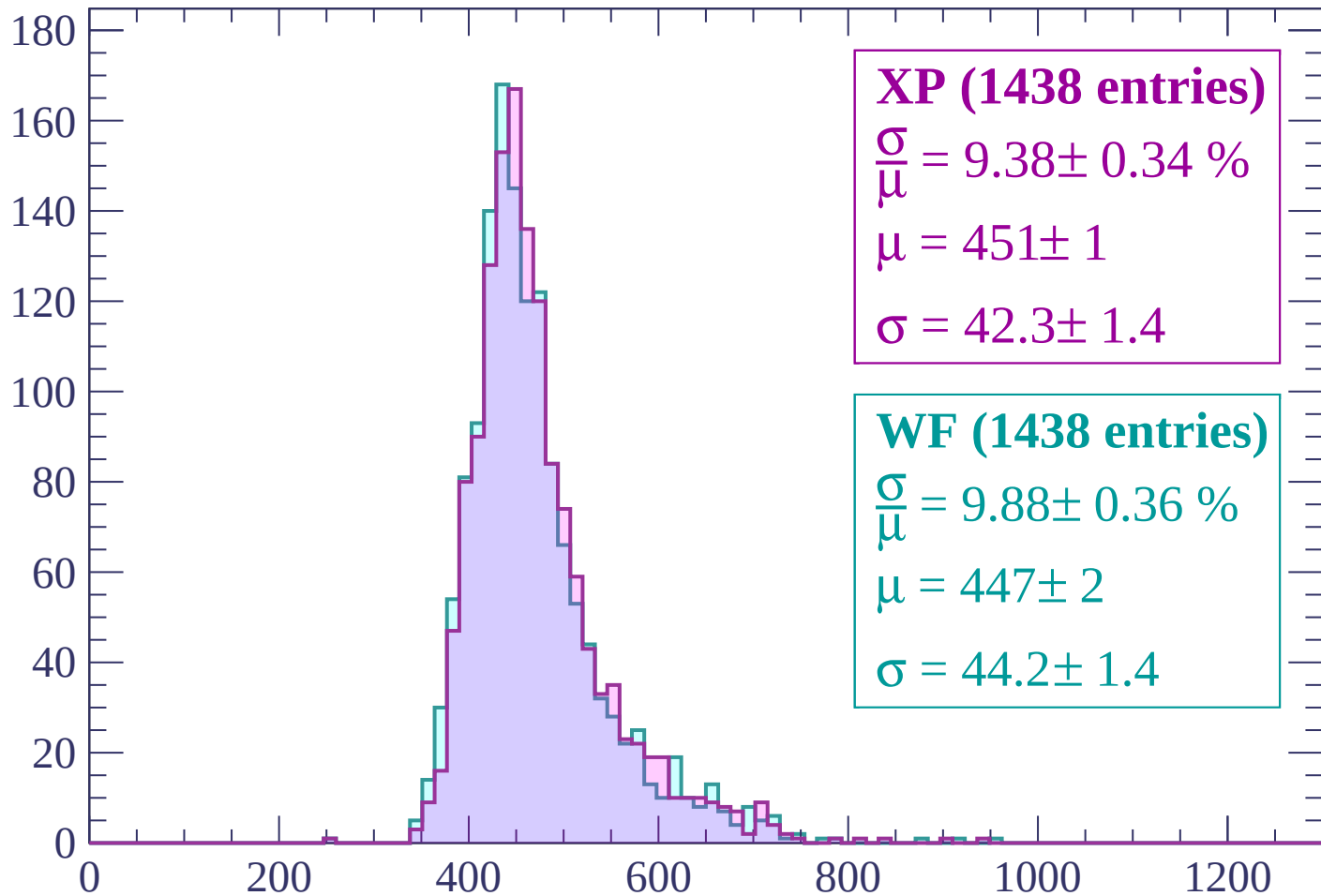
Energy loss | $-720 < p < -660$

Count



Energy loss | $-660 < p < -600$

Count



XP (1438 entries)

$\frac{\sigma}{\mu} = 9.38 \pm 0.34 \%$

$\mu = 451 \pm 1$

$\sigma = 42.3 \pm 1.4$

WF (1438 entries)

$\frac{\sigma}{\mu} = 9.88 \pm 0.36 \%$

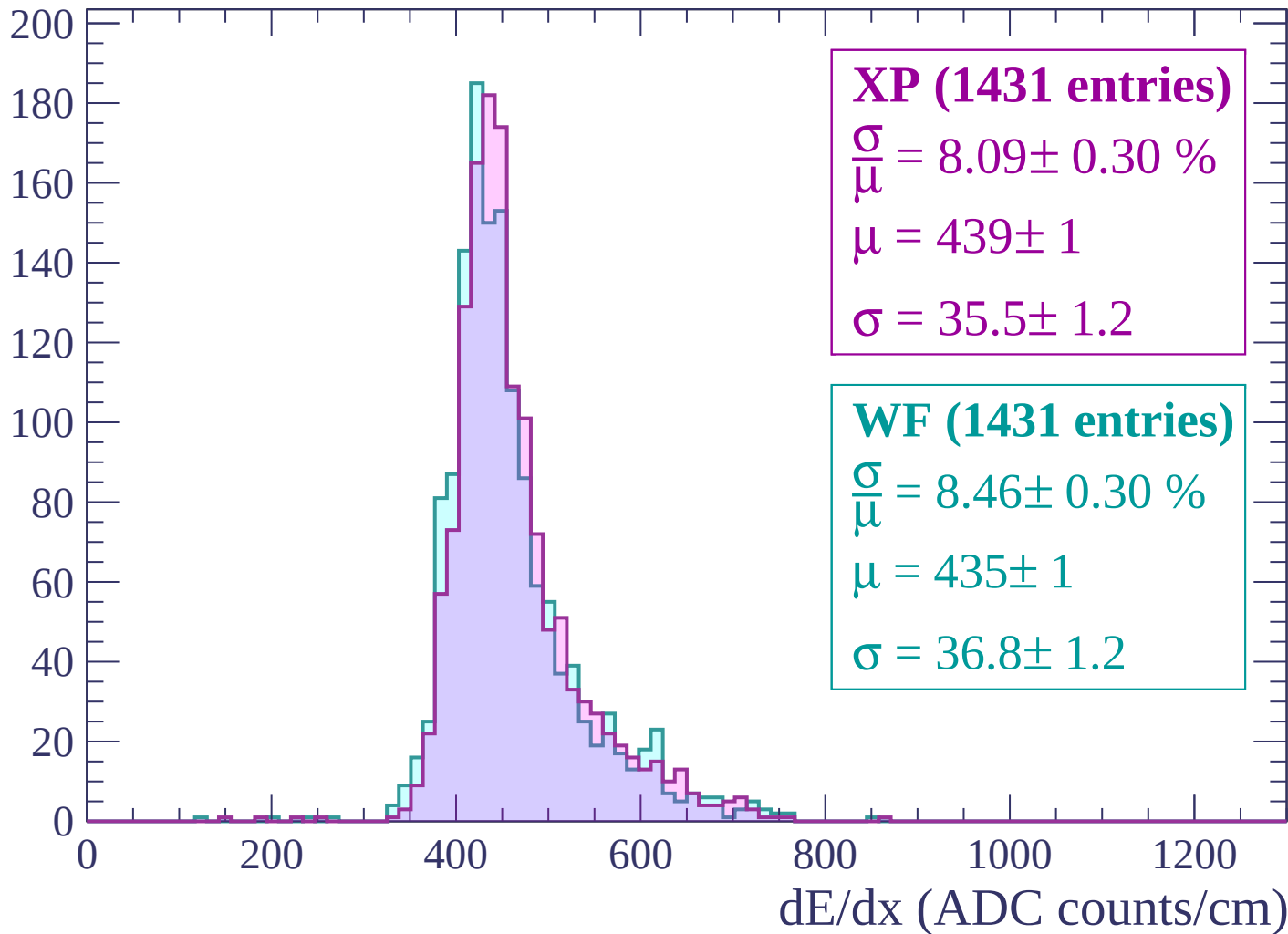
$\mu = 447 \pm 2$

$\sigma = 44.2 \pm 1.4$

dE/dx (ADC counts/cm)

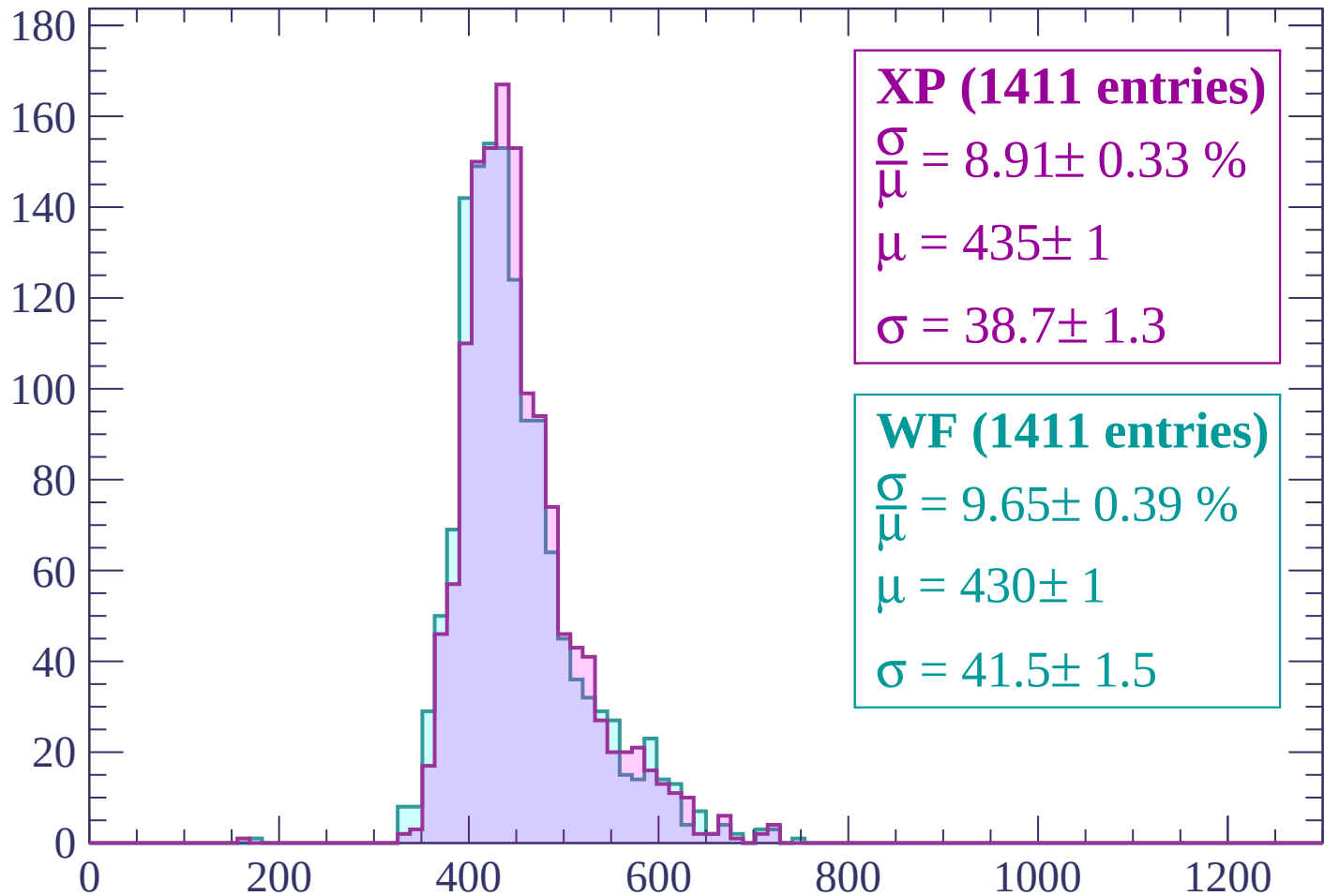
Energy loss | $-600 < p < -540$

Count



Energy loss | $-540 < p < -480$

Count



XP (1411 entries)

$$\frac{\sigma}{\mu} = 8.91 \pm 0.33 \%$$

$$\mu = 435 \pm 1$$

$$\sigma = 38.7 \pm 1.3$$

WF (1411 entries)

$$\frac{\sigma}{\mu} = 9.65 \pm 0.39 \%$$

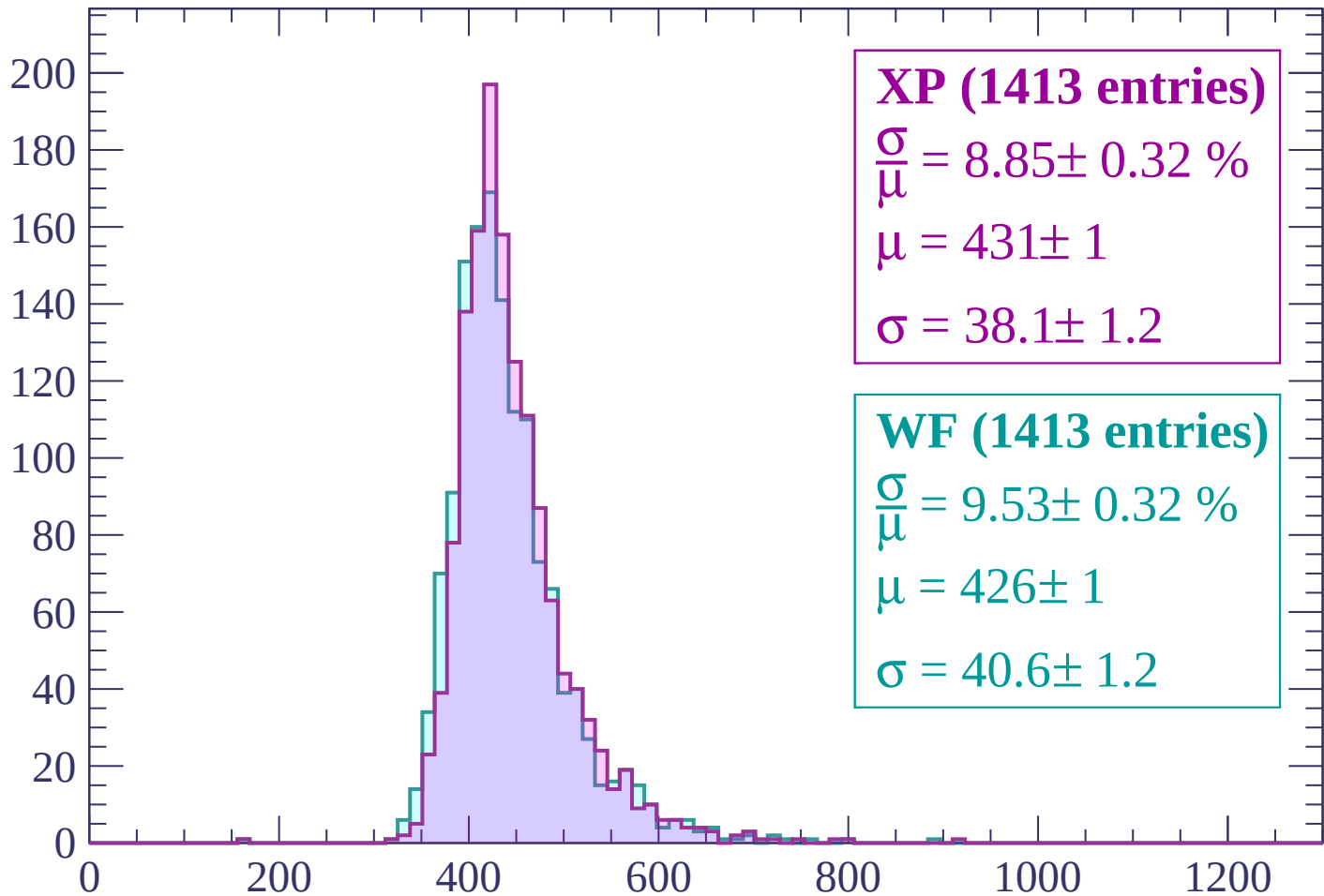
$$\mu = 430 \pm 1$$

$$\sigma = 41.5 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -420$

Count



XP (1413 entries)

$$\frac{\sigma}{\mu} = 8.85 \pm 0.32 \%$$

$$\mu = 431 \pm 1$$

$$\sigma = 38.1 \pm 1.2$$

WF (1413 entries)

$$\frac{\sigma}{\mu} = 9.53 \pm 0.32 \%$$

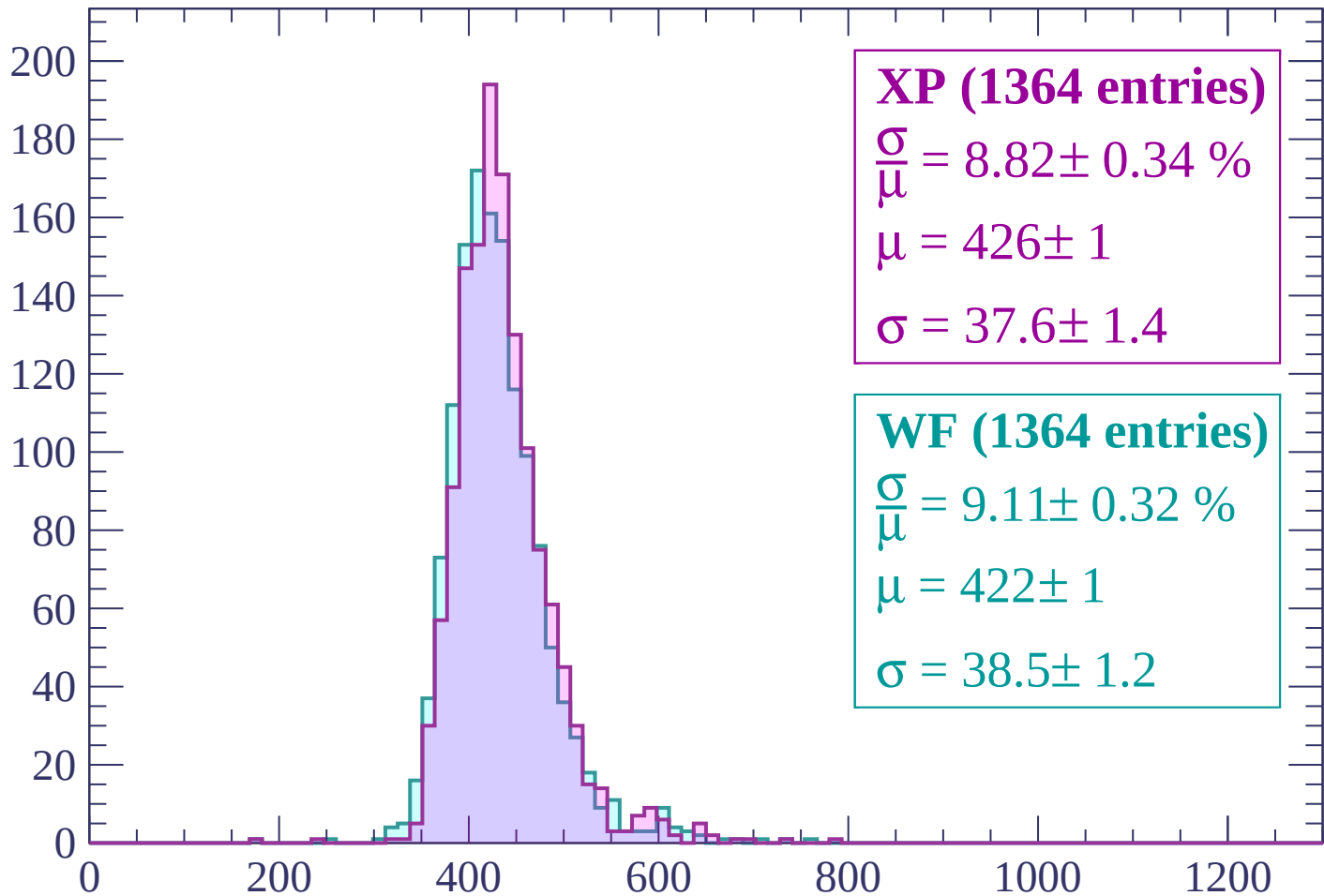
$$\mu = 426 \pm 1$$

$$\sigma = 40.6 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-420 < p < -360$

Count



XP (1364 entries)

$\frac{\sigma}{\mu} = 8.82 \pm 0.34 \%$

$\mu = 426 \pm 1$

$\sigma = 37.6 \pm 1.4$

WF (1364 entries)

$\frac{\sigma}{\mu} = 9.11 \pm 0.32 \%$

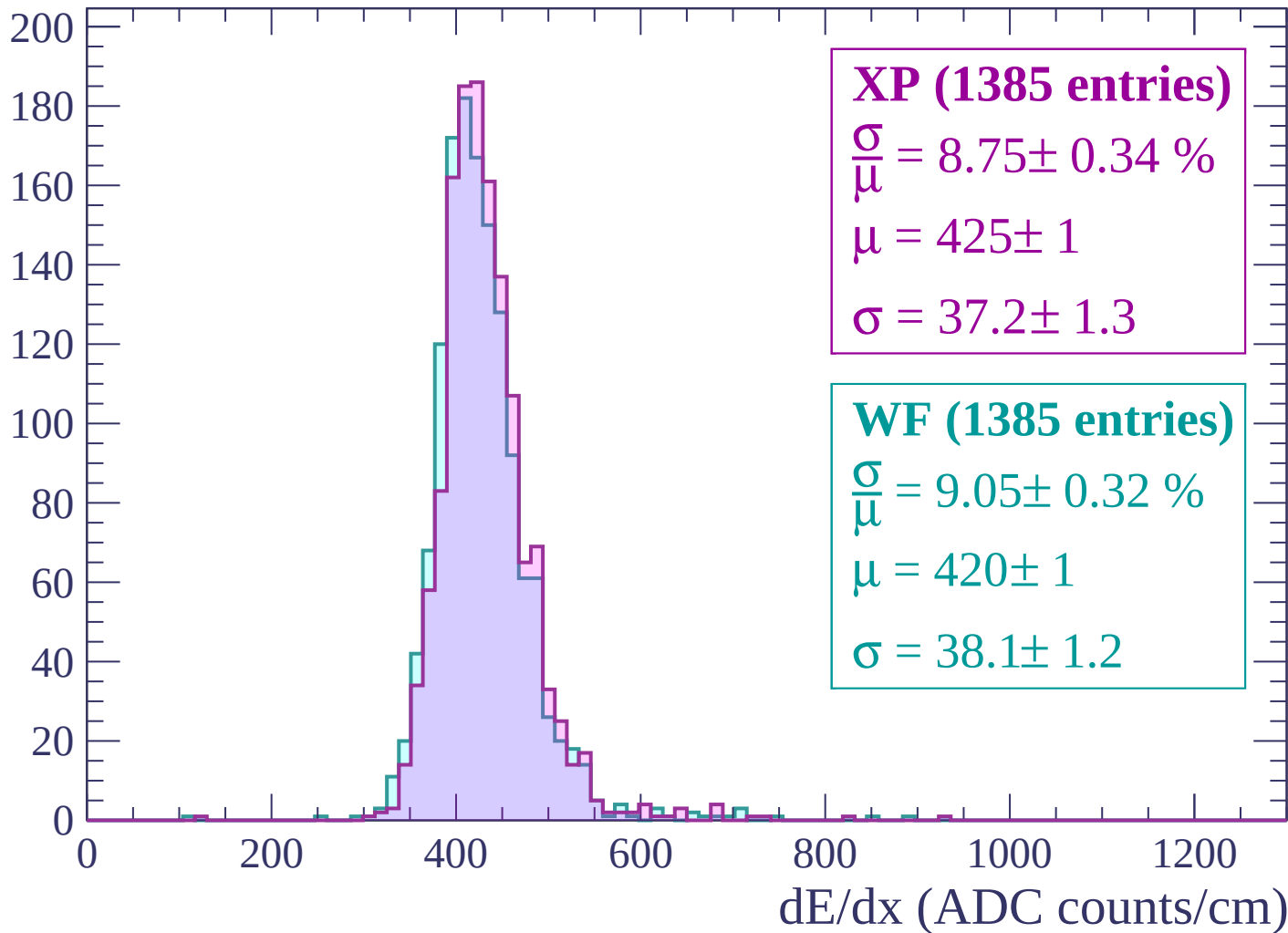
$\mu = 422 \pm 1$

$\sigma = 38.5 \pm 1.2$

dE/dx (ADC counts/cm)

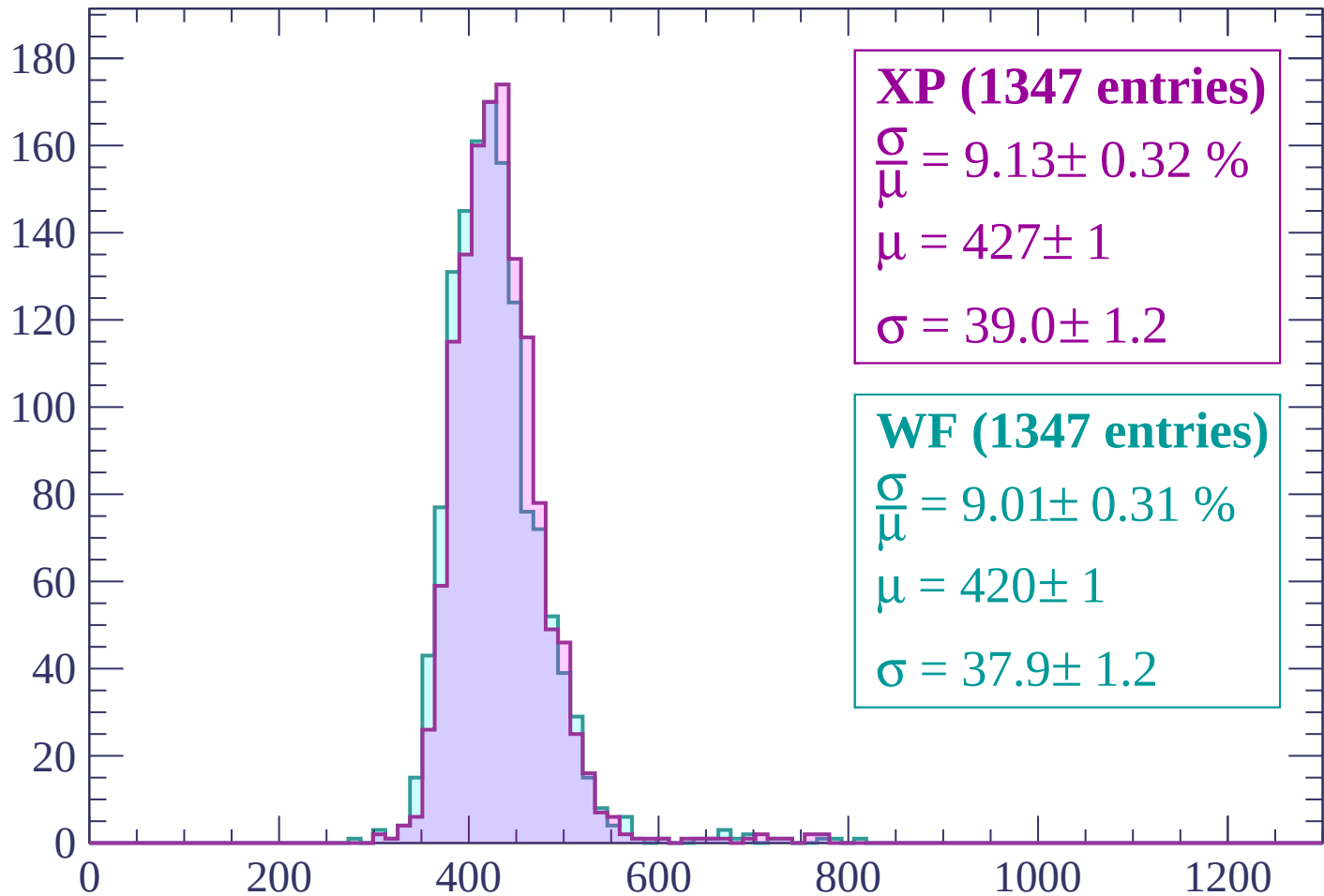
Energy loss | $-360 < p < -300$

Count



Energy loss | $-300 < p < -240$

Count



XP (1347 entries)

$\frac{\sigma}{\mu} = 9.13 \pm 0.32 \%$

$\mu = 427 \pm 1$

$\sigma = 39.0 \pm 1.2$

WF (1347 entries)

$\frac{\sigma}{\mu} = 9.01 \pm 0.31 \%$

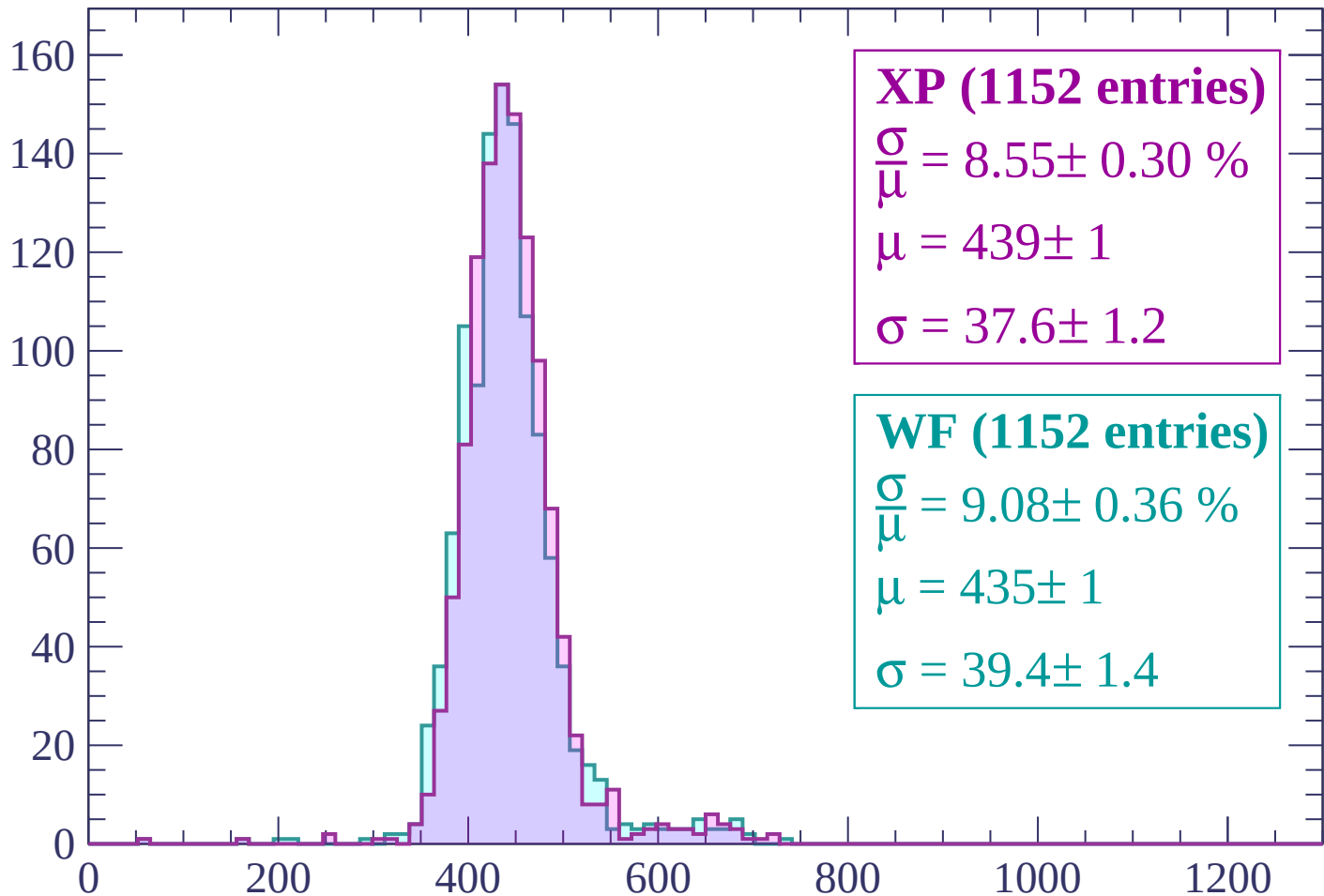
$\mu = 420 \pm 1$

$\sigma = 37.9 \pm 1.2$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count



XP (1152 entries)

$\frac{\sigma}{\mu} = 8.55 \pm 0.30 \%$

$\mu = 439 \pm 1$

$\sigma = 37.6 \pm 1.2$

WF (1152 entries)

$\frac{\sigma}{\mu} = 9.08 \pm 0.36 \%$

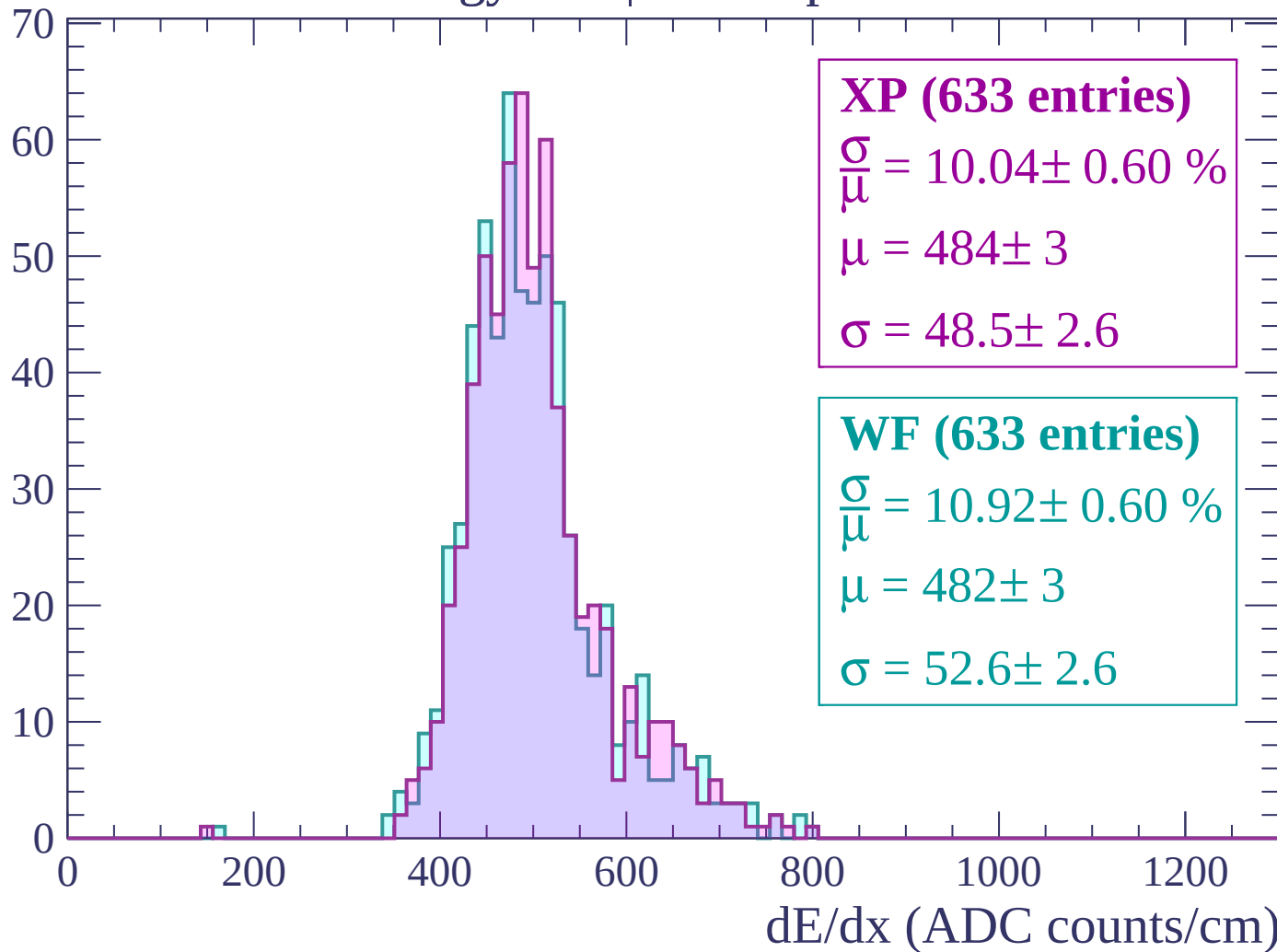
$\mu = 435 \pm 1$

$\sigma = 39.4 \pm 1.4$

dE/dx (ADC counts/cm)

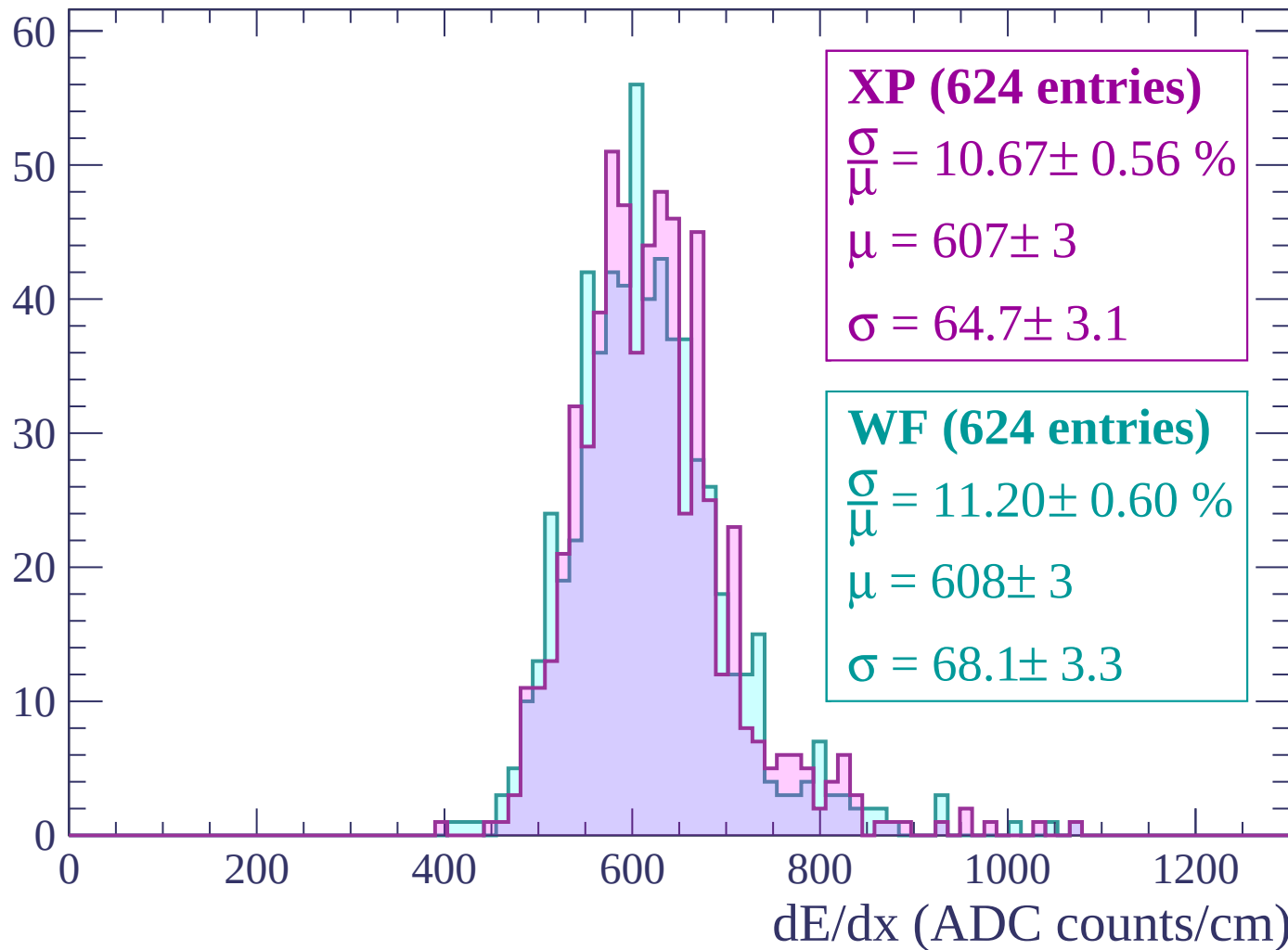
Energy loss | -180 < p < -120

Count



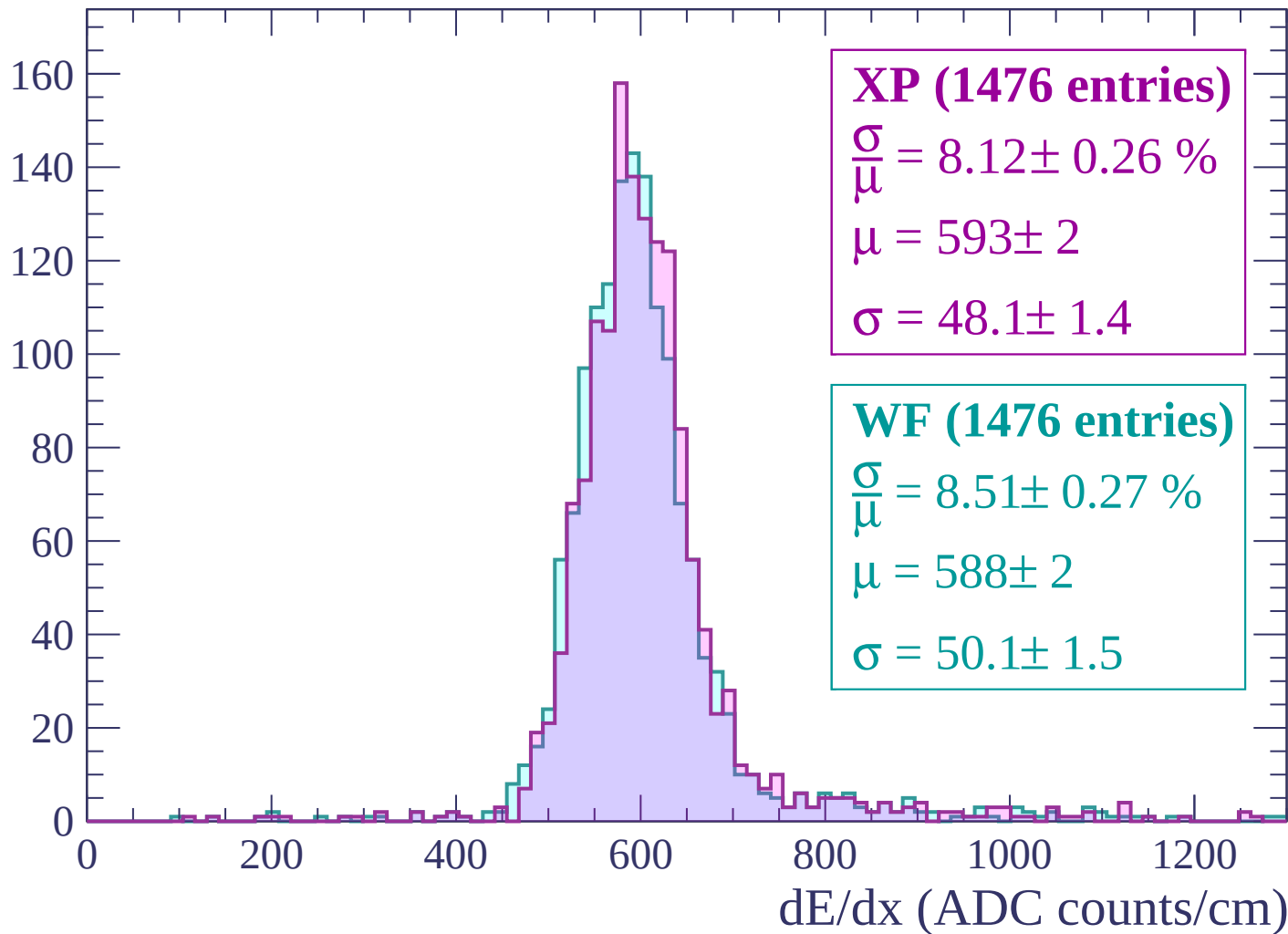
Energy loss | $-120 < p < -60$

Count



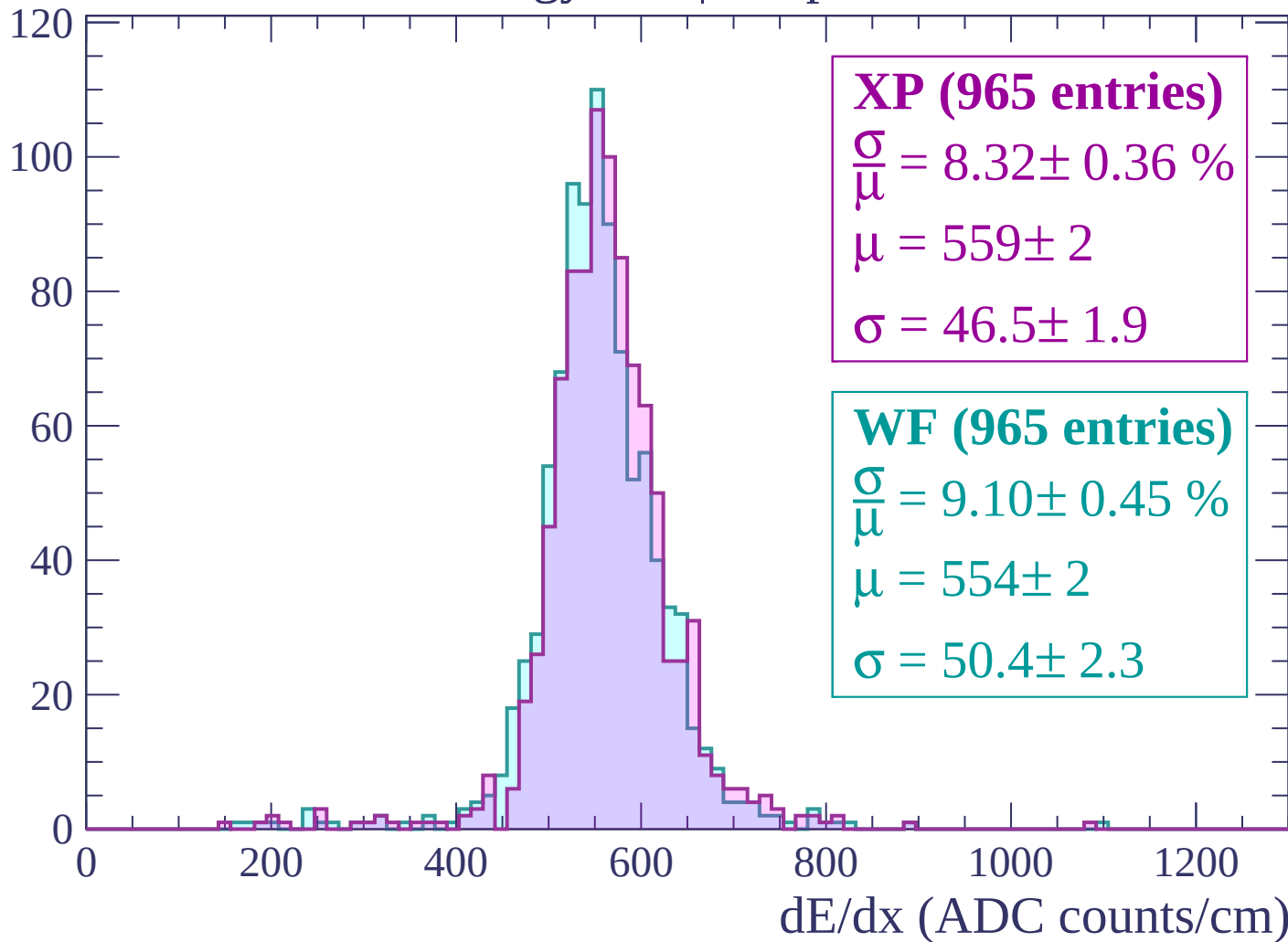
Energy loss | $-60 < p < 0$

Count



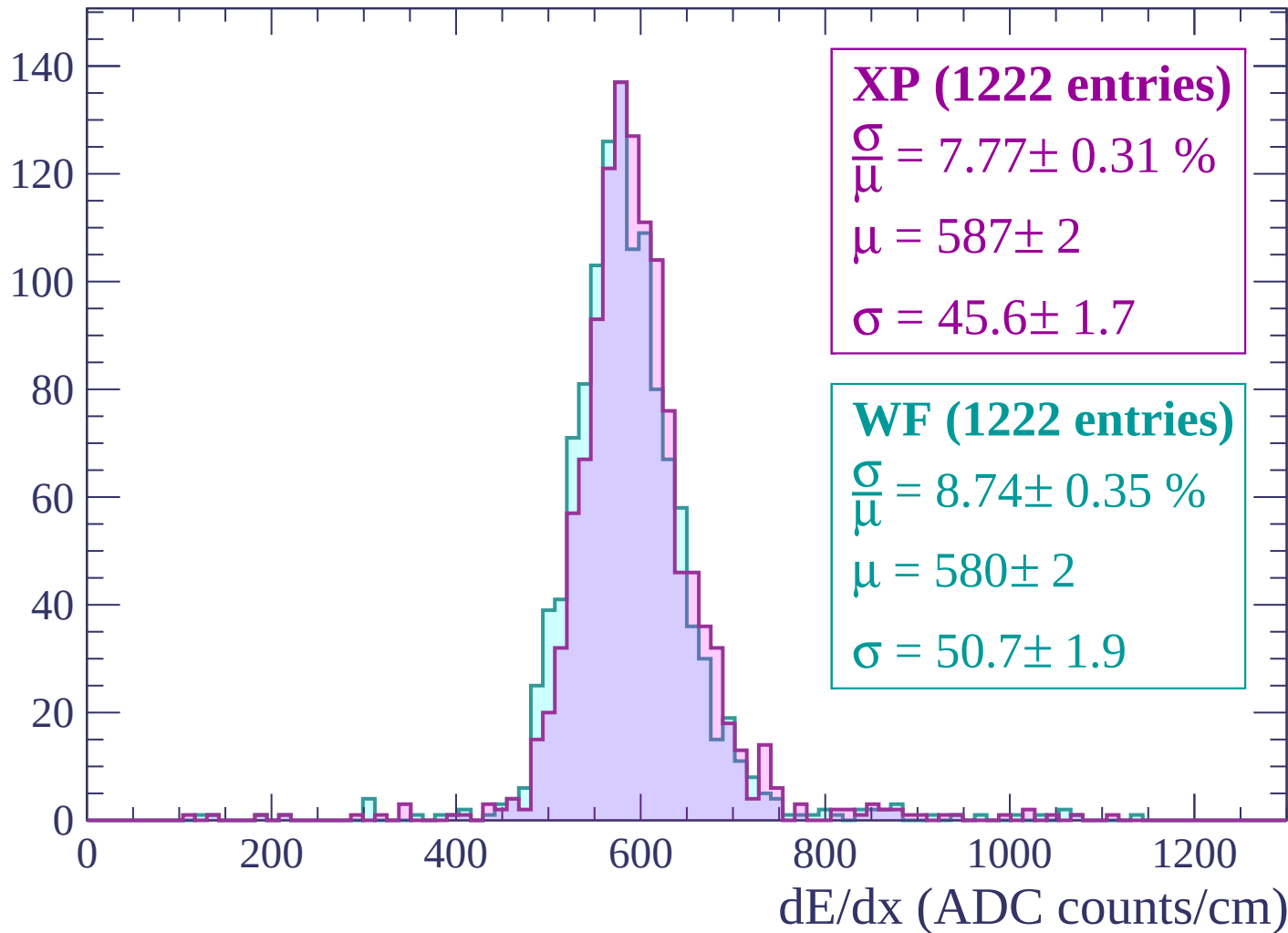
Energy loss | $0 < p < 60$

Count



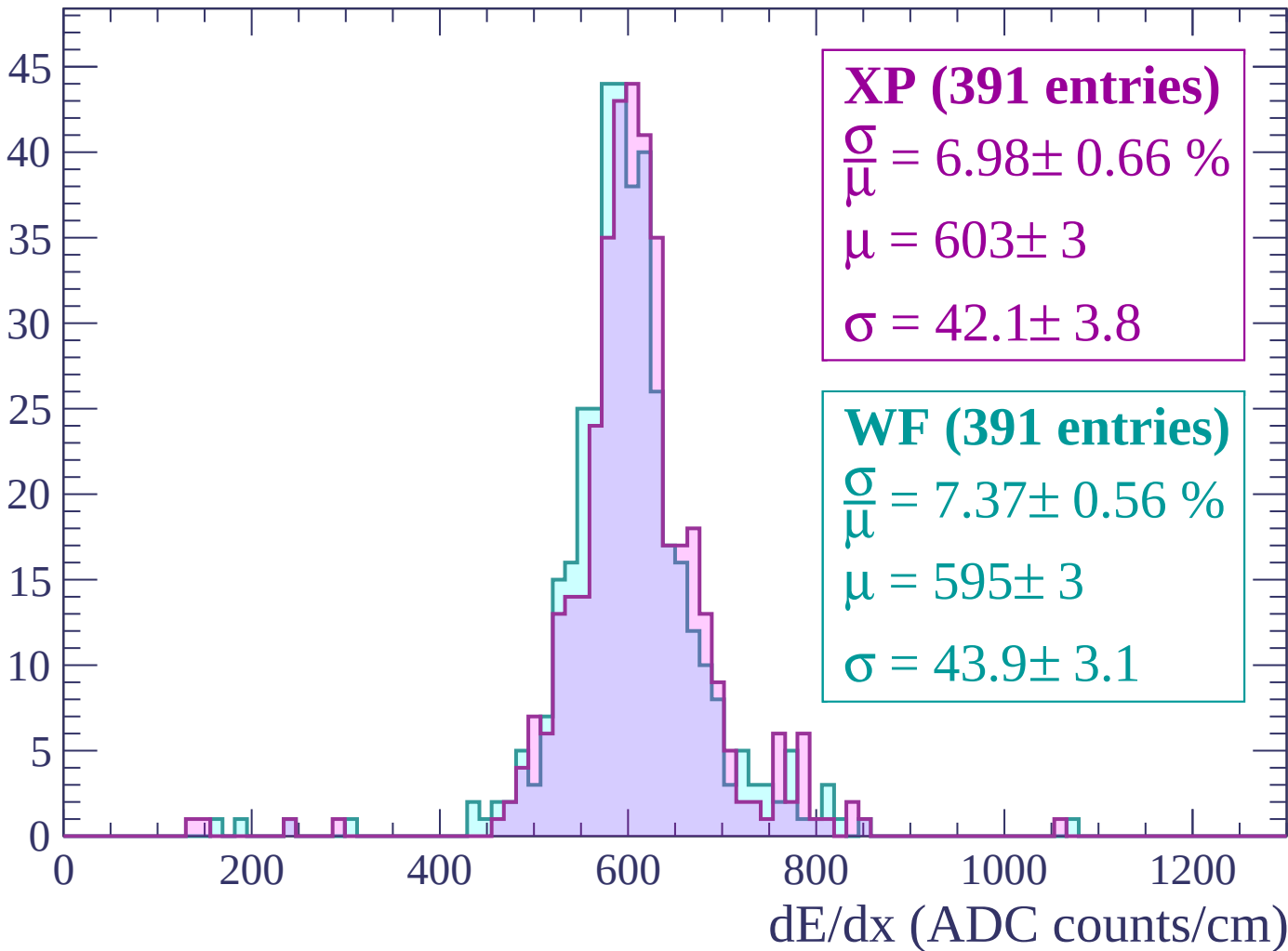
Energy loss | $60 < p < 120$

Count



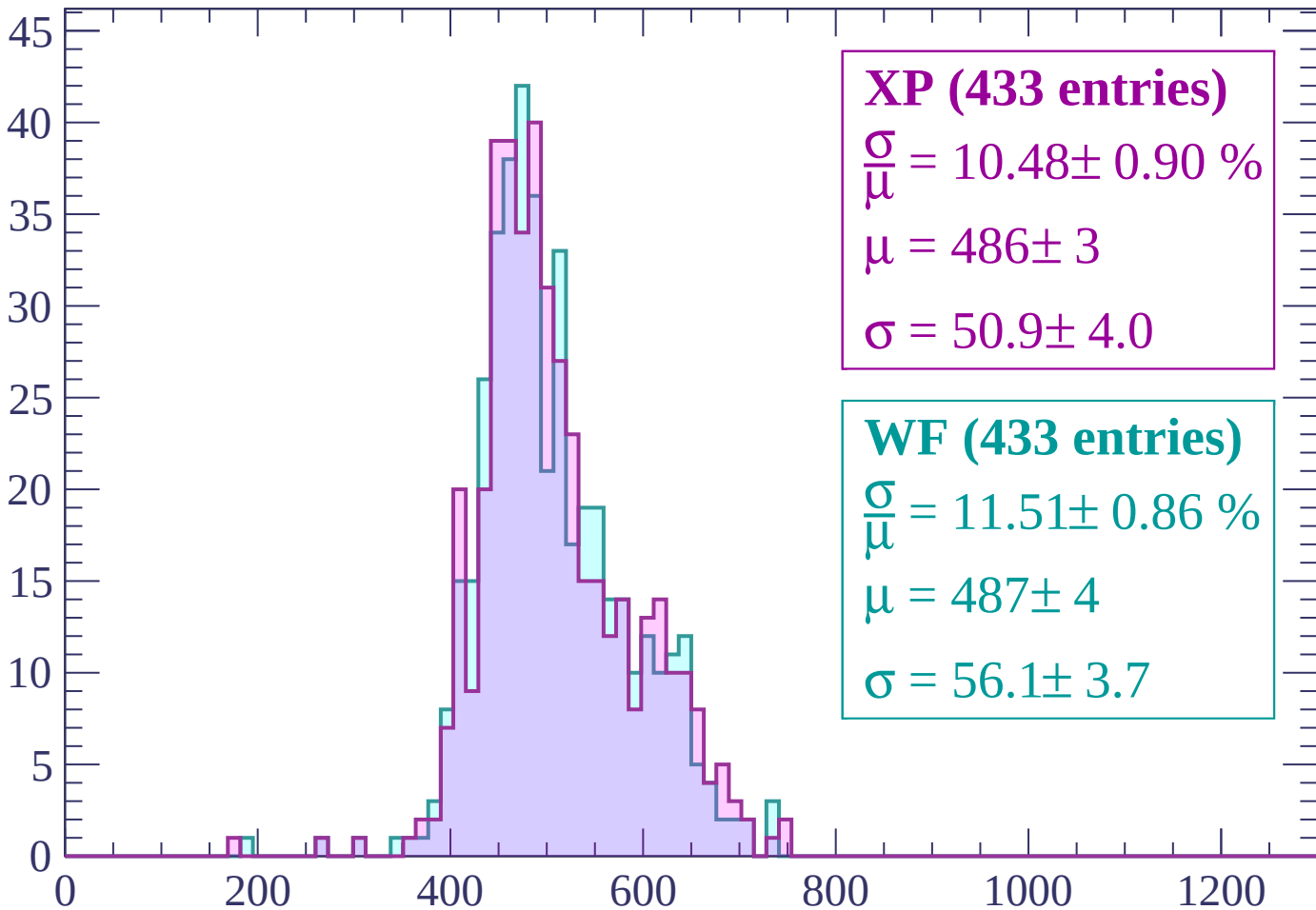
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP (433 entries)

$$\frac{\sigma}{\mu} = 10.48 \pm 0.90 \%$$

$$\mu = 486 \pm 3$$

$$\sigma = 50.9 \pm 4.0$$

WF (433 entries)

$$\frac{\sigma}{\mu} = 11.51 \pm 0.86 \%$$

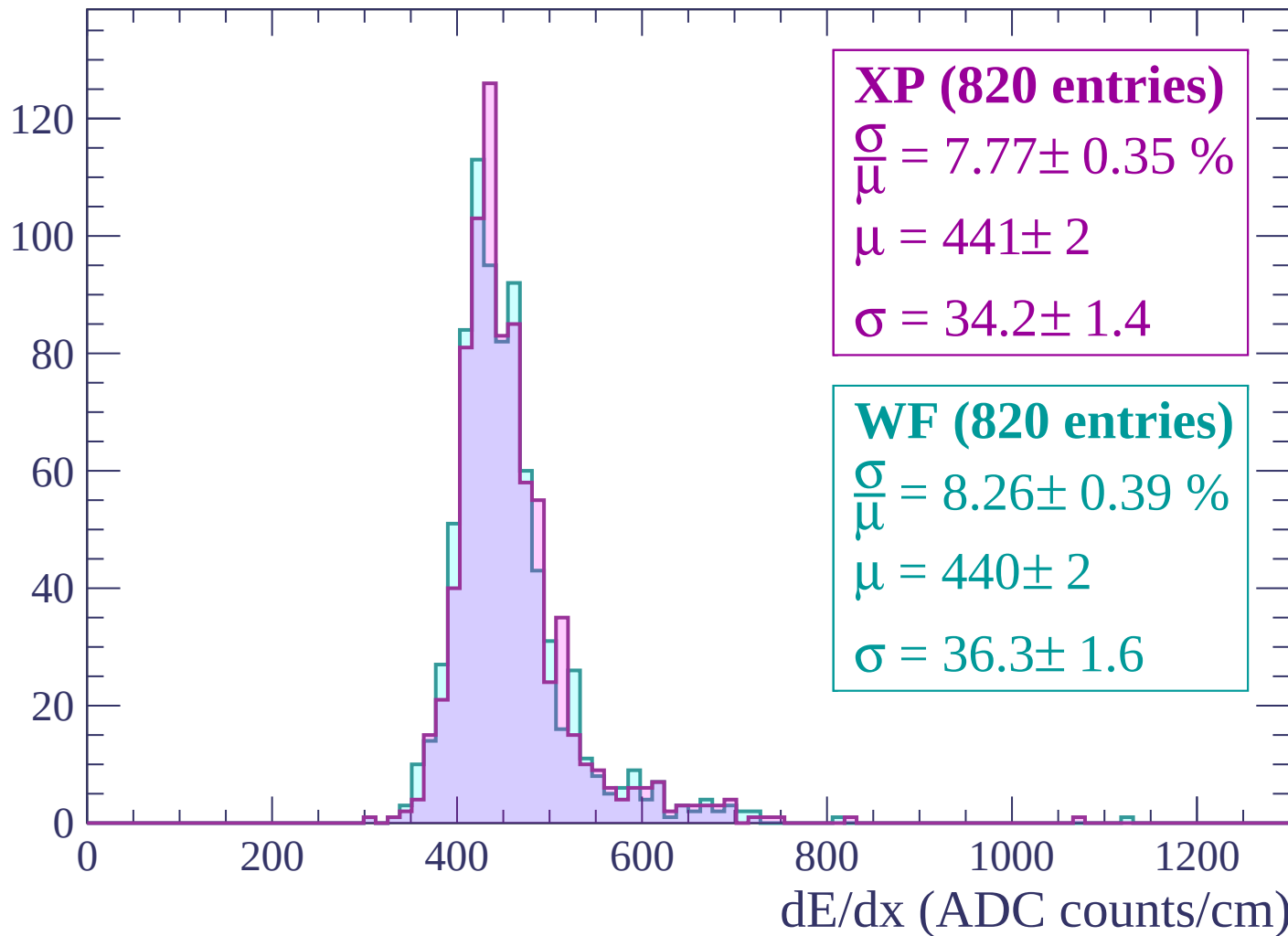
$$\mu = 487 \pm 4$$

$$\sigma = 56.1 \pm 3.7$$

dE/dx (ADC counts/cm)

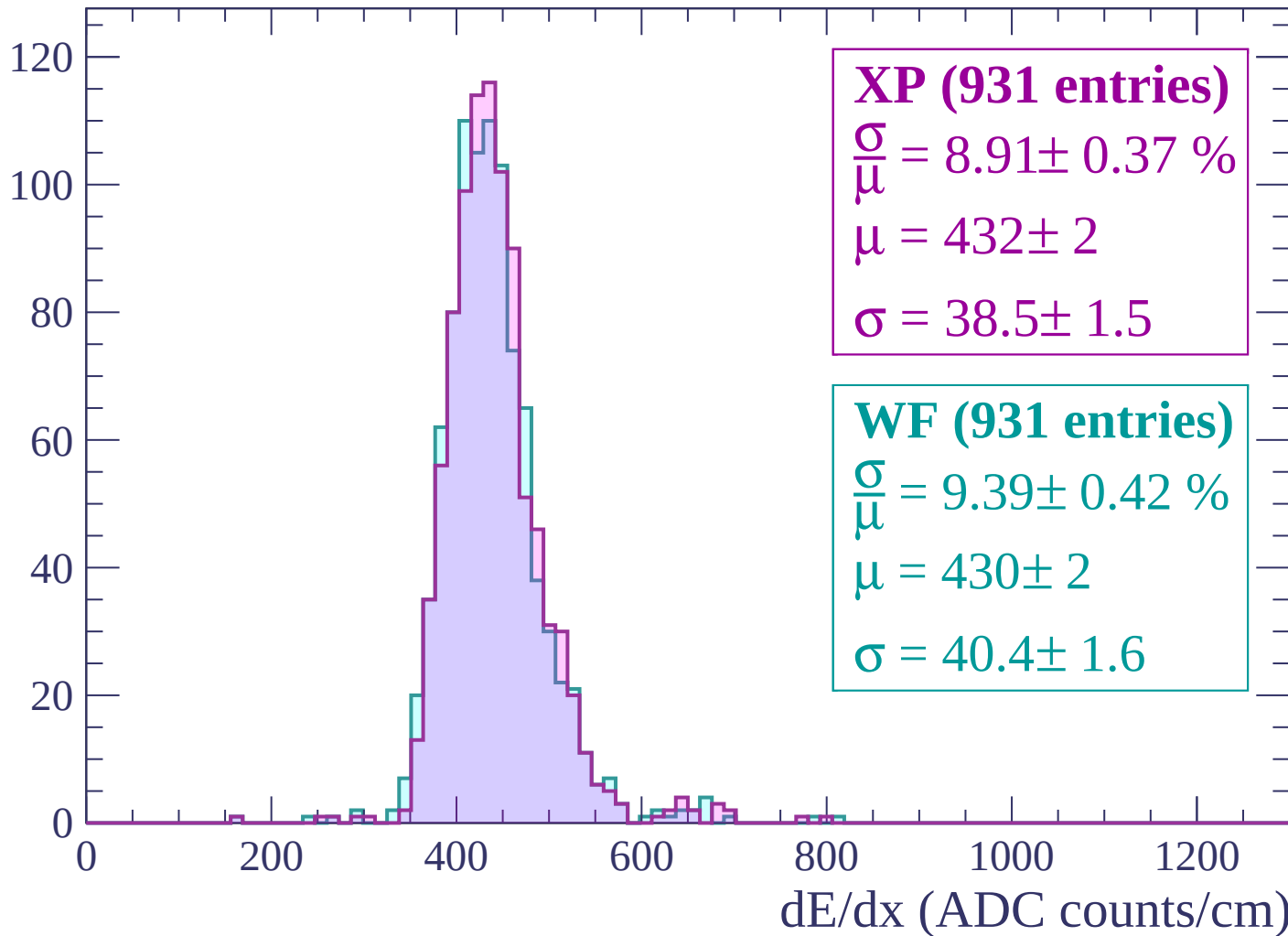
Energy loss | $240 < p < 300$

Count



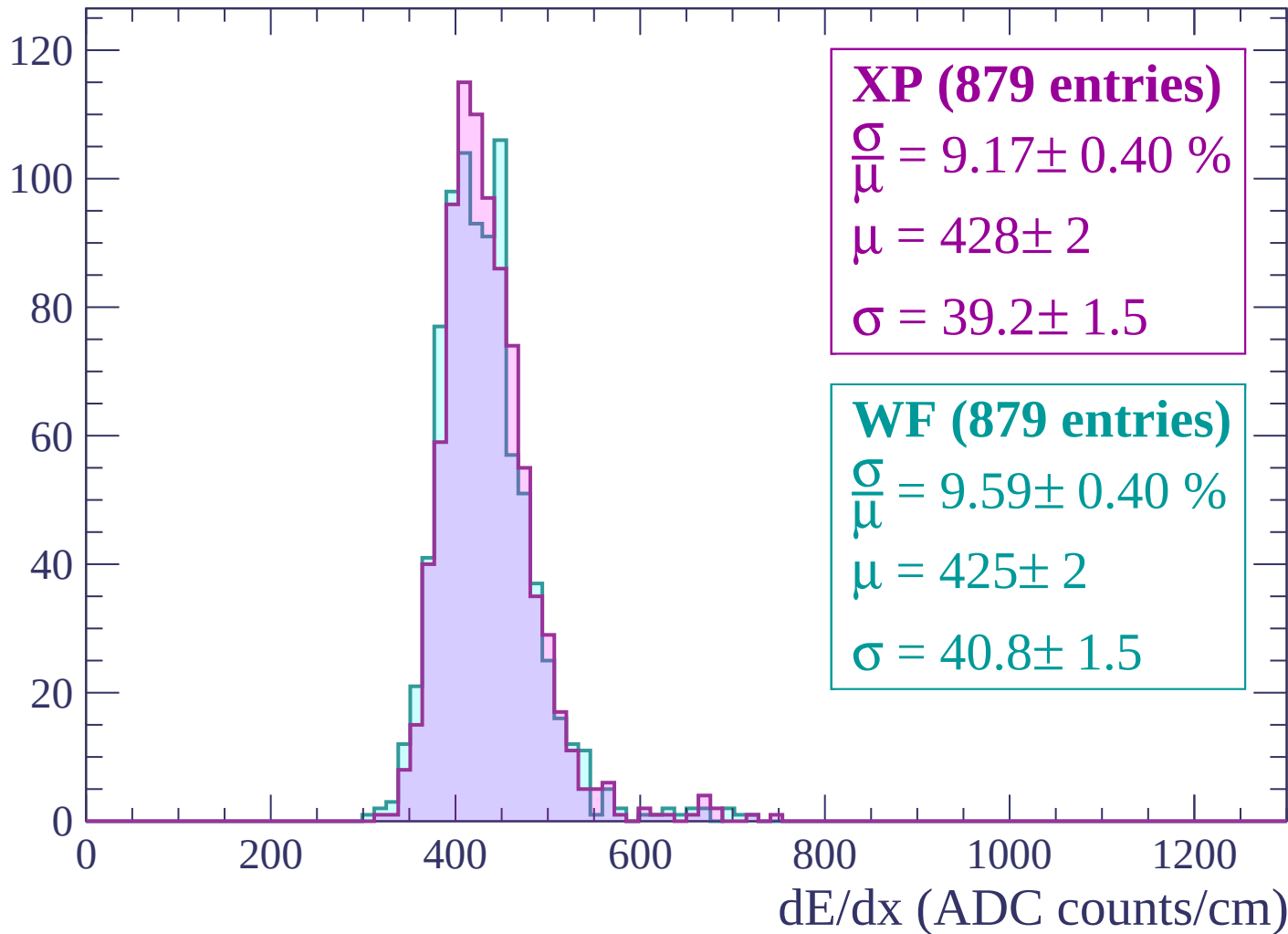
Energy loss | $300 < p < 360$

Count



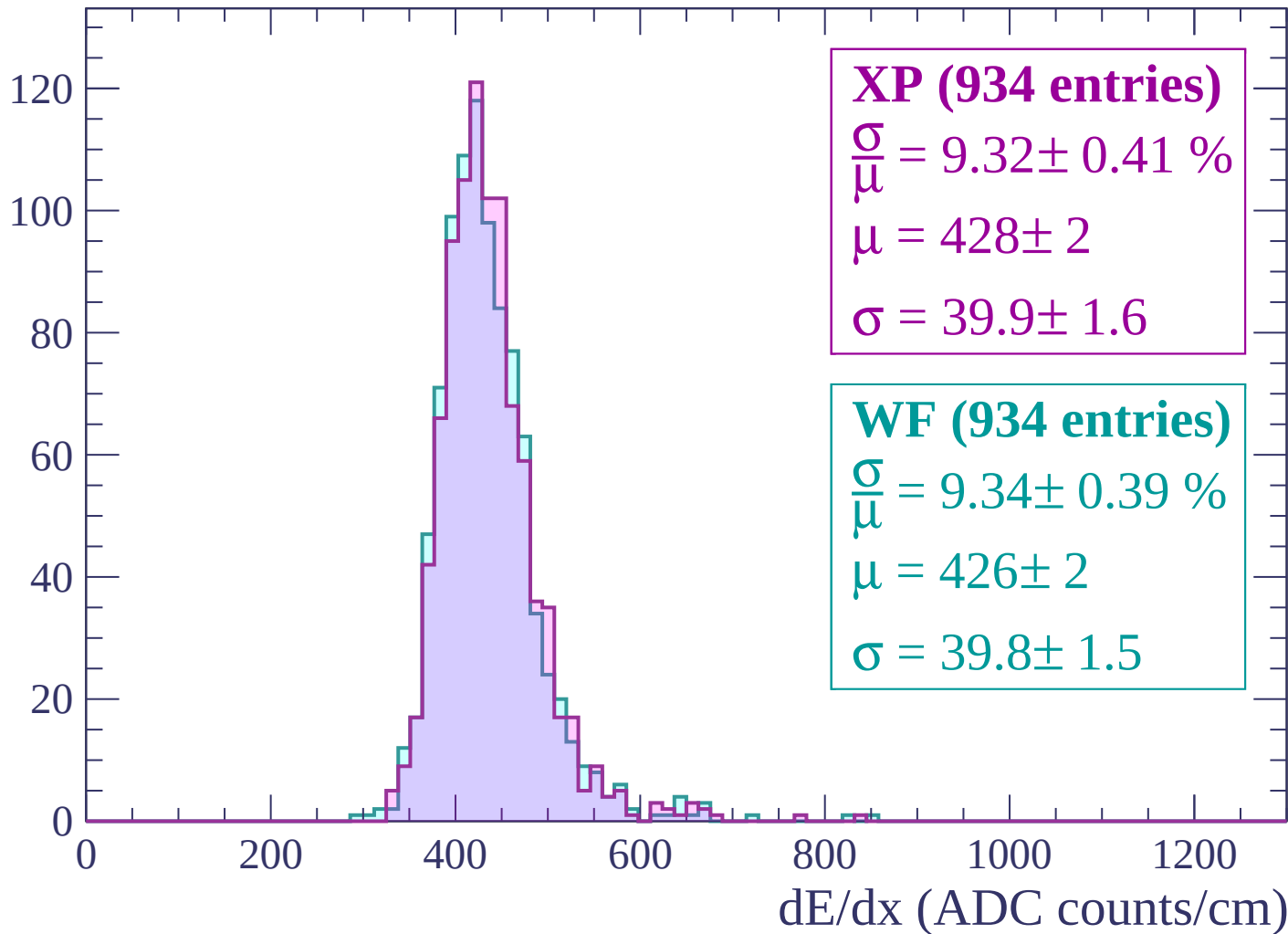
Energy loss | $360 < p < 420$

Count



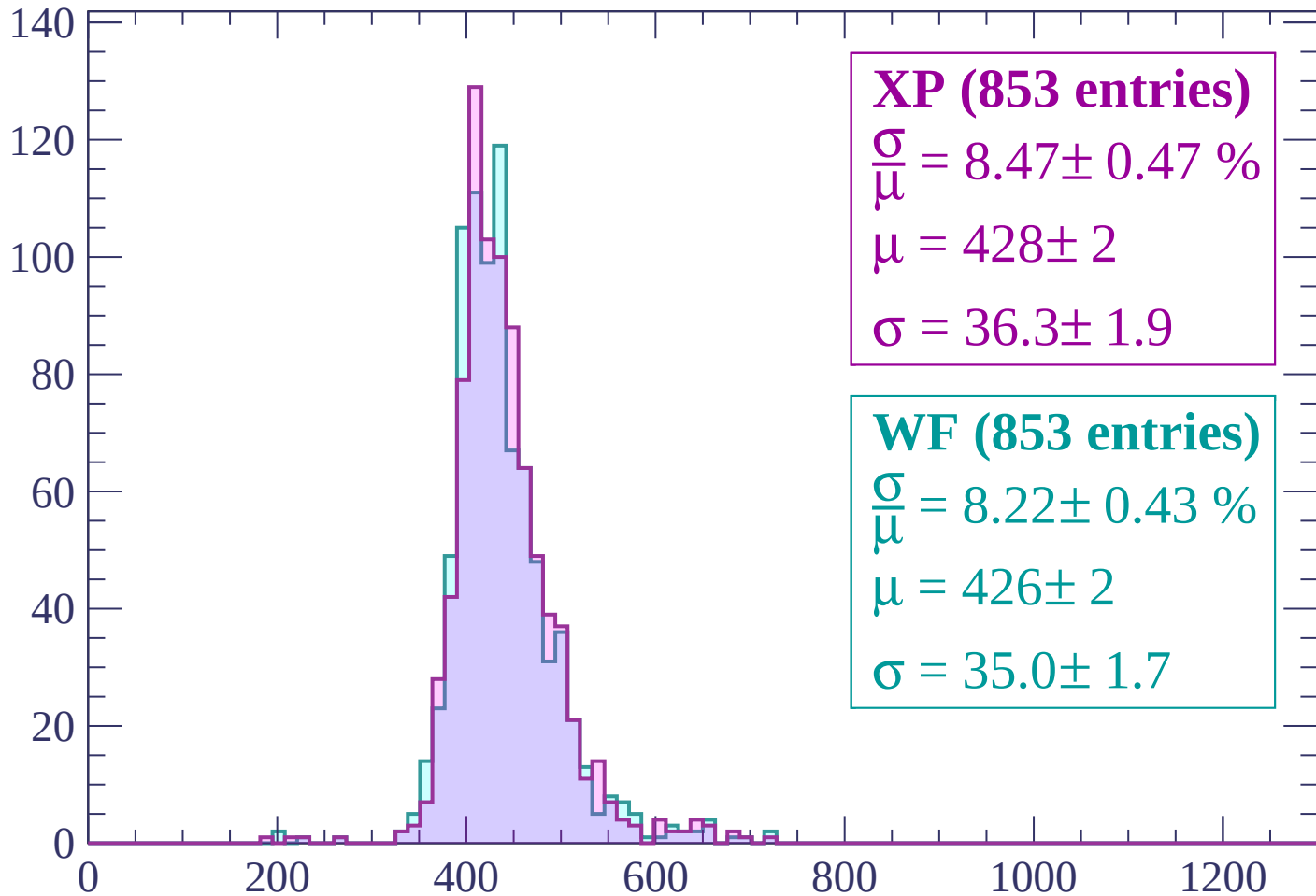
Energy loss | $420 < p < 480$

Count



Energy loss | $480 < p < 540$

Count



XP (853 entries)

$$\frac{\sigma}{\mu} = 8.47 \pm 0.47 \%$$

$$\mu = 428 \pm 2$$

$$\sigma = 36.3 \pm 1.9$$

WF (853 entries)

$$\frac{\sigma}{\mu} = 8.22 \pm 0.43 \%$$

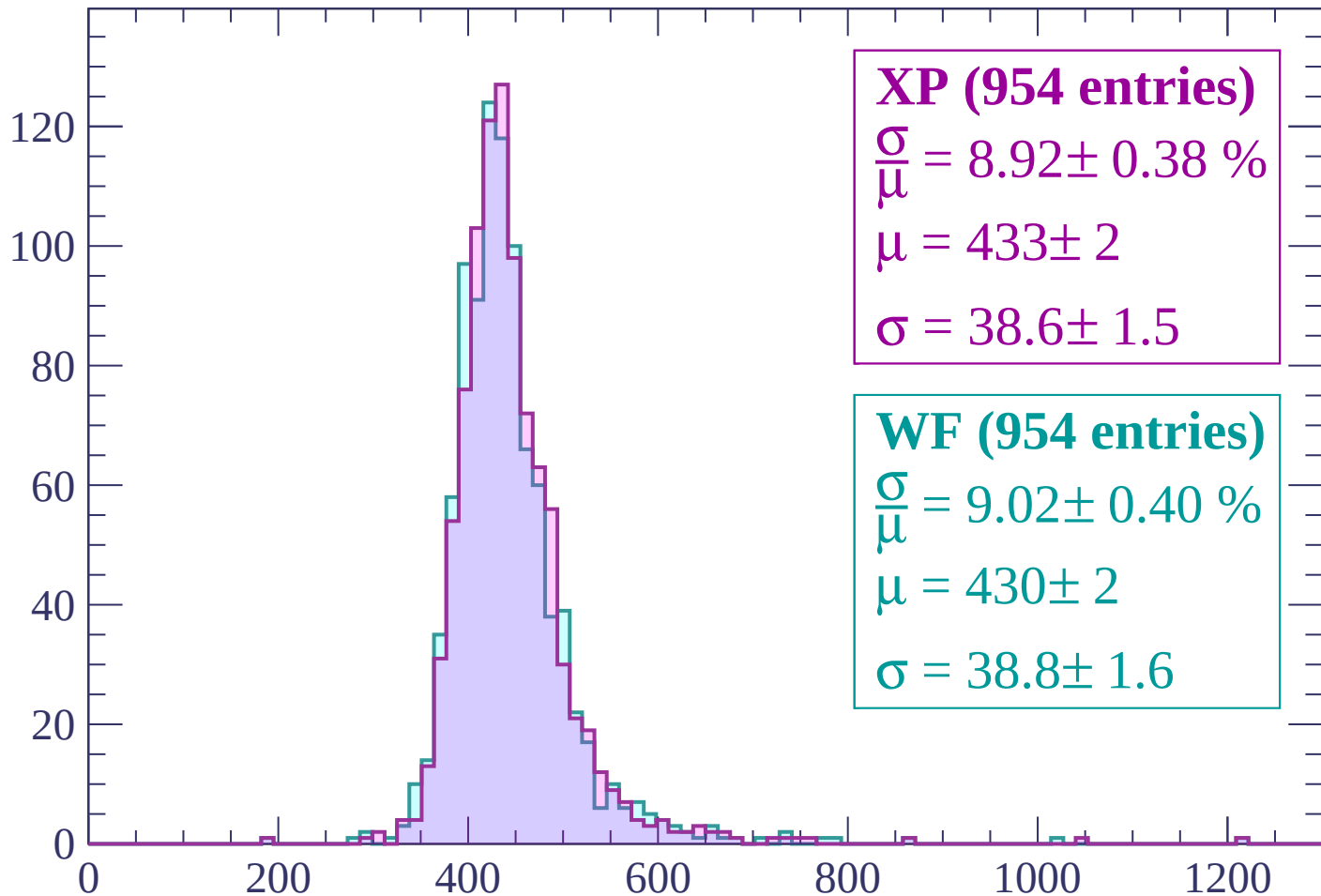
$$\mu = 426 \pm 2$$

$$\sigma = 35.0 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$

Count



XP (954 entries)

$\frac{\sigma}{\mu} = 8.92 \pm 0.38 \%$

$\mu = 433 \pm 2$

$\sigma = 38.6 \pm 1.5$

WF (954 entries)

$\frac{\sigma}{\mu} = 9.02 \pm 0.40 \%$

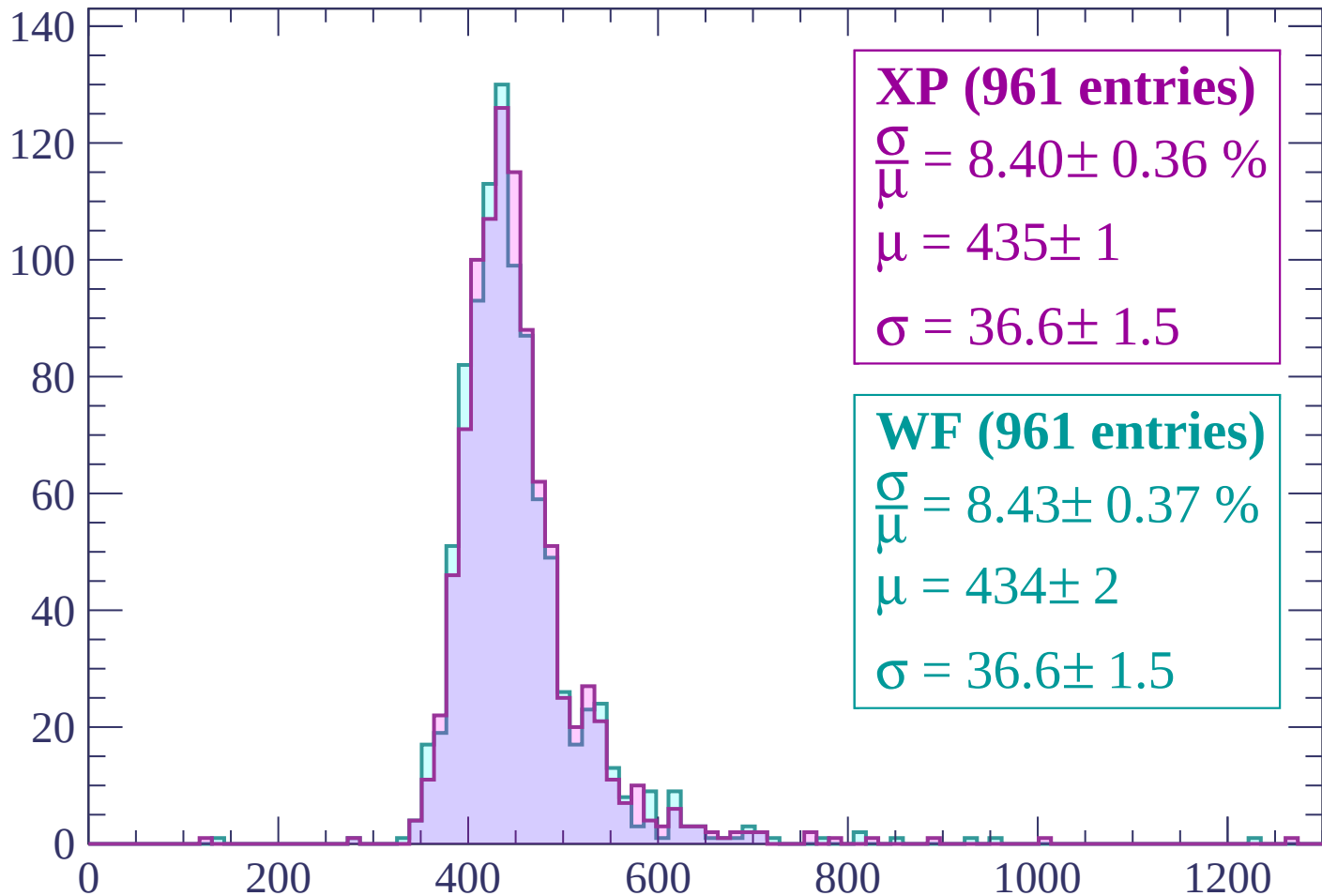
$\mu = 430 \pm 2$

$\sigma = 38.8 \pm 1.6$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$

Count



XP (961 entries)

$\frac{\sigma}{\mu} = 8.40 \pm 0.36 \%$

$\mu = 435 \pm 1$

$\sigma = 36.6 \pm 1.5$

WF (961 entries)

$\frac{\sigma}{\mu} = 8.43 \pm 0.37 \%$

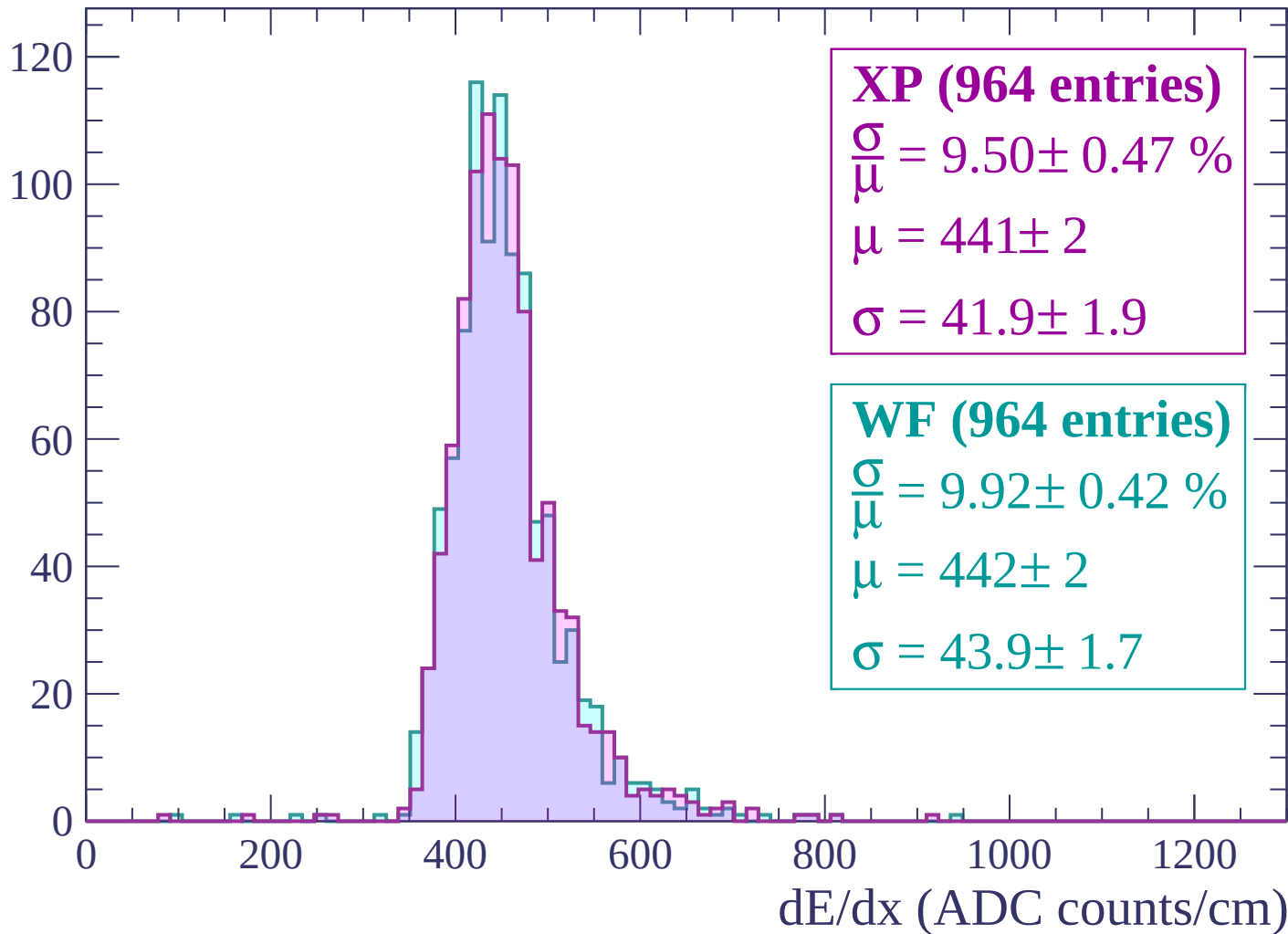
$\mu = 434 \pm 2$

$\sigma = 36.6 \pm 1.5$

dE/dx (ADC counts/cm)

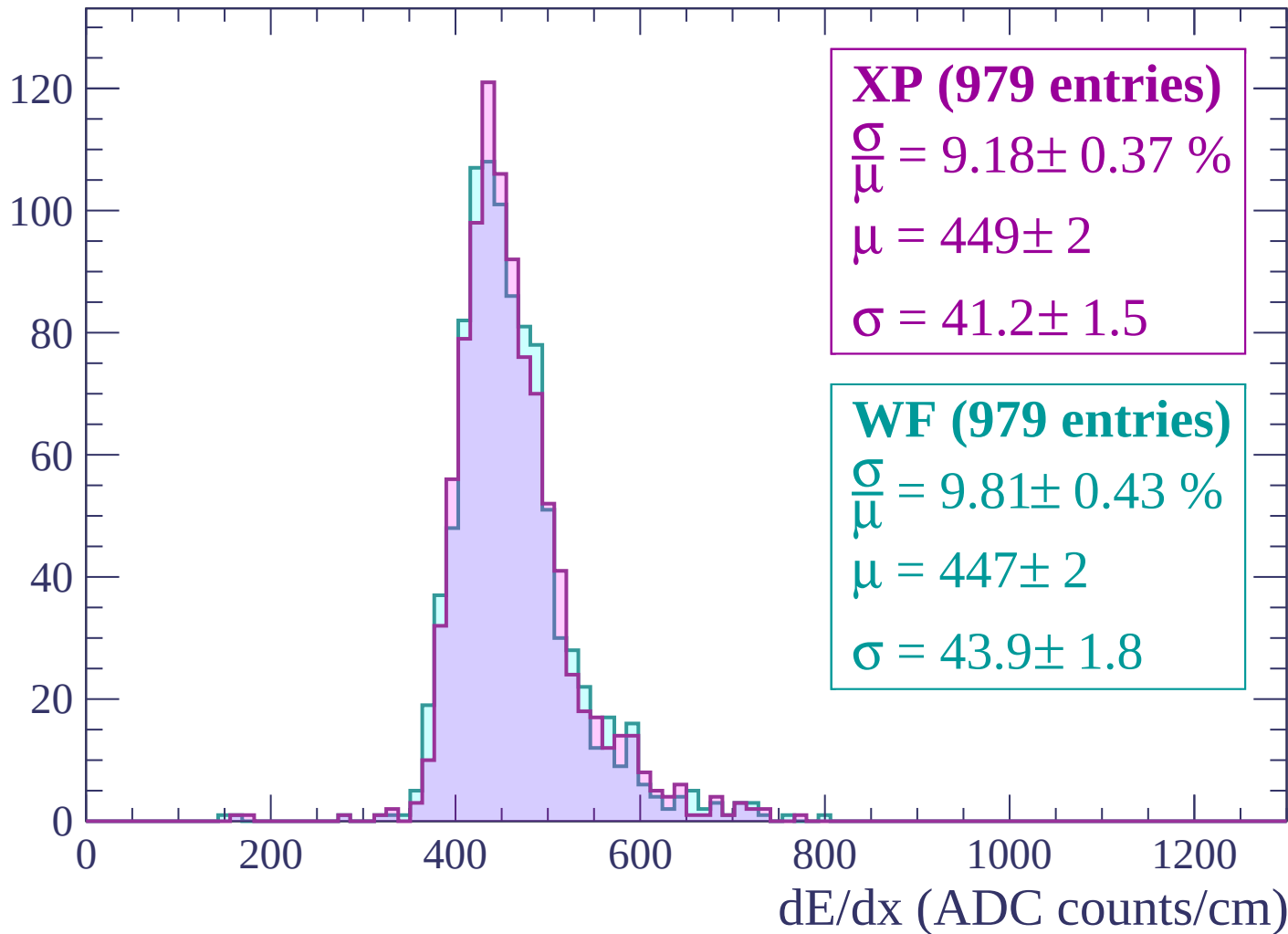
Energy loss | $660 < p < 720$

Count



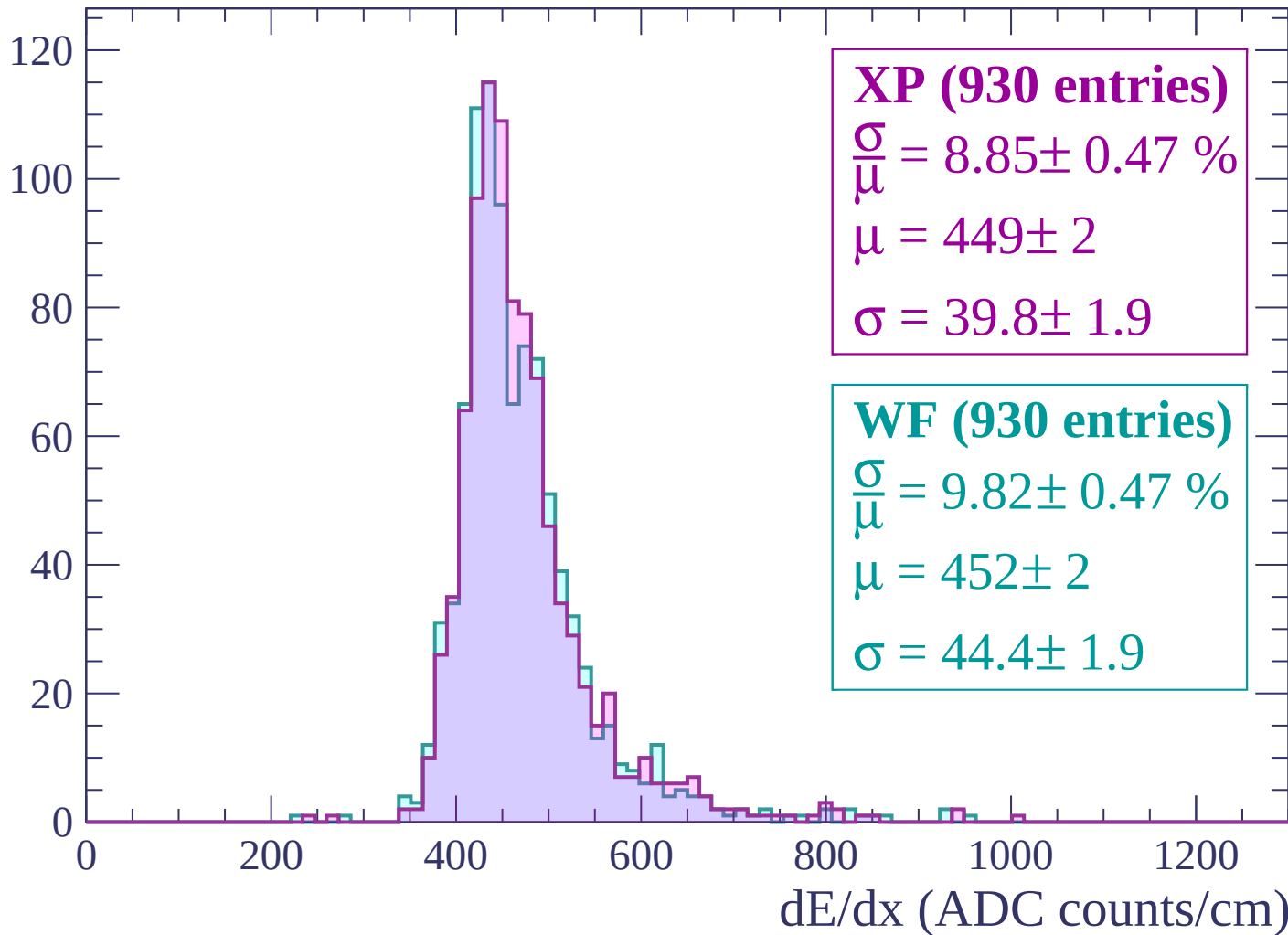
Energy loss | $720 < p < 780$

Count



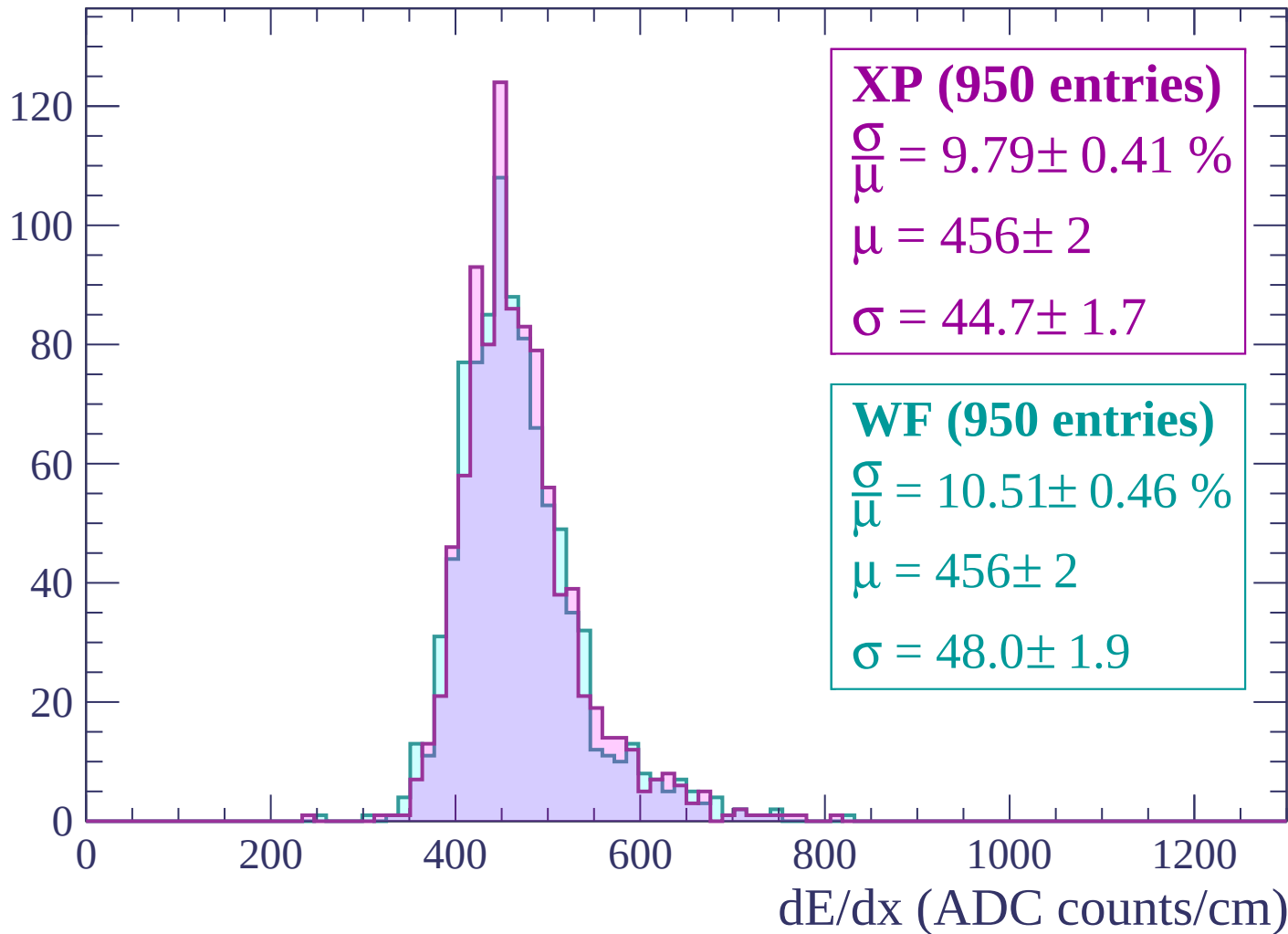
Energy loss | $780 < p < 840$

Count



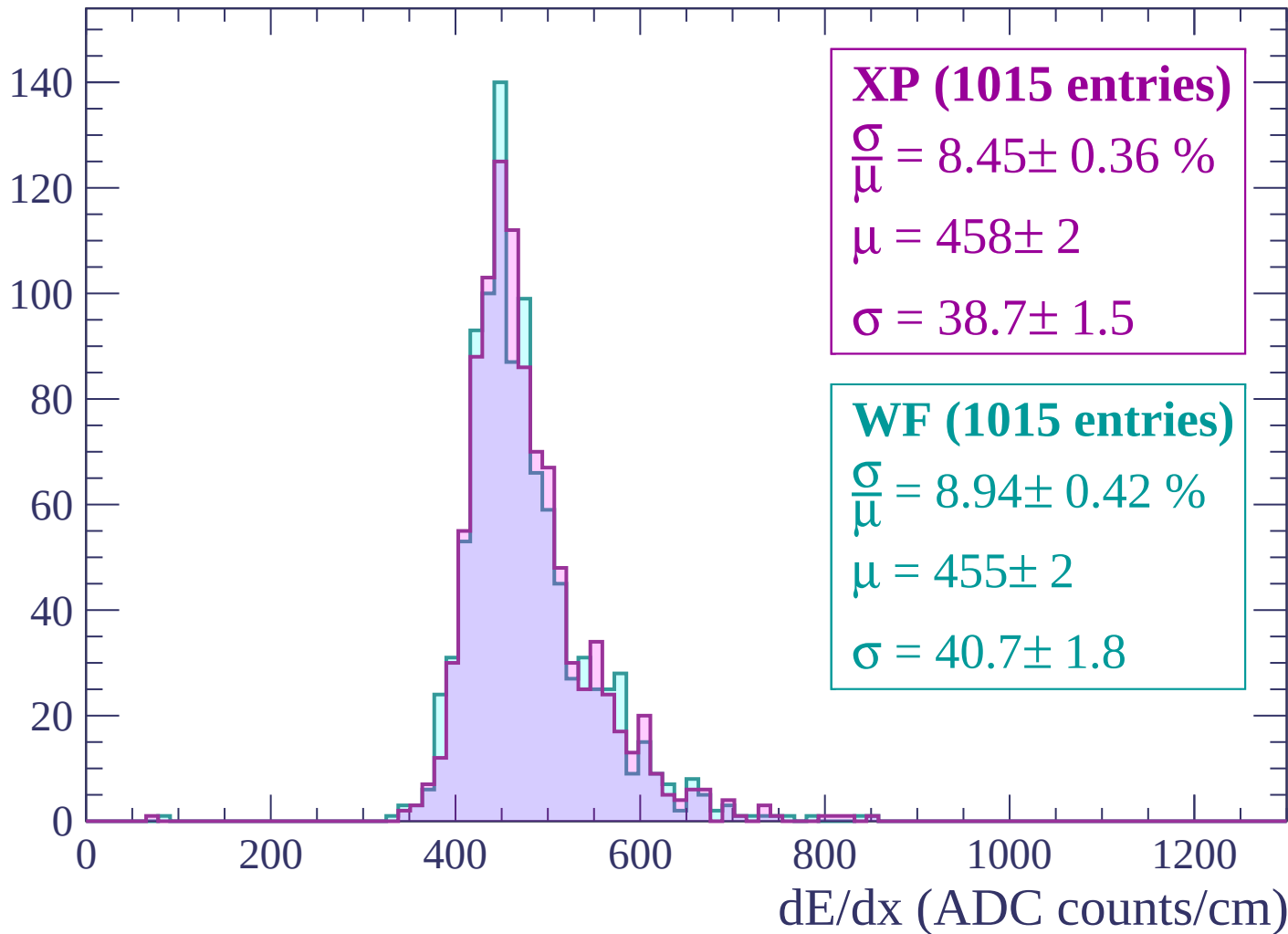
Energy loss | $840 < p < 900$

Count



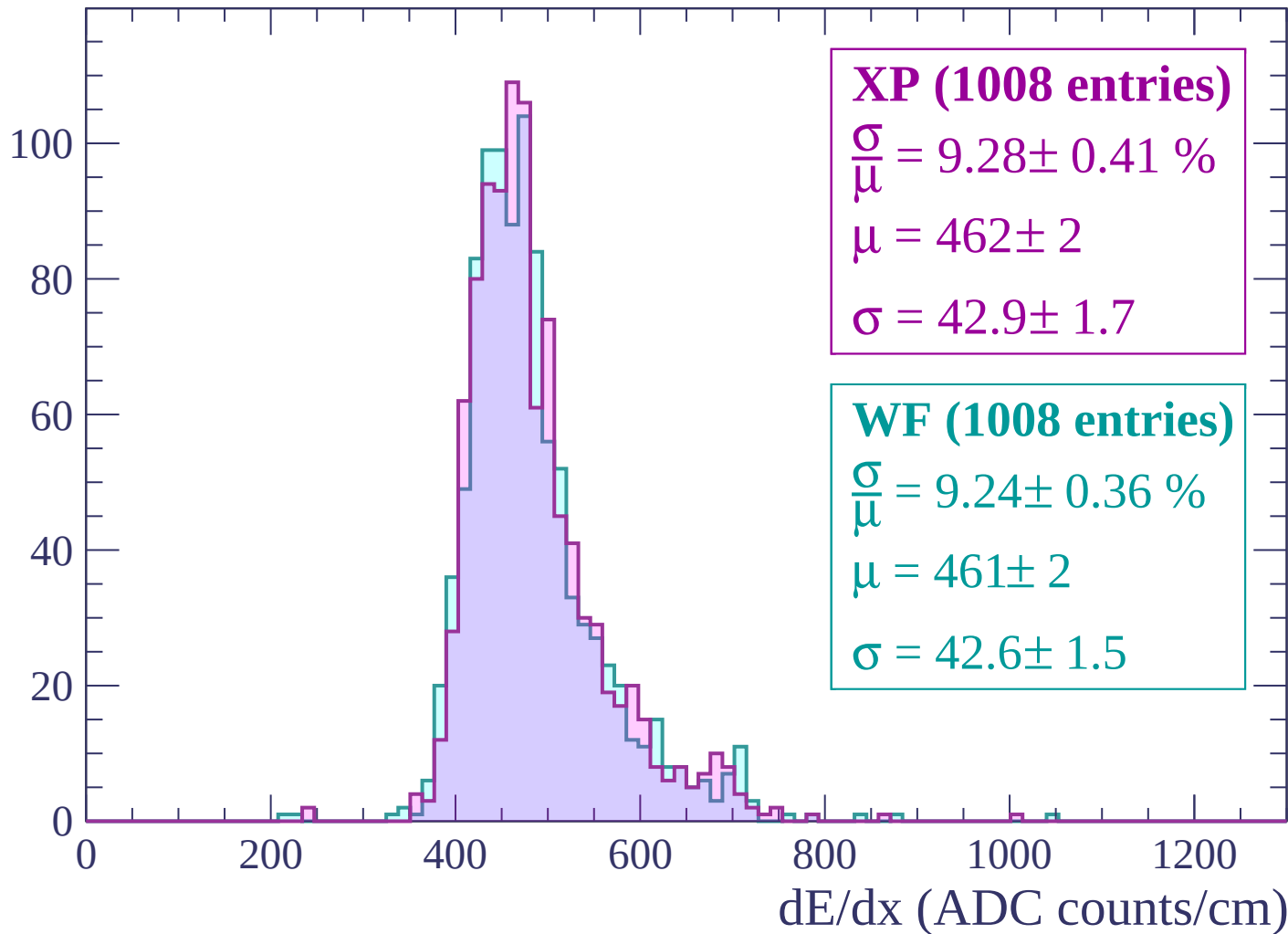
Energy loss | $900 < p < 960$

Count



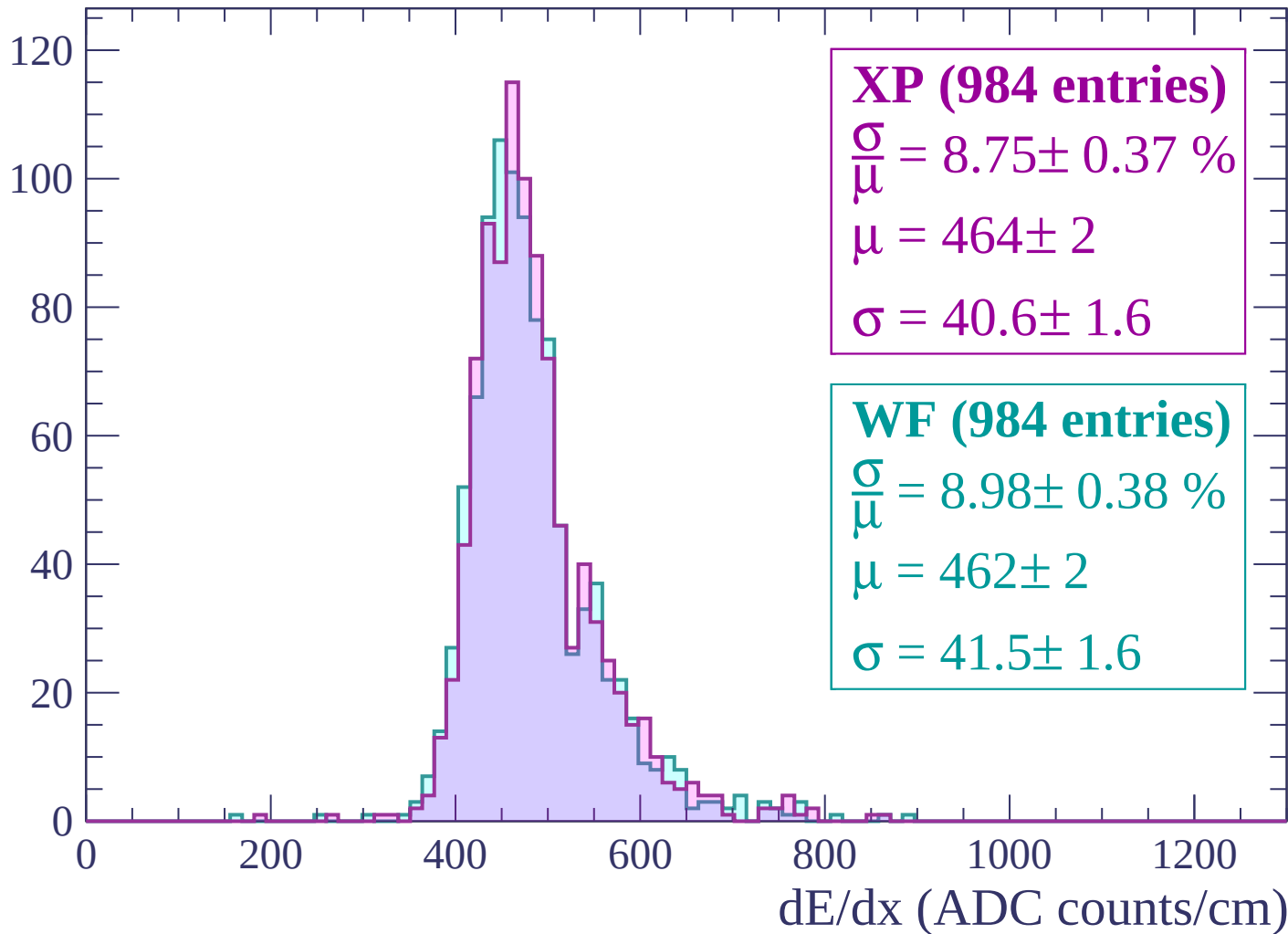
Energy loss | $960 < p < 1020$

Count



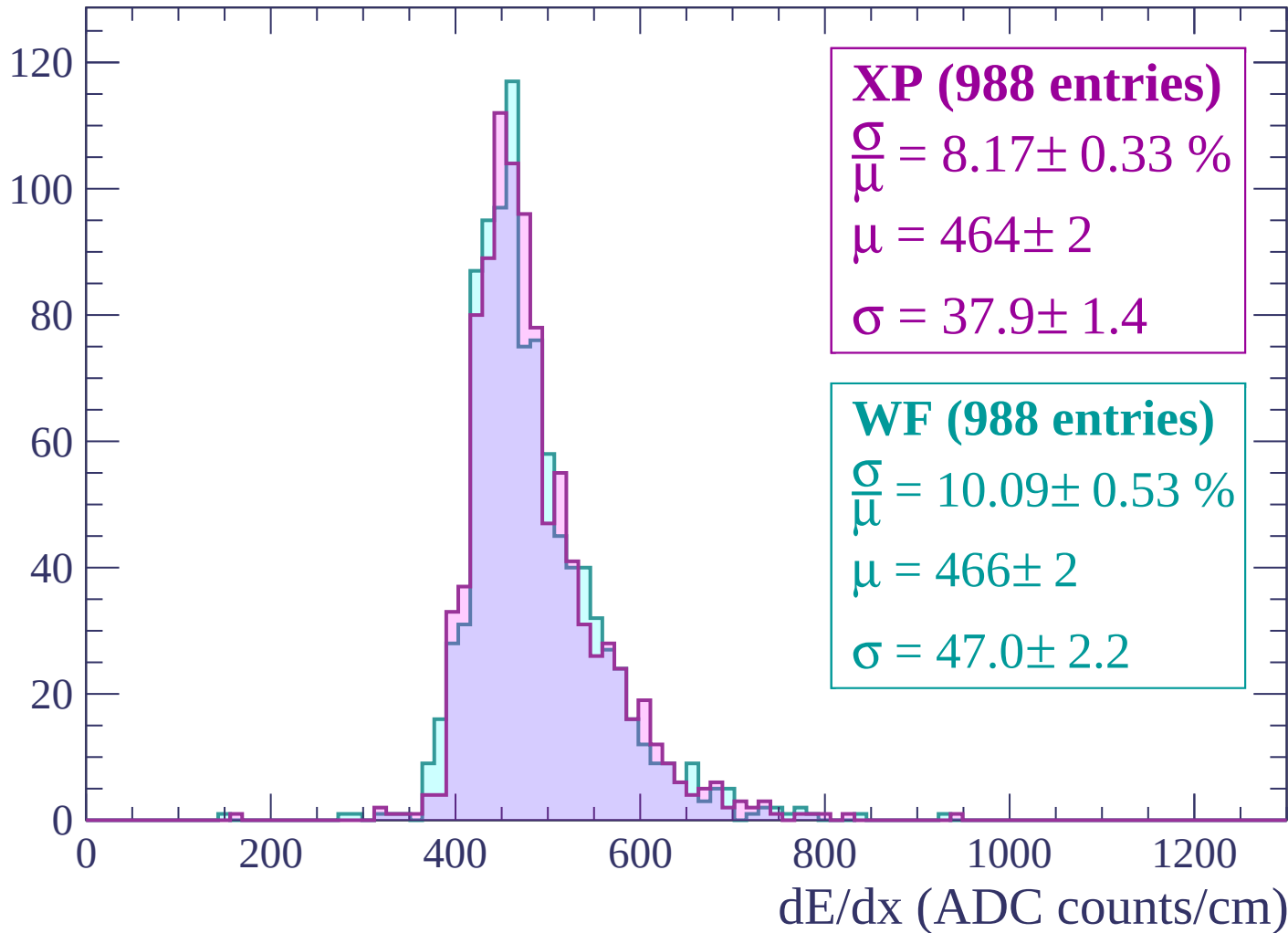
Energy loss | $1020 < p < 1080$

Count



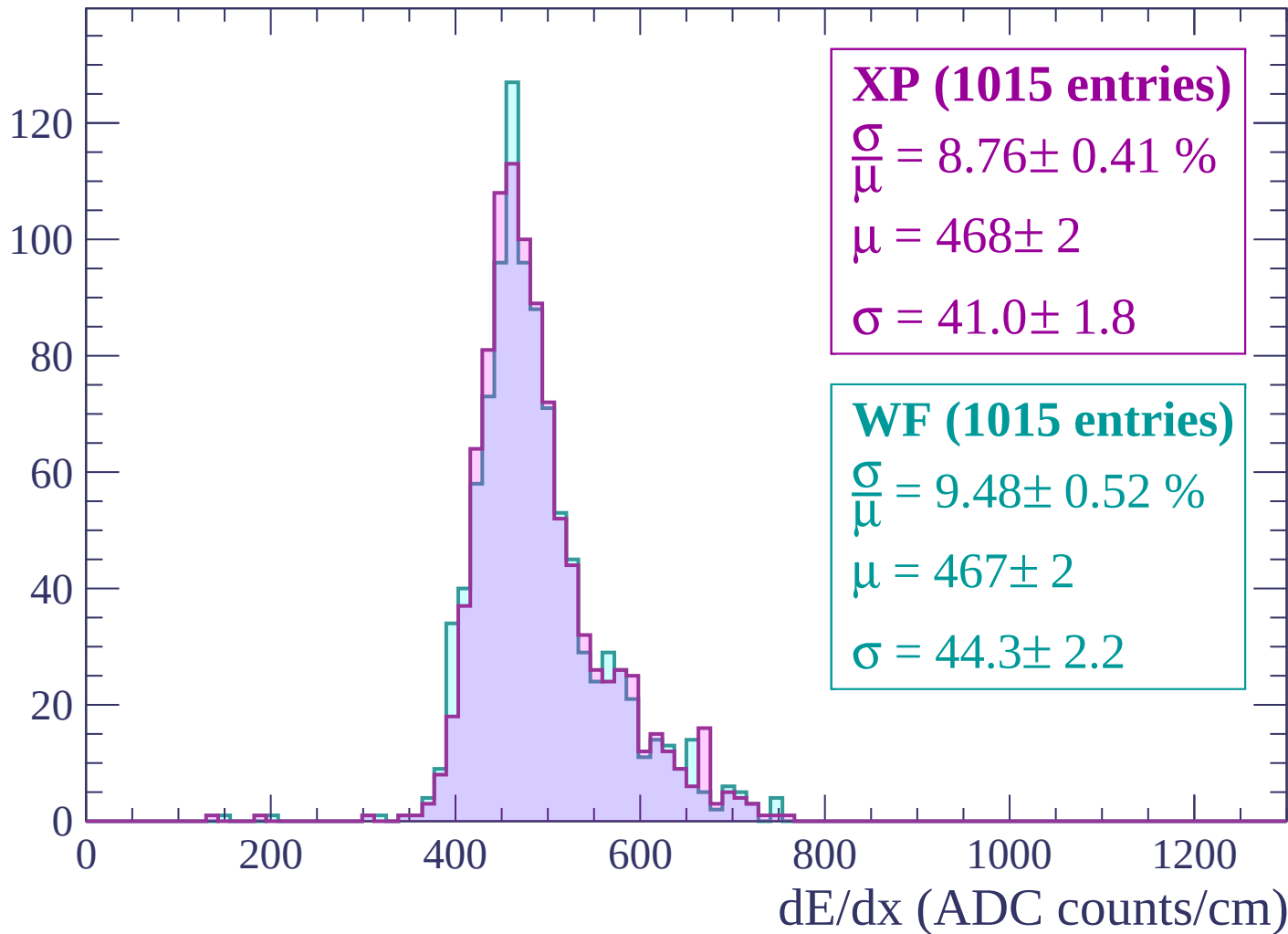
Energy loss | $1080 < p < 1140$

Count



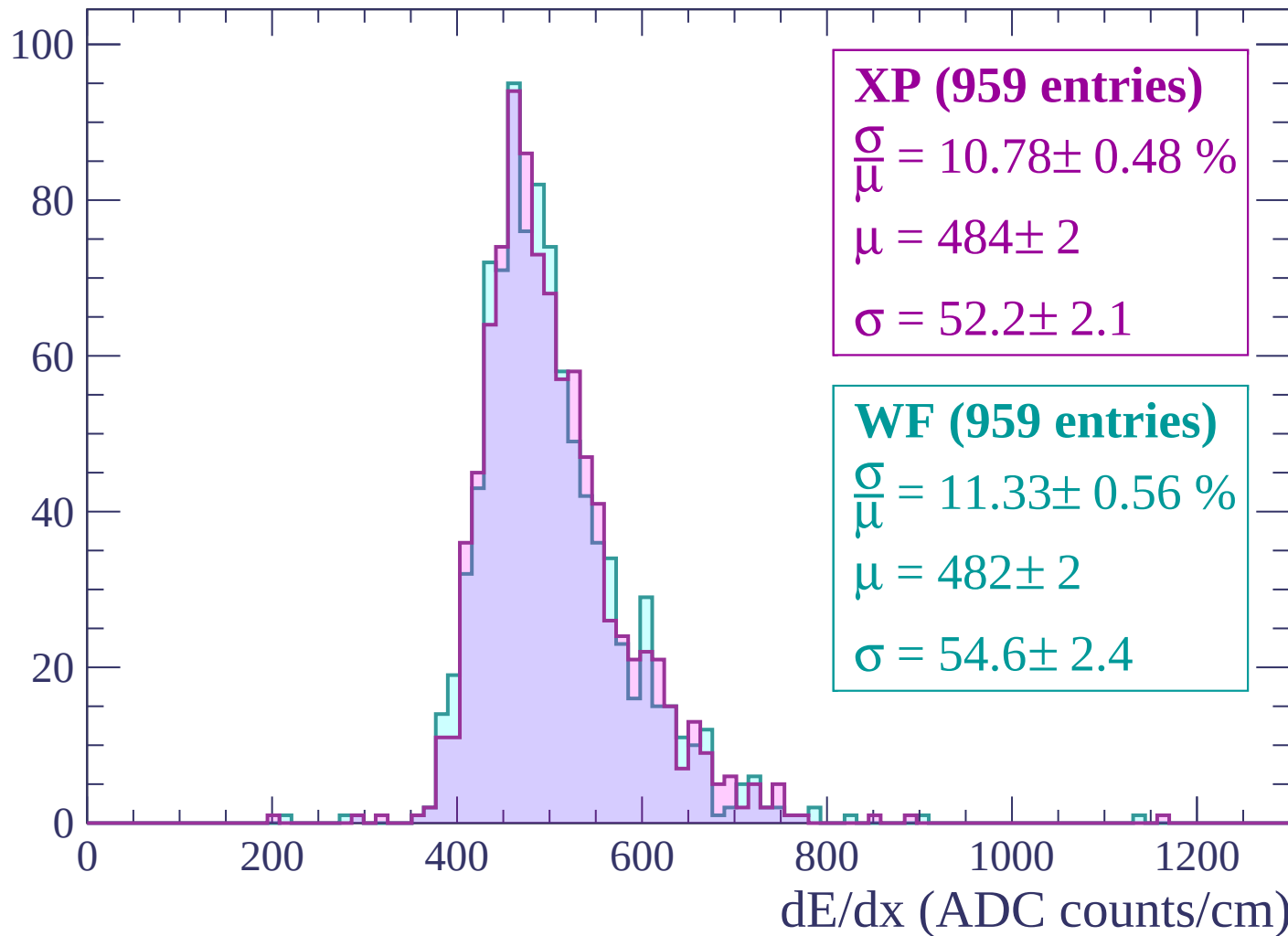
Energy loss | $1140 < p < 1200$

Count



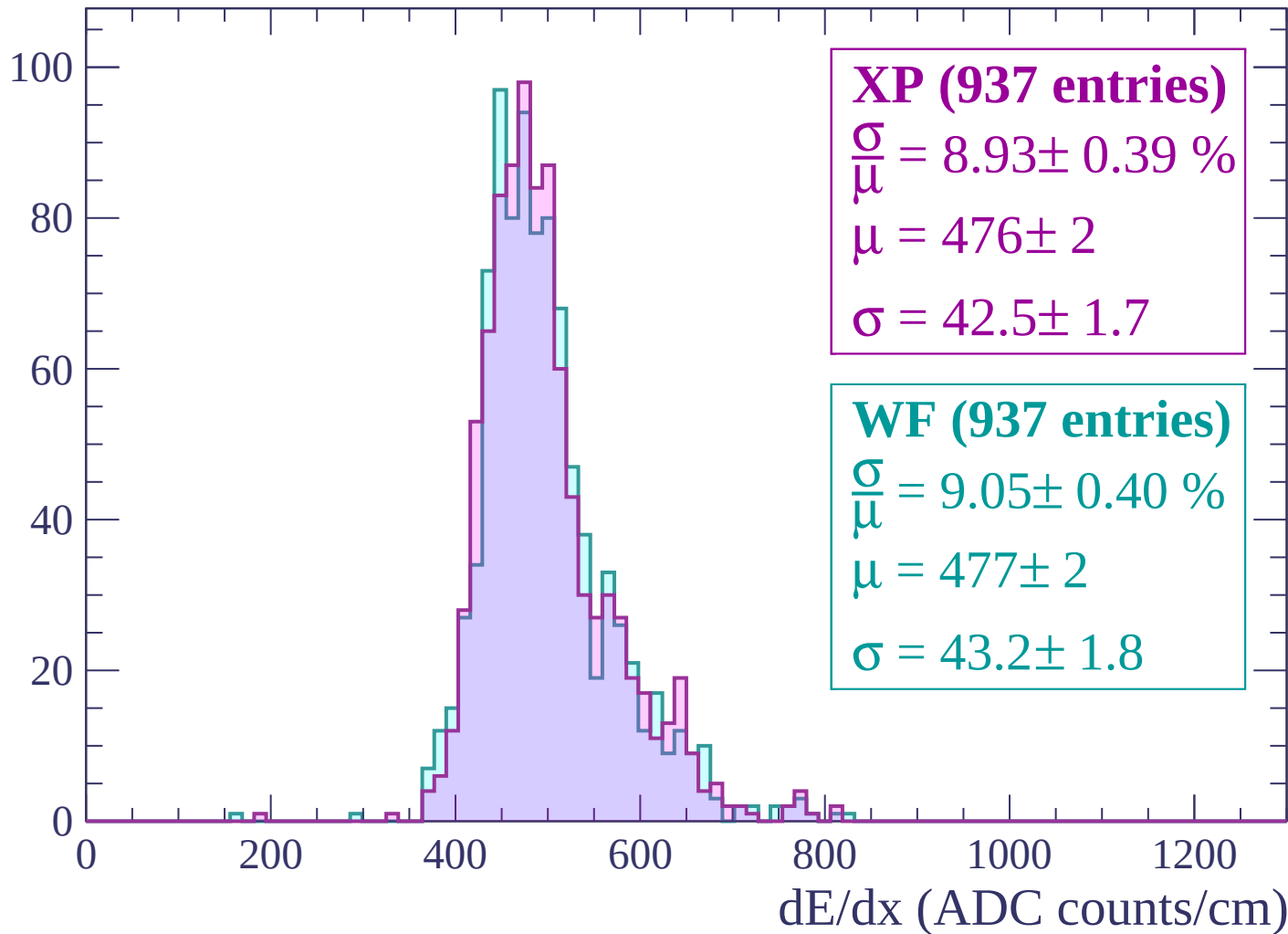
Energy loss | $1200 < p < 1260$

Count



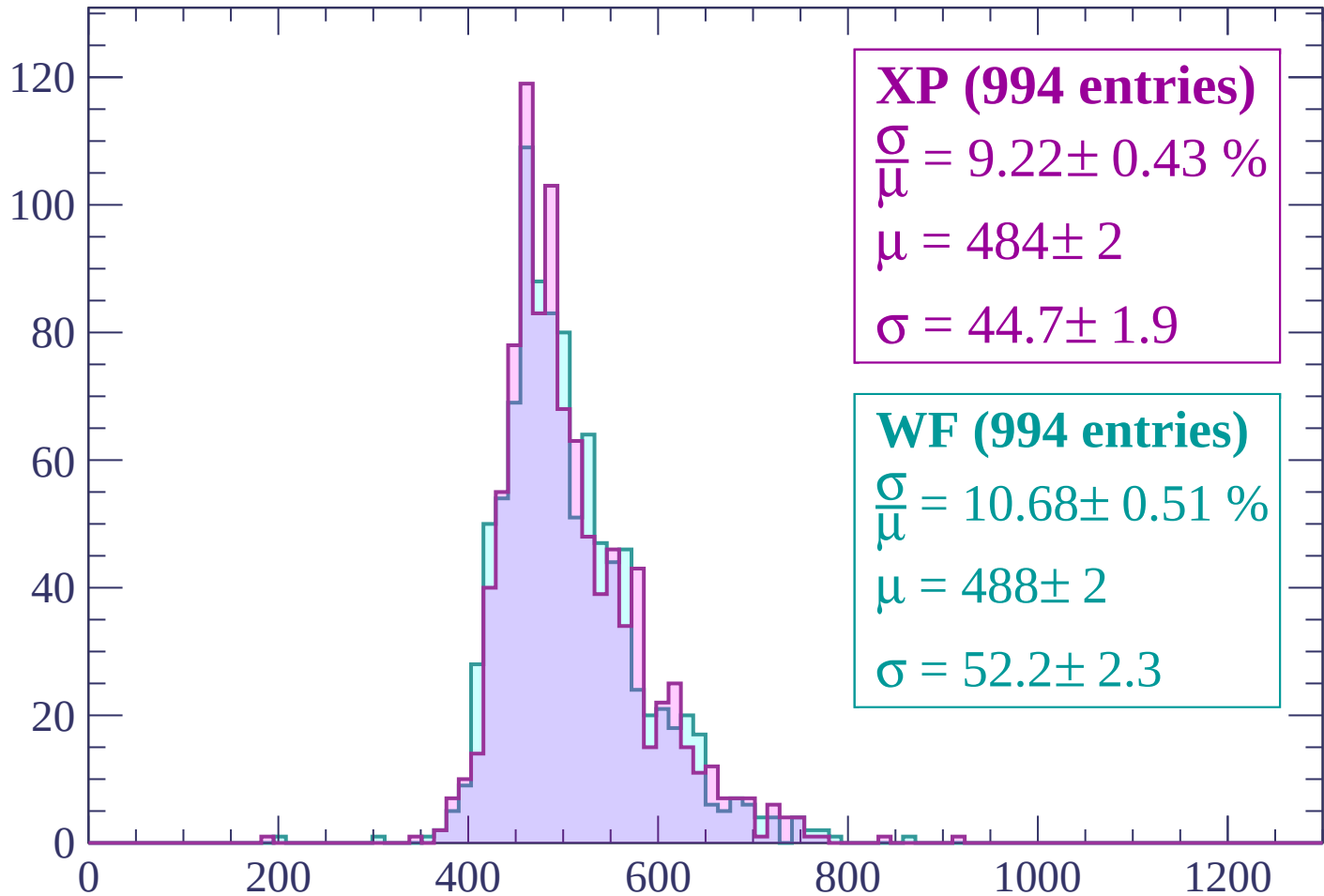
Energy loss | $1260 < p < 1320$

Count



Energy loss | $1320 < p < 1380$

Count



XP (994 entries)

$\frac{\sigma}{\mu} = 9.22 \pm 0.43 \%$

$\mu = 484 \pm 2$

$\sigma = 44.7 \pm 1.9$

WF (994 entries)

$\frac{\sigma}{\mu} = 10.68 \pm 0.51 \%$

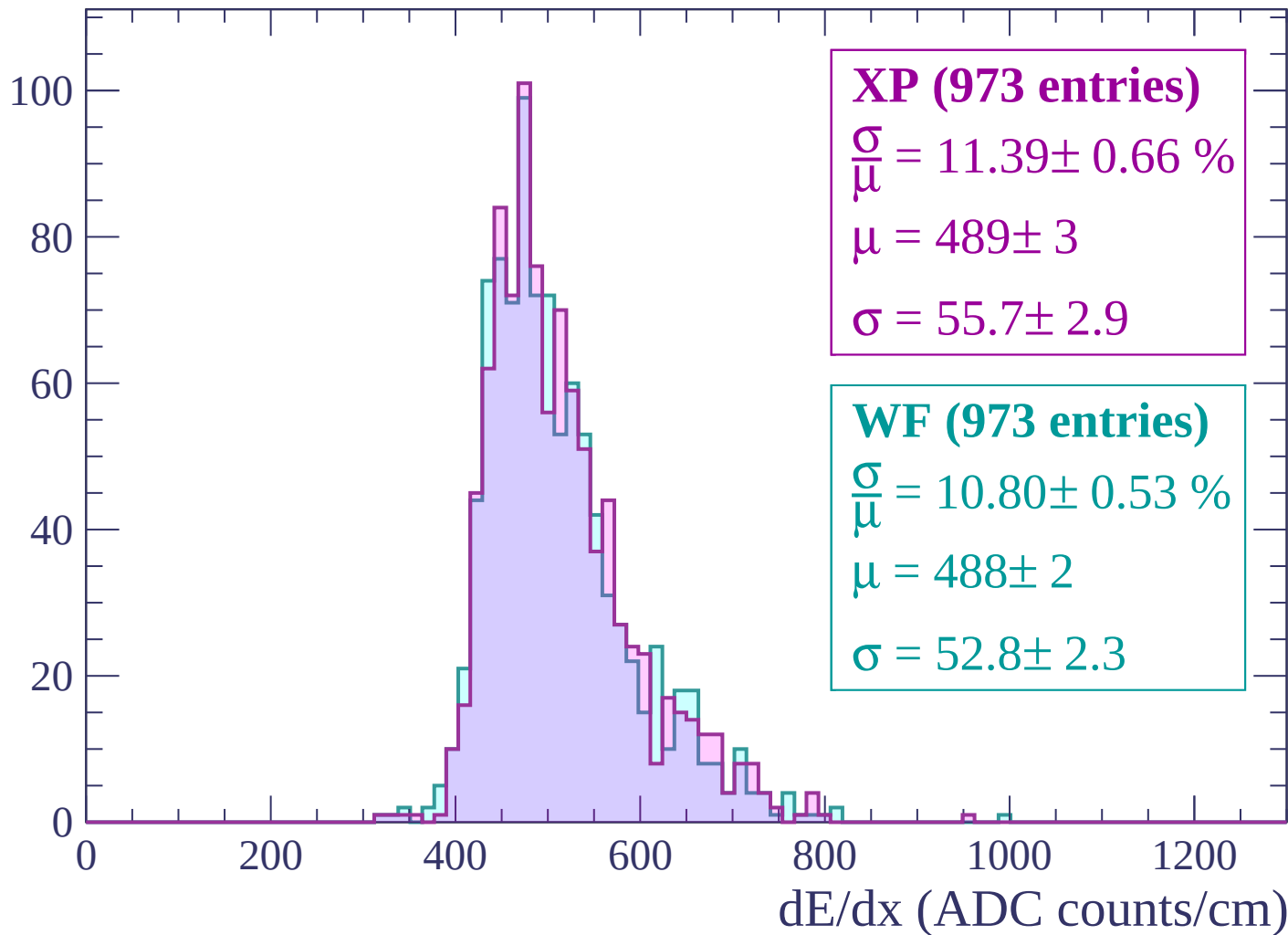
$\mu = 488 \pm 2$

$\sigma = 52.2 \pm 2.3$

dE/dx (ADC counts/cm)

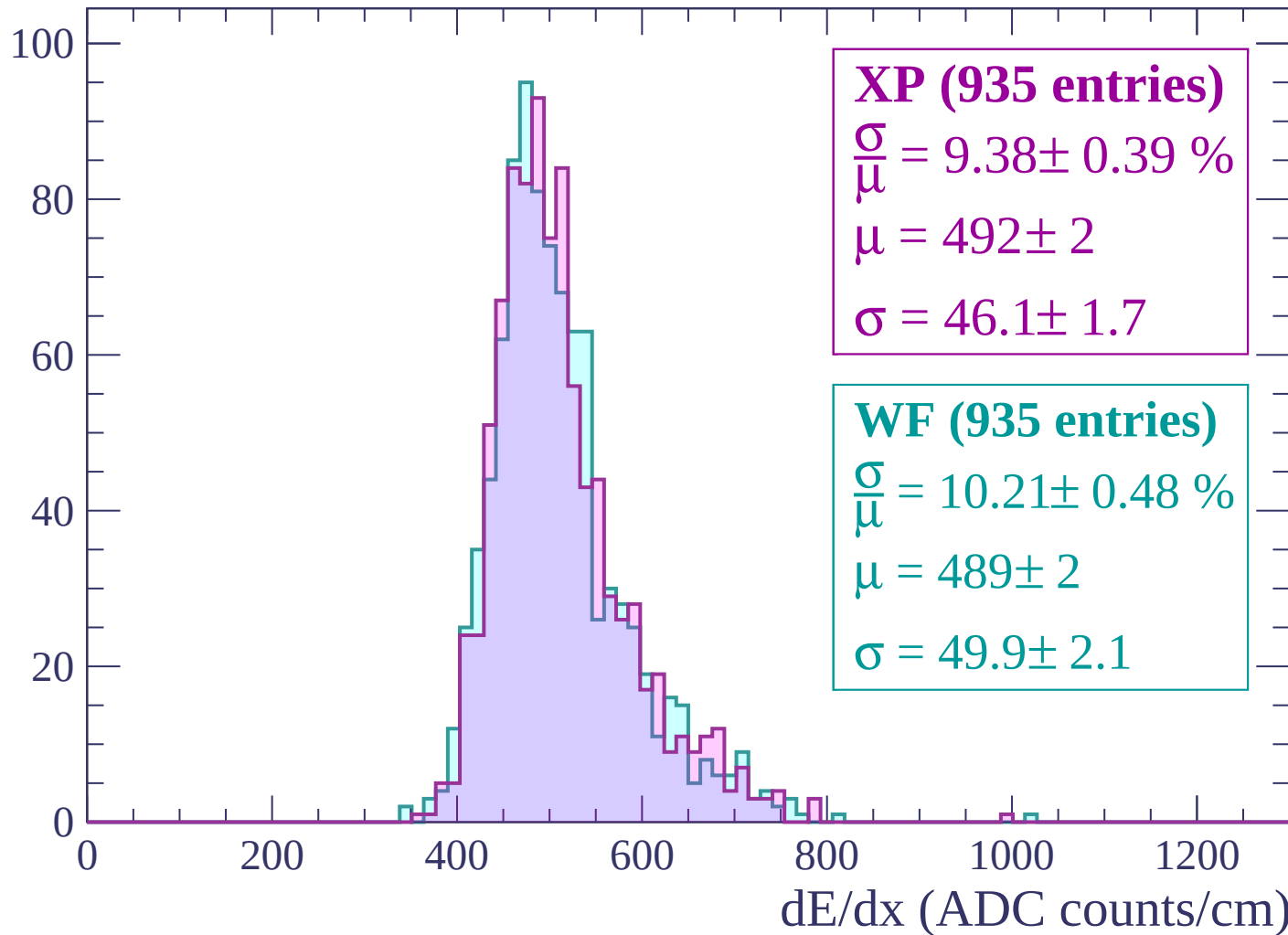
Energy loss | $1380 < p < 1440$

Count



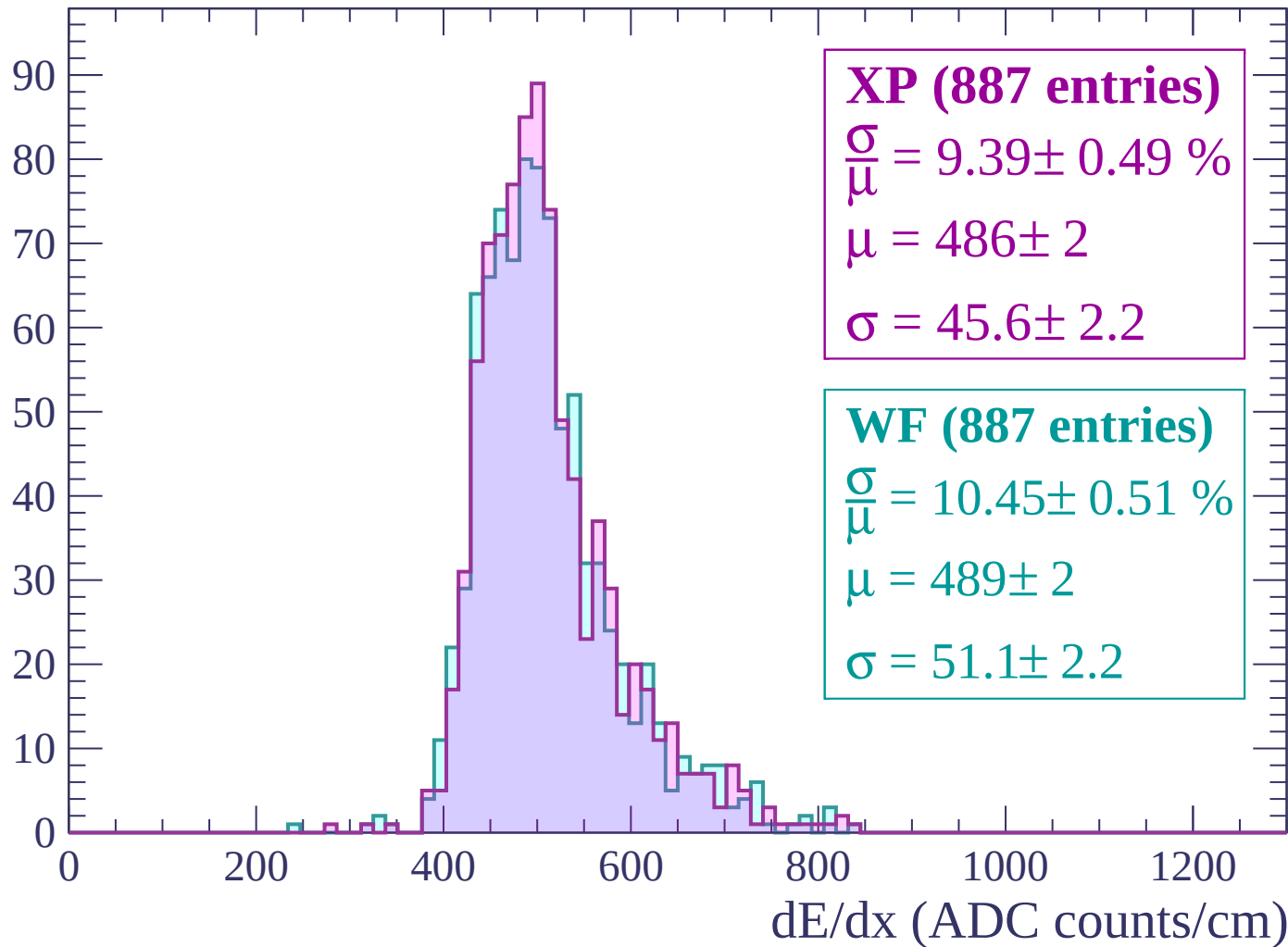
Energy loss | $1440 < p < 1500$

Count



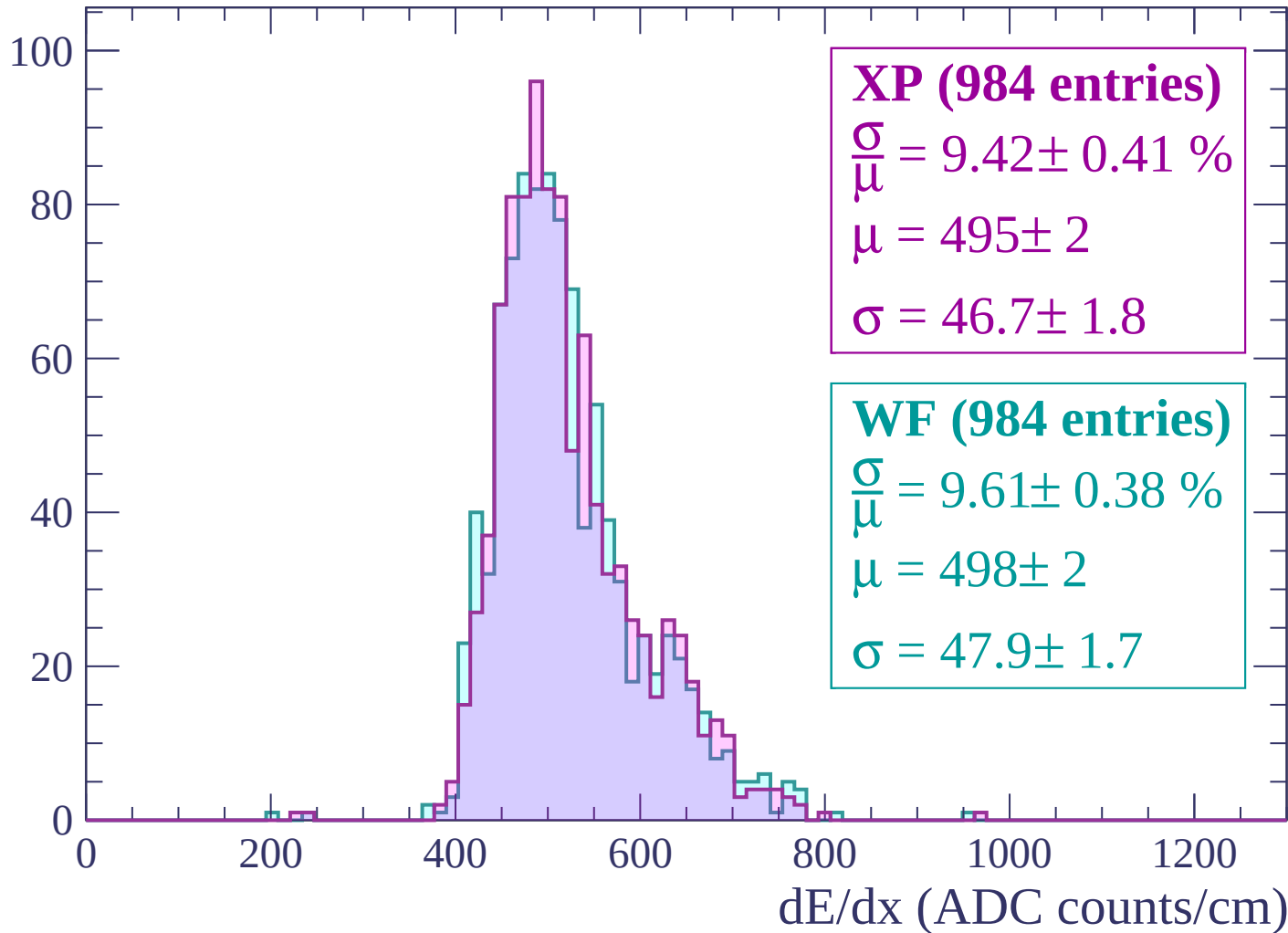
Energy loss | $1500 < p < 1560$

Count



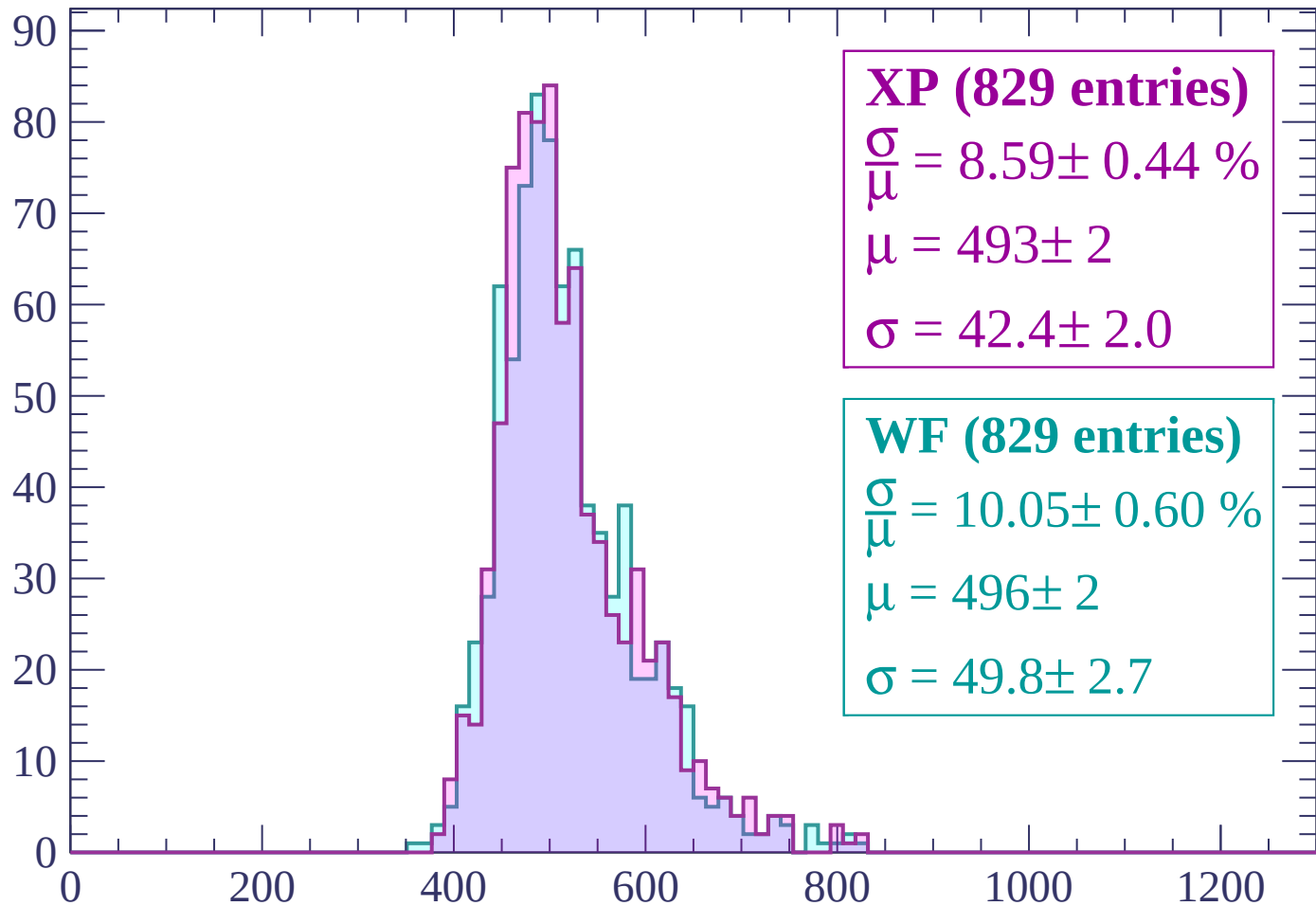
Energy loss | $1560 < p < 1620$

Count



Energy loss | $1620 < p < 1680$

Count



XP (829 entries)

$$\frac{\sigma}{\mu} = 8.59 \pm 0.44 \%$$

$$\mu = 493 \pm 2$$

$$\sigma = 42.4 \pm 2.0$$

WF (829 entries)

$$\frac{\sigma}{\mu} = 10.05 \pm 0.60 \%$$

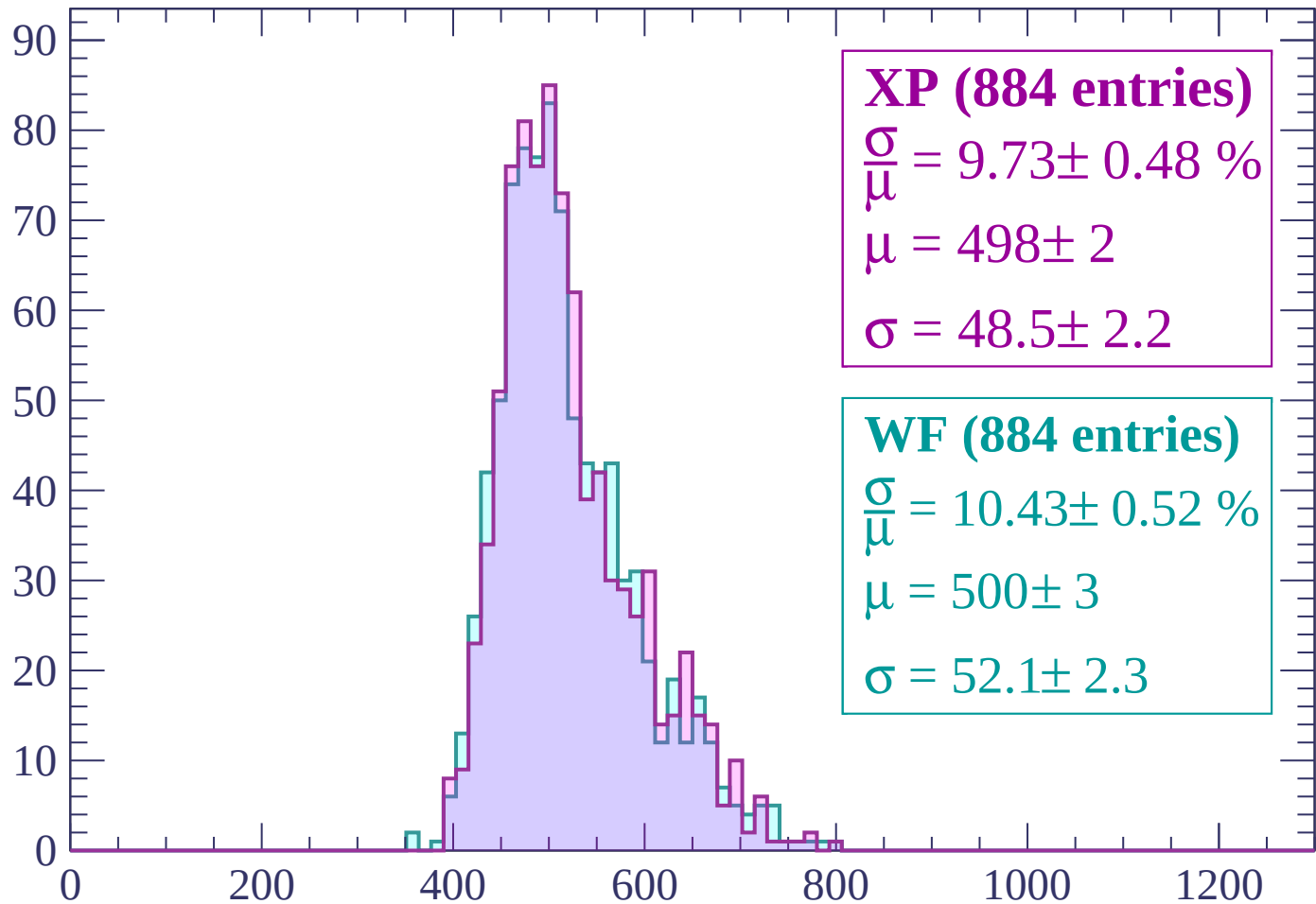
$$\mu = 496 \pm 2$$

$$\sigma = 49.8 \pm 2.7$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$

Count



XP (884 entries)

$$\frac{\sigma}{\mu} = 9.73 \pm 0.48 \%$$

$$\mu = 498 \pm 2$$

$$\sigma = 48.5 \pm 2.2$$

WF (884 entries)

$$\frac{\sigma}{\mu} = 10.43 \pm 0.52 \%$$

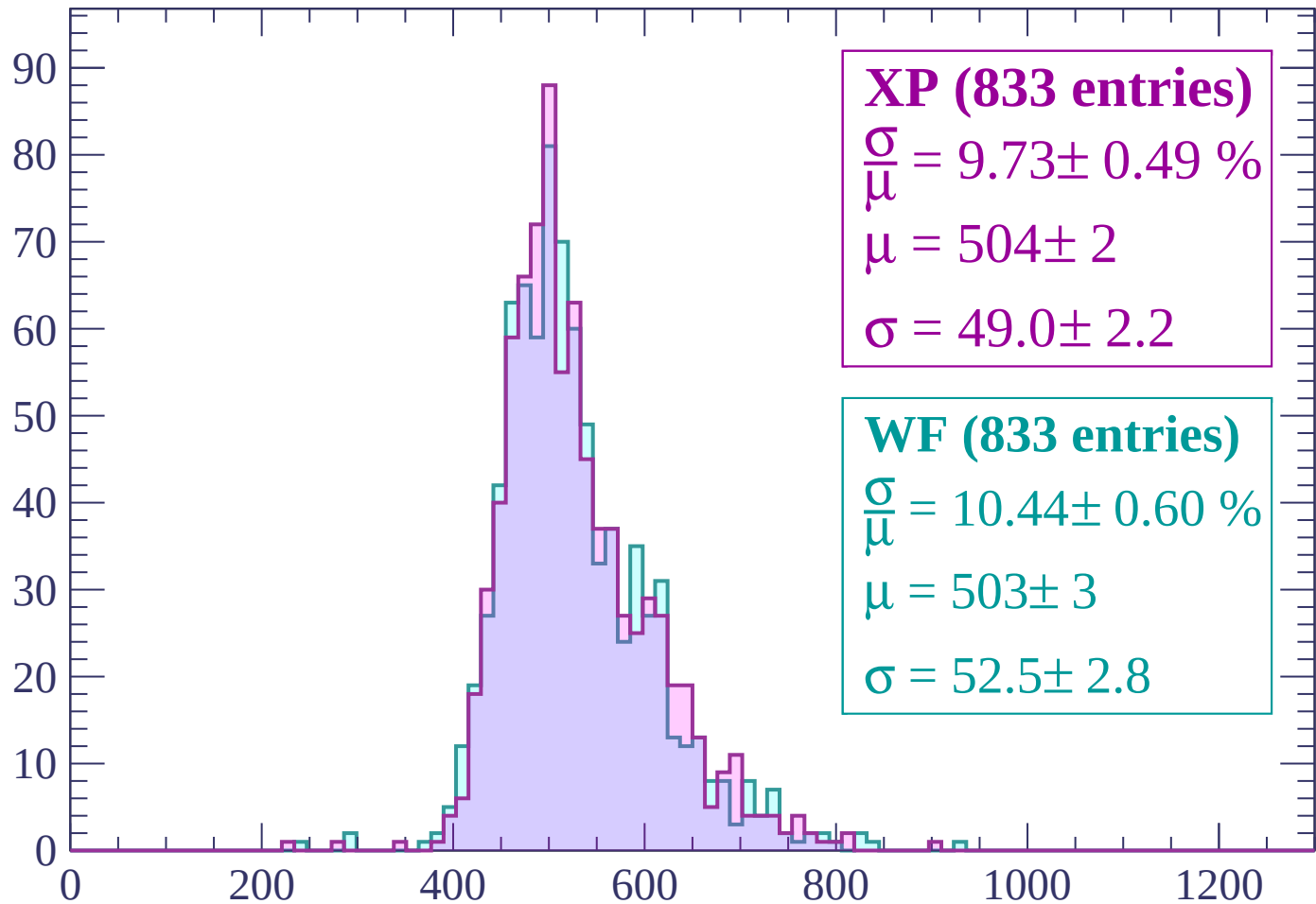
$$\mu = 500 \pm 3$$

$$\sigma = 52.1 \pm 2.3$$

dE/dx (ADC counts/cm)

Energy loss | $1740 < p < 1800$

Count



XP (833 entries)

$$\frac{\sigma}{\mu} = 9.73 \pm 0.49 \%$$

$$\mu = 504 \pm 2$$

$$\sigma = 49.0 \pm 2.2$$

WF (833 entries)

$$\frac{\sigma}{\mu} = 10.44 \pm 0.60 \%$$

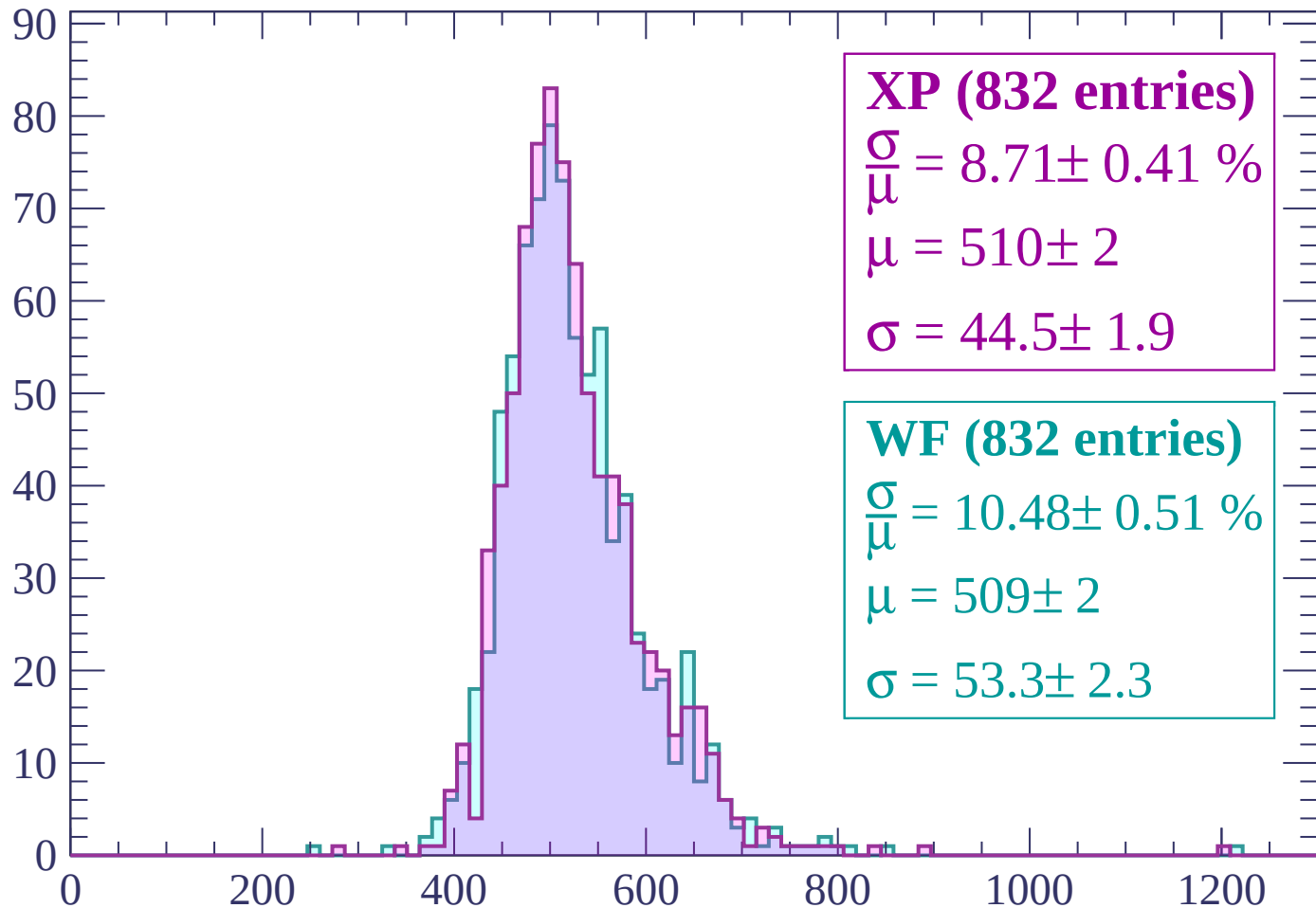
$$\mu = 503 \pm 3$$

$$\sigma = 52.5 \pm 2.8$$

dE/dx (ADC counts/cm)

Energy loss | $1800 < p < 1860$

Count



XP (832 entries)

$$\frac{\sigma}{\mu} = 8.71 \pm 0.41 \%$$

$$\mu = 510 \pm 2$$

$$\sigma = 44.5 \pm 1.9$$

WF (832 entries)

$$\frac{\sigma}{\mu} = 10.48 \pm 0.51 \%$$

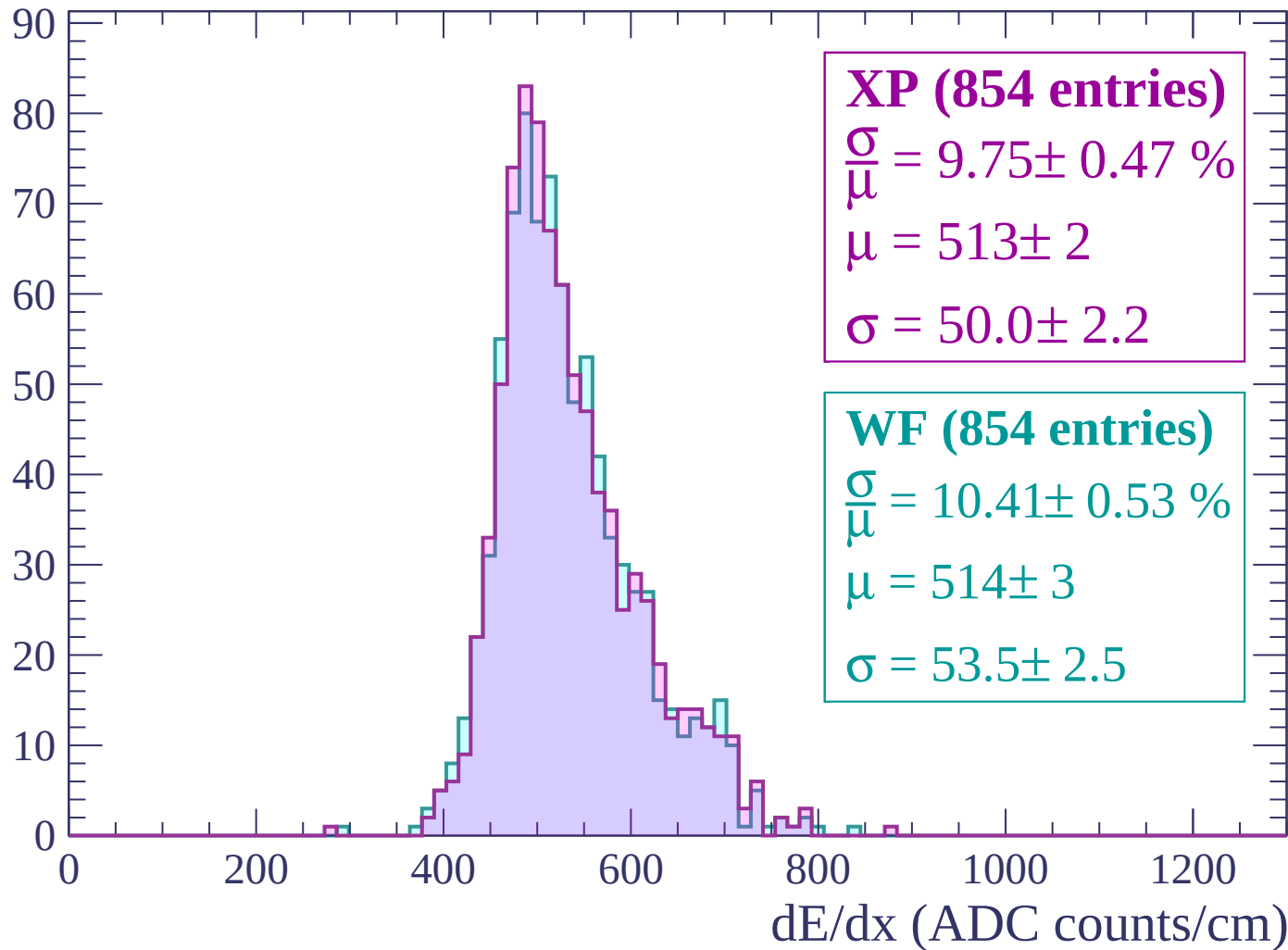
$$\mu = 509 \pm 2$$

$$\sigma = 53.3 \pm 2.3$$

dE/dx (ADC counts/cm)

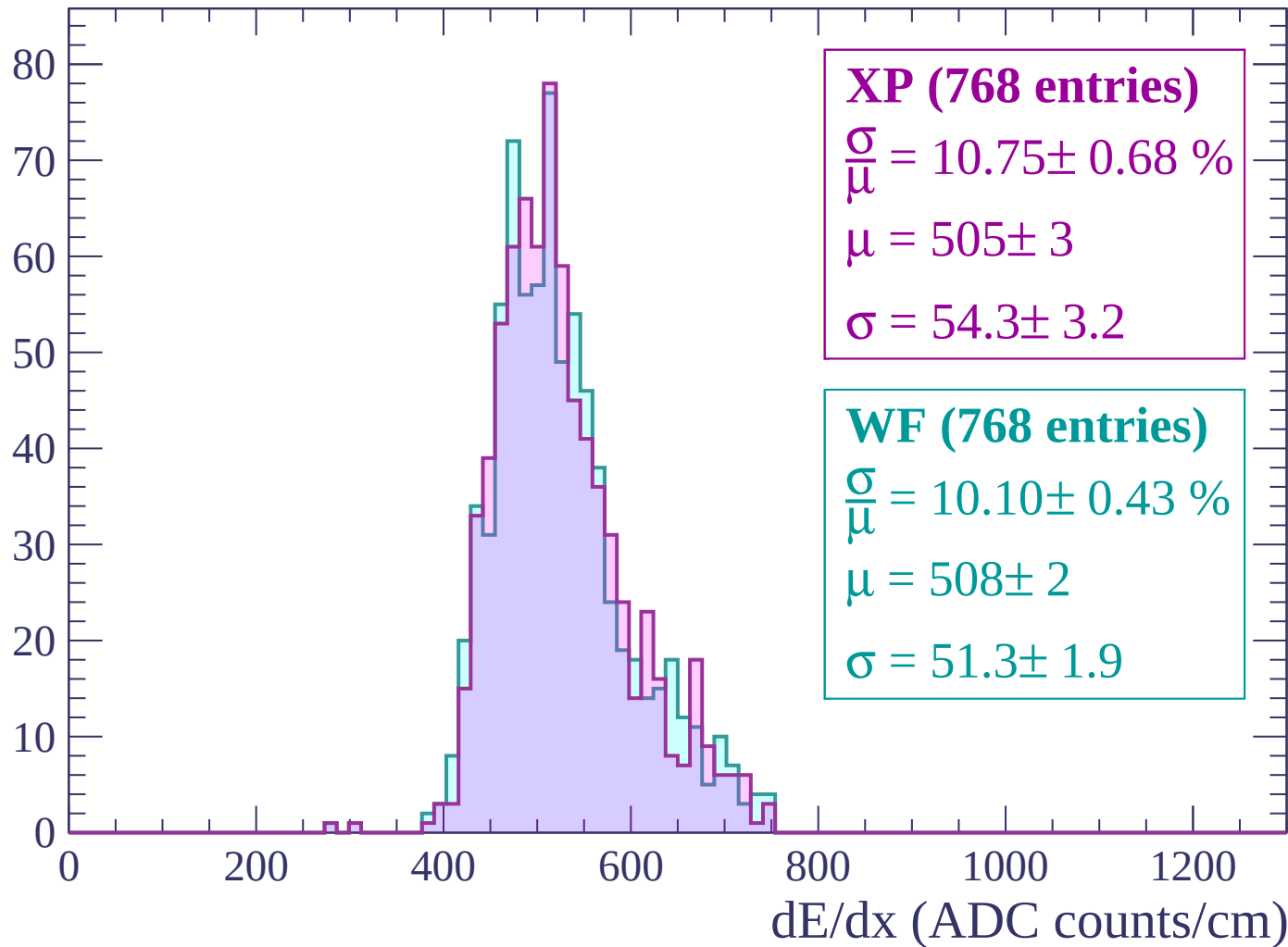
Energy loss | $1860 < p < 1920$

Count



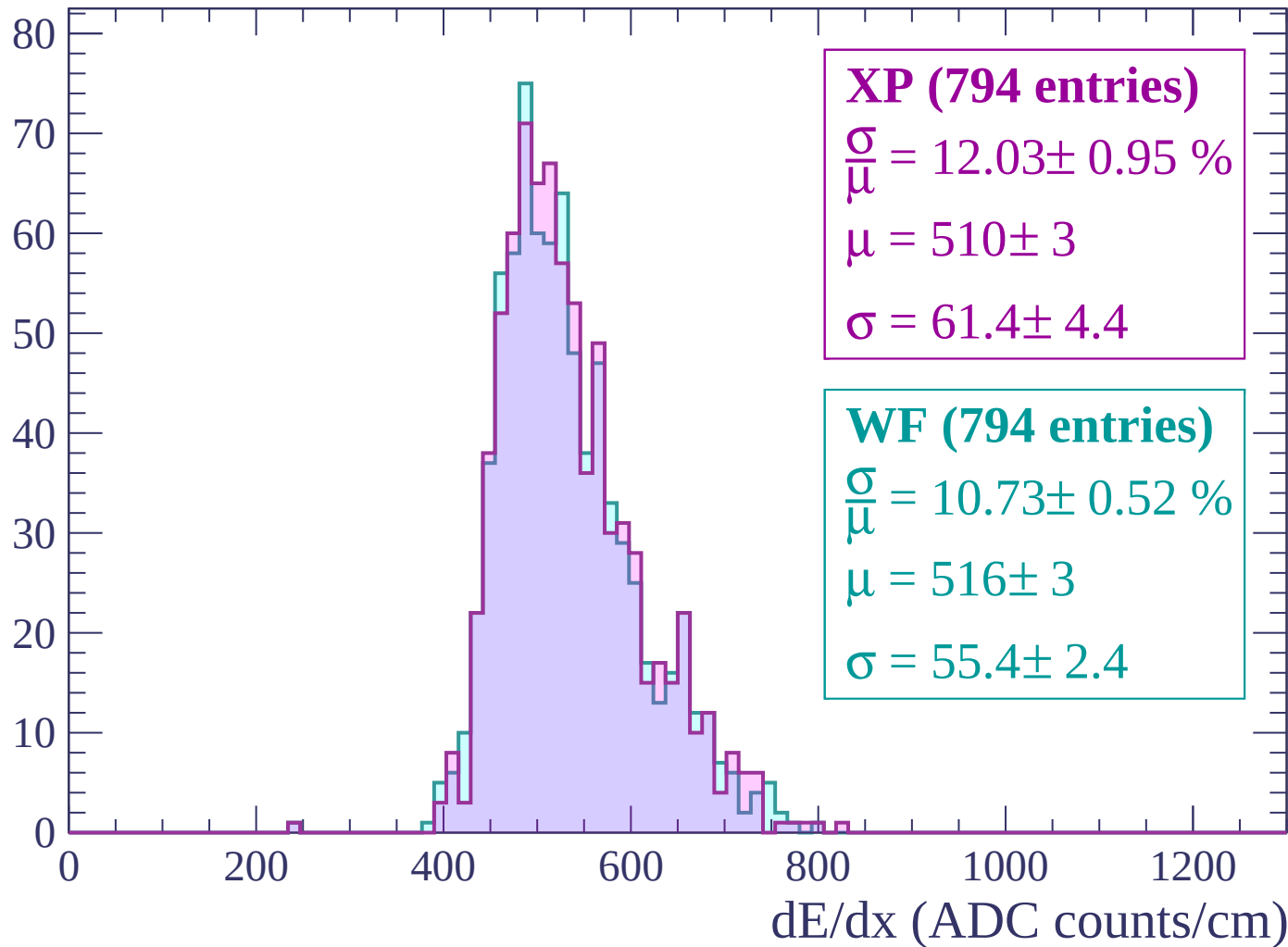
Energy loss | $1920 < p < 1980$

Count



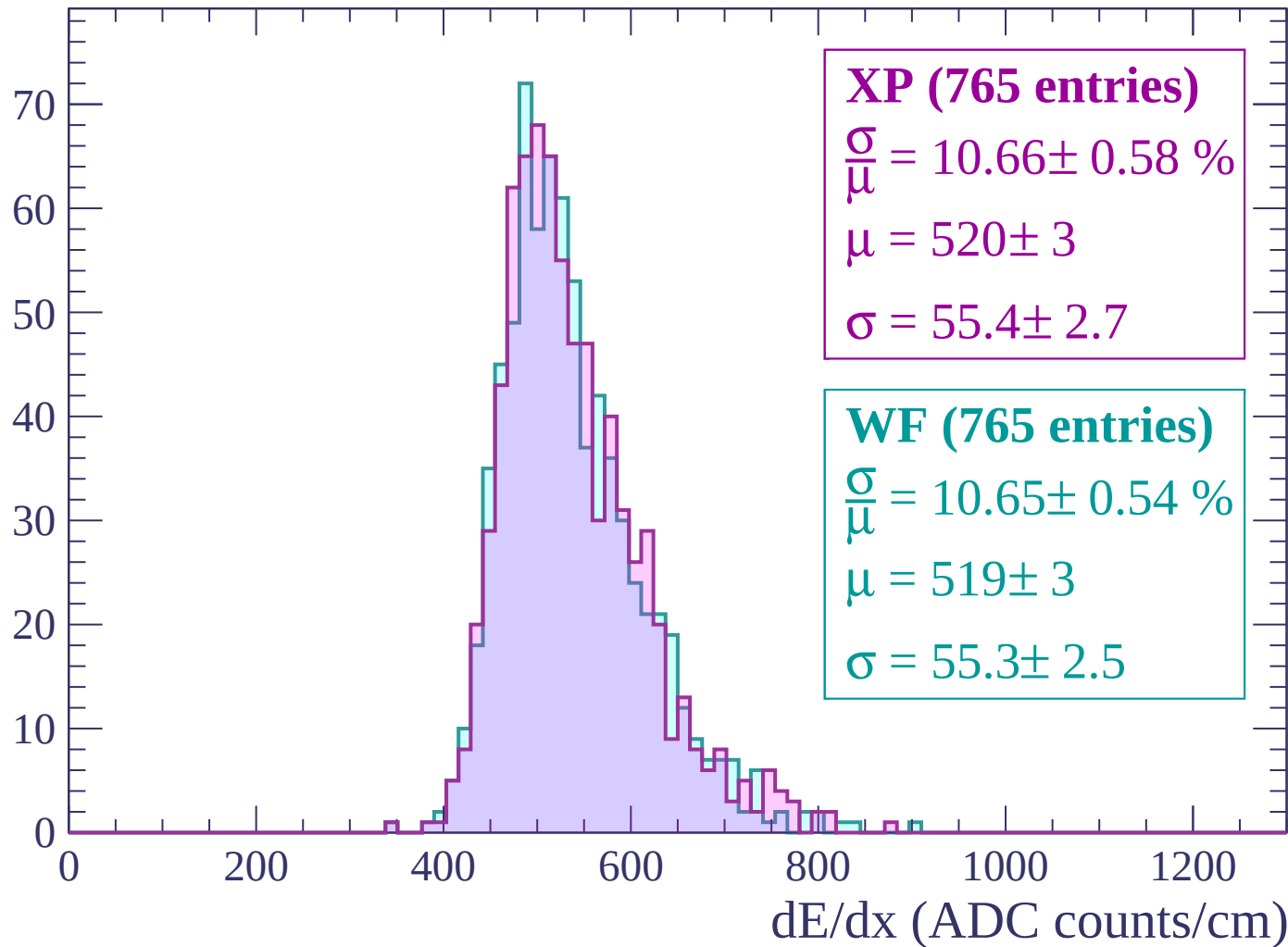
Energy loss | $1980 < p < 2040$

Count



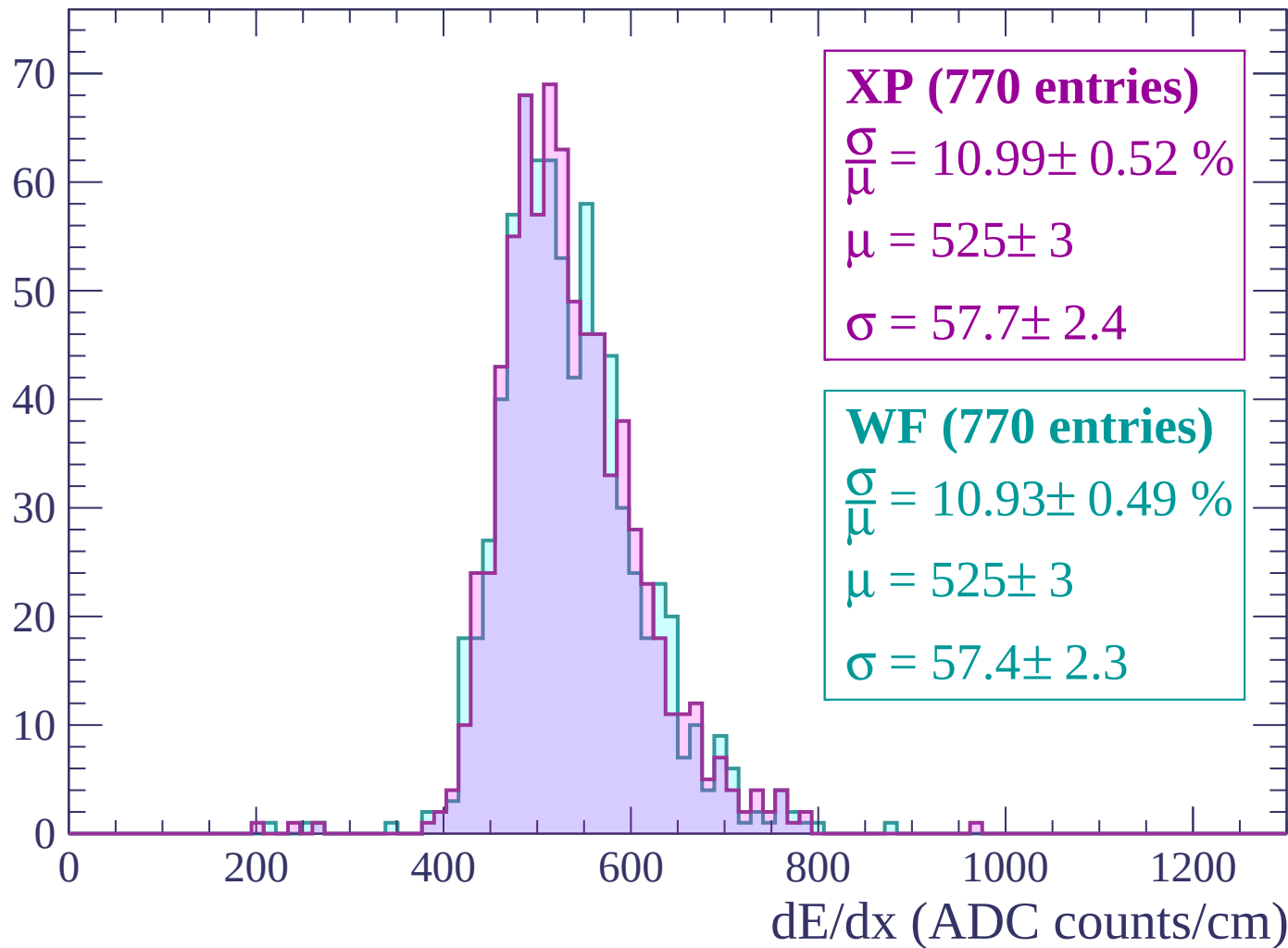
Energy loss | $2040 < p < 2100$

Count



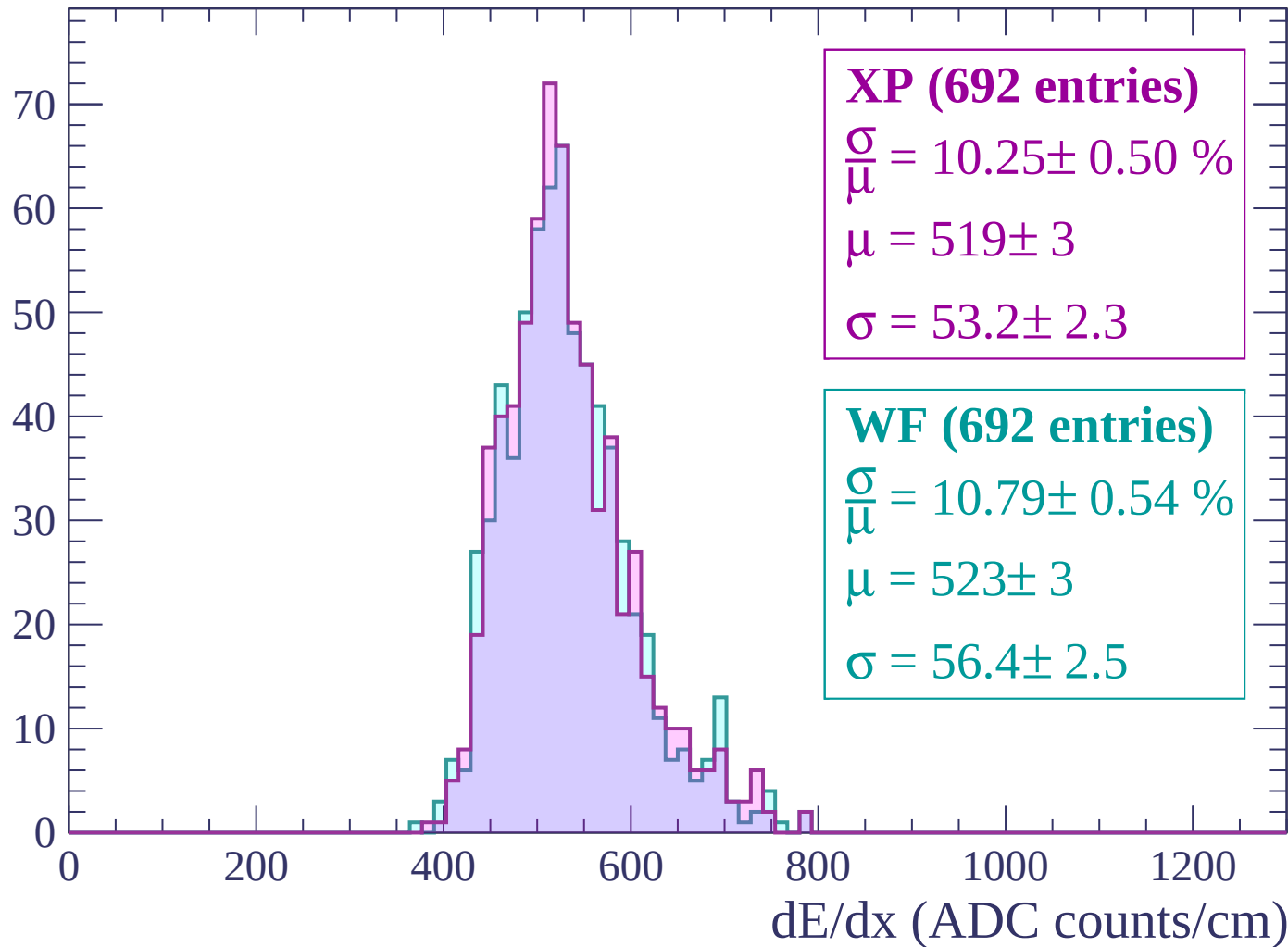
Energy loss | $2100 < p < 2160$

Count



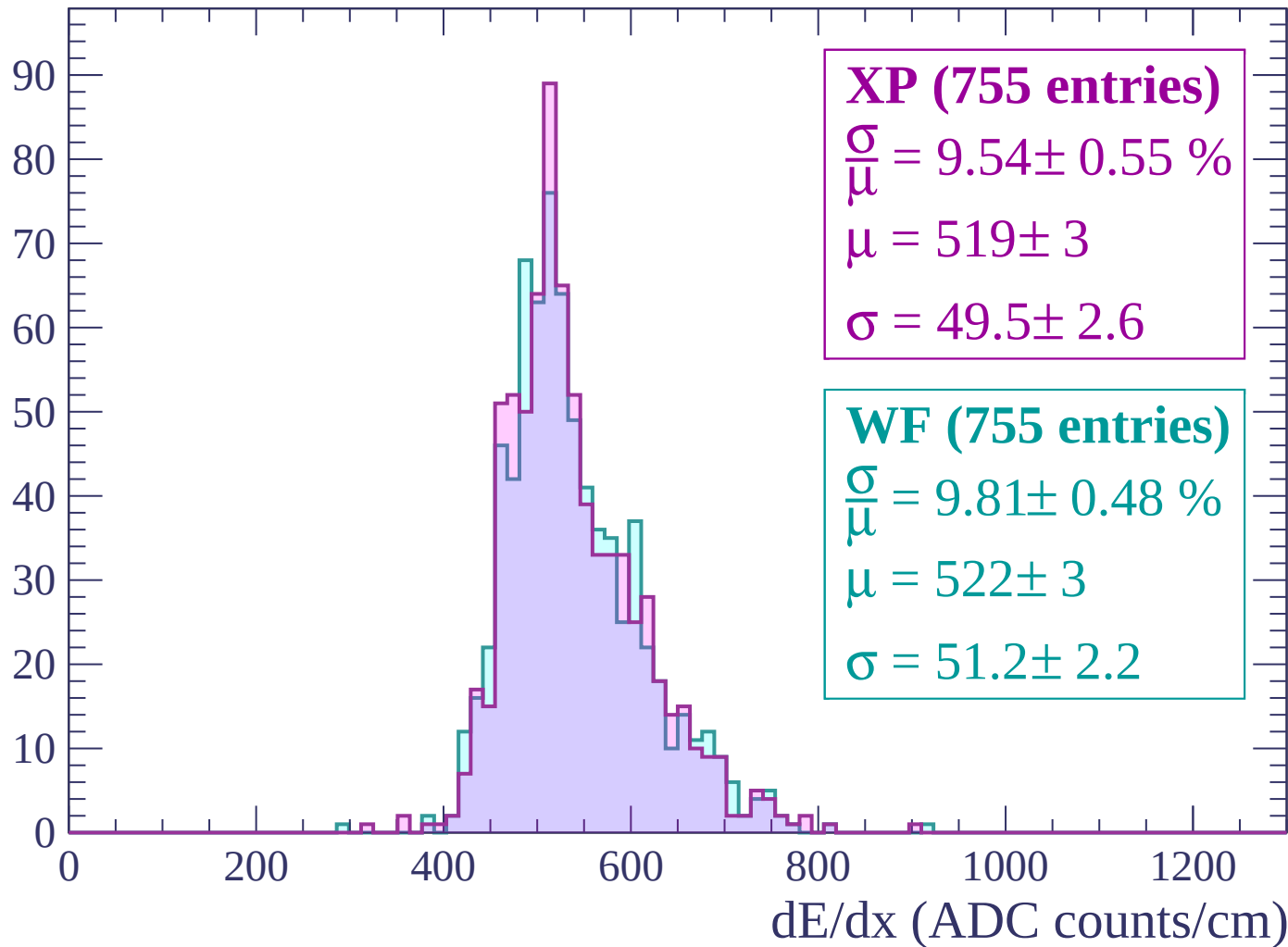
Energy loss | $2160 < p < 2220$

Count



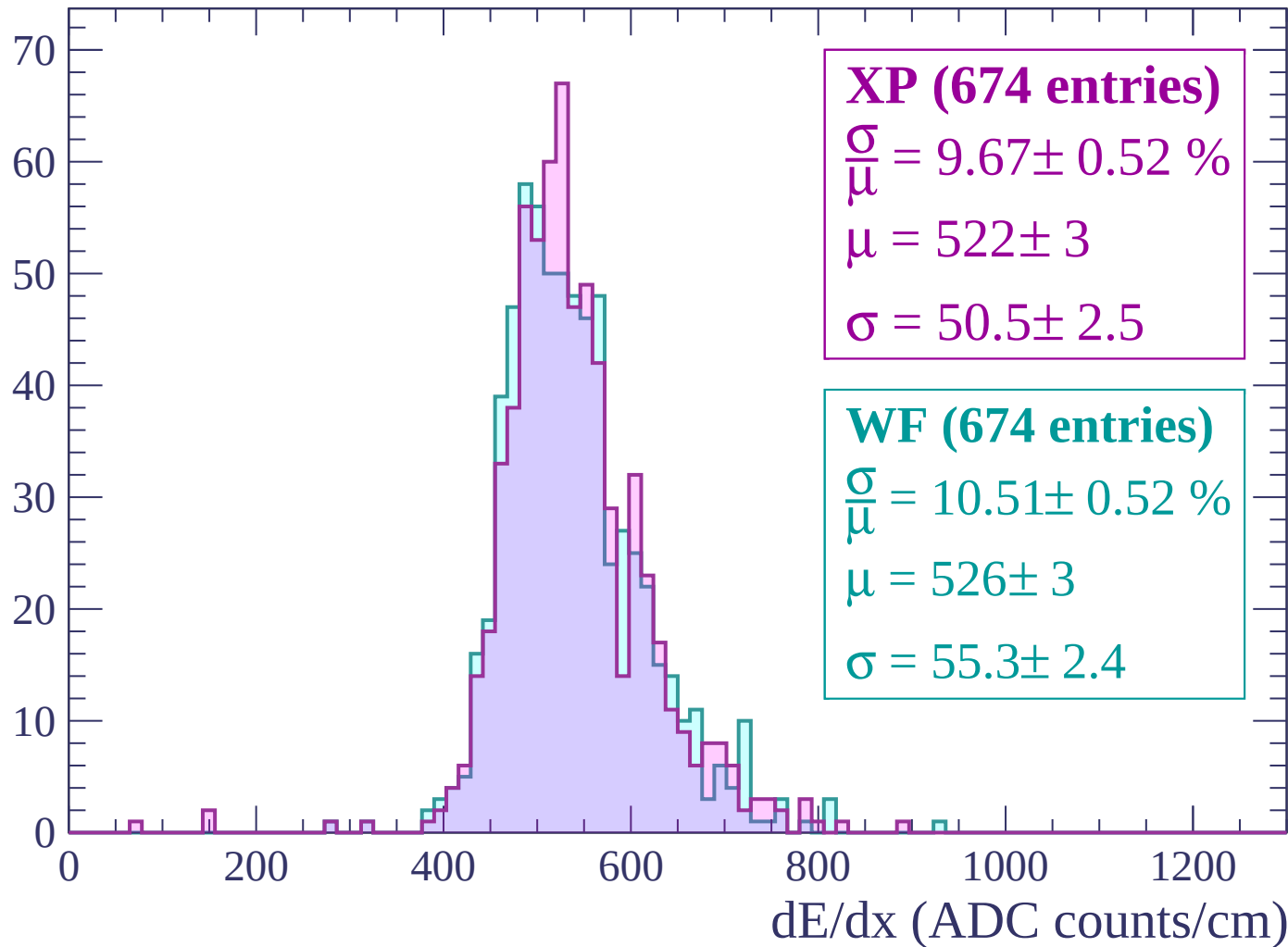
Energy loss | $2220 < p < 2280$

Count



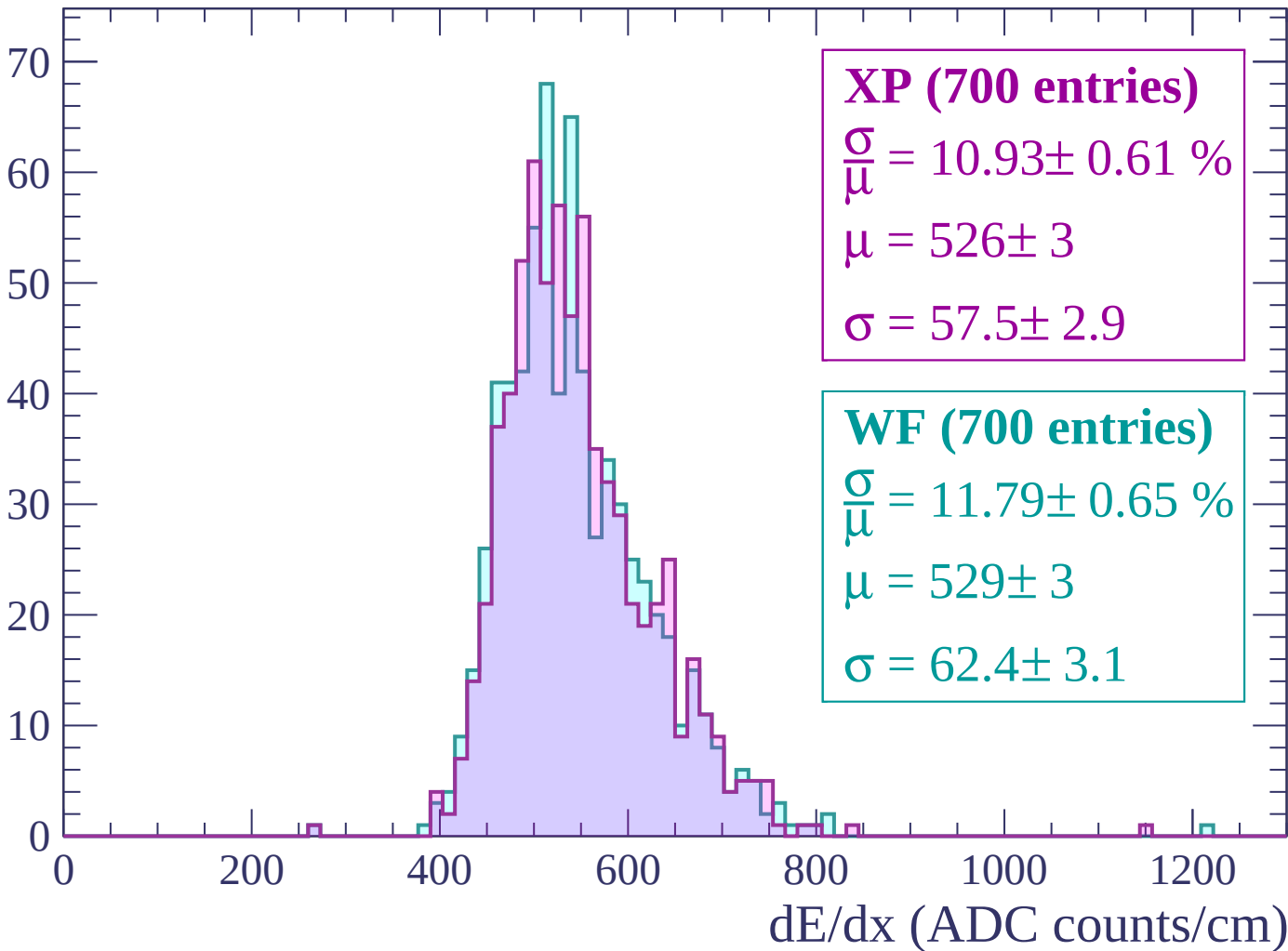
Energy loss | $2280 < p < 2340$

Count



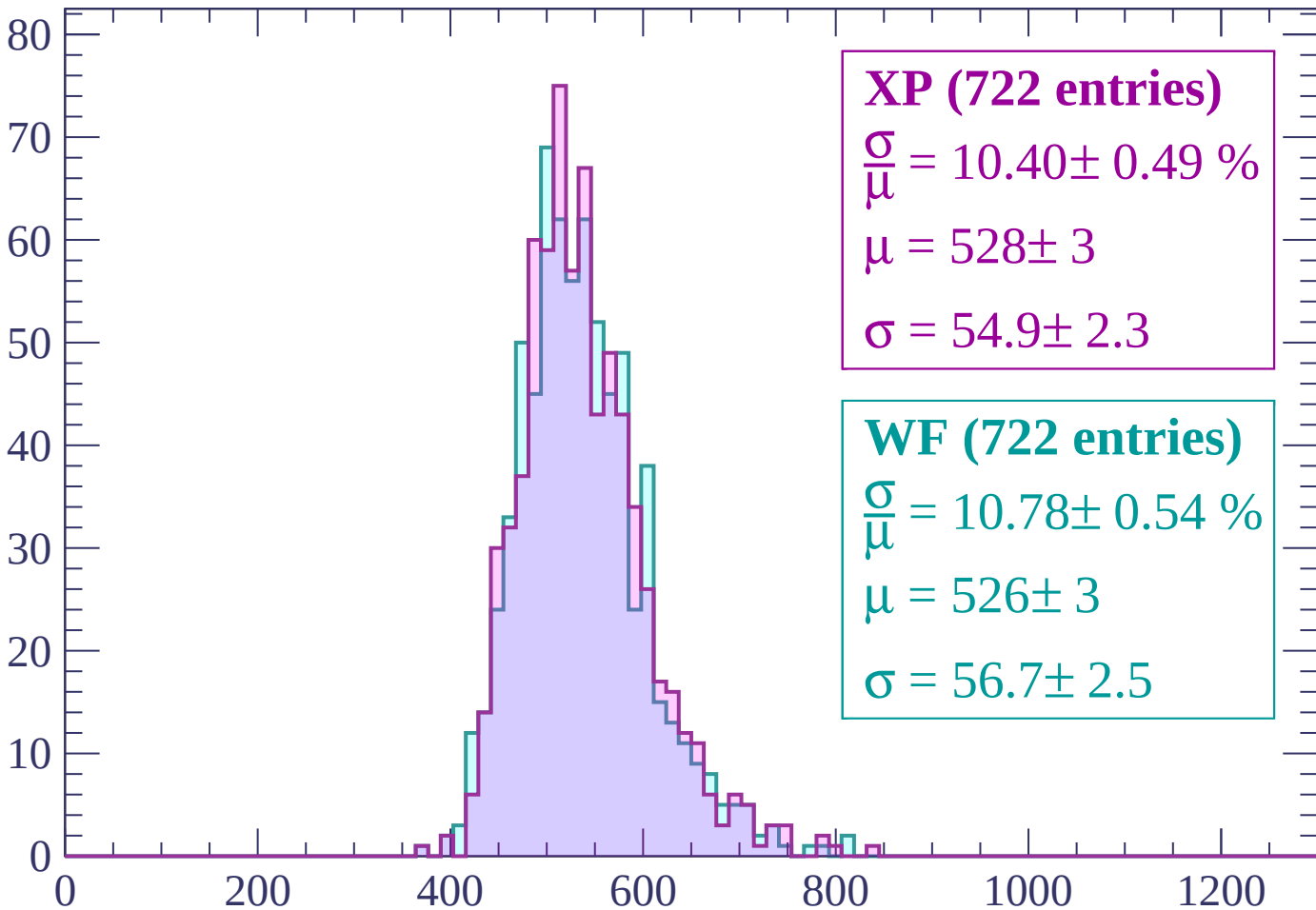
Energy loss | $2340 < p < 2400$

Count



Energy loss | $2400 < p < 2460$

Count



XP (722 entries)

$$\frac{\sigma}{\mu} = 10.40 \pm 0.49 \%$$

$$\mu = 528 \pm 3$$

$$\sigma = 54.9 \pm 2.3$$

WF (722 entries)

$$\frac{\sigma}{\mu} = 10.78 \pm 0.54 \%$$

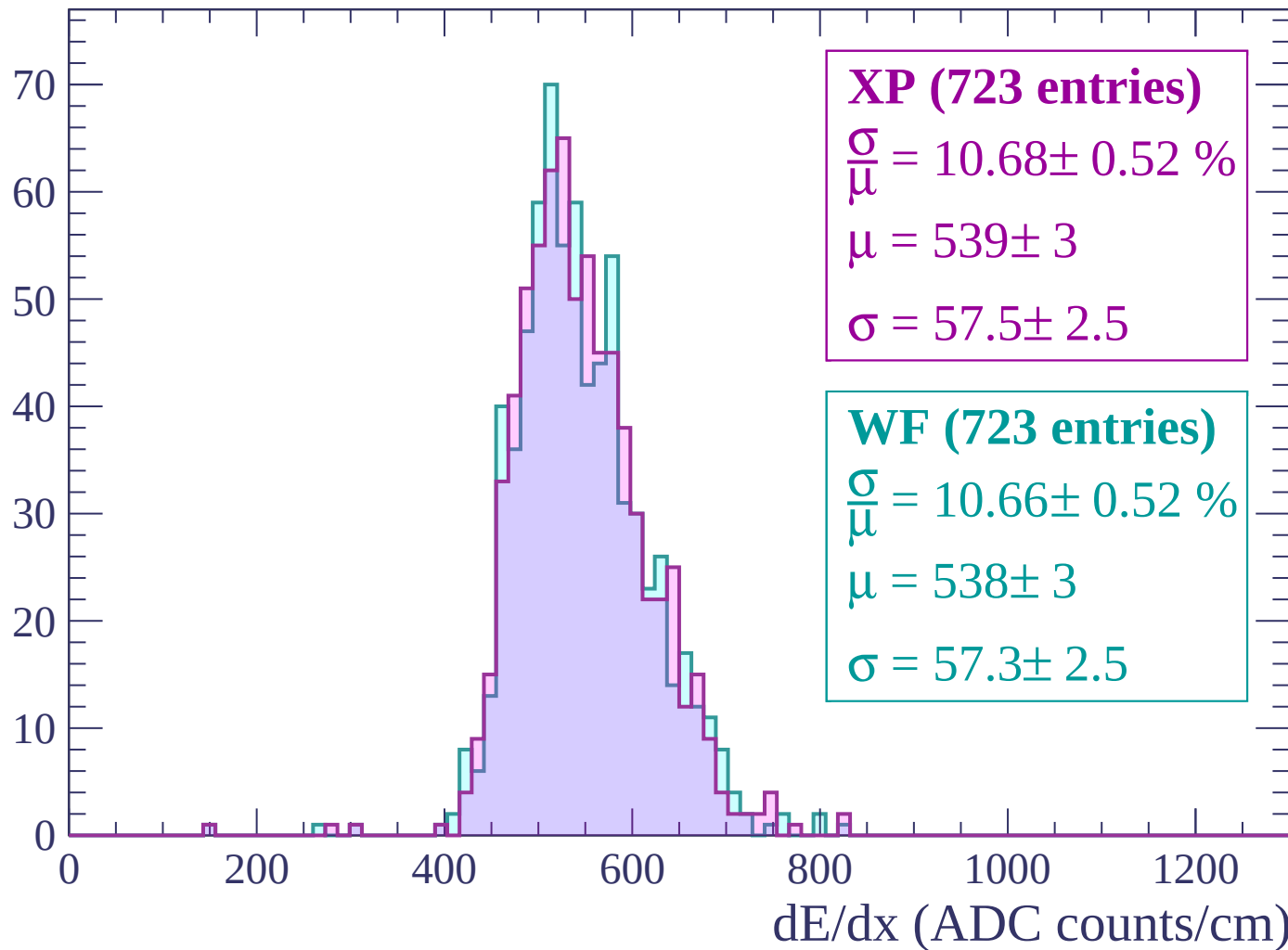
$$\mu = 526 \pm 3$$

$$\sigma = 56.7 \pm 2.5$$

dE/dx (ADC counts/cm)

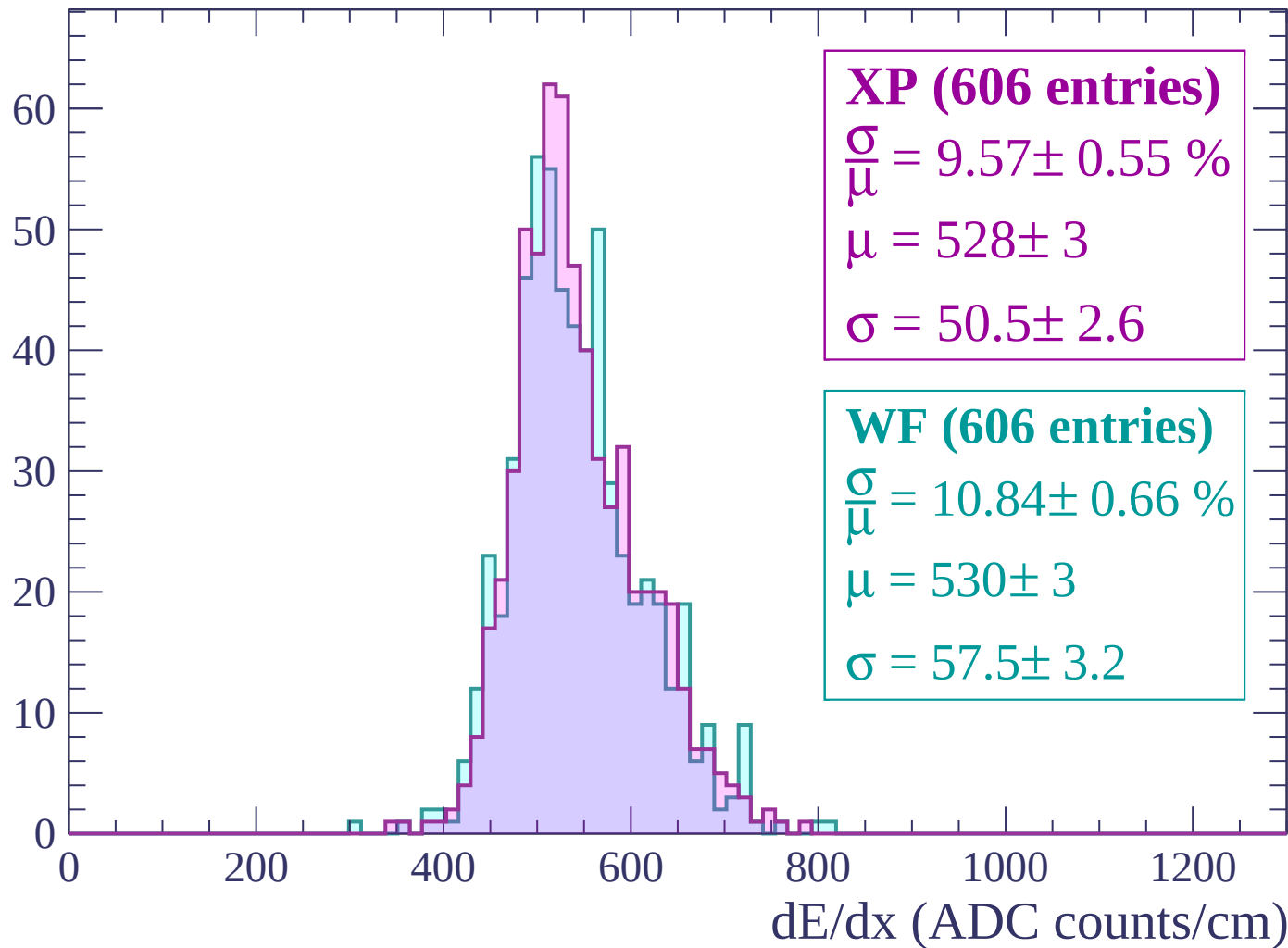
Energy loss | $2460 < p < 2520$

Count



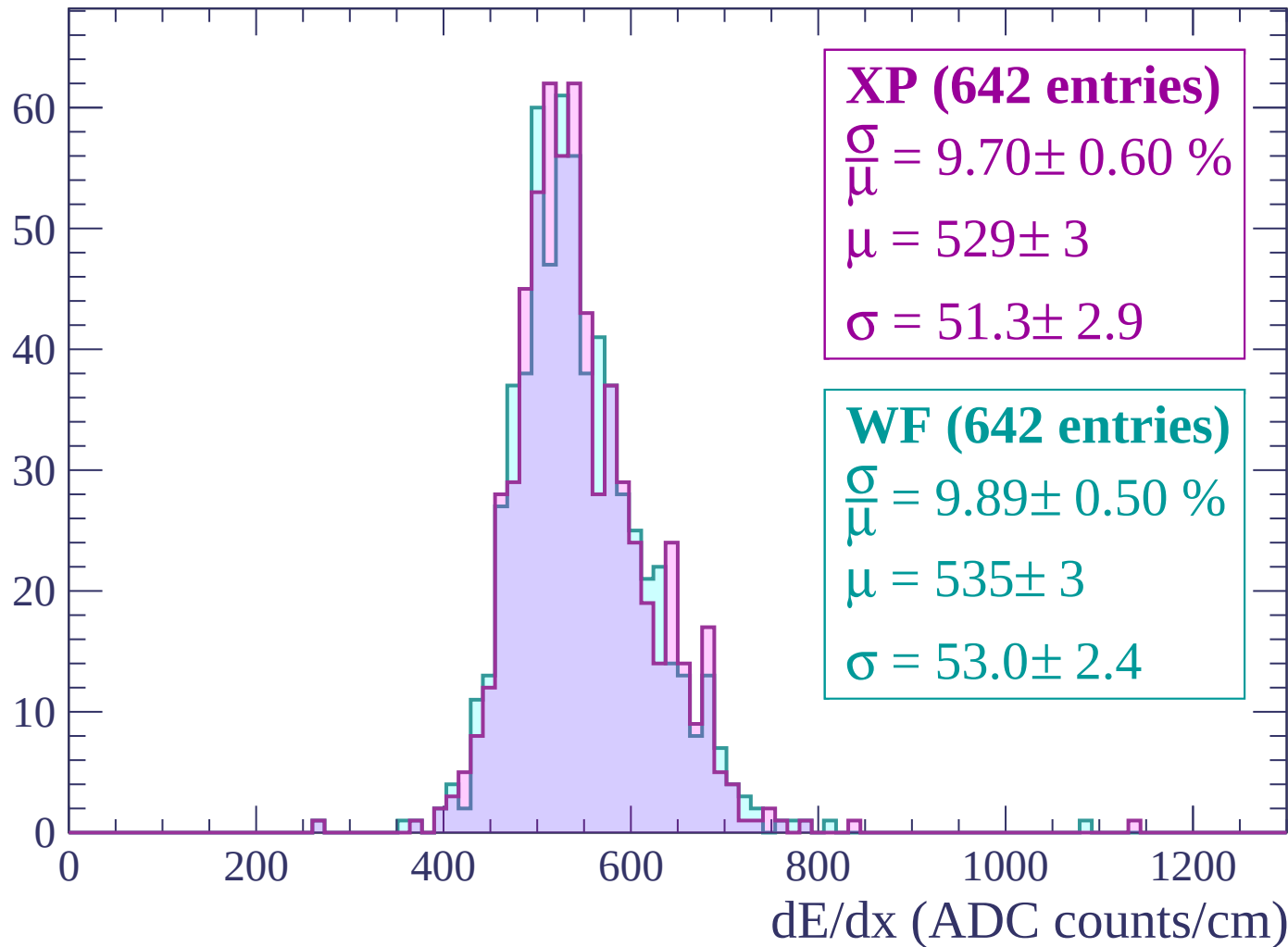
Energy loss | $2520 < p < 2580$

Count



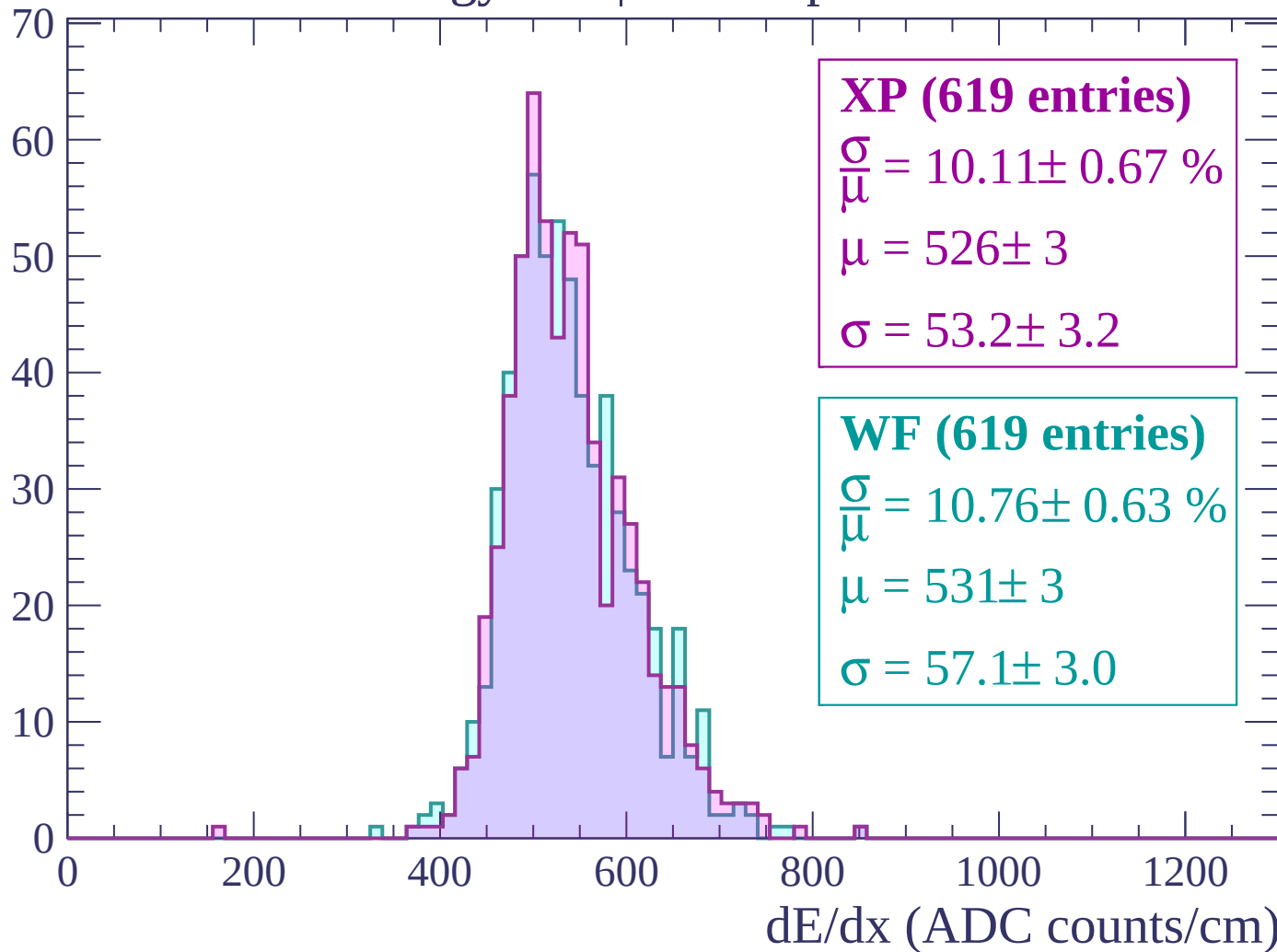
Energy loss | $2580 < p < 2640$

Count



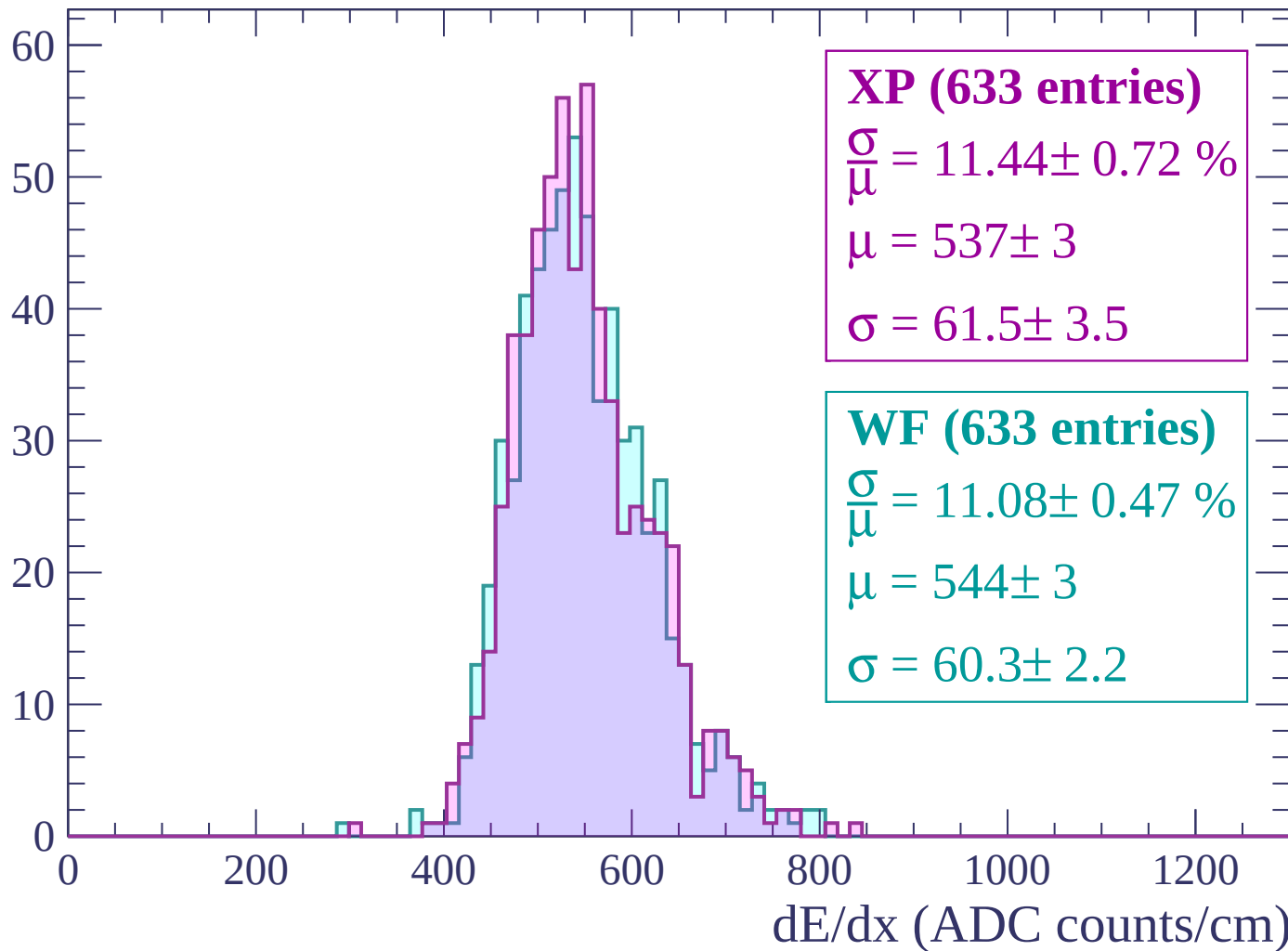
Energy loss | $2640 < p < 2700$

Count



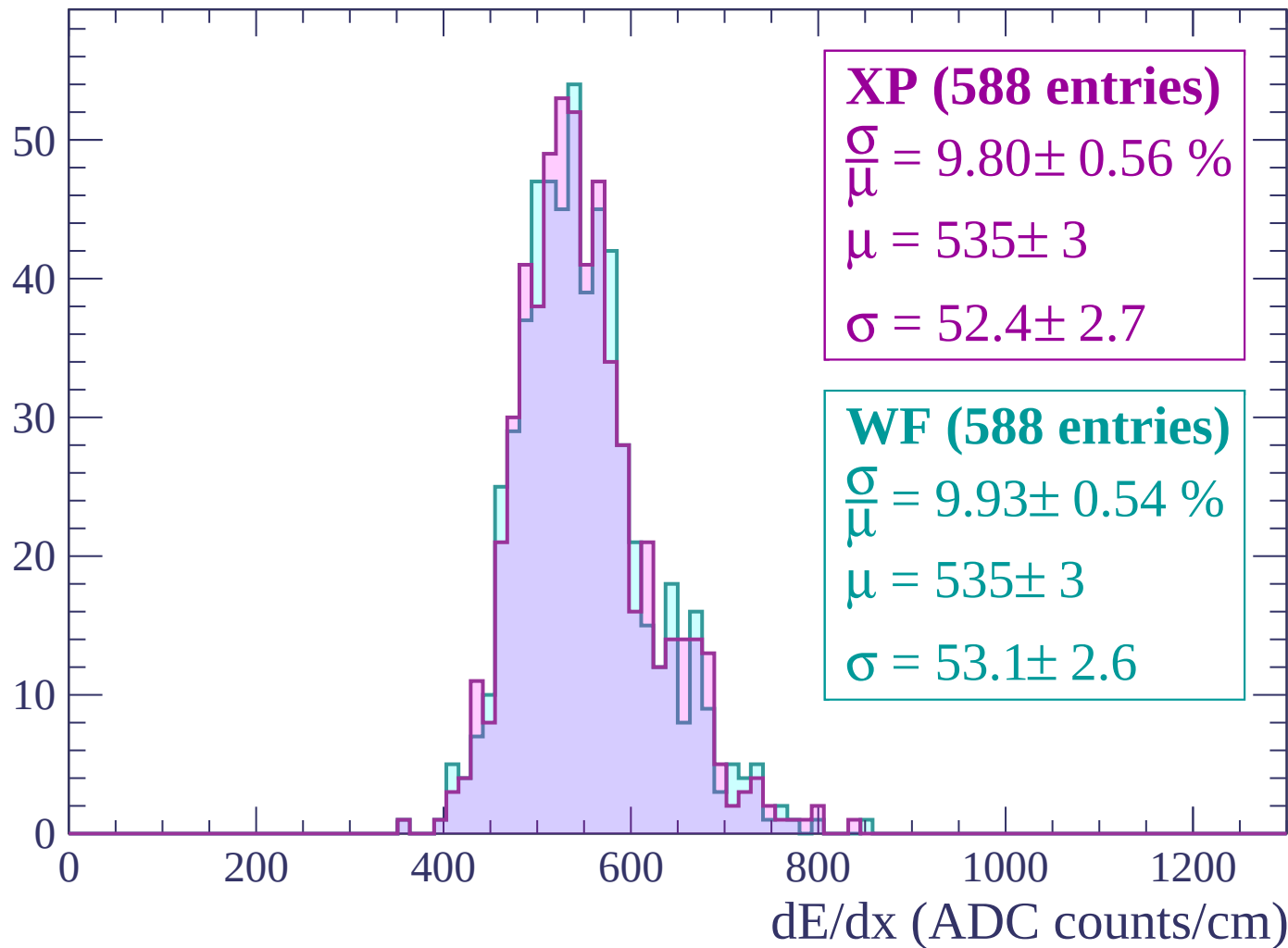
Energy loss | $2700 < p < 2760$

Count



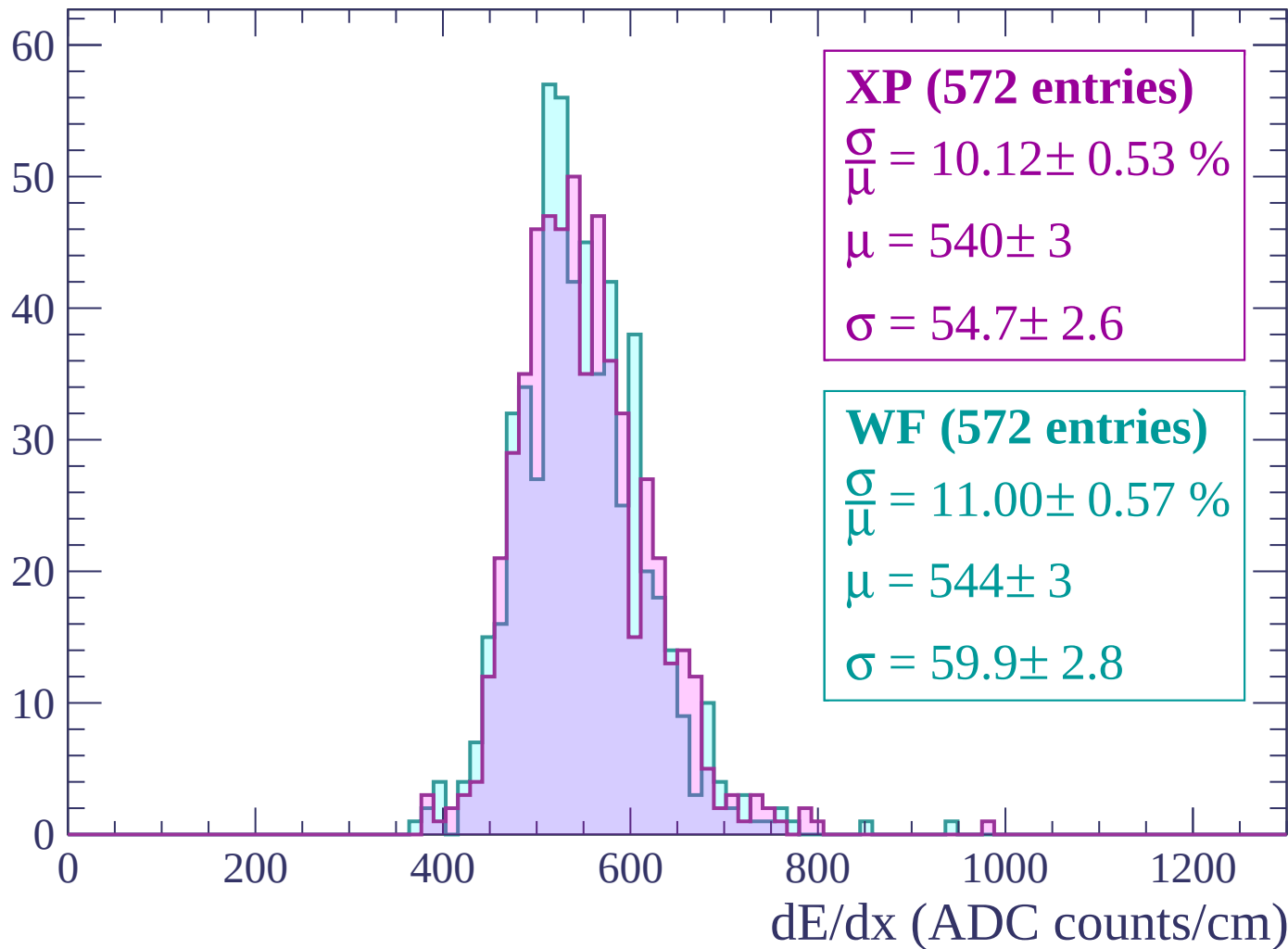
Energy loss | $2760 < p < 2820$

Count



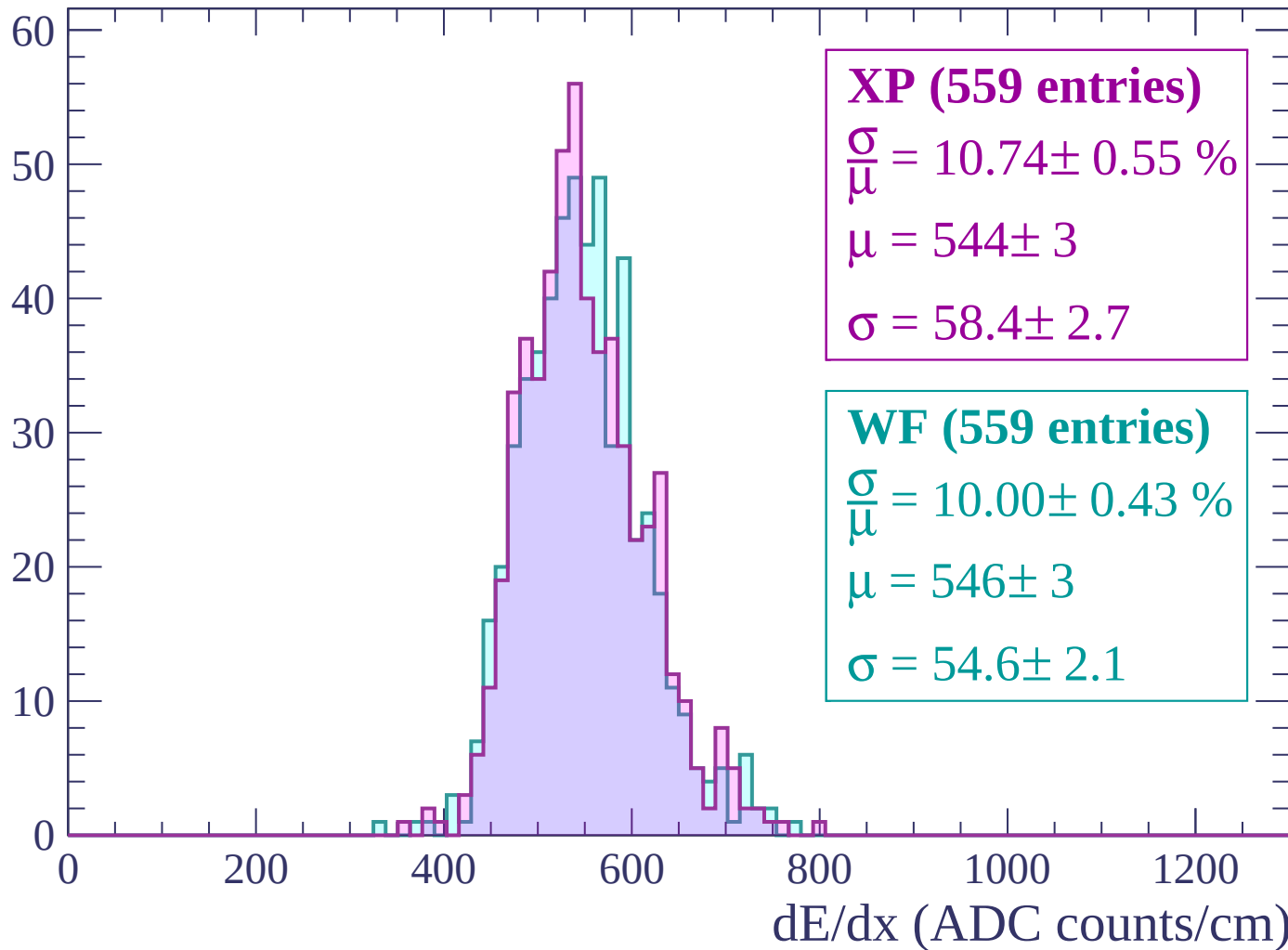
Energy loss | $2820 < p < 2880$

Count



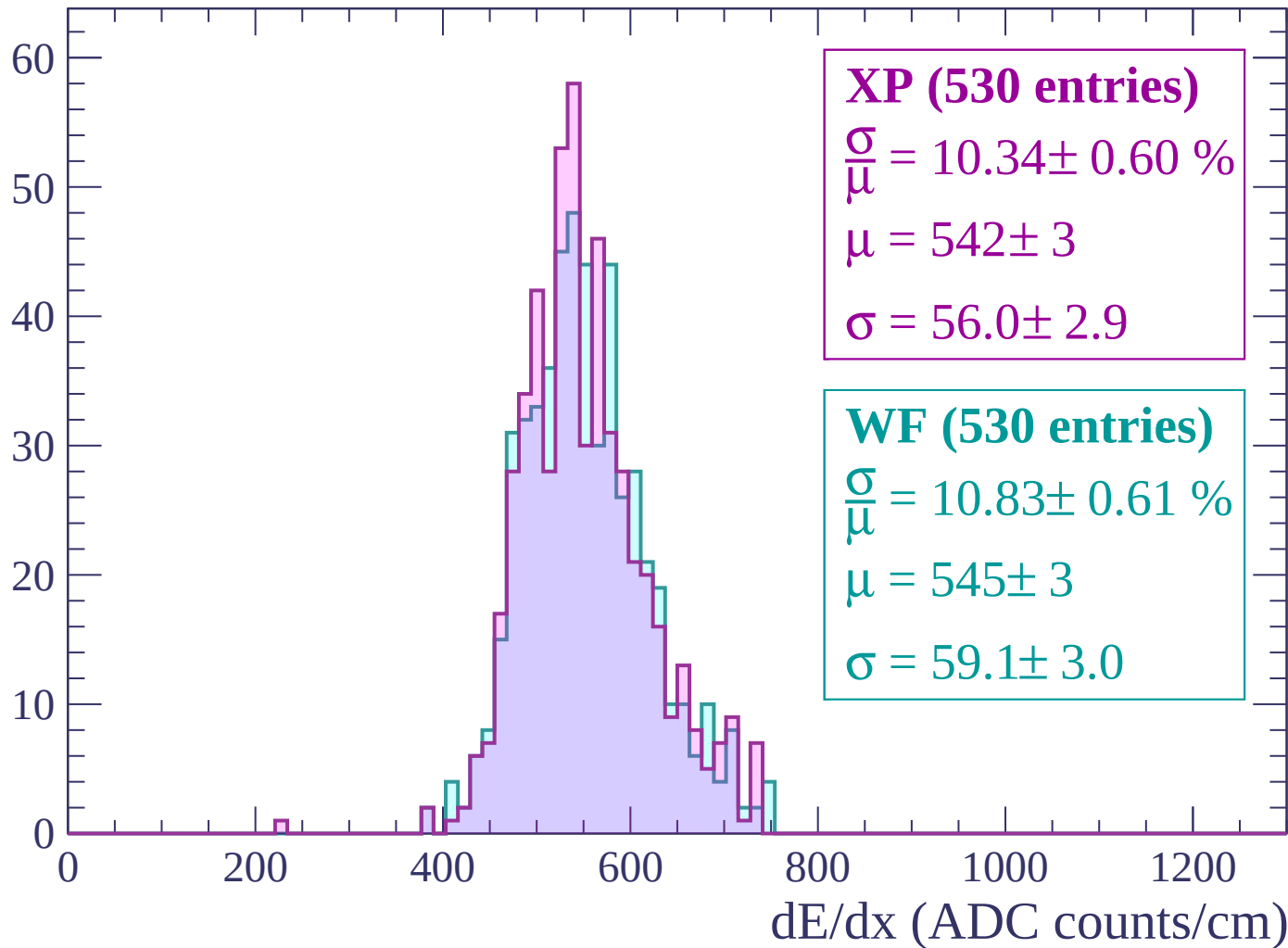
Energy loss | $2880 < p < 2940$

Count



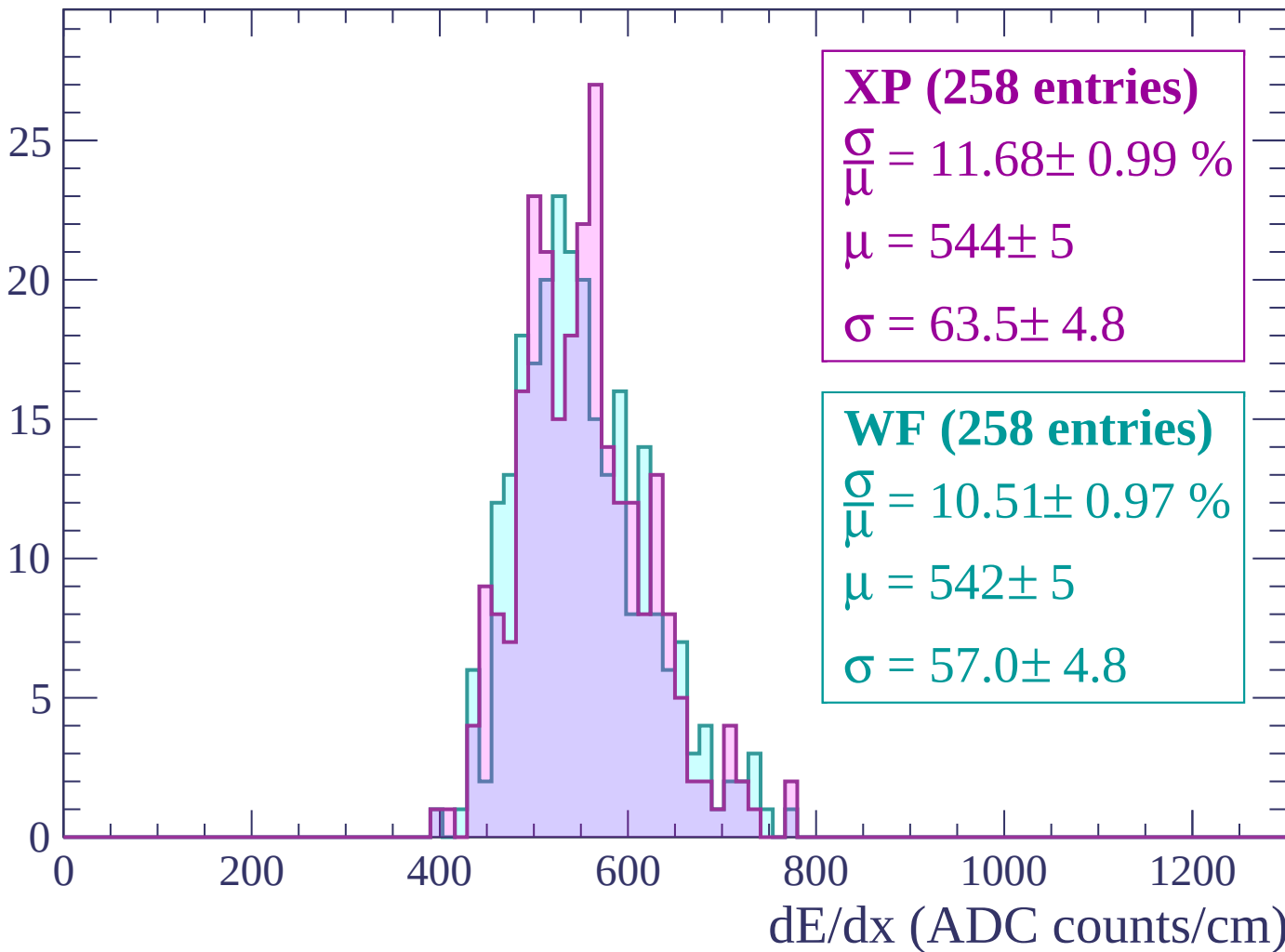
Energy loss | $2940 < p < 3000$

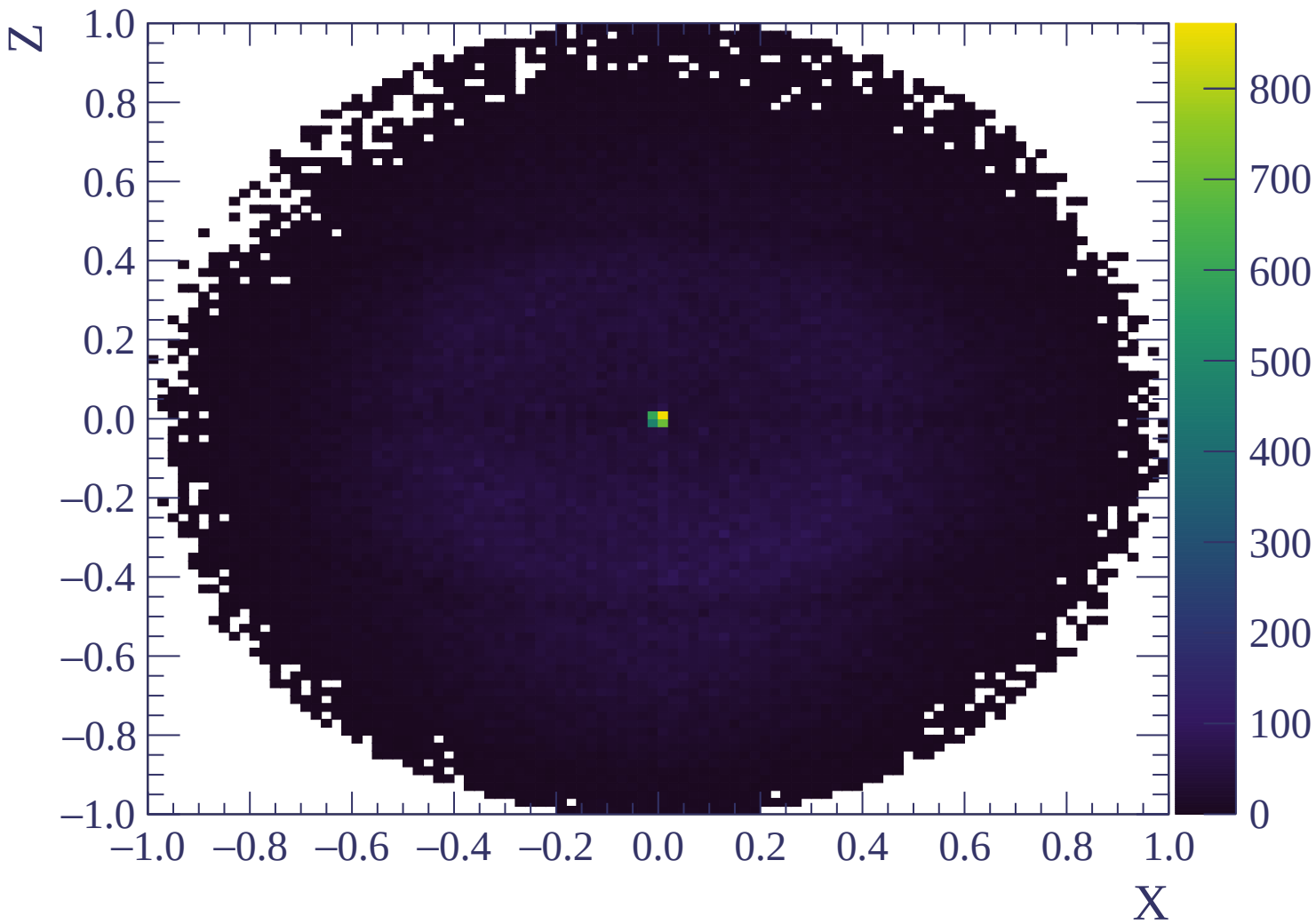
Count

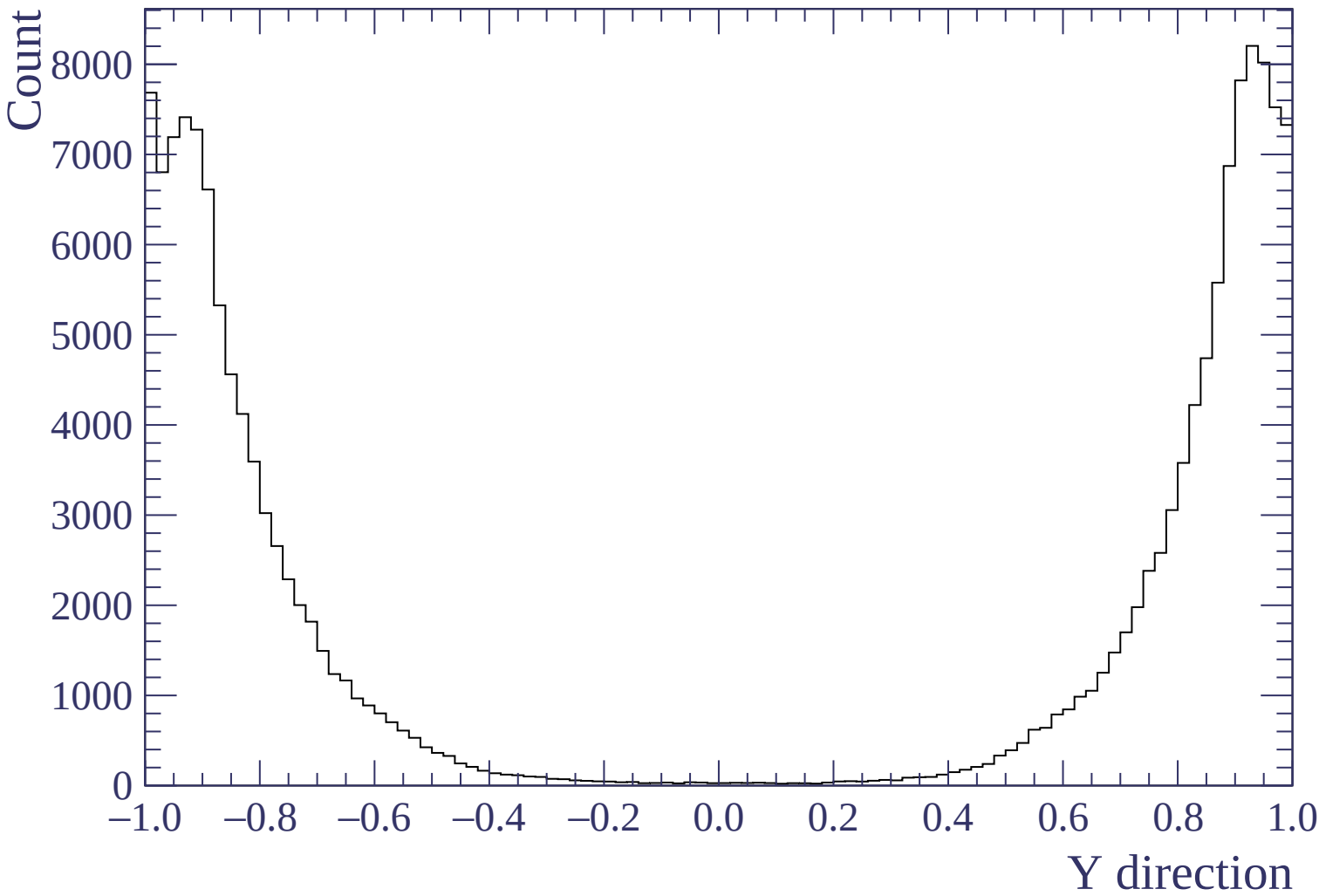


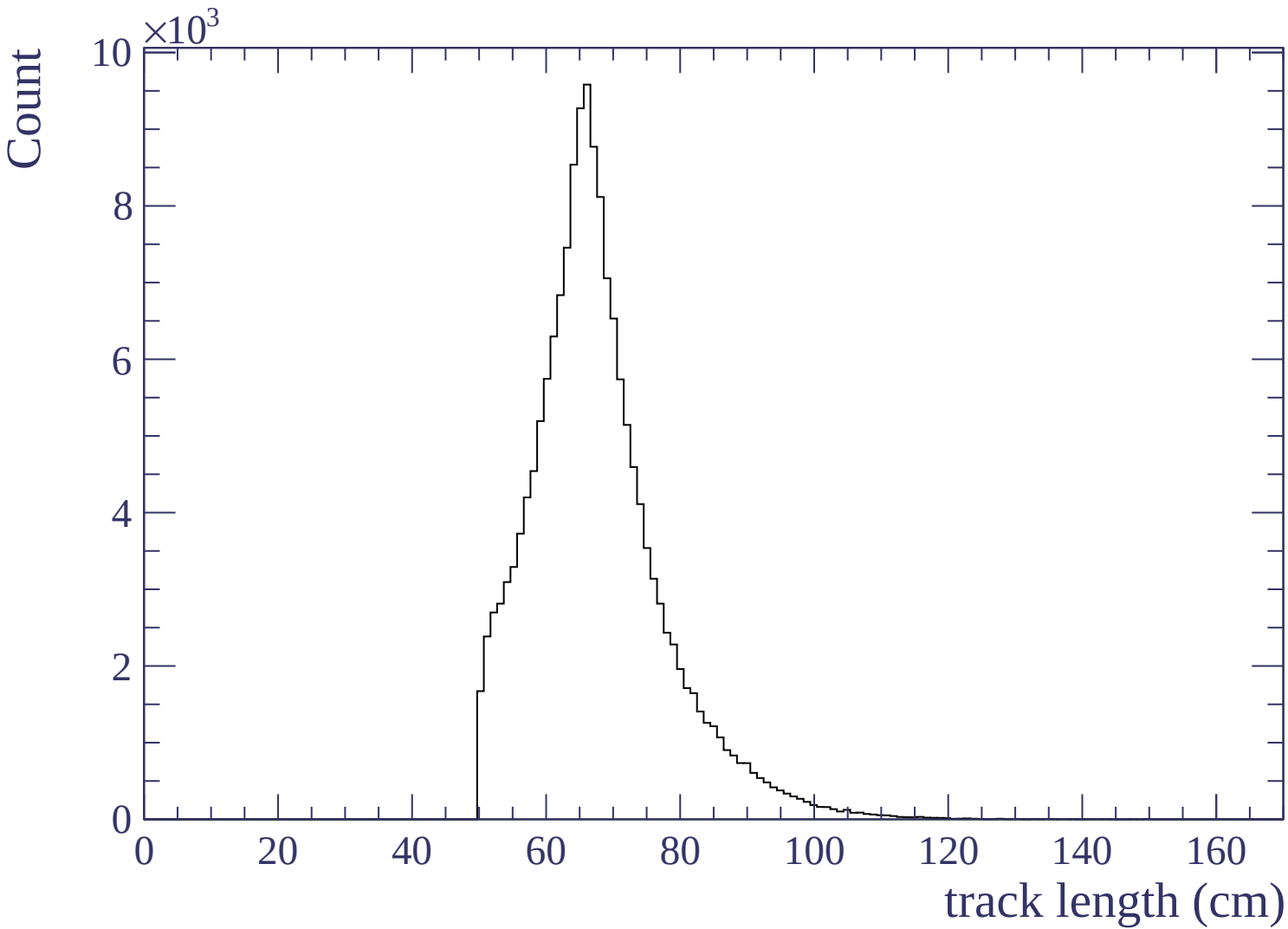
Energy loss | $3000 < p < 3060$

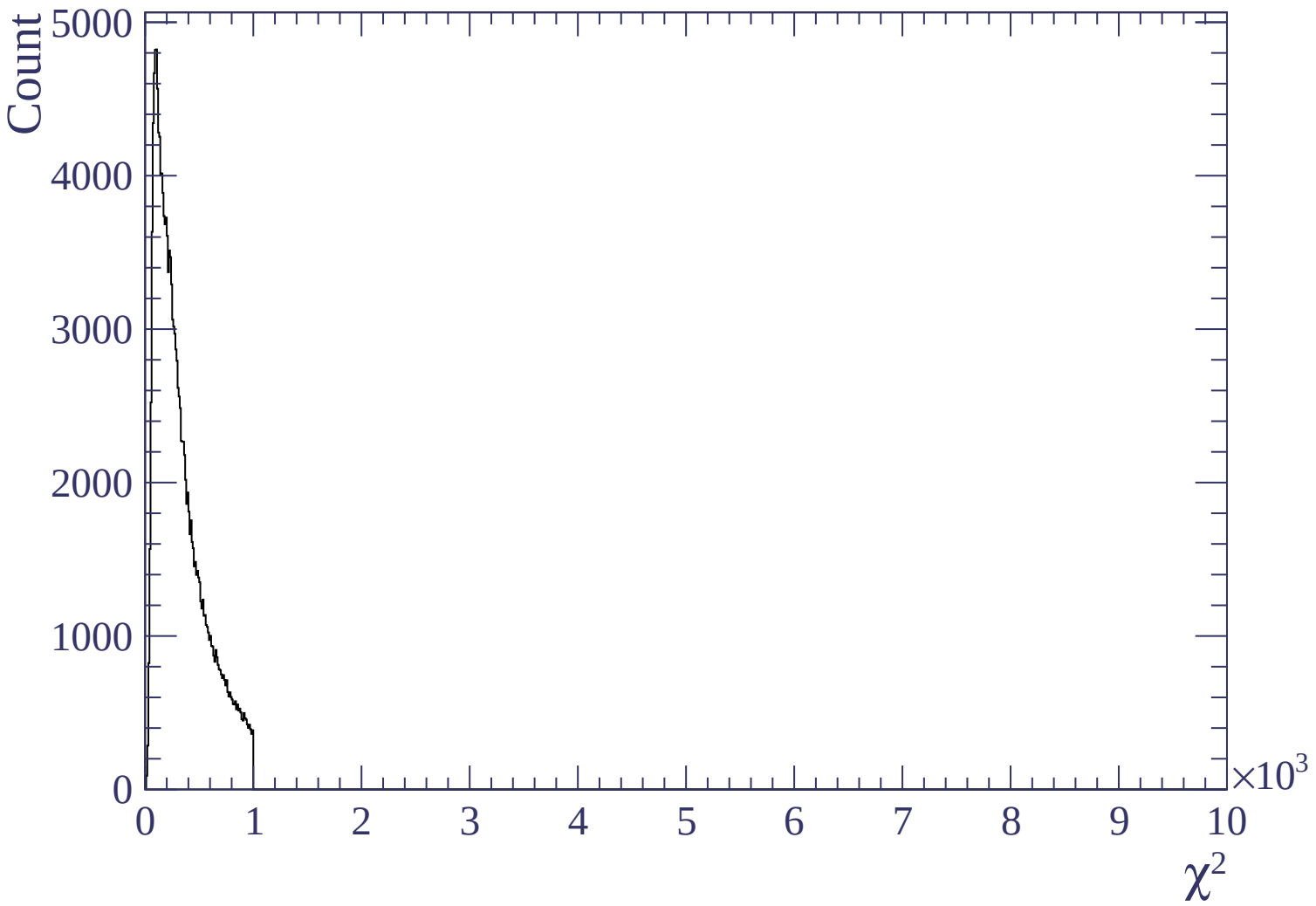
Count

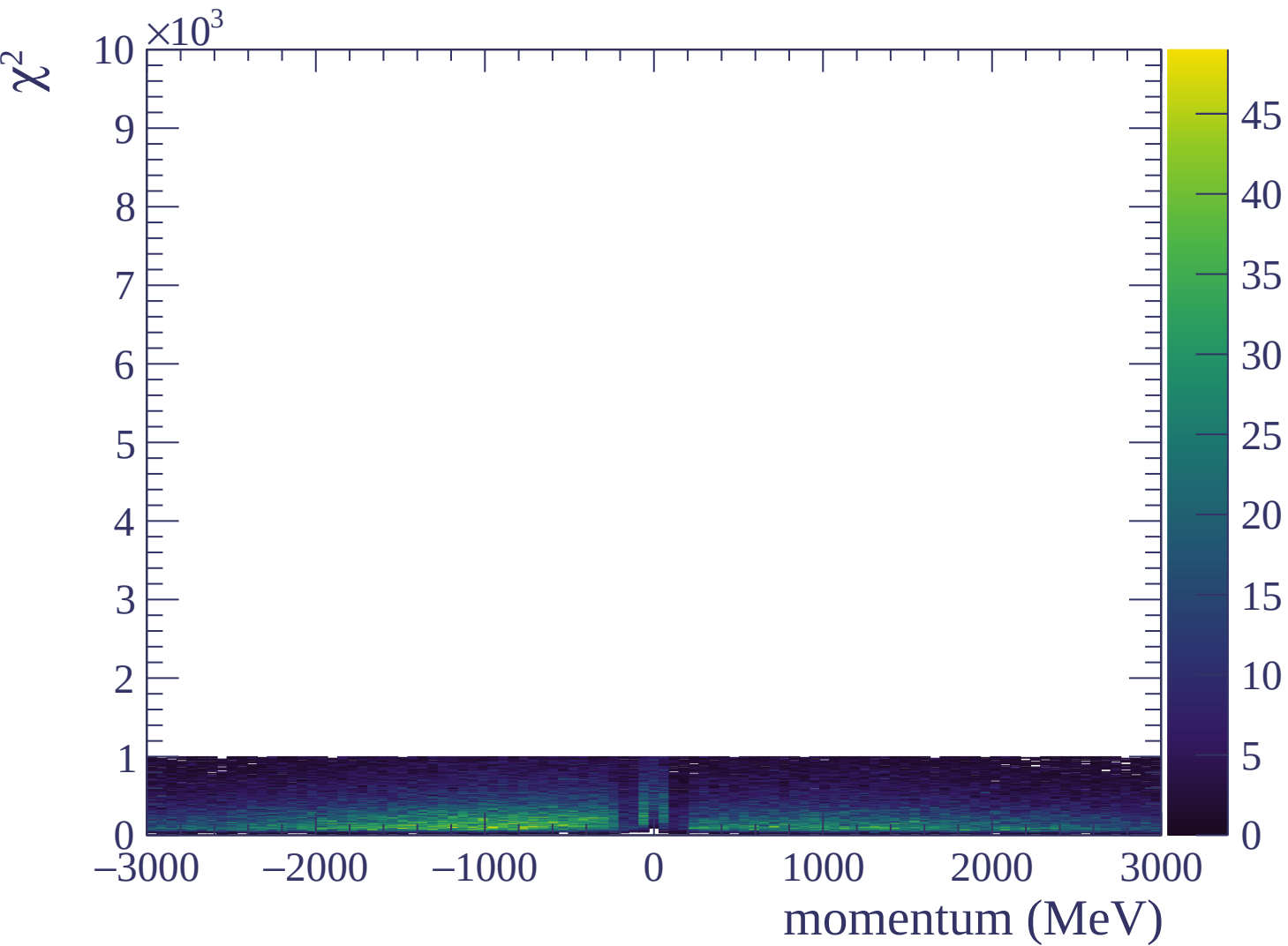












Count

